

Appendix 3

[List of SCOMET Items]

To

Schedule- 2 of ITC (HS)

Classification of Export and

Import Items, 2018

Note:

1. This is Updated and Revised SCOMET List 2023 (as notified on **28.08.2023**)
2. The entries shown in **Magenta** are updated entries as per DGFT Notification No. 27/2023 dated 28.08.2023 which shall come into force after 30 days of issue of Notification, i.e 28.09.2023.
3. The entries shown in **Maroon** are updated entries as per the DGFT Notification No. 46/2015-20 dated 30.11.2022, which came into force after 30 days of the issue of the Notification, i.e 30th December 2022.

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APPENDIX-3

Special Chemicals, Organisms, Materials, Equipment and Technologies (SCOMET) export of which is regulated

Export of Special Chemicals, Organisms, Materials, Equipment and Technologies (SCOMET) listed below shall be permitted only against an export authorisation issued in this behalf unless export is prohibited or is permitted without authorisation subject to fulfillment of conditions, if any, as indicated under/against any specific category or item. Provisions of Chapter IV A of the FT(D&R) Act, 1992 as amended in 2010 shall apply to the goods, services and technologies as specified in the list below.

Chapter 10 of the Foreign Trade Policy 2023 specifies the policy related to export of items covered under the SCOMET listed in this document. Chapter 10 of Handbook of Procedures 2023 details the procedures related to export of items in SCOMET.

Supply of SCOMET Items from DTA to SEZ/EoU and outside the country

- No export authorisation is required for supply of SCOMET items from DTA to SEZ/EoU.
- Export Authorisation, however, required if the SCOMET items are to be physically exported outside the country from SEZ/EoU i.e. to another country (Refer Rule 26 of the SEZ Rules, 2006). All supplies of SCOMET items from DTA to SEZ/EoU will be reported to the Development Commissioner (DC) of the respective SEZ/EoU by the supplier in the prescribed proforma [Annexure 1 to Appendix-3 to Schedule 2 of ITC (HS) Classifications of Export and Import Items] within one week of the supplies getting effected.
- An annual report of such supplies from DTA to SEZ/EoU shall be reported to SCOMET Section, DGFT (Hqs), Department of Commerce, Vanijya Bhawan, Akbar Road, New Delhi-110011, by the Development Commissioner (DC) of respective SEZ/EoU in the prescribed proforma [Annexure 2 to Appendix-3 to Schedule 2 of ITC (HS) Classifications of Export and Import Items]. Report by the DC, SEZ/EoU is to be filed by 15th May of every financial year, in respect of supplies effected from DTA to SEZ/EoU during the preceding financial year.

Export of SCOMET Items from SEZ to outside the country

Export permission is required if the SCOMET items are to be physically exported outside the country from SEZ i.e. to another country (Refer Rule 26 of the SEZ Rules, 2006).

Entering into an Arrangement or Understanding for Site Visits, On-site Verification and Access to Records / Documentation

It is mandatory for all companies and their subsidiaries registered in India and all other business entities operating in India and involved in the manufacture, processing and use of Special Chemicals, Organisms, Materials, Equipment and Technologies (SCOMET) listed below to obtain permission of the DGFT before entering into any arrangement or understanding that involves an obligation to facilitate or undertake site visits, on-site verification or access to records/ documentation, by foreign Governments or foreign third parties, either acting directly or through an Indian party or parties. Requests for such permissions shall be considered in the manner in which requests for export/import licence are considered.

Provided that where obligations involving site visits, on-site verification or access to records/ documentation by foreign governments or foreign third parties are to be undertaken pursuant to a bilateral agreement or a multilateral treaty to which India is a party, the provisions of the relevant agreement or treaty shall apply.

Exporters are advised to refer to the relevant guidelines relating to the export of SCOMET items in the Handbook of Procedures, as issued from time to time. [Para 10.08 of the Handbook of Procedures, 2023](#) delineates the Procedure/Guidelines for filing / Evaluation of Applications for Entering into an Arrangement or Understanding for Site Visits, On-site Verification and Access to Records / Documentation.

Note The DGFT Notification No. 5/2015-2020 dated 24.04.2017 is issued to amend Table A of Schedule 2 and Appendix 3 of ITC(HS) Classification of Export and Import Items in supersession of DGFT Notification Nos.38 (RE-2010) /2009-2014 dated 31st March, 2011, No. 93(RE-2012)/2009-14 dated 6th January 2012, No. 37(RE-

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2012}/2009-14 dated 14th March 2013, No. 26(RE-2013}/2009-14 dated 3rd July 2013, No. 56 (RE - 2013}/2009-2014 dated 12th December 2013, No. 115 (RE - 2013}/2009-2014 dated 13th March 2015, No. 116(RE-2013}/2009-14 dated 13th March 2015, No. 05/(2015-2020) dated 29th April 2016 and No. 38/(2015-2020} dated 17th February 2017, except as respects acts done before such supersession. Annexure to Notification No. 5/2015-2020 dated 24.04.2017 will replace the existing 'Appendix 3' to Schedule 2 of ITC (HS) Classification of Export and Import Items.

Items on the SCOMET List are organized in the following categories.

Category 0 Nuclear materials, nuclear-related other materials, equipment and technology

- 0A Prescribed Substances
- 0A1 Source Material
- 0A2 Special Fissionable Material
- 0A3 Other Materials
- 0B Prescribed Equipment
- 0C Technology

Category 1 Toxic chemical agents and other chemicals

- 1A Prohibited chemicals, related technology and software
- 1B Chemicals permitted only to States party to the Chemical Weapons Convention
- 1C Chemicals permitted also to States not party to the Chemical Weapons Convention
- 1D Other Chemicals
- 1E¹ Other Chemicals

Category 2² Micro-organisms, toxins

- 2A Bacteria, related technology and software
- 2B Fungi, related technology and software
- 2C Parasites³
- 2D Viruses, related technology and software
- 2E [Reserved]
- 2F Toxins, related technology and software
- 2G Plant pathogens, related technology and software
- 2H Genetic Elements and Genetically-modified Organisms, related technology and software

Category 3 Materials, Materials Processing Equipment and related technologies

- 3A Materials
- 3A1 Special materials
- 3A2 Structural materials
- 3A3 Rocket propellants and constituent chemicals
- 3A4 High explosives
- 3A5 Stealth materials
- 3B Materials processing and production equipment, related technology and specially designed components and accessories therefor.
- 3C [Reserved]
- 3D⁴ Chemical and biomaterial manufacturing and handling equipment and related technology and software

Category 4 Nuclear-related other equipment and technology, not controlled under Category 0

- 4A Equipment, assemblies, components including test and production equipment

¹ Added vide Notification No. 02/2015-20 dated 24.04.2019

² Para 1(i) Notification No.17/2015-20 dated 03.07.2018

³ No change

⁴ Para 1(ii) Notification No.17/2015-20 dated 03.07.2018

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4B Equipment, assemblies, components including test and measurement equipment usable in development of nuclear explosive devices

4C Technology

Category 5 Aerospace systems, equipment, including production and test equipment, related technology and specially designed components and accessories therefore.

5A Rocket systems

5A1 Systems

5A2 Production and test equipment

5A3 Technology

5B Unmanned aerial vehicles

5C Avionics and navigation systems

5D⁶ [Reserved]

5E Micro-light aircraft and powered 'hang-gliders'

Category 6 Munitions List

Category 7

7A [Reserved]

7B [Reserved]

7C [Reserved]

7D [Reserved]

7E [Reserved]

Category 8 Special Materials and Related Equipment, Material Processing, Electronics, Computers, Telecommunications, Information Security, Sensors and Lasers, Navigation and Avionics, Marine, Aerospace and Propulsion

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COMMODITY IDENTIFICATION NOTE TO SCOMET

Note 1 If items are *prima facie*, classifiable under two or more headings, the heading which provides the most specific description shall be preferred to heading providing a more general description. The end-use of the item would be a relevant criteria in determining the classification.

Note 2 Notwithstanding anything contained in Note 1, the following items, will be classified under the relevant description in Category 0:

- a) Radioactive materials covered under 6A007.;
- b) Any material containing Beryllium or "Zirconium with Hafnium content less than 2000 ppm" as the major constituent, or more than 60% Hafnium by weight, or "Boron enriched in Boron-10 isotope" or Niobium or tantalum covered under 6A008.;
- c) Nuclear power generating equipment or propulsion equipment, including "nuclear reactors", and components therefor covered under 6A009. and 6A017.;
- d) Simulators specially designed for military "nuclear reactors" covered under 6A017.;
- e) Any alloy with niobium as a major constituent in solid or powder form covered under 8C102.;
- f) Uranium-titanium alloys covered under 8C104.;
- g) Any material containing "Zirconium with Hafnium content less than 2000 ppm" or "Boron enriched in Boron-10 isotope" as the major constituent covered under 8C111.;
- h) Plutonium and Neptunium covered under 8C112.

Note 3 Items specified in Category 6A007.b which correspond to items specified in Category 1A are prohibited for exports.

Note 4 Licence applications for items in Categories 6A008 a.13 and 6A008.a 21, which correspond to items specified in 3A401.a, 3A401.b respectively, would normally be denied.

GENERAL NOTES

1. Terms in "quotations" are defined terms. Refer to 'Glossary' in SCOMET list.
2. In some instances chemicals are listed by name and CAS number. The list applies to chemicals of the same structural formula (including hydrates) regardless of name or CAS number. CAS numbers are shown to assist in identifying a particular chemical or mixture, irrespective of nomenclature. CAS numbers cannot be used as unique identifiers because some forms of the listed chemical have different CAS numbers, and mixtures containing a listed chemical may also have different CAS numbers.
3. The object of the controls contained in this Schedule should not be defeated by the export of any non-controlled goods containing one or more controlled components when the controlled component or components are the principal element of the goods and can feasibly be removed or used for other purposes.

N.B. In judging whether the controlled component or components are to be considered the principal element, it is necessary to weigh the factors of quantity, value and technological know-how involved and other special circumstances which might establish the controlled component or components as the principal element of the goods being procured.
4. Goods specified in this Schedule include both new and used goods.

GENERAL TECHNOLOGY NOTE

1. The export of "technology" which is "required" for the "development", "production" or "use" of items controlled in Category 8 is controlled according to the provisions in each sub-category. This "technology" remains under control even when applicable to any uncontrolled item.
2. Controls do not apply to that "technology" which is the minimum necessary for the installation, operation, maintenance (checking) or repair of those items which are not controlled or whose export has been authorised.

Note This does not release such "technology" controlled in entries 8E102.e. & 8E102.f. and 8E802.a. & 8E802.b.

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3. Controls do not apply to "technology" "in the public domain", to "basic scientific research" or to the minimum necessary information for patent applications.

GENERAL SOFTWARE NOTE

The Lists do not control "software" which is any of the following:

1. Generally available to the public by being:
 - a. Sold from stock at retail selling points without restriction, by means of:
 1. Over-the-counter transactions;
 2. Mail order transactions;
 3. Electronic transactions; or
 4. Telephone call transactions; and
 - b. Designed for installation by the user without further substantial support by the supplier;
Note Entry 1 of the General Software Note does not release "software" controlled by (8A5, 8B5, 8C5, 8D5, 8E5)-Part-II (Information Security)
2. "In the public domain"; or
3. The minimum necessary "object code" for the installation, operation, maintenance (checking) or repair of those items whose export has been authorised.
Note Entry 3 of the General Software Note does not release "software" controlled by (8A5, 8B5, 8C5, 8D5, 8E5)-Part-II (Information Security).

GENERAL "INFORMATION SECURITY" NOTE

"Information security" items or functions should be considered against the provisions in (8A5, 8B5, 8C5, 8D5, 8E5)-Part-II (Information Security), even if they are components, "software" or functions of other items.

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GLOSSARY

Expressions used in the SCOMET List have the following meanings:

“Accuracy”

(Usually measured in terms of inaccuracy) is the maximum deviation, positive or negative, of an indicated value from an accepted standard or true value.

“Active flight control systems”

Function to prevent undesirable aircraft and missile motions or structural loads by autonomously processing outputs from multiple sensors and then providing necessary preventive commands to effect automatic control.

“Active pixel”

A minimum (single) element of the solid state array which has a photoelectric transfer function when exposed to light (electromagnetic) radiation.

“Additives”

Substances used in explosive formulations to improve their properties.

“Aircraft”

A fixed wing, swivel wing, rotary wing (helicopter), tilt rotor or tilt-wing airborne vehicle.

“Airship”

A power-driven airborne vehicle that is kept buoyant by a body of gas (usually helium, formerly hydrogen) which is lighter than air.

“All compensations available”

"All compensations available" means after all feasible measures available to the manufacturer to minimise all systematic positioning errors for the particular machine-tool model or measuring errors for the particular coordinate measuring machine are considered.

“Allocated by the ITU”

The allocation of frequency bands according to the current edition of the ITU Radio Regulations for primary, permitted and secondary services.

N.B. Additional and alternative allocations are not included.

“Angle random walk”

The angular error build up with time that is due to white noise in angular rate. (IEEE STD 528-2001)

“Angular position deviation”

The maximum difference between angular position and the actual, very accurately measured angular position after the work piece mount of the table has been turned out of its initial position.

“Asymmetric algorithm”

A cryptographic algorithm using different, mathematically-related keys for encryption and decryption.

Technical Note

A common use of asymmetric algorithms is key management.

“Authentication”

Verifying the identity of a user, process or device, often as a prerequisite to allowing access to resources in an information system. This includes verifying the origin or content of a message or other information, and all

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aspects of access control where there is no encryption of files or text except as directly related to the protection of passwords, Personal Identification Numbers (PINs) or similar data to prevent unauthorized access.

“Automated Command and Control Systems”

Electronic systems, through which information essential to the effective operation of the grouping, major formation, tactical formation, unit, ship, subunit or weapons under command is entered, processed and transmitted. This is achieved by the use of computer and other specialised hardware designed to support the functions of a military command and control organisation. The main functions of an automated command and control system are: the efficient automated collection, accumulation, storage and processing of information; the display of the situation and the circumstances affecting the preparation and conduct of combat operations; operational and tactical calculations for the allocation of resources among force groupings or elements of the operational order of battle or battle deployment according to the mission or stage of the operation; the preparation of data for appreciation of the situation and decision-making at any point during operation or battle; computer simulation of operations.

“Average output power”

The total laser output energy, in joules, divided by the period over which a series of consecutive pulses is emitted, in seconds. For a series of uniformly-spaced pulses it is equal to the total laser output energy in a single pulse, in joules, multiplied by the pulse frequency of the laser, in Hertz.

“Basic gate propagation delay time”

The propagation delay time value corresponding to the basic gate used in a monolithic integrated circuit. For a 'family' of monolithic integrated circuits, this may be specified either as the propagation delay time per typical gate within the given 'family' or as the typical propagation delay time per gate within the given 'family'.

Technical Notes

1. *Basic gate propagation delay time is not to be confused with the input/output delay time of a complex monolithic integrated circuit.*
2. *'Family' consists of all integrated circuits to which all of the following are applied as their manufacturing methodology and specifications except their respective functions:*
 - a. *The common hardware and software architecture;*
 - b. *The common design and process technology; and*
 - c. *The common basic characteristics.*

“Basic scientific research”

Experimental or theoretical work undertaken principally to acquire new knowledge of the fundamental principles of phenomena or observable facts, not primarily directed towards a specific practical aim or objective.

“Bias (accelerometer)”

The average over a specified time of accelerometer output, measured at specified operating conditions that has no correlation with input acceleration or rotation. Bias is expressed in g or in metres per second² (g or m/s²). (IEEE Std 528-2001) (Micro g equals 1x10⁻⁶ g).

“Bias (gyro)”

The average over a specified time of gyro output measured at specified operating conditions that have no correlation with input rotation or acceleration. Bias is typically expressed in degrees per hour (deg/hr). (IEEE Std 528-2001).

“Biocatalysts”

'Enzymes' for specific chemical or biochemical reactions or other biological compounds which bind to and accelerate the degradation of CW agents.

Technical Note

'Enzymes' means biocatalysts for specific chemical or biochemical reactions.

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“Biological agents”

Pathogens or toxins, selected or modified (such as altering purity, shelf life, virulence, dissemination characteristics, or resistance to UV radiation) to produce casualties in humans or animals, degrade equipment or damage crops or the environment.

“Biopolymers”

Biological macromolecules as follows:

- a. Enzymes for specific chemical or biochemical reactions;
- b. 'Anti-idiotypic', 'monoclonal' or 'polyclonal' 'antibodies';
- c. Specially designed or specially processed 'receptors'

Technical Notes

1. 'Anti-idiotypic antibodies' means antibodies which bind to the specific antigen binding sites of other antibodies;
2. 'Monoclonal antibodies' means proteins which bind to one antigenic site and are produced by a single clone of cells;
3. 'Polyclonal antibodies' means a mixture of proteins which bind to the specific antigen and are produced by more than one clone of cells;
4. 'Receptors' means biological macromolecular structures capable of binding ligands, the binding of which affects physiological functions.

“Camming (axial displacement)”

Axial displacement in one revolution of the main spindle measured in a plane perpendicular to the spindle faceplate, at a point next to the circumference of the spindle faceplate (Reference: ISO 230/1 1986, paragraph 5.63 as applicable).

“Chemical Laser”

A laser in which the excited species is produced by the output energy from a chemical reaction.

“Circuit element”

A single active or passive functional part of an electronic circuit, such as one diode, one transistor, one resistor, one capacitor, etc.

“Circular Error Probable (CEP) or Circle of Equal Probability”

In a circular normal distribution, the radius of the circle containing 50% of the individual measurements being made, or the radius of the circle within which there is a 50% probability of being located.

“Circulation-controlled anti-torque or circulation-controlled direction control systems”

Control systems using air blown over aerodynamic surfaces to increase or control the forces generated by the surfaces.

“Civil aircraft”

Those aircraft listed by designation in published airworthiness certification lists by the competent civil aviation authorities to fly commercial civil internal and external routes or for legitimate civil, private or business use.

“Comminution”

A process to reduce a material to particles by crushing or grinding.

“Communications channel controller”

The physical interface which controls the flow of synchronous or asynchronous digital information. It is an assembly that can be integrated into computer or telecommunications equipment to provide communications access.

“Compensation systems”

Consist of the primary scalar sensor, one or more reference sensors (e.g., vector “magnetometers”) together with software that permit reduction of the rigid body rotation noise of the platform.

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“Composite”

A matrix and an additional phase or additional phases consisting of particles, whiskers, fibres or any combination thereof, present for a specific purpose or purposes.

“Composite theoretical performance (CTP)”

A measure of computational performance given in millions of theoretical operations per second (Mtops).

“Compound rotary table”

A table allowing the workpiece to rotate and tilt about two non-parallel axes, which can be coordinated simultaneously for contouring control.

“III/V compounds”

Polycrystalline or binary or complex monocrystalline products consisting of elements of groups IIIA and VA of Mendeleyev's periodic classification table (e.g., gallium arsenide, gallium-aluminium arsenide, indium phosphide).

“Contouring control”

Two or more “numerically controlled” motions operating in accordance with instructions that specify the next required position and the required feed rates to that position. These feed rates are varied in relation to each other so that a desired contour is generated. (Ref. International Organization for Standardization (ISO) 2806-(1994) as amended)

“Critical temperature”

(sometimes referred to as the transition temperature) of a specific “superconductive” material is the temperature at which the material loses all resistance to the flow of direct electrical current.

“Cryptographic activation”

Any technique that specifically activates or enables cryptographic capability of an item, by means of a mechanism implemented by the manufacturer of the item, where this mechanism is uniquely bound to any of the following:

1. A single instance of the item; or
2. One customer, for multiple instances of the item.

Technical Notes

1. *"Cryptographic activation" techniques and mechanisms may be implemented as hardware, "software" or "technology".*
2. *Mechanisms for cryptographic activation can, for example, be serial number-based licence keys or authentication instruments such as digitally signed certificates'.*

“Cryptography”

The discipline which embodies principles, means and methods for the transformation of data in order to hide its information content, prevent its undetected modification or prevent its unauthorized use. "Cryptography" is limited to the transformation of information using one or more 'secret parameters' (e.g., crypto variables) or associated key management.

Notes

1. *"Cryptography" does not include 'fixed' data compression or coding techniques.*
2. *"Cryptography" includes decryption.*

Technical Notes

1. *'Secret parameter': a constant or key kept from the knowledge of others or shared only within a group.*
2. *'Fixed': the coding or compression algorithm cannot accept externally supplied parameters (e.g., cryptographic or key variables) and cannot be modified by the user.*

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“CTP”

It is equivalent to composite theoretical performance.

“CW Laser”

A laser that produces nominally constant output energy for greater than 0.25 seconds.

“Cyber incident response”

The process of exchanging necessary information on a cybersecurity incident with individuals or organizations responsible for conducting or coordinating remediation to address the cybersecurity incident.

“Data-Based Referenced Navigation (DBRN) Systems”

Systems which use various sources of previously measured geo-mapping data integrated to provide accurate navigation information under dynamic conditions. Data sources include bathymetric maps, stellar maps, gravity maps, magnetic maps or 3-D digital terrain maps.

“Deformable Mirrors”

Mirrors

- a. Having a single continuous optical reflecting surface which is dynamically deformed by the application of individual torques or forces to compensate for distortions in the optical waveform incident upon the mirror;
or
- b. Having multiple optical reflecting elements that can be individually and dynamically repositioned by the application of torques or forces to compensate for distortions in the optical waveform incident upon the mirror.

Deformable mirrors are also known as adaptive optic mirrors.

“Designed or modified”

describes equipment, parts or components which, as a result of development, or modification, have become endowed with specified properties that make them fit for a particular application.

“Development”

It is related to all stages prior to serial production, such as: design, design research, design analyses, design concepts, assembly and testing of prototypes, pilot production schemes, design data, process of transforming design data into a product, configuration design, integration design, layouts.

“Diffusion bonding”

A solid state joining of at least two separate pieces of metals into a single piece with a joint strength equivalent to that of the weakest material, wherein the principal mechanism is interdiffusion of atoms across the interface.

“Digital computer”

Equipment which can, in the form of one or more discrete variables, perform all of the following:

- a. Accept data;
- b. Store data or instructions in fixed or alterable (writable) storage devices;
- c. Process data by means of a stored sequence of instructions which is modifiable; and
- d. Provide output of data.

Technical Note

Modifications of a stored sequence of instructions include replacement of fixed storage devices, but not a physical change in wiring or interconnections.

“Digital transfer rate”

The total bit rate of the information that is directly transferred on any type of medium. (See also total digital transfer rate).

“Direct-acting hydraulic pressing”

A deformation process which uses a fluid-filled flexible bladder in direct contact with the workpiece.

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“Discrete component”

A separately packaged circuit element with its own external connections.

“Effective gram”

Effective gram for plutonium isotope is defined as the isotope weight in grams.

“Electronically steerable phased array antenna”

An antenna which forms a beam by means of phase coupling, (i.e., the beam direction is controlled by the complex excitation coefficients of the radiating elements) and the direction of that beam can be varied (both in transmission and reception) in azimuth or in elevation, or both, by application of an electrical signal.

“Electronic assembly”

A number of electronic components (i.e., circuit elements, discrete components, integrated circuits, etc.) connected together to perform (a) specific function(s), replaceable as an entity and normally capable of being disassembled.

“End-effectors”

Grippers, 'active tooling units' and any other tooling that is attached to the baseplate on the end of a robot manipulator arm.

Technical Note

'Active tooling units' are devices for applying motive power, process energy or sensing to a workpiece.

“Energetic materials”

Substances or mixtures that react chemically to release energy required for their intended application. Explosives, pyrotechnics and propellants are subclasses of energetic materials.

“Equivalent Density”

The mass of an optic per unit optical area projected onto the optical surface.

“Equivalent standards”

Comparable national or international standards recognised by one or more Wassenaar Arrangement Participating States and applicable to the relevant entry.

“Explosives”

Solid, liquid or gaseous substances or mixtures of substances which, in their application as primary, booster, or main charges in warheads, demolition and other applications, are required to detonate.

“Expression Vectors”

Carriers (e.g., plasmid or virus) used to introduce genetic material into host cells.

“FADEC Systems”

Full Authority Digital Engine Control Systems – A digital electronic control system for a gas turbine engine that is able to autonomously control the engine throughout its whole operating range from demanded engine start until demanded engine shut-down, in both normal and fault conditions.

“Fibrous or filamentary materials”

Include

- a. Continuous monofilaments;
- b. Continuous yarns and rovings;
- c. Tapes, fabrics, random mats and braids;
- d. Chopped fibres, staple fibres and coherent fibre blankets;
- e. Whiskers, either monocrystalline or polycrystalline, of any length;
- f. Aromatic polyamide pulp.

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“Film type integrated circuit”

An array of circuit elements and metallic interconnections formed by deposition of a thick or thin film on an insulating substrate.

“First generation image intensifier tubes”

Electrostatically focused tubes, employing input and output fibre optic or glass face plates, multi-alkali photocathodes (S-20 or S-25), but not microchannel plate amplifiers.

“Flight control optical sensor array”

A network of distributed optical sensors, using laser beams, to provide real-time flight control data for on-board processing.

“Flight path optimization”

A procedure that minimizes deviations from a four-dimensional (space and time) desired trajectory based on maximizing performance or effectiveness for mission tasks.

“Fly-by-light system”

A primary digital flight control system employing feedback to control the aircraft during flight, where the commands to the effectors/actuators are optical signals.

“Fly-by-wire system”

A primary digital flight control system employing feedback to control the aircraft during flight, where the commands to the effectors/actuators are electrical signals.

“Focal plane array”

A linear or two-dimensional planar layer, or combination of planar layers, of individual detector elements, with or without readout electronics, which work in the focal plane.

Note This definition does not include a stack of single detector elements or any two, three or four element detectors provided time delay and integration is not performed within the element.

“Fractional bandwidth”

The instantaneous bandwidth divided by the centre frequency, expressed as a percentage.

“Frequency hopping”

A form of spread spectrum in which the transmission frequency of a single communication channel is made to change by a random or pseudo-random sequence of discrete steps.

“Frequency mask trigger”

For signal analysers a mechanism where the trigger function is able to select a frequency range to be triggered on as a subset of the acquisition bandwidth while ignoring other signals that may also be present within the same acquisition bandwidth. A frequency mask trigger may contain more than one independent set of limits.

“Frequency switching time”

The time (i.e., delay) taken by a signal when switched from an initial specified output frequency, to arrive at or within any of the following:

- a. ± 100 Hz of a final specified output frequency of less than 1 GHz; or
- b. ± 0.1 part per million of a final specified output frequency equal to or greater than 1 GHz.

“Frequency synthesizer”

Any kind of frequency source, regardless of the actual technique used, providing a multiplicity of simultaneous or alternative output frequencies, from one or more outputs, controlled by, derived from or disciplined by a lesser number of standard (or master) frequencies.

“Fuel cell”

An electrochemical device that converts chemical energy directly into Direct Current (DC) electricity by consuming fuel from an external source.

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“Fusible”

Capable of being cross-linked or polymerized further (cured) by the use of heat, radiation, catalysts, etc., or that can be melted without pyrolysis (charring).

“Gas atomization”

A process to reduce a molten stream of metal alloy to droplets of 500 µm diameter or less by a high pressure gas stream.

“Geographically dispersed”

Sensors are considered geographically dispersed when each location is distant from any other more than 1,500 m in any direction. Mobile sensors are always considered geographically dispersed.

"Gate-Around Field-Effect Transistor" ("GAAFET")

A device having a single or multiple semiconductor conduction channel element(s) with a common gate structure that surrounds and controls current in all of the semiconductor conduction channel elements. Note This definition includes nanosheet or nanowire field-effect and surrounding gate transistors and other "GAAFET" semiconductor channel element structures.

"Hard selectors"

Data or set of data, related to an individual (e.g., family name, given name, e-mail, street address, phone number or group affiliations).

“Hot isostatic densification”

A process of pressurising a casting at temperatures exceeding 375 K (102°C) in a closed cavity through various media (gas, liquid, solid particles, etc.) to create equal force in all directions to reduce or eliminate internal voids in the casting.

“Hybrid computer”

Equipment which can perform all of the following:

- a. Accept data;
- b. Process data, in both analogue and digital representations; and
- c. Provide output of data.

“Hybrid integrated circuit”

Any combination of integrated circuit(s), or integrated circuit with circuit elements or discrete components connected together to perform (a) specific function(s), and having all of the following characteristics:

- a. Containing at least one unencapsulated device;
- b. Connected together using typical IC production methods;
- c. Replaceable as an entity; and
- d. Not normally capable of being disassembled.

“Image enhancement”

The processing of externally derived information-bearing images by algorithms such as time compression, filtering, extraction, selection, correlation, convolution or transformations between domains (e.g., fast Fourier transform or Walsh transform). This does not include algorithms using only linear or rotational transformation of a single image, such as translation, feature extraction, registration or false coloration.

“Information security”

All the means and functions ensuring the accessibility, confidentiality or integrity of information or communications, excluding the means and functions intended to safeguard against malfunctions. This includes cryptography, cryptographic activation, 'cryptanalysis', protection against compromising emanations and computer security.

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Technical Note

'Cryptanalysis': the analysis of a cryptographic system or its inputs and outputs to derive confidential variables or sensitive data, including clear text. (ISO 7498-2-1988 (E), paragraph 3.3.18 as applicable).

“Instantaneous bandwidth”

The bandwidth over which output power remains constant within 3 dB without adjustment of other operating parameters.

“Insulation”

It is applied to the components of a rocket motor, i.e. the case, nozzle, inlets, case closures, and includes cured or semi-cured compounded rubber sheet stock containing an insulating or refractory material. It may also be incorporated as stress relief boots or flaps.

“Interior lining”

It is suited for the bond interface between the solid propellant and the case or insulating liner. Usually a liquid polymer based dispersion of refractory or insulating materials, e.g. carbon filled hydroxyl terminated polybutadiene (HTPB) or other polymer with added curing agents sprayed or screeded over a case interior.

“Interleaved Analogue-to-Digital Converter (ADC)”

Devices that have multiple ADC units that sample the same analogue input at different times such that when the outputs are aggregated, the analogue input has been effectively sampled and converted at a higher sampling rate.

“Intrinsic magnetic gradiometer”

A single magnetic field gradient sensing element and associated electronics the output of which is a measure of magnetic field gradient.

“Intrusion software”

Software specially designed or modified to avoid detection by 'monitoring tools', or to defeat 'protective countermeasures', of a computer or network-capable device, and performing any of the following:

- a. The extraction of data or information, from a computer or network-capable device, or the modification of system or user data; or
- b. The modification of the standard execution path of a “program” or process in order to allow the execution of externally provided instructions.

Notes

1. Intrusion software does not include any of the following:
 - a. Hypervisors, debuggers or Software Reverse Engineering (SRE) tools;
 - b. Digital Rights Management (DRM) software; or
 - c. Software designed to be installed by manufacturers, administrators or users, for the purposes of asset tracking or recovery.
2. Network-capable devices include mobile devices and smart meters.

Technical Notes

1. 'Monitoring tools': software or hardware devices, that monitor system behaviours or processes running on a device. This includes antivirus (AV) products, end point security products, Personal Security Products (PSP), Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS) or firewalls.
2. 'Protective countermeasures': techniques designed to ensure the safe execution of code, such as Data Execution Prevention (DEP), Address Space Layout Randomisation (ASLR) or sandboxing.

“Isolated live cultures”

Includes live cultures in dormant form and in dried preparations.

“Isostatic presses”

Equipment capable of pressurising a closed cavity through various media (gas, liquid, solid particles, etc.) to create equal pressure in all directions within the cavity upon a workpiece or material.

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“Laser”

An item that produces spatially and temporally coherent light through amplification by stimulated emission of radiation.

“Library (parametric technical database)”

A collection of technical information, reference to which may enhance the performance of relevant systems, equipment or components.

“Lighter-than-air vehicles”

Balloons and airships that rely on hot air or on lighter-than-air gases such as helium or hydrogen for their lift.

“Linearity (Usually measured in terms of non-linearity)”

It is the maximum deviation of the actual characteristic (average of upscale and downscale readings), positive or negative, from a straight line so positioned as to equalise and minimise the maximum deviations.

“Local area network”

A data communication system having all of the following characteristics:

- a. Allows an arbitrary number of independent 'data devices' to communicate directly with each other; and
- b. Is confined to a geographical area of moderate size (e.g., office building, plant, campus, warehouse).

Technical Note

'Data device' means equipment capable of transmitting or receiving sequences of digital information.

“Magnetic gradiometers”

Are designed to detect the spatial variation of magnetic fields from sources external to the instrument. They consist of multiple magnetometers and associated electronics the output of which is a measure of magnetic field gradient. (See also Intrinsic Magnetic Gradiometer)

“Magnetometers”

Are designed to detect magnetic fields from sources external to the instrument. They consist of a single magnetic field sensing element and associated electronics the output of which is a measure of the magnetic field.

“Main storage”

The primary storage for data or instructions for rapid access by a central processing unit. It consists of the internal storage of a “digital computer” and any hierarchical extension thereto, such as cache storage or non-sequentially accessed extended storage.

“Matrix”

A substantially continuous phase that fills the space between particles, whiskers or fibres.

“Measurement uncertainty”

The characteristic parameter which specifies in what range around the output value the correct value of the measurable variable lies with a confidence level of 95%. It includes the uncorrected systematic deviations, the uncorrected backlash and the random deviations (Reference: ISO 10360-2 as applicable).

“Mechanical alloying”

An alloying process resulting from the bonding, fracturing and rebonding of elemental and master alloy powders by mechanical impact. Non-metallic particles may be incorporated in the alloy by addition of the appropriate powders.

“Melt extraction”

A process to solidify rapidly and extract a ribbon-like alloy product by the insertion of a short segment of a rotating chilled block into a bath of a molten metal alloy.

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“Melt spinning”

A process to solidify rapidly a molten metal stream impinging upon a rotating chilled block, forming a flake, ribbon or rod-like product.

“Microcircuit”

A device in which a number of passive and/or active elements are considered as indivisibly associated on or within a continuous structure to perform the function of a circuit

“Microcomputer microcircuit”

A monolithic integrated circuit or multichip integrated circuit containing an arithmetic logic unit (ALU) capable of executing general purpose instructions from an internal storage, on data contained in the internal storage.

Technical Note

The internal storage may be augmented by an external storage.

“Microprocessor microcircuit”

A “monolithic integrated circuit” or “multichip integrated circuit” containing an arithmetic logic unit (ALU) capable of executing a series of general purpose instructions from an external storage.

Technical Note

The microprocessor microcircuit normally does not contain integral user-accessible storage, although storage present on-the-chip may be used in performing its logic function.

Note *This definition includes chip sets which are designed to operate together to provide the function of a microprocessor microcircuit.*

“Microprogram”

A sequence of elementary instructions maintained in a special storage, the execution of which is initiated by the introduction of its reference instruction into an instruction register.

“Military use”

shall mean incorporation into military items listed in SCOMET 6 or for the use, development, or production of military items listed in this category.

“Monolithic Microwave Integrated Circuit (MMIC)”

A monolithic integrated circuit that operates at microwave or millimeter wave frequencies.

“Missiles”

means complete rocket systems and unmanned aerial vehicle systems.

“Modified in the context of software”

describes software which has been intentionally changed such that it has properties that make it fit for specified purposes or applications. Its properties may also make it suitable for purposes or applications other than those for which it was modified.

“Monofilament or filament”

is the smallest increment of fibre, usually several micrometres in diameter.

“Monolithic integrated circuit”

A combination of passive or active circuit elements or both which:

- Are formed by means of diffusion processes, implantation processes or deposition processes in or on a single semiconducting piece of material, a so-called 'chip';
- Can be considered as indivisibly associated; and
- Perform the function(s) of a circuit.

Technical Note *‘Circuit element’ is a single active or passive functional part of an electronic circuit, such as one diode, one transistor, one resistor, one capacitor, etc.*

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“Monospectral imaging”

sensors are capable of acquisition of imaging data from one discrete spectral band.

“Multichip integrated circuit”

Two or more monolithic integrated circuits bonded to a common substrate.

“Multiple channel analogue-to-Digital Converter (ADC)”

Devices that integrate more than one ADC, designed so that each ADC has a separate analogue input.

“Multispectral imaging sensors”

are capable of simultaneous or serial acquisition of imaging data from two or more discrete spectral bands. Sensors having more than twenty discrete spectral bands are sometimes referred to as hyperspectral imaging sensors.

“Network access controller”

A physical interface to a distributed switching network. It uses a common medium which operates throughout at the same digital transfer rate using arbitration (e.g., token or carrier sense) for transmission. Independently from any other, it selects data packets or data groups (e.g., IEEE 802) addressed to it. It is an assembly that can be integrated into computer or telecommunications equipment to provide communications access.

“Neural computer”

A computational device designed or modified to mimic the behaviour of a neuron or a collection of neurons, i.e., a computational device which is distinguished by its hardware capability to modulate the weights and numbers of the interconnections of a multiplicity of computational components based on previous data.

“Nuclear reactor”

includes the items within or attached directly to the reactor vessel, the equipment which controls the level of power in the core, and the components which normally contain or come into direct contact with or control the primary coolant of the reactor core.

“Numerical control”

The automatic control of a process performed by a device that makes use of numeric data usually introduced as the operation is in progress. (Ref. ISO 2382 (2015).

“Object code”

An equipment executable form of a convenient expression of one or more processes (source code (or source language)) which has been compiled by a programming system.

“Operations, Administration or Maintenance (OAM)”

Means performing one or more of the following tasks:

- a. Establishing or managing any of the following:
 - 1. Accounts or privileges of users or administrators;
 - 2. Settings of an item; or
 - 3. Authentication data in support of the tasks described in a.1. or a.2 above;
- b. Monitoring or managing the operating condition or performance of an item; or
- c. Managing logs or audit data in support of any of the tasks described in a. or b above.

Note “OAM” does not include any of the following tasks or their associated key management functions:

- a. Provisioning or upgrading any cryptographic functionality that is not directly related to establishing or managing authentication data in support of the tasks described in a.1. or a.2. above; or
- b. Performing any cryptographic functionality on the forwarding or data plane of an item.

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“Optical computer”

A computer designed or modified to use light to represent data and whose computational logic elements are based on directly coupled optical devices.

“Optical integrated circuit”

A monolithic integrated circuit or a hybrid integrated circuit, containing one or more parts designed to function as a photosensor or photoemitter or to perform (an) optical or (an) electro-optical function(s).

“Optical switching”

The routing of or switching of signals in optical form without conversion to electrical signals.

“Overall current density”

The total number of ampere-turns in the coil (i.e., the sum of the number of turns multiplied by the maximum current carried by each turn) divided by the total cross-section of the coil (comprising the superconducting filaments, the metallic matrix in which the superconducting filaments are embedded, the encapsulating material, any cooling channels, etc.).

“Peak power”

The highest power attained in the pulse duration.

“Personal area network”

A data communication system having all of the following characteristics:

- a. Allows an arbitrary number of independent or interconnected 'data devices' to communicate directly with each other; and
- b. Is confined to the communication between devices within the immediate physical vicinity of an individual person or device controller (e.g., single room, office or automobile).

Technical Notes

1. *'Data device' means equipment capable of transmitting or receiving sequences of digital information.*
2. *The "local area network" extends beyond the geographical area of the "personal area network".*

“Plasma atomization”

A process to reduce a molten stream or solid metal to droplets of 500 µm diameter or less, using plasma torches in an inert gas environment.

“Power management”

Changing the transmitted power of the altimeter signal so that received power at the aircraft altitude is always at the minimum necessary to determine the altitude.

“Precursors”

Speciality chemicals used in the manufacture of explosives.

“Pressure transducers”

are devices that convert pressure measurements into an electrical signal.

“Previously separated”

The application of any process intended to increase the concentration of the controlled isotope.

“Primary flight control”

Aircraft stability or manoeuvring control using force/moment generators, i.e. aerodynamic control surfaces or propulsive thrust vectoring.

“Principal element”

An element is a principal element when its replacement value is more than 35% of the total value of the system of which it is an element. Element value is the price paid for the element by the manufacturer of the system, or by the system integrator. Total value is the normal international selling price to unrelated parties at the point of

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manufacture or consolidation of shipment.

“Production”

Means all production stages, such as: product engineering, manufacture, integration, assembly (mounting), inspection, testing, quality assurance.

“Production equipment”

Tooling, templates, jigs, mandrels, moulds, dies, fixtures, alignment mechanisms, test equipment, other machinery and components therefor, limited to those specially designed or modified for development or for one or more phases of production.

“Production facilities”

Equipment and specially designed software therefor integrated into installations for development or for one or more phases of production.

“Program”

A sequence of instructions to carry out a process in, or convertible into, a form executable by an electronic computer.

“Propellants”

Substances or mixtures that react chemically to produce large volumes of hot gases at controlled rates to perform mechanical work.

“Public domain”

means a domain that has no restrictions upon dissemination of information within or from it; the existence of any legal rights to the intellectual property in that information does not remove the information from being in public domain.

Note Copyright restrictions do not remove “technology” or “software” from being in the public domain.

“Pulse compression”

The coding and processing of a radar signal pulse of long time duration to one of short time duration, while maintaining the benefits of high pulse energy.

“Pulse duration”

Duration of a laser pulse is the time between the half-power points on the leading edge and trailing edge of an individual pulse.

“Pulsed laser”

A laser having a pulse duration that is less than or equal to 0.25 seconds.

“Pyrotechnic(s)”

Mixtures of solid or liquid fuels and oxidizers which, when ignited, undergo an energetic chemical reaction at a controlled rate intended to produce specific time delays, or quantities of heat, noise, smoke, visible light or infrared radiation. Pyrophorics are a subclass of pyrotechnics, which contain no oxidizers but ignite spontaneously on contact with air.

“Quantum cryptography”

A family of techniques for the establishment of a shared key for cryptography by measuring the quantum-mechanical properties of a physical system (including those physical properties explicitly governed by quantum optics, quantum field theory, or quantum electrodynamics).

“Radar frequency agility”

Any technique which changes, in a pseudo-random sequence, the carrier frequency of a pulsed radar transmitter between pulses or between groups of pulses by an amount equal to or larger than the pulse bandwidth.

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“Radar spread spectrum”

Any modulation technique for spreading energy originating from a signal with a relatively narrow frequency band, over a much wider band of frequencies, by using random or pseudo-random coding.

“Radiant sensitivity”

Radiant sensitivity (mA/W) = $0.807 \times (\text{wavelength in nm}) \times \text{Quantum Efficiency (QE)}$.

Technical Note

QE is usually expressed as a percentage; however, for the purposes of this formula QE is expressed as a decimal number less than one, e.g., 78% is 0.78.

“Radiation hardened”

means that the component or equipment is designed or rated to withstand radiation levels which meet or exceed a total radiation dose of 5×10^3 Gy or 5×10^5 rads (Si).

“Real-time bandwidth”

For signal analysers, the widest frequency range for which the analyser can continuously transform time-domain data entirely into frequency-domain results using a Fourier or other discrete time transform that processes every incoming time point, without a reduction of measured amplitude of more than 3 dB below the actual signal amplitude caused by gaps or windowing effects, while outputting or displaying the transformed data.

“Real-time processing”

The processing of data by a computer system providing a required level of service, as a function of available resources, within a guaranteed response time, regardless of the load of the system, when stimulated by an external event.

“Repeatability”

The closeness of agreement among repeated measurements of the same variable under the same operating conditions when changes in conditions or non-operating periods occur between measurements. (Reference: IEEE STD 528-2001 (one sigma standard deviation))

“Required”

As applied to technology, refers to only that portion of technology which is peculiarly responsible for achieving or exceeding the controlled performance levels, characteristics or functions. Such required technology may be shared by different products.

“Riot control agents”

Substances which, under the expected conditions of use for riot control purposes produce rapidly in humans sensory irritation or disabling physical effects which disappear within a short time following termination of exposure. (Tear gases are a subset of “riot control agents”).

“Robot”

A manipulation mechanism, which may be of the continuous path or of the point-to-point variety, may use sensors, and has all the following characteristics:

- a) Is multifunctional;
- b) Is capable of positioning or orienting material, parts, tools or special devices through variable movements in three dimensional space;
- c) Incorporates three or more closed or open loop servo-devices which may include stepping motors; and
- d) Has user-accessible programmability by means of the teach/playback method or by means of an electronic computer which may be a programmable logic controller, i.e., without mechanical intervention.

Note The above definition does not include the following devices:

1. Manipulation mechanisms which are only manually/tele-operator controllable;
2. Fixed sequence manipulation mechanisms which are automated moving devices, operating according to mechanically fixed programmed motions. The programme is mechanically limited by fixed stops, such as pins or cams. The sequence of motions and the selection of paths or angles are not variable or changeable

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by mechanical, electronic or electrical means;

3. *Mechanically controlled variable sequence manipulation mechanisms which are automated moving devices, operating according to mechanically fixed programmed motions. The programme is mechanically limited by fixed, but adjustable stops, such as pins or cams. The sequence of motions and the selection of paths or angles are variable within the fixed programme pattern. Variations or modifications of the programme pattern (e.g., changes of pins or exchanges of cams) in one or more motion axes are accomplished only through mechanical operations;*
4. *Non-servo-controlled variable sequence manipulation mechanisms which are automated moving devices, operating according to mechanically fixed programmed motions. The programme is variable but the sequence proceeds only by the binary signal from mechanically fixed electrical binary devices or adjustable stops;*
5. *Stacker cranes defined as Cartesian coordinate manipulator systems manufactured as an integral part of a vertical array of storage bins and designed to access the contents of those bins for storage or retrieval.*

“Rotary atomization”

A process to reduce a stream or pool of molten metal to droplets to a diameter of 500 µm or less by centrifugal force.

“Roving”

is a bundle (typically 12-120) of approximately parallel ‘strands’.

N.B. ‘Strand’ is a bundle of monofilaments (typically over 200) arranged approximately parallel.

“Run-out (out-of-true running)”

Radial displacement in one revolution of the main spindle measured in a plane perpendicular to the spindle axis at a point on the external or internal revolving surface to be tested (Reference: ISO 230/1-1986, paragraph 5.61 as applicable).

“Sample rate”

For an Analogue-to-Digital Converter (ADC) the maximum number of samples that are measured at the analogue input over a period of one second, except for oversampling ADCs. For oversampling ADCs the "sample rate" is taken to be its output word rate. "Sample rate" may also be referred to as sampling rate, usually specified in Mega Samples Per Second (MSPS) or Giga Samples Per Second (GSPS), or conversion rate, usually specified in Hertz (Hz).

"Satellite navigation system"

A system consisting of ground stations, a constellation of satellites, and receivers, that enables receiver locations to be calculated on the basis of signals received from the satellites. It includes Global Navigation Satellite Systems and Regional Navigation Satellite Systems.

“Scale factor (gyro or accelerometer)”

The ratio of change in output to a change in the input intended to be measured. Scale factor is generally evaluated as the slope of the straight line that can be fitted by the method of least squares to input-output data obtained by varying the input cyclically over the input range.

“Signal analysers”

Apparatus capable of measuring and displaying basic properties of the single-frequency components of multi-frequency signals.

“Signal processing”

The processing of externally derived information-bearing signals by algorithms such as time compression, filtering, extraction, selection, correlation, convolution or transformations between domains (e.g., fast Fourier transform or Walsh transform).

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“Software”

A collection of one or more programs, or micro-programs, fixed in any tangible medium of expression. However, unless otherwise provided for against any item on the SCOMET List, the List does not control software which is either in the public domain or is generally available to the public by being:

- a. Sold from stock at retail selling points without restriction, by means of:
 1. Over-the-counter transactions;
 2. Mail order transactions;
 3. Electronic transactions; or
 4. Telephone call transactions; and
- b. Designed for installation by the user without further substantial support by the supplier.

“Solidify rapidly”

A process involving the solidification of molten material at cooling rates exceeding 1000 K/sec.

“Source code”

A convenient expression of one or more processes which may be turned by a programming system into equipment executable form (“object code” (or object language)).

“Spacecraft”

Active and passive satellites and space probes

“Spacecraft bus”

Equipment that provides the support infrastructure of the “spacecraft” and location for the “spacecraft payload”.

“Spacecraft payload”

Equipment, attached to the spacecraft bus, designed to perform a mission in space (e.g., communications, observation, science).

“Space-qualified”

Designed, manufactured, or qualified through successful testing, for operation at altitudes greater than 100 km above the surface of the Earth.

Note A determination that a specific item is “space-qualified” by virtue of testing does not mean that other items in the same production run or model series are “space-qualified” if not individually tested.

“Space qualified”

Products designed, manufactured and tested to meet the special electrical, mechanical or environmental requirements for use in the launch and deployment of satellites or high altitude flight systems operating at altitudes of 100 km or higher.

Note A determination that a specific item is “space-qualified” by virtue of testing does not mean that other items in the same production run or model series are “space-qualified” if not individually tested.

“Specially designed”

qualifies the description of equipment, parts, components or software which, as a result of development, have unique properties that distinguish them for certain predetermined purposes. For example, a piece of equipment that is specially designed will only be considered so if it has no other function or use. Thus a piece of manufacturing equipment that is specially designed to produce a certain type of component will only be considered such if it is not capable of producing other types of components.

“Specific modulus”

Young's modulus in pascals, equivalent to N/m², divided by specific weight in N/m³, measured at a temperature of 296 ± 2 K (23 ± 2°C) and a relative humidity of (50 ± 5)%.

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“Specific tensile strength”

Ultimate tensile strength in pascals, equivalent to N/m², divided by specific weight in N/m³, measured at a temperature of 296 ± 2 K ($23 \pm 2^\circ\text{C}$) and a relative humidity of $(50 \pm 5)\%$.

“Spinning mass gyros”

Spinning mass gyros are gyros which use a continually rotating mass to sense angular motion.

“Splat quenching”

A process to ‘solidify rapidly’ a molten metal stream impinging upon a chilled block, forming a flake-like product.

“Spread spectrum”

The technique whereby energy in a relatively narrow-band communication channel is spread over a much wider energy spectrum.

“Spread spectrum radar”

see Radar spread spectrum

"Stability"

Standard deviation (1 sigma) of the variation of a particular parameter from its calibrated value measured under stable temperature conditions. This can be expressed as a function of time.

Statement of Understanding

For gyroscopes and accelerometers, "stability" can be estimated by determining the Allan variance noise-analysis value at the integration period (i.e., sample time) consistent with the stated measurement period, which may include extrapolating the Allan variance noise analysis beyond the instability point into the rate/acceleration-random walk or rate/acceleration ramp regions to an integration period consistent with the stated measurement period (Reference: IEEE Std 952-1997 [R2008] or IEEE Std 1293-1998 [R2008]).

"Steady State Mode"

The term "steady state mode" defines engine operation conditions, where the engine parameters, such as thrust/power, rpm and others, have no appreciable fluctuations, when the ambient air temperature and pressure at the engine inlet are constant.

"Sub-orbital craft"

A craft having an enclosure designed for the transport of people or cargo, which is designed to:

- a. Operate above the stratosphere;
- b. Perform a non-orbital trajectory; and
- c. Land back on Earth with the people or cargo intact.

“Substrate”

A sheet of base material with or without an interconnection pattern and on which or within which discrete components or integrated circuits or both can be located.

“Substrate blanks”

Monolithic compounds with dimensions suitable for the production of optical elements such as mirrors or optical windows.

“Superalloy”

Nickel-, cobalt- or iron-base alloys having a stress rupture life greater than 1000 hours at 400 MPa and an ultimate tensile strength greater than 850 MPa, at 922 K (649°C) or higher.

“Superconductive”

Refers to materials, (i.e., metals, alloys or compounds) which can lose all electrical resistance (i.e., which can attain infinite electrical conductivity and carry very large electrical currents without Joule heating).

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Technical Note

The superconductive state of a material is individually characterised by a critical temperature, a critical magnetic field, which is a function of temperature, and a critical current density which is, however, a function of both magnetic field and temperature.

“Super High Power Laser (SHPL)”

A “laser” capable of delivering (the total or any portion of) the output energy exceeding 1 kJ within 50 ms or having an average or CW power exceeding 20 kW.

“Superplastic forming”

A deformation process using heat for metals that are normally characterised by low values of elongation (less than 20%) at the breaking point as determined at room temperature by conventional tensile strength testing, in order to achieve elongations during processing which are at least 2 times those values.

“Symmetric algorithm”

A cryptographic algorithm using an identical key for both encryption and decryption.

Technical Note

A common use of symmetric algorithms is confidentiality of data.

“Systolic array computer”

A computer where the flow and modification of the data is dynamically controllable at the logic gate level by the user.

“Tape”

It is a material constructed of interlaced or unidirectional monofilaments, ‘strands’, rovings, tows, or yarns, etc., usually preimpregnated with resin.

N.B. ‘Strand’ is a bundle of monofilaments (typically over 200) arranged approximately parallel

“Technology means”,

except as otherwise provided for against any item in the SCOMET List, information (including information embodied in software) other than information in the public domain, that is capable of being used in:

- a. the development, production or use of any goods or software;
- b. the development of, or the carrying out of, an industrial or commercial activity or the provision of a service of any kind.

Explanation 1: When technology is described wholly or partly by reference to the uses to which it (or the goods to which it relates) may be put, it shall include services which are provided or used, or which are capable of being used, in the development, production or use of such technology or goods.

Explanation 2: The information takes the form of ‘technical data’ or ‘technical assistance’. Specified technology is defined in the General Technology Note to the SCOMET Category 8. Specified technology for the Munitions List is defined in 6A022.

Technical Notes

1. ‘Technical data’ may take forms such as blueprints, plans, diagrams, models, formulae, algorithms, tables, engineering designs and specifications, manuals, instructions, etc., written or recorded on other media or devices such as disk, tape, read-only memories.
2. ‘Technical assistance’ may take forms such as instruction, skills, training, working knowledge, consulting services. ‘Technical assistance’ may involve transfer of ‘technical data’.

“Three dimensional integrated circuit”

A collection of semiconductor dies or active device layers, integrated together, and having through semiconductor via connections passing completely through an interposer, substrate, die or layer to establish interconnections between the device layers. An interposer is an interface that enables electrical connections.

“Tilting spindle”

A tool-holding spindle which alters, during the machining process, the angular position of its centre line with respect to any other axis.

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“Time constant”

The time taken from the application of a light stimulus for the current increment to reach a value of $1-1/e$ times the final value (i.e., 63% of the final value).

“Tip shroud”

A stationary ring component (solid or segmented) attached to the inner surface of the engine turbine casing or a feature at the outer tip of the turbine blade, which primarily provides a gas seal between the stationary and rotating components.

“Total control of flight”

Automated control of aircraft state variables and flight path to meet mission objectives responding to real time changes in data regarding objectives, hazards or other aircraft.

“Total digital transfer rate”

The number of bits, including line coding, overhead and so forth per unit time passing between corresponding equipment in a digital transmission system. (See also digital transfer rate)

“Tow”

is a bundle of monofilaments, usually approximately parallel.

“Toxins”

means toxins in the form of deliberately isolated preparations or mixtures, no matter how produced, other than toxins present as contaminants of other materials such as pathological specimens, crops, foodstuffs or seed stocks of microorganisms.

“Transfer laser”

A laser in which the lasing species is excited through the transfer of energy by collision of a non-lasing atom or molecule with a lasing atom or molecule species.

“Tunable”

The ability of a “laser” to produce a continuous output at all wavelengths over a range of several “laser” transitions. A line selectable “laser” produces discrete wavelengths within one “laser” transition and is not considered “tunable”.

“Unidirectional positioning repeatability”

The smaller of values $R_{\uparrow}$ and $R_{\downarrow}$ (forward and backward), as defined by 3.21 of ISO 230-2:2014 or national equivalents, of an individual machine tool axis as applicable.

“Unmanned aerial vehicle (UAV)”

Any “aircraft” capable of initiating flight and sustaining controlled flight and navigation without any human presence on board.

“Usable in, usable for, usable as or capable of”

It qualifies the description of equipment, parts, components, materials, technology or software which are suitable for a particular purpose. There is no requirement that the equipment, parts, components, technology or software should have been configured, modified or specified for that particular purpose. (Contrast with specially designed – see above).

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“Use”

Operation, installation (including on-site installation), maintenance (checking), repair, overhaul and refurbishing.

“User-accessible programmability”

The facility allowing a user to insert, modify or replace program by means other than:

- a. A physical change in wiring or interconnections; or
- b. The setting of function controls including entry of parameters.

“Vaccine”

It is a medicinal product in a pharmaceutical formulation licensed by, or having marketing or clinical trial authorisation from, the regulatory authorities of either the country of manufacture or of use, which is intended to stimulate a protective immunological response in humans or animals in order to prevent disease in those to whom or to which it is administered.

“Vacuum atomisation”

A process to reduce a molten stream of metal to droplets of a diameter of 500 µm or less by the rapid evolution of a dissolved gas upon exposure to a vacuum.

“Variable geometry airfoils”

Use trailing edge flaps or tabs, or leading edge slats or pivoted nose droop, the position of which can be controlled in flight.

"Vulnerability disclosure"

The process of identifying, reporting or communicating a vulnerability to, or analysing a vulnerability with, individuals or organisations responsible for conducting or coordinating remediation for the purpose of resolving the vulnerability.

“Yarn”

It is a bundle of twisted ‘strands’ *N.B.* ‘Strand’ is a bundle of monofilaments (typically over 200) arranged approximately parallel.

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ACRONYMS AND ABBREVIATIONS

Acronym Or Abbreviation	Meaning
ADC	Analogue-To-Digital Converter
AGMA	American Gear Manufacturers' Association
AHRS	Attitude And Heading Reference Systems
AISI	American Iron And Steel Institute
ALE	Atomic Layer Epitaxy
AMPS	Aircraft Missile Protection System
APP	Adjusted Peak Performance
APU	Auxiliary Power Unit
ASTM	American Society For Testing And Materials
ATC	Air Traffic Control
BJT	Bipolar Junction Transistors
BPP	Beam Parameter Product
BSC	Base Station Controller
C3I	Command, Communications, Control & Intelligence
CAD	Computer-Aided-Design
CAS	Chemical Abstracts Service
CCD	Charge Coupled Device
CDU	Control And Display Unit
CEP	Circular Error Probable
CMM	Coordinate Measuring Machine
CMOS	Complementary Metal Oxide Semiconductor
CNTD	Controlled Nucleation Thermal Deposition
CPLD	Complex Programmable Logic Device
CPU	Central Processing Unit
CVD	Chemical Vapour Deposition
CW	Chemical Warfare
CW (for lasers)	Continuous Wave
DAC	Digital-To-Analogue Converter
DANL	Displayed Average Noise Level
DBRN	Data-Base Referenced Navigation
DDS	Direct Digital Synthesizer
DEW	Directed Energy Weapon Systems
DMA	Dynamic Mechanical Analysis
DME	Distance Measuring Equipment
DMOSFET	Diffused Metal Oxide Semiconductor Field Effect Transistor
DS	Directionally Solidified
EB	Exploding Bridge
EB-PVD	Electron Beam Physical Vapour Deposition
EBW	Exploding Bridge Wire
ECM	Electro-Chemical Machining
ECAD	Electronic Computer-Aided Design

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EDM	Electrical Discharge Machines
EFI	Exploding Foil Initiators
EIRP	Effective Isotropic Radiated Power
EMC	Electromagnetic Compatibility
EMCDB	Elastomer Modified Cast Double Based Propellants
EMP	Electromagnetic Pulse
ERF	Electrorheological Finishing
ERP	Effective Radiated Power
ESD	Electrostatic Discharge
ETO	Emitter Turn-Off Thyristor
ETT	Electrical Triggering Thyristor
FADEC	Full Authority Digital Engine Control
FFT	Fast Fourier Transform
FPGA	Field Programmable Gate Array
FPIC	Field Programmable Interconnect
FPLA	Field Programmable Logic Array
FPO	Floating Point Operation
FWHM	Full-Width Half-Maximum
GAAFET	Gate-All-Around Field-Effect Transistor
GISN	General "Information Security" Note
GNSS	Global Navigation Satellite System
GSM	Global System For Mobile Communications
GST	General Software Note
GTN	General Technology Note
GTO	Gate Turn-Off Thyristor
GLONASS	Global Navigation Satellite System
GPS	Global Positioning System
HBT	Hetero-Bipolar Transistors
HDMI	High-Definition Multimedia Interface
HEMT	High Electron Mobility Transistors
ICAO	International Civil Aviation Organisation
IEC	International Electro-Technical Commission
IED	Improvised Explosive Device
IEEE	Institute Of Electrical And Electronic Engineers
IFOV	Instantaneous-Field-Of-View
IGBT	Insulated Gate Bipolar Transistor
IGCT	Integrated Gate Commutated Thyristor
IHO	International Hydrographic Organization
ILS	Instrument Landing System
IMU	Inertial Measurement Unit
INS	Inertial Navigation System
IP	Internet Protocol
IRS	Inertial Reference System
IRU	Inertial Reference Unit
ISA	International Standard Atmosphere

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ISAR	Inverse Synthetic Aperture Radar
ISO	International Organization For Standardization
ITU	International Telecommunication Union
JT	Joule-Thomson
LIDAR	Light Detection And Ranging
LIDT	Laser Induced Damage Threshold
LOA	Length Overall
LRU	Line Replaceable Unit
LTT	Light Triggering Thyristor
Mach	Ratio Of Speed Of An Object To Speed Of Sound (After Ernst Mach)
MLS	Microwave Landing Systems
MMIC	Monolithic Microwave Integrated Circuit
MOCVD	Metal Organic Chemical Vapour Deposition
MOSFET	Metal-Oxide-Semiconductor Field Effect Transistor
MPM	Microwave Power Module
MRF	Magnetorheological Finishing
MRF	Minimum Resolvable Feature Size
MRI	Magnetic Resonance Imaging
MTBF	Mean-Time-Between-Failures
Mtops	Million Theoretical Operations Per Second
MTTF	Mean-Time-To-Failure
NA	Numerical Aperture
NDT	Non-Destructive Test
NEQ	Net Explosive Quantity
NIJ	National Institute of Justice
OAM	Operations, Administration Or Maintenance
OSI	Open Systems Interconnection
PAI	Polyamide-Imides
PAR	Precision Approach Radar
PCL	Passive Coherent Location
PIN	Personal Identification Number
PMR	Private Mobile Radio
PPM	Parts Per Million
PVD	Physical Vapour Deposition
QAM	Quadrature-Amplitude-Modulation
RAP	Reactive Atom Plasmas
RF	Radio Frequency
RNC	Radio Network Controller
RNSS Regional Navigation Satellite System e.g. 'NavIC'	Indian Regional Navigation Satellite System 'QZSS' – Quasi Zenith Satellite System
RPV	Remotely Piloted Air Vehicles
S-FIL	Step And Flash Imprint Lithography
SAR	Synthetic Aperture Radar

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SAS	Synthetic Aperture Sonar
SC	Single Crystal
SCR	Silicon Controlled Rectifier
SFDR	Spurious Free Dynamic Range
SHPL	Super High Powered Laser
SLAR	Sidelooking Airborne Radar
SOI	Silicon-On-Insulator
SQUID	Superconducting Quantum Interference Device
SRA	Shop Replaceable Assembly
SRAM	Static Random Access Memory
SSB	Single Sideband
SSR	Secondary Surveillance Radar
SSS	Side Scan Sonar
TE-PVD	Thermal Evaporation-Physical Vapour Deposition
TIR	Total Indicated Reading
TVR	Transmitting Voltage Response
UTS	Ultimate Tensile Strength
VJFET	Vertical Junction Field Effect Transistor
VOR	Very High Frequency Omni-Directional Range
WLAN	Wireless Local Area Network

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Category 0: Nuclear materials, nuclear-related other materials, equipment and technology

Note Export of these items is regulated under the Atomic Energy Act, 1962 and rules framed, and notifications/orders issued there under from time-to-time by the Department of Atomic Energy. The licensing authority for items in this category is the Department of Atomic Energy. An application for licence to export shall be made in writing to the Joint Secretary (I&M), Department of Atomic Energy, Anushakti Bhavan, CSM Marg, Mumbai 400 001

0A PRESCRIBED SUBSTANCES

Note Any radioactive material in Category 0A shall additionally attract the provisions of Radiation Protection Rules, 2004 made under the Atomic Energy Act, 1962 and the provisions of Section-16 of the Atomic Energy Act, 1962.

0A1 Source Material

- 0A101** Uranium containing the mixture of isotopes occurring in nature.
- 0A102** Uranium depleted in the isotope 235.
- 0A103** Thorium.
- 0A104** Any of the materials specified above in 0A101, 0A102, or 0A103 in the form of metal, alloy, chemical compound, or concentrate.
- 0A105** Any other material containing one or more of the foregoing.

Note 1

Source material includes uranium and thorium ores or concentrates.

Note 2

Exports of following items, for the use only in non-nuclear activities, to a given recipient country, within a period of one calendar year, not exceeding the limits specified below, are not controlled:

- | | |
|---|-----------------|
| a. Uranium (containing the mixture of isotopes in nature): | 100 kilograms. |
| b. Depleted uranium (uranium depleted in the isotope 235 below that occurring in nature): | 1000 kilograms. |
| c. Thorium: | 1000 kilograms. |

Note 3 0A1 does not control following –

- i. Uranium and thorium ores, mineral concentrates or other materials that contain less than 300 parts per million (ppm) of uranium or/and thorium;
- ii. Alloys containing less than 5 % thorium;
- iii. Ceramic products containing thorium, which have been manufactured for non-nuclear use.

0A2 Special Fissionable Material

- 0A201** Plutonium-239.
- 0A202** Uranium-233.
- 0A203** Uranium enriched in the isotopes 235 or 233.
- 0A204** Neptunium.
- 0A205** Any material containing one or more of the foregoing.
- 0A206** Such other fissionable material determined by the Central Government from time to time.

Technical note

The term “uranium enriched in the isotopes 235 or 233” means uranium containing the isotopes 235 or 233 or both in an amount such that the abundance ratio of the sum of these isotopes to the

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isotope 238 is greater than the ratio of the isotope 235 to the isotope 238 occurring in nature.

Note

1. The term “special fissionable material” does not include source material.
2. Any quantity of special fissionable material is prescribed substance.
3. 0A2 does not control -
 - a. Plutonium with an isotopic concentration of plutonium-238 exceeding 80%, and
 - b. Special fissionable material when used in gram quantities or less as a sensing component in instruments.

0A3 Other Materials

‘Other Materials’ means non-nuclear materials for reactors, nuclear related dual-use materials indicated below and such materials as determined by the Central Government from time to time.

0A301 Deuterium and heavy water
Deuterium, heavy water (deuterium oxide) and any other deuterium compound, in which the ratio of deuterium to hydrogen atoms exceeds 1:5000,
for use in a nuclear reactor in quantities exceeding 5 kilograms of deuterium atoms in one consignment or 25 kilograms of deuterium atoms, for any one recipient country within a period of one calendar year.

0A302 Nuclear grade graphite
Nuclear grade graphite having a purity level better than 5 parts per million (ppm) boron equivalent and with a density greater than 1.5 gram/cc -
for use in a nuclear reactor or any other nuclear activities in quantities exceeding 1 kilogram.

Note 1 The item 0A302 does not cover graphite powder.

Note 2

For the purpose of export control, the Government will determine whether or not the exports of graphite meeting the above specifications are for nuclear reactor use. Graphite having a purity level better than 5 ppm (parts per million) boron equivalent and with a density greater than 1.50 g/cm³ not for use in a nuclear reactor or any other nuclear activities is not covered by this paragraph.

Boron Equivalent (BE) may be determined experimentally or is calculated as the sum of BE_Z for impurities (excluding BE_{carbon} since carbon is not considered an impurity) including boron, where: BE_Z ppm = CF x concentration of element Z (in ppm); B x A_Z); σ_Z x A_B) divided by (σ_{CF} is the conversion factor: (z are the thermal neutron capture cross sections (in barns) for naturally σ_B and σ occurring boron and element Z respectively; and A_B and A_Z are the atomic masses of naturally occurring boron and element Z respectively

0A303 Zirconium with hafnium content of less than 1 part to 500 parts of zirconium by weight (i.e. less than 2000 ppm) in the form of metal, alloys containing more than 50% zirconium by weight, compounds, manufactures thereof, waste or scrap of any of the foregoing.

0A304 Beryllium metal, its compounds, alloys containing more than 50% beryllium by weight, manufactures thereof, and waste or scrap of any of the foregoing and its minerals / concentrates including beryl but excluding:
a. beryllium windows used for x-ray machines or for bore-hole logging devices, and
b. beryl in the form of emerald, aquamarine or ‘cut & polished’ semi-precious stones for use in jewellery.

0A305 Lithium enriched in the Lithium-6 (⁶Li) isotope to greater than its natural isotopic abundance (i.e. more than 7.5%) and the products or devices containing enriched lithium such as elemental lithium, alloys, compounds, mixtures containing lithium, manufactures thereof, waste or scrap of any of the foregoing.

0A306 Niobium and Tantalum, their metals, alloys and minerals including columbite and tantalite.

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- 0A307** [Reserved]
- 0A308** Tritium, tritium compounds or mixtures containing tritium in which the ratio of tritium to hydrogen atoms exceeds 1 part in 1000, except when utilized in such quantities and for such purposes as for organic labelled compounds, Gas Filled Light Sources and as Tritiated Water for radiotracer studies.
- 0A309** Hafnium
Hafnium metal, alloys containing more than 60% hafnium by weight, hafnium compounds containing more than 60% hafnium by weight, manufactures thereof, and waste or scrap of any of the foregoing.
- 0A310** Radium-226
Radium-226 (²²⁶Ra), radium-226 alloys, radium-226 compounds, mixtures containing radium-226, manufactures thereof, and products or devices containing any of the foregoing, except medical applicators and a product or device containing less than 0.37 GBq (10mCi) of Ra-226 in any form.
- 0A311** Boron
Boron enriched in the Boron-10 (¹⁰B) isotope to greater than its natural isotopic abundance as follows:
Elemental boron, compounds, mixtures containing boron, manufactures thereof, waste or scrap of any of the foregoing.
- 0A312** Helium-3
Helium-3 (³He), mixtures containing helium-3, and products or devices containing any of the foregoing.
Note A product or device containing less than 1gm of Helium-3 is excluded.
- 0A313** ‘Radionuclides’ appropriate for making neutron sources based on alpha-n reaction, in the following forms:
- Elemental;
 - Compounds having a total activity of 37 GBq per kg or greater;
 - Mixtures having a total activity of 37 GBq per kg or greater;
 - Products or devices containing any of the foregoing.

Radionuclides controlled by this item include:

Actinium-225	Actinium-227	Californium-253
Curium-240	Curium-241	Curium-242
Curium-243	Curium-244	Einsteinium-253
Einsteinium-254	Gadolinium-148	Plutonium-236
Plutonium-238	Polonium-209	Polonium-210
Polonium-208	Radium-223	Thorium-228
Thorium-227	Uranium-230	Uranium-232

0B Prescribed Equipment

- 0B001** Nuclear Reactors; associated equipment, components, and systems especially designed, prepared, or adapted or used or intended to be used in such reactors including but not limited to:-
- Complete nuclear reactors
 - Nuclear reactor vessels
 - Nuclear reactor fuel charging and discharging machines
 - Nuclear reactor control rods and equipment
 - Nuclear reactor pressure tubes
 - Nuclear fuel cladding: Zirconium metal tubes or zirconium alloy tubes (or assemblies of

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- tubes), in which hafnium to zirconium ratio is 1:500 or less, for use as nuclear fuel cladding
- g. Primary coolant pumps or circulators
- h. Nuclear reactor internals
- i. Heat exchangers (steam generators) for use in the primary or intermediate coolant circuit of a nuclear reactor
- j. Neutron detectors
- k. External thermal shields.

0B002

Plants for processing, production, concentration, conversion or recovery of Prescribed Substances (such as uranium, plutonium, thorium, deuterium, heavy water, tritium, lithium); associated equipment, components and systems especially designed, prepared or adapted or used or intended to be used in such plants including but not limited to:

- a. Plants for production or concentration of deuterium, heavy water or deuterium compounds-
 - 1. Water - Hydrogen Sulphide Exchange Towers with diameters of 1.5 m or greater and capable of operating at pressures greater than or equal to 2 MPa (300 psi), especially designed or prepared for heavy water production.
 - 2. Especially designed or prepared blowers and compressors for hydrogen-sulphide gas circulation. These blowers or compressors have a throughput capacity greater than or equal to 56 m³/second (120,000 SCFM) while operating at pressures greater than or equal to 1.8 MPa (260 psi) suction and have seals designed for wet H₂S service
 - 3. Ammonia-Hydrogen Exchange Towers greater than or equal to 35 m in height with diameters of 1.5 m to 2.5 m capable of operating at pressures greater than 15 MPa especially designed or prepared for heavy water production
 - 4. Tower Internals and Stage Pumps: Tower internals and stage pumps especially designed or prepared for heavy water production. Tower internals include especially designed stage contactors which promote intimate gas/liquid contact. Stage pumps include especially designed submersible pumps for circulation of liquid ammonia within a contacting stage internal to the stage towers.
 - 5. Ammonia Crackers with operating pressures greater than or equal to 3 MPa especially designed or prepared for heavy water production.
 - 6. Infrared Absorption Analyzers capable of 'on-line' hydrogen/deuterium ratio analysis
 - 7. Catalytic Burners for conversion of enriched deuterium gas into heavy water
 - 8. Complete heavy water upgrade systems or columns therefor
 - 9. Ammonia synthesis converters or synthesis units for heavy water production utilizing the ammonia-hydrogen exchange process.
- b. Plants for the conversion of uranium
 - 1. Systems for the conversion of uranium ore concentrates to UO₃;
 - 2. Systems for the conversion of UO₃ to UF₆;
 - 3. Systems for the conversion of UO₃ to UO₂;
 - 4. Systems for the conversion of UO₂ to UF₄;
 - 5. Systems for the conversion of UF₄ to UF₆;
 - 6. Systems for the conversion of UF₄ to uranium metal;
 - 7. Systems for the conversion of UF₆ to UO₂;
 - 8. Systems for the conversion of UF₆ to UF₄;
 - 9. Systems for the conversion of UO₂ to UCl₄.
- c. Plants for the conversion of plutonium
 - 1. Systems for the conversion of plutonium nitrate to oxide
 - 2. Systems for plutonium metal production
- d. Tritium facilities or plants, and equipment therefor, as follows:
 - 1. Facilities or plants for the production, recovery, extraction, concentration or handling of tritium;
 - 2. Equipment for tritium facilities or plants, as follows:
 - i. Hydrogen or helium refrigeration units capable of cooling to 23 K (-250 °C) or less, with heat removal capacity greater than 150 W;
 - ii. Hydrogen isotope storage or hydrogen isotope purification systems using metal hydrides as the storage or purification medium.

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- e. Lithium isotope separation facilities or plants, and systems and equipment therefor as follows -
 - 1. Facilities or plants for the separation of lithium isotopes;
 - 2. Equipment for the separation of lithium isotopes based on the lithium-mercury amalgam process, as follows:
 - a) Packed liquid-liquid exchange columns especially designed for lithium amalgams;
 - b) Mercury or lithium amalgam pumps;
 - c) Lithium amalgam electrolysis cells;
 - d) Evaporators for concentrated lithium hydroxide solution;
 - 3. Ion exchange systems especially designed for lithium isotope separation, and especially designed component parts therefor;
 - 4. Chemical exchange systems (employing crown ethers, cryptands, or lariat ethers) especially designed for lithium isotope separation, and especially designed component parts therefor.

0B003 Plants for reprocessing of irradiated nuclear fuel and equipment, components and systems especially designed, prepared or adapted or used or intended to be used in such plants, including but not limited to:

- a. Irradiated fuel element decladding equipment and chopping machines: Remotely operated equipment especially designed or prepared for use in a reprocessing plant as identified above and intended to expose or prepare the irradiated nuclear material in fuel assemblies, bundles or rods for processing.
- b. Dissolvers: Dissolver vessels or dissolvers employing mechanical devices especially designed or prepared for use in a reprocessing plant as identified above, intended for dissolution of irradiated nuclear fuel and which are capable of withstanding hot, highly corrosive liquid, and which can be remotely loaded, operated, and maintained.
- c. Solvent extractors and solvent extraction equipment: Especially designed or prepared solvent extractors (such as packed or pulse columns, mixer settlers or centrifugal contactors) for use in a plant for the reprocessing of irradiated fuel. Solvent extractors must be resistant to the corrosive effect of nitric acid. Solvent extractors are normally fabricated to extremely high standards (special welding and inspection and quality assurance and quality control techniques) out of low carbon stainless steels, titanium, zirconium, or other high quality materials.
- d. Chemical holding or storage vessels: Especially designed or prepared holding or storage vessels for use in a plant for the reprocessing of irradiated fuel. The holding or storage vessels must be resistant to the corrosive effect of nitric acid. The holding or storage vessels are normally fabricated of materials such as low carbon stainless steels, titanium or zirconium, or other high quality materials. Holding or storage vessels may be designed for remote operation and maintenance and may have the following features for control of nuclear criticality:
 - 1. Walls or internal structures with a boron equivalent of at least 2%;
 - 2. A maximum diameter of 175 mm for cylindrical vessels; or
 - 3. A maximum width of 75 mm for either a slab or annular vessel.
- e. Neutron measurement systems for process control Neutron measurement systems especially designed or prepared for integration and use with automated process control systems in a plant for the reprocessing of irradiated fuel elements.

0B004 Plants for treatment, handling, storage and transportation of radioactive wastes from nuclear reactors or from plants for processing Source Materials or Special Fissionable Materials or from nuclear reprocessing plants; irradiated nuclear fuel; Special Fissionable Materials, and equipment especially designed, prepared, adapted, or intended to be used therefor.

0B005 All systems, associated equipment, components for separation or enrichment of isotopes of uranium, plutonium, lithium, boron or other elements, other than analytical instruments, especially designed, prepared, adapted, used or intended to be used therefor as follows:

- a. Gas centrifuges and assemblies and components especially designed or prepared for use in gas Centrifuges

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1. Gas centrifuges;
2. Complete rotor assemblies; Thin-walled cylinders, or a number of interconnected thin-walled cylinders, manufactured from one or more of the high strength-to-density ratio materials described in the Note-1 in 0B005.a. If interconnected, the cylinders are joined together by flexible bellows or rings as described in 0B005.a.4. The rotor is fitted with an internal baffle(s) and end caps, as described in 0B005.a.5 and 0B005.a.6.
3. Rotor tube cylinders: Especially designed or prepared thin-walled cylinders with thickness of 12 mm or less, a diameter of between 75 mm and 650 mm, and manufactured from one or more of 'high strength-to-density ratio materials' described in the Note-1 in 0B005.a;
4. Rings or bellows: Rings or bellows with wall thickness of 3 mm or less and a diameter of between 75 mm and 650 mm especially designed to give local support to a rotor tube or to join together a number of rotor tubes, made from 'high strength-to-density ratio materials' described in the Note-1 in 0B005.a.
5. Baffles: Disc-shaped components of between 75 mm and 650 mm diameter especially designed or prepared for mounting inside a rotor tube, in order to isolate the take-off chamber from the main separation chamber and manufactured from 'high strength-to-density ratio materials' described in the Note-1 in 0B005.a.
6. Top or bottom caps: Especially designed or prepared disc-shaped components of between 75 mm and 400 mm diameter especially designed or prepared to fit the ends of a rotor tube, and so contain the UF₆ within the rotor tube, and in some cases to support, retain or contain as an integrated part an element of the upper bearing (top cap) or to carry the rotating elements of the motor and lower bearing (bottom cap), and manufactured from 'high strength-to-density ratio materials' described in the Note-1 in 0B005.a;
7. Especially prepared Magnetic Suspension Bearings with both of the following attributes:
 - a. Bearing assemblies consisting of an annular magnet suspended within a housing made of or protected by "materials resistant to corrosion by UF₆" (see Note 3 of 0B005) containing a damping medium and having the magnet coupling with a pole piece or second magnet fitted to the top cap of the rotor;
 - b. Active magnetic bearings especially designed or prepared for use with gas centrifuges. These bearings usually have the following characteristics: i) Designed to keep centred a rotor spinning at 600 Hz or more; and ii) Associated to a reliable electrical power supply and/or to an uninterruptible power supply (UPS) unit in order to function for more than one hour.
8. Bearings / Dampers: Especially designed or prepared bearings comprising a pivot/cup assembly mounted on a damper. The pivot is normally a hardened steel shaft with a hemisphere at one end with a means of attachment to the bottom cap described in 0B005.a.6 at the other. The shaft may however have a hydrodynamic bearing attached. The cup is pellet-shaped with a hemispherical indentation in one surface. These components are often supplied separately to the damper.
9. Molecular pumps: Molecular pumps are high vacuum pumps consisting of especially designed or prepared cylinders having internally machined or extruded helical grooves and internally machined bores. Typical dimensions are as follows: 75 mm to 650 mm internal diameter, 10 mm or more wall thickness, with the length equal to or greater than the diameter. The grooves are typically rectangular in cross-section and 2 mm or more in depth.
10. Ring-shaped motor stators: Especially designed or prepared ring-shaped stators for high speed multiphase AC hysteresis (or reluctance) motors for synchronous operation within a vacuum at a frequency of 600 Hz or greater and a power of 40 VA or greater. The stators may consist of multi-phase windings on a laminated low loss iron core comprised of thin layers typically 2.0 mm thick or less.
11. Centrifuge housing/recipients to contain the rotor tube assembly of a gas centrifuge consisting of rigid cylinder of wall thickness up to 30 mm with precision machined ends that are parallel to each other and perpendicular to the cylinder's longitudinal axis to within 0,05 degrees or less.
12. Scoops consisting of tubes for the extraction of UF₆ gas from within the rotor tube by a Pitot tube action and capable of being fixed to the central gas extraction system.

Note 1 The high strength-to-density ratio materials used for centrifuge rotating components include the following:

- (a) Maraging steel capable of an ultimate tensile strength of 1.95 GPa or more;
- (b) Aluminium alloys capable of an ultimate tensile strength of 0.46 GPa or more;
- (c) Filamentary materials suitable for use in composite structures and having a specific modulus of 3.18×10^6 m or greater and a specific ultimate tensile strength of 7.62×10^4 m or greater.

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Note 2 'Specific Modulus' is the Young's Modulus in N/m^2 divided by the specific weight in N/m^3 ; 'Specific Ultimate Tensile Strength' is the ultimate tensile strength in N/m^2 divided by the specific weight in N/m^3 .

b. Especially designed or prepared auxiliary systems, equipment and components for gas centrifuge enrichment plants

1. Machine header piping systems for handling UF_6 within the centrifuge cascades;
2. Frequency changers (converters or inverters) especially designed or prepared to supply motor stators for gas centrifuge enrichment, having all of the following characteristics, and especially designed components therefor:
 - a. A multiphase frequency output of 600 Hz or greater; and
 - b. High stability (with frequency control better than 0.2 %).

c. Especially designed or prepared assemblies and components for use in gaseous diffusion enrichment

1. Gaseous diffusion barriers and barrier materials resistant to corrosion by UF_6 described in the Note-3 in 0B005;
2. Gaseous diffuser housings made of or protected by materials resistant to corrosion by UF_6 described in the Note-3 in 0B005;
3. Compressors (positive displacement, centrifugal and axial flow types) or gas blowers with a suction volume capacity of $1\text{ m}^3/\text{min}$ or more of UF_6 , discharge pressure up to 500 kPa and having a pressure ratio of 10:1 or less designed for long term operation in the UF_6 environment and made of or protected by materials resistant to corrosion by UF_6 described in the Note-3 in 0B005.

d. Especially designed or prepared auxiliary systems, equipment and components for use in gaseous diffusion enrichment:

Piping systems and header systems for handling UF_6 within the gaseous diffusion cascades.

e. Especially designed or prepared systems, equipment and components for use in aerodynamic enrichment plants:

1. Especially designed or prepared separation nozzles and assemblies thereof. The separation nozzles consist of slit-shaped, curved channels having a radius of curvature less than 1 mm, made of materials resistant to corrosion by UF_6 described in the Note-3 in 0B005 and having a knife-edge within the nozzle that separates the gas flowing through the nozzle into two fractions;
2. Especially designed or prepared vortex tubes and assemblies thereof. The vortex tubes are cylindrical or tapered, made of or protected by materials resistant to corrosion by UF_6 described in the Note-3 in 0B005 and with one or more tangential inlets. The tubes may be equipped with nozzle type appendages at either or both ends;
3. Especially designed or prepared compressors or gas-blowers made of or protected by materials resistant to corrosion by the UF_6 (see the Note-3 in 0B005) / carrier gas (hydrogen or helium) mixture;
4. Especially designed or prepared separation element housings made of or protected by materials resistant to corrosion by UF_6 described in the Note-3 in 0B005, for containing vortex tubes or separation nozzles;
5. Especially designed or prepared header-piping systems, made of or protected by materials resistant to corrosion by UF_6 described in the Note-3 in 0B005, for handling UF_6 within the aerodynamic cascades;
6. UF_6 /carrier gas separation systems for separating UF_6 from carrier gas (hydrogen or helium).

f. Especially designed or prepared systems, equipment and components for use in chemical exchange or ion exchange enrichment plants.

1. Countercurrent Liquid-liquid exchange columns (Chemical exchange), having mechanical power input, especially designed or prepared for uranium enrichment using the chemical exchange process. For corrosion resistance to concentrated hydrochloric acid solutions, these columns and their internals are normally made of or protected by materials resistant to corrosion by concentrated hydrochloric acid solutions. The stage residence time of the columns is normally designed to be 30 seconds or less.

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2. Liquid-liquid centrifugal contactors (Chemical exchange), especially designed or prepared for uranium enrichment using the chemical exchange process. Such contractors are made of or protected by materials resistant to corrosion by concentrated hydrochloric acid solutions. The stage residence time of the columns is normally designed to be 30 seconds or less.
3. Uranium reduction systems and equipment (Chemical exchange):
 - a. Especially designed or prepared electrochemical reduction cells to reduce uranium from one valence state to another for uranium enrichment using the chemical exchange process. The cell materials in contact with process solutions must be corrosion resistant to concentrated hydrochloric acid solutions;
 - b. Especially designed or prepared systems consisting of solvent extraction equipment and pumps or other transfer devices at the product end of the cascade for taking the U^{+4} out of the organic stream.
4. Feed preparation systems (Chemical exchange) consisting of dissolution, solvent extraction and/or ion exchange equipment for producing high-purity uranium chloride.
5. Uranium oxidation systems (Chemical exchange) for oxidation of U^{+3} to U^{+4}
6. Fast-reacting ion exchange resins/adsorbents (Ion exchange):

Fast-reacting ion-exchange resins or adsorbents, especially designed or prepared for uranium enrichment using the chemical exchange process, including porous macroreticular resins, and/or pellicular structures and other composite structures in any suitable form including particles or fibres chemically resistant to concentrated hydrochloric acid solutions.
7. Ion exchange columns (Ion exchange):

Cylindrical columns for containing and supporting packed beds of ion exchange resin/adsorbent and made of or protected by materials resistant to corrosion by concentrated hydrochloric acid solutions.
8. Ion exchange reflux systems (Ion exchange):

Chemical or electrochemical oxidation or reduction systems for regeneration of the chemical oxidizing or reducing agent(s) used in ion exchange enrichment cascades.
- g. Especially designed or prepared systems, equipment and components for use in laser-based enrichment plants.
 1. Uranium vaporization systems (atomic vapour based methods)
 2. Liquid or vapour uranium metal handling systems and components (atomic vapour based methods)
 3. Uranium metal 'product' and 'tails' collector assemblies (atomic vapour based methods)
 4. Separator module housings (atomic vapour based methods)
 5. Supersonic expansion nozzles (molecular based methods)
 6. 'Product' or 'tails' collectors (molecular based methods)
 7. UF_6 /carrier gas compressors (molecular based methods)
 8. Rotary shaft seals (molecular based methods)
 9. Fluorination systems (molecular based methods)
 10. UF_6 /carrier gas separation systems (molecular based methods)
 11. 'Lasers' or 'laser systems or components' for the separation of uranium isotopes.
- h. Especially designed or prepared systems, equipment and components for use in plasma separation enrichment plants
 1. Microwave power sources and antennae: Especially designed or prepared microwave power sources and antennae for producing or accelerating ions and having the following characteristics: greater than 30 GHz frequency and greater than 50 kW mean power output for ion production.
 2. Radio frequency ion excitation coils for frequencies of more than 100 kHz
 3. Uranium plasma generation systems
 4. Uranium metal 'product' and 'tails' collector assemblies made of or protected by materials resistant to the heat and corrosion of uranium metal vapour.
 5. Separator module housings (cylindrical vessels) for containing the uranium plasma source, radio-frequency drive coil and the 'product' and 'tails' collectors.

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i. Especially designed or prepared systems, equipment and components for use in electromagnetic enrichment plants.

1. Electromagnetic isotope separators for separation of uranium isotopes and equipment and components therefor, including ion sources (consisting of a vapour source, ionizer, and beam accelerator), ion collectors (consisting of collector plates), vacuum housings and magnet pole pieces;
2. High voltage power supplies for ion sources: Especially designed or prepared high-voltage power supplies for ion sources, having all of the following characteristics: capable of continuous operation, output voltage of 20,000 V or greater, output current of 1 A or greater, and voltage regulation of better than 0.01% over a time period of 8 hours
3. High-power, direct current magnet power supplies: Especially designed or prepared high-power, direct current magnet power supplies having all of the following characteristics: capable of continuously producing a current output of 500 A or greater at a voltage of 100 V or greater and with a current or voltage regulation better than 0.01% over a period of 8 hours.

j. Especially designed or prepared other equipment and components for use in enrichment plants:

1. Feed systems / product and tails withdrawal systems such as feed autoclaves, ovens, or systems, desublimers, cold traps or pumps, solidification or liquefaction stations, 'product' or 'tails' stations used for handling UF₆;
2. Special shut-off valves, control valves, bellow sealed valves, manual or automated, shut-off or control, made of or protected by materials resistant to corrosion by UF₆;
3. UF₆ mass spectrometers / ion sources capable of taking on-line samples from UF₆ gas stream; ;
4. Rotary shaft seals for compressors or blowers;
5. Heat exchangers made of or protected by "materials resistant to corrosion by UF₆";
6. Vacuum systems including vacuum manifolds, vacuum headers and vacuum pumps made of, or protected by, materials resistant to corrosion by UF₆.

Notes to 0B005:

1: Controls under 0B005 also apply to the plants and equipment that are intended for isotope separation of other elements.

2: "Other elements" means all elements other than hydrogen, uranium and plutonium.

3: Materials resistant to corrosion by UF₆ include copper, copper alloys, stainless steel, aluminium, aluminium oxide, aluminium alloys, nickel or alloys containing 60% or more nickel and fluorinated hydrocarbon polymers.

0B006 Plants for the fabrication of nuclear reactor fuel elements, and equipment especially designed or prepared therefor including but not limited to:

- a. fully automatic pellet inspection stations especially designed or prepared for checking final dimensions and surface defects of the fuel pellets;
- b. automatic welding machines especially designed or prepared for welding end caps onto the fuel pins (or rods);
- c. automatic test and inspection stations especially designed or prepared for checking the integrity of completed fuel pins (or rods);
- d. systems especially designed or prepared to manufacture nuclear fuel cladding.

Item 'c' typically includes equipment for: 1) x-ray examination of pin (or rod) end cap welds, 2) helium leak detection from pressurized pins (or rods), and 3) gamma-ray scanning of the pins (or rods) to check for correct loading of the fuel pellets inside.

0B007 Plants or systems for production, handling, storage and transportation of Radioisotopes in quantities exceeding 100 Curies (3.7 X 10¹² Becquerel).

0B008 Neutron generators including neutron chain reacting assemblies and fusion assemblies of all kinds for producing fissile materials.

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0C Technology and software

Technology and software for the development, production or use of prescribed substances or prescribed equipment specified in 0A or 0B. ”

Category 1 Toxic chemical agents and other chemicals

1A Export of the following chemicals, related technology and software is prohibited:

(This corresponds to Schedule 1 to the Chemical Weapons Convention (CWC))

Note: Where reference is made below to groups of di-alkylated chemicals, followed by a list of alkyl groups in parentheses, all chemicals possible by all possible combinations and alkyl groups listed in parentheses are considered prohibited unless explicitly exempted.

Sl. No.	Schedule 1 Toxic Chemicals of CWC	CAS number
(1)	O-Alkyl (<C10 , incl. cycloalkyl) alkyl1 (Me, Et,n-Pr or i-Pr) phosphonofluoridates	
	e.g. Sarin: O-Isopropyl methylphosphonofluoridate	(107-44-8)
	Soman: O-Pinacolyl methylphosphonofluoridate	(96-64-0)
(2)	O-Alkyl, (<C10, incl. cycloalkyl) N,N-dialkyl1 (Me, Et, n-Pr or i-Pr) phosphoramidocyanidates	
	e.g. Tabun: O-Ethyl N,N-dimethyl phosphoramidocyanidate	(77-81-6)
(3)	O-Alkyl (H or < C10, incl. cycloalkyl) S-2-Dialkyl (Me, Et, n-Pr or i -Pr)-aminoethyl alkyl (Me, Et, n-Pr or i-Pr) phosphonothiolates and corresponding alkylated or protonated salts	
	e.g. VX: O-Ethyl S- 2 diisopropylaminoethyl methyl phosphonothiolate	(50782-69-9)
(4)	Sulphur mustards:	
	2-Chloroethylchloromethylsulphide	(2625-76-5)
	Mustard gas: Bis (2-chloroethyl) sulphide	(505-60-2)
	Bis (2-chloroethylthio) methane	(63869-13-6)
	Sesquimustard:1,2-Bis (2-chloroethylthio) ethane	(3563-36-8)
	1,3-Bis (2-chloroethylthio)-n-propane	(63905-10-2)
	1,4-Bis (2-chloroethylthio)-n-butane	(142868-93-7)
	1,5-Bis (2-chloroethylthio)-n-Pentane	(142868-94-8)
	Bis (2-Chloroethylthiomethyl) ether	(63918-90-1)
	O-Mustard: Bis (2-Chloroethylthiomethyl) ether	(63918-89-8)
(5)	Lewisites:	
	Lewisite 1: 2-Chlorovinylchloroarsine	(541-25-3)
	Lewisite 2: Bis (2-Chlorovinyl) chloroarsine	(40334-69-8)
	Lewisite 3: Tris (2-Chlorovinyl) arsine	(40334-70-1)
(6)	Nitrogen mustards:	
	HN1: Bis (2-chloroethyl) ethylamine	(538-07-8)
	HN2: Bis (2-chloroethyl) methylamine	(51-75-2)
	HN3: Tris (2-chloroethyl) amine	(555-77-1)
(7)	Saxitoxin	(35523-89-8)
(8)	Ricin	(9009-86-3)
(13)	P-alkyl (H or ≤C10, incl. cycloalkyl) N-(1-(dialkyl(≤C10, incl. cycloalkyl)amino)alkylidene(H or ≤C10, incl. cycloalkyl) phosphonamidic fluorides and corresponding alkylated or protonated salts	
	e.g. N-(1-(di-n-decylamino)-n-decylidene)-P-decylphosphonamidic fluoride	(2387495-99-8)
	Methyl-(1-(diethylamino)ethylidene)phosphonamidofluoridate	(2387496-12-8)
(14)	O-alkyl (H or ≤C10, incl. cycloalkyl) N-(1-(dialkyl(≤C10, incl. cycloalkyl)amino)alkylidene(H or ≤C10, incl. cycloalkyl) phosphoramidofluoridates and corresponding alkylated or protonated salts	
	e.g. O-n-Decyl N-(1-(di-n-decylamino)-n-decylidene)phosphoramidofluoridate	(2387496-00-4)
	Methyl (1-(diethylamino)ethylidene)phosphoramidofluoridate	(2387496-04-8)
	Ethyl (1-(diethylamino)ethylidene)phosphoramidofluoridate	(2387496-06-0)

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(15)	Methyl-(bis(diethylamino)methylene)phosphonamidofluoridate	(2387496-14-0)
(16)	Carbamates (quaternaries and bisquaternaries of dimethylcarbamoyloxypyridines) Quaternaries of dimethylcarbamoyloxypyridines: 1-[N,N-dialkyl($\leq C10$)-N-(n-(hydroxyl, cyano, acetoxy)alkyl($\leq C10$)) ammonio]-n-[N-(3-dimethylcarbamoxy-α-picolinyl)-N,N-dialkyl($\leq C10$) ammonio]decane dibromide (n=1-8) e.g. 1-[N,N-dimethyl-N-(2-hydroxy)ethylammonio]-10-[N-(3-dimethylcarbamoxy-α-picolinyl)-N,N-dimethylammonio]decane dibromide Bisquaternaries of dimethylcarbamoyloxypyridines: 1,n-Bis[N-(3-dimethylcarbamoxy-α-picolyl)-N,N-dialkyl($\leq C10$) ammonio]-alkane-(2,(n-1)-dione) dibromide (n=2-12) e.g. 1,10-Bis[N-(3-dimethylcarbamoxy-α-picolyl)-N-ethyl-N-methylammonio]decane-2,9-dione dibromide	(77104-62-2)
		(77104-00-8)
Sl. No.	Schedule 1 CWC Precursors	CAS number
(9)	Alkyl (Me, Et, n-Pr or i-Pr) phosphonyldifluorides	
	e.g. DF: Methyl phosphonyldifluoride	(676-99-3)
(10)	O-Alkyl (H or $< C10$, incl. cycloalkyl) O-2 dialkyl (Me, Et, n-Pr or i-Pr)-aminoethylalkyl (Me, Et N-Pr or i -Pr) phosphonites and corresponding alkylated or protonated salts	
	e.g. QL: O-Ethyl O-2-diisopropylaminoethyl methyl phosphonite	(57856-11-8)
(11)	Chlorosarin: O-Isopropyl methylphosphonochloridate	(1445-76-7)
(12)	Chlorosoman: O-Pinacolyl methylphosphonochloridate	(7040-57-5)*

Note⁷: Notwithstanding above, export of remnant of chemicals, after testing / analysis in the OPCW designated laboratories, falling under Category 1A, will be allowed by way of being taken back by the officials of Organisation for the Prohibition of Chemical Weapons (OPCW) to the OPCW Secretariat or any designated authority. Such export shall be subject to the condition that for each export, officials of OPCW shall notify the details, prior to actual export, to the National Authority, Chemical Weapons Convention, Cabinet Secretariat, Ministry of External Affairs (D&ISA) and the Directorate General of Foreign Trade.

1B Export of chemicals listed in 1B below is permitted only to States party to the Chemical Weapons Convention, against an export authorisation

(This corresponds to Schedule 2 to the Chemicals Weapons Convention)

Note to exporters:

- A list of States Parties can be found at the official website of the Organization for the Prohibition of Chemical Weapons at www.opcw.org.
- A prior authorization will be required for export of chemicals in this category. This permission shall be subject to the condition that for each export consignment, exporters shall, within 30 days of exports, notify the details to the National Authority, Chemical Weapons Convention, Cabinet Secretariat; Ministry of External Affairs (D&ISA); Department of Chemicals and Petrochemicals and the Directorate General of Foreign Trade and submit to DGFT, a copy of Bill of Entry into the destination State Party within 30 days of delivery’.

Note: Where reference is made below to groups of dialkylated chemicals, followed by a list of alkyl groups in parentheses, all chemicals possible by all possible combinations and alkyl groups listed in parentheses are included unless explicitly exempted.

⁷ Notification No. 34 dated 25.09.2018

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Sl. No.	Schedule 2 CWC Precursors	CAS number
(1)	Amiton 0,0-Diethyl S-[2-(diethylamino) ethyl] phosphorothiolate and corresponding alkylated or protonated salts	(78-53-5)
(2)	PFIB: 1,1,3,3,3-Pentafluoro-2-(trifluoromethyl)1-propene	(382-21-8)
(3)	BZ: 3-Quinuclidinyl benzilate	(6581-06-2)
(4)	Chemicals, except for those listed in Schedule 1, containing a phosphorus atom to which is bonded one methyl, ethyl or propyl (normal or iso) group but not further carbon atoms	
	e.g. Methylphosphonyl dichloride	(676-97-1)
	Dimethyl methylphosphonate	(756-79-6)
	Exemption:- Fonofos: O-Ethyl S-phenyl ethylphosphonothiolothionate	(944-22-9)
(5)	N, N-Dialkyl (ME, Et, n-Pr or i-Pr) phosphoramidic dihalides	
(6)	Dialkyl (Me, Et, n- Pr or i-Pr) N, N- dialkyl (Me, Et, n-Pr or i-Pr)-phosphoramidates	
(7)	Arsenic trichloride	(7784-34-1)
(8)	2,2-Diphenyl-2 hydroxyacetic acid	(76-93-7)
(9)	Quinuclidine-3- ol	(1619-34-7)
(10)	N,H-Dialkyl (Me, Et, n-Pr or i-Pr) aminoethyl-2 -chlorides and corresponding protonated salts	
(11)	N, N-Dialkyl (Me, Et, n-Pr or i-Pr) aminoethane-2-ols and corresponding protonated salts	
	Exemptions: N,N-Dimethylaminoethanol and corresponding protonated salts	(108-01-0)
	N,N-Diethylaminoethanol and corresponding protonated salts	(100-37-8)
(12)	N, N-Dialkyl (Me, Et, n-Pr or i-Pr) aminoethane-2-thiols and corresponding protonated salts	
(13)	Thiodiglycol: Bis(2-hydroxyethyl) sulphide	(111-07-3)
(14)	Pinacolyl alcohol: 3,3-Dimethylbutane-2-ol	(464-07-3)'

A list of commercially important Schedule-2 Chemicals of CWC is given below:

Sl.No.	SCOMET Entry	Name of Chemical	Entry into Schedule	CAS number	ITC (HS) code
1	1B001	2-Chloro N, N -Di-isopropyl ethylamine	2B10	(96-79-7)	29211910
2	1B002	2-(N,N-Diethylamino)ethanethiol	2B12	(100-38-9)	29306000
3	1B003	Dimethyl Methylphosphonate	2B04	(756-79-6)	29314100
4	1B004	2-(N,N-Diisopropylamino)ethanol	2B11	(96-80-0)	29221800
5	1B005	N, N-Diethyl Amino ethyl Chloride Hydrochloride	2B10	(869-24-9)	29211300
6	1B006	Di-ethyl Amino ethanethiol Hydrochloride	2B12	(1942-52-5)	29309095
7	1B007	Di-Methyl Amino ethyl chloride Hydrochloride	2B10	(4584-46- 7)	29211200
8	1B008	Di-Methyl Amino ethanethiol	2B12	(108-02-1)	29309092
9	1B009	Di-Methyl Amino ethanethiol Hydrochloride	2B12	(13242-44- 9)	29309093
10	1B010	Phosphorothioic acid, S [2-(diethylamino) ethyl]	2A01	(78-53-5)	29309097
		O, O – diethyl ester			
11	1B011	1-Propene, 1,1, 3, 3, 3, - Pentafluoro – 2-(trifluoromethyl) (PFIB)	2A02	(382-21-8)	29035910

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12	1B012	phenyl, 1 – azabicyclo [2.2.2.] oct-3-yl ester	2A03	(1709855)	29333940
13	1B013	Phosphonic Acid, Methyl- compound with (aminoimino methyl) urea (1: 1)	2B04	(84402-58-4)	29314500
14	1B014	1-Propanaminium N, N, N-trimethyl – 3-[1- oxo-9 octadecenyl) amino]-. (Z)-methyl methylphosphonate	2B04	(70055-71-9)	29313900
15	1B015	Phosphonic acid, [methyl bis (5-ethyl-2- methyl-2-oxido-1, 3, 2- dioxaphosphorinan- 5-yl) methyl] ester	2B04	(42595-45-9)	29314920
16	1B016	Phosphonic acid, [methyl-(5-ethyl-2- methyl 2-oxido-1,3,2-dioxaphosphorinan- 5-yl) methyl] ester	2B04	(41203-81-0)	29314700
17	1B017	Phosphonic acid, propyl-dimethyl ester	2B04	(18755-43- 6)	29314200
18	1B018	Phosphonous acid, methyl-diethyl ester	2B04	(15715-41-0)	29313900
19	1B019	Phosphonic acid, ethyl-	2B04	(1782267)	29313900
20	1B020	Phosphonic acid, propyl-	2B04	(4672-38-2)	29313900
21	1B021	Phosphinic acid, methyl-	2B04	(4206-94-4)	29313900
22	1B022	Phosphonochloridic acid, methyl-, methyl ester	2B04	(1066-52- 0)	29313900
23	1B023	Phosphonothioic dichloride, ethyl-	2B04	(993-43-1)	29313900
24	1B024	Phosphonic acid methyl-	2B04	(993-13-5)	29314400
25	1B025	‘(Reserved)’	-	-	-
26	1B026	Phosphonic dichloride, methyl-	2B04	(676-97-1)	29315100
27	1B027	Phosphonous dichloride, methyl -	2B04	(676-83- 5)	29313900
28	1B028	Phosphonic acid, ethyl-, diethyl ester	2B04	(78-38-6)	29314300
29	1B029	Arsenous trichloride	2B07	(7784-34-1)	28121930
30	1B030	Benzeneacetic acid, alpha-hydroxy-alpha-phenyl	2B08	(76-93-7)	29333940
31	1B031	1-Azabicyclo (2.2.2.) octan-3-ol	2B09	(1619-34-7)	29333500
32	1B032	Ethanamine, 2-Chloro-N, N-dimethyl-	2B10	(107-99-3)	29211920
33	1B033	‘(Reserved)’	-	-	-
34	1B034	‘(Reserved)’	-	-	-
35	1B035	Ethanol, 2, 2’-thiobis-	2B13	(111-48-8)	29309091 29307000
36	1B036	2-Butanol, 3, 3-dimethyl-	2B14	(464-07-3)	29051910
37	1B037	Diphenyl Methyl Phosphonate	2B04	(7526-26-3)	29313900
38	1B038	2-(N,N-Diethylamino)ethylchloride	2B10	(100-35-6)	29211990
39	1B039	Mixture of (5-ethyl-2-methyl-2-oxido-1,3,2-dioxaphosphinan-5-yl)methyl methyl methylphosphonate (CAS RN 41203-81-0) and Bis[(5-Ethyl-2-methyl-2- oxido-1,3,2-dioxaphosphinan-5-yl)methyl] methylphosphonate (CAS RN 42595-45-9)	2B10	(170836-68-7)	38249100
40	1B040	3-Quinuclidinyl benzilate	2A03	(6581-06-2)	29333940
41	1B041	Others	-	-	-

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1C⁸ Export of chemicals (Excluding Software and Technology) listed in 1C below is allowed to the countries specified in Table 1 on the basis of a one time General authorization for export of Chemicals and related equipments (GAEC) issued by DGFT, subject to the following conditions(including those below) and the procedure as prescribed from time to time:

- I. The exporter is required to register and obtain General authorization for export of Chemicals and related equipments only once during the validity period. Subsequent export/re-export is subject to post reporting(s) on quarterly basis to relevant Govt. authorities;
- II. General authorization for export of Chemicals and related equipments issued for export / re-export of SCOMET items under the above Categories / Sub Categories (excluding software and technology) shall be valid for a period of **five years** from the date of issue of General authorization for export of Chemicals and related equipments subject to subsequent post reporting(s) on quarterly basis to be reported within 30 days from the last quarter;

Table 1

Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Republic of Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Republic of Korea, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom, United States.
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Note : Export of items under the 1C Category to the countries other than those specified in Table 1 and export of 1C category items related Software and Technology to any country shall be allowed only against an export authorisation and in compliance to the conditions and documentation specified under the respective Categories of SCOMET List. Further, applications for export of chemicals in this category to countries not Party to the Chemical Weapons Convention (CWC) shall be submitted along with a Government signed End-Use-Certificate (EUC).

List of 1C chemicals:

S. No.	SCOMET Entry	Name of the Chemical	Entry in CWC Schedule 3	CAS Numbers	ITC (HS) Codes
		<u>A. Toxic Chemicals</u>			
1	1C001	Phosgene : (Carbonyl dichloride)	3A01	75-44-5	28121100
2	1C002	Cyanogen chloride [(CN) C1]	3A02	506-77-4	285310 00
3	1C003	Hydrocyanic acid	3A03	74-90-8	28111200
4	1C004	‘(Reserved)’	-	-	-
		<u>B. Precursors</u>			
5	1C005	Phosphorus Oxychloride	3B05	10025-87-3	28121200
6	1C006	Phosphorus trichloride	3B06	7719-12-2	28121300
7	1C007	Phosphorous Pentachloride	3B07	10026-13-8	28121400
8	1C008	Trimethyl Phosphite	3B08	121-45-9	29202300

⁸ Notification No. 13/2015-20 dated 28.06.2017

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9	1C009	Triethyl Phosphite	3B09	122-52-1	29202400
10	1C010	Dimethyl Phosphite	3B10	868-85-9	29202100
11	1C011	Diethyl Phosphite	3B11	762-04-9	29202200
12	1C012	Sulphur monochloride	3B12	10025-67-9	28121500
13	1C013	Sulphur dichloride	3B13	10545-99-0	28121600
14	1C014	Thionyl Chloride	3B14	7719-09-7	28121700
15	1C015	Ethyl-diethanolamine	3B15	139-87-7	29221720
16	1C016	(Reserved)	-	-	-
17	1C017	Triethanolamine	3B17	102-71-6	29221500

1D Export of chemicals (Excluding Software and Technology) listed in 1D below is allowed to the countries specified in Table 1 on the basis of a one time General authorization for export of Chemicals and related equipments (GAEC) issued by DGFT, subject to the following conditions(including those below) and the procedure as prescribed from time to time:

- I. The exporter is required to register and obtain General authorization for export of Chemicals and related equipments only once during the validity period. Subsequent export/re-export is subject to post reporting(s) on quarterly basis to relevant Govt. authorities;
- II. General authorization for export of Chemicals and related equipments issued for export / re-export of SCOMET items under the above Categories / Sub Categories (excluding software and technology) shall be valid for a period of **five years** from the date of issue of General authorization for export of Chemicals and related equipments subject to subsequent post reporting(s) on quarterly basis to be reported within 30 days from the last quarter;

Table 1

Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Republic of Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Republic of Korea, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom, United States.
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Note : Export of items under 1D Category to the countries other than those specified in Table 1 and export of 1D category items related Software and Technology to any country shall be allowed only against an export authorisation and in compliance to the conditions and documentation specified under the respective Categories of SCOMET List.

Export of chemicals, related technology and software in this category to other countries shall be allowed only against an export authorisation, and in that case the exporter shall submit to the DGFT a copy of the bill of entry into the destination country within 30 days of delivery.

Sl.No.	SCOMET Entry	Chemical	CAS Number
1	1D001	2-Chloroethanol	(107-07-3)
2	1D002	3-Hydroxy-1-methylpiperidine	(3554-74-3)
3	1D003	3-Quinuclidone	(3731-38-2)
4	1D004	Ammonium bifluoride	(1341-49-7)
5	1D005	Diethylaminoethanol	(100-37-8)
6	1D006	Diisopropylamine	(108-18-9)
7	1D007	Dimethylamine	(124-40-3)

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8	1D008	Dimethylamine hydrochloride	(506-59-2)
9	1D009	Hydrogen fluoride	(7664-39-3)
10	1D010	Methyl benzilate	(76-89-1)
11	1D011	O,O-Diethyl phosphorothioate	(2465-65-8)
12	1D012	O,O-Diethyl phosphorodithioate	(298-06-6)
13	1D013	Pinacolone	(75-97-8)
14	1D014	Phosphorus pentasulphide	(1314-80-3)
15	1D015	Potassium bifluoride	(7789-23-3)
16	1D016	Potassium cyanide	(151-50-8)
17	1D017	Potassium fluoride	(7789-23-3)
18	1D018	Sodium bifluoride	(1333-83-1)
19	1D019	Sodium cyanide	(143-33-9)
20	1D020	Sodium fluoride	(7681-49-4)
21	1D021	Sodium hexafluorosilicate	(16893-85-9)
22	1D022	Sodium sulphide	(1313-82-2)
23	1D023	Triethanolamine hydrochloride	(637-39-8)
24	1D024	Triisopropyl phosphate	(116-17-6)
25	1D025	Diethylamine	(109-89-7)
26	1D026	Methyl dichlorophosphate	(677-24-7)
27	1D027	Ethyl dichlorophosphate	(1498-51-7)
28	1D028	Methyl difluorophosphate	(22382-13-4)
29	1D029	Ethyl difluorophosphate	(460-52-6)
30	1D030	Diethyl chlorophosphite	(589-57-1)
31	1D031	Methyl chlorofluorophosphate	(754-01-8)
32	1D032	Ethyl chlorofluorophosphate	(762-77-6)
33	1D033	N,N-Dimethylformamidine	(44205-42-7)
34	1D034	N,N-Diethylformamidine	(90324-67-7)
35	1D035	N,N-Dipropylformamidine	(48044-20-8)
36	1D036	N,N-Diisopropylformamidine	(857522-08-8)
37	1D037	N,N-Dimethylacetamidine	(2909-14-0)
38	1D038	N,N-Diethylacetamidine	(14277-06-6)
39	1D039	N,N-Dipropylacetamidine	(1339586-99-0)
40	1D040	N,N-Dimethylpropanamidine	(56776-14-8)
41	1D041	N,N-Diethylpropanamidine	(84764-73-8)
42	1D042	N,N-Diprophylpropanamidine	(1341496-89-6)
43	1D043	N,N-Dimethylbutanamidine	(1340437-35-5)
44	1D044	N,N-Diethylbutanamidine	(53510-30-8)
45	1D045	N,N-Diprophylbutanamidine	(1342422-35-8)
46	1D046	N,N-Diisopropylbutanamidine	(1315467-17-4)
47	1D047	N,N-Dimethylisobutanamidine	(321881-25-8)
48	1D048	N,N-Diethylisobutanamidine	(1342789-47-2)
49	1D049	N,N-Dipropylisobutanamidine	(13422700-45-1)

1E⁹ Export of Chemicals as specified below is allowed to State Parties to the Chemical Weapons Convention (CWC) on the basis of a one time General authorization for export of Chemicals and related equipments (GAEC) issued by DGFT, subject to the following conditions (including those below) and the procedure as prescribed from time to time:

⁹ Notification No. 02 dated 24.04.2019

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- I. The exporter is required to register and obtain General authorization for export of Chemicals and related equipments only once during the validity period. Subsequent export/re-export is subject to post reporting(s) on quarterly basis to relevant Govt. authorities;
- II. General authorization for export of Chemicals and related equipments issued for export / re-export of SCOMET items under the above Categories / Sub Categories (excluding software and technology) shall be valid for a period of **five years** from the date of issue of General authorization for export of Chemicals and related equipments subject to subsequent post reporting(s) on quarterly basis to be reported within 30 days from the last quarter;

Note1: List of CWC State parties can be found on www.opcw.org

Note 2: Export of items under the 1E Category to the countries other than the countries not party to CWC and export of 1E category items related Software and Technology to any country shall be allowed only against an export authorisation and in compliance to the conditions and documentation specified under the respective Categories of SCOMET List.

S. No.	SCOMET Entry	Name of the Chemical	Entry in CWC Schedule 3	CAS Numbers	ITC (HS) Codes
1.	1E 004	Chloropicrin : Trichloronitro-Methane	3A04	76-06-2	29049100
2.	1E 016	Methyldiethanolamine	3B16	105-59-9	29221710

Export of chemicals as specified above to States not Party to the CWC shall continue to be restricted and will be allowed only against an export authorisation, and in that case also exporters shall submit to the DGFT a copy of the bill of entry into the destination country within 30 days of export. Further, applications for export of chemicals in this category to countries not Party to the CWC shall be submitted along with a Government signed End-Use Certificate in the prescribed format [Appendix 2S(ii)].

Technical Note to Category 1:

Chemicals are listed by name, Chemical Abstract Service (CAS) number and CWC Schedule (where applicable). Chemicals of the same structural formula (e.g., hydrates, **isotopically-labelled forms or all possible stereoisomers**) are controlled regardless of name or CAS number. CAS numbers are shown to assist in identifying whether a particular chemical or mixture is controlled, irrespective of nomenclature. However, CAS numbers cannot be used as unique identifiers in all situations because some forms of the listed chemical have different CAS numbers, and mixtures containing a listed chemical may also have different CAS numbers.

Note: The following Chemicals under the relevant sub-categories of Category 1 and sub-category 3A306 are added under the purview of GAEC.

Description	SCOMET sub-category	Item Name	List of applicable Countries for export
Mixture of Chemicals containing 1% or less of chemicals under 1B010, 1B011, 1B012 and 1B040 of SCOMET List.	1B010	As per the relevant entries of SCOMET List	Please refer to Table 1 below at Note 1.
	1B011		
	1B012		
	1B040		
Mixture of Chemicals containing 30% or less of chemicals listed under 1B (except 1B010, 1B011, 1B012 and 1B040) of SCOMET List.	1B (except 1B010, 1B011, 1B012 and 1B040)		

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In addition to existing procedure under 1C, mixture of Chemicals containing 30% or less of chemicals listed under 1C of SCOMET List.	1C	As per the relevant entries of SCOMET List	Please refer to Table 1 below at Note 1.
In addition to existing procedure under 1E, mixture of Chemicals containing 30% or less of chemicals listed under 1E of SCOMET List.	1E	As per the relevant entries of SCOMET List	• States party to the Chemical Weapons Convention. • List of States Parties can be found at the official website of the Organization for the Prohibition of Chemical Weapons at www.opcw.org. Note: For more details, please refer Category 1E of the SCOMET List.
Sodium Perchlorate	3A306	Sodium Perchlorate	Please refer to Table 2 below at Note 1.

Note 1:

Export of chemicals/Mixture of Chemicals (excluding Software and Technology) listed above is allowed to the countries specified in the relevant column above on the basis of a one-time General Authorization for Export of Chemicals and related equipment (GAEC) issued by DGFT, subject to the following conditions (in addition to the existing procedure) and the procedure as prescribed from time to time:

I. The exporter is required to register and obtain General Authorization for Export of Chemicals/Mixture of Chemicals and related equipment only once during the validity period. Subsequent transfer/export/re-export to the countries specified below is subject to post reporting(s) on quarterly basis to relevant Government authorities;

II. General Authorization for Export of Chemicals/Mixture of Chemicals and related equipment issued for export / re-export of SCOMET items under the above Categories / Sub Categories (excluding software and technology) shall be valid for a period of **five years** from the date of issue of General Authorization for Export of Chemicals and related equipment subject to subsequent post reporting(s) on quarterly basis to be reported within 30 days from the last quarter.

Table 1

Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Republic of Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Republic of Korea, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom, United States.

Table 2

Argentina, Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Poland, Portugal, Republic of Korea, Russian Federation, South Africa, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom, United States of America.

Category 2 Micro-organisms, toxins

2A Bacteria (including Rickettsials), whether natural, enhanced or modified, either in the form of isolated live cultures or as material including living material which has been deliberately inoculated or contaminated with such cultures, related technology and software:’

2A001 Bacillus anthracis

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2A002	Bordetella bronchiseptica
2A003	Brucella abortus
2A004	Brucella melitensis
2A005	Brucella suis
2A006	Chlamydia psittaci (Chlamydophila psittaci)
2A007	Clostridium botulinum
2A008	Clostridium perfringens, epsilon toxin producing types <i>Note Limiting this control to epsilon toxin-producing strains of Clostridium perfringens therefore exempts from control the transfer of other Clostridium perfringens strains to be used as positive control cultures for food testing and quality control.</i>
2A009	Corynebacterium diphtheriae
2A010	Francisella tularensis
2A011	Klebsiella pneumoniae
2A012	Legionella pneumophila
2A013	Leptospira interrogans - all serotypes reported in India
2A014	Mycobacterium bovis
2A015	Mycobacterium tuberculosis
2A016	Mycoplasma mycoides - var mycoides
2A017	Mycoplasma mycoides - var Capri
2A018	Neisseria meningitidis
2A019	Pasteurella multocida type B
2A020	Burkholderia mallei (Pseudomonas mallei)
2A021	Burkholderia pseudomallei (Pseudomonas pseudomallei)
2A022	Salmonella paratyphi
2A023	Shigella dysenteriae
2A024	Staphylococcus aureus
2A025	Streptococcus pneumoniae
2A026	(Reserved)
2A027	Vibrio cholera
2A028	Yersinia pestis
2A029	Shiga toxin producing Escherichia coli (STEC) of serogroups O26, O45, O103, O104, O111, O121, O145, O157, and other shiga toxin producing serogroups <i>Note Shiga toxin producing Escherichia (EHEC coli (STEC) includes inter alia enterohaemorrhagic E. coli), verotoxin producing E. coli (VTEC) or verocytotoxin producing E. coli (VTEC).</i>
2A030	Mycoplasma capricolum subspecies capripneumoniae ('strain F38')
2A031	Salmonella enterica subspecies enterica serovar Typhi (Salmonella typhi)
2A032	Clostridium argentinense (formerly known as Clostridium botulinum Type G), botulinum neurotoxin producing strains
2A033	Clostridium baratii, botulinum neurotoxin producing strains
2A034	Clostridium butyricum, botulinum neurotoxin producing strains
2A035	Coxiella burnetii
2A036	Mycoplasma mycoides subspecies mycoides SC (small colony)
2A037	Rickettsia rickettsii
2A038	Rickettsia quintana
2A039	Rickettsia prowazekii

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2B Fungi, whether natural, enhanced or modified, either in the form of isolated live cultures or as material including living material which has been deliberately inoculated or contaminated with such cultures, related technology and software:

2B001	Blastomyces dermatitidis
2B002	Coccidioides immitis
2B003	Histoplasma capulatum
2B004	Nocardia asteroides
2B005	Paracoccidioides braziliensis
2B006	Coccidioides posadasii
2B007	Pneumocystis carinii
2B008	Bipolaris oryzae (Cochliobolus miyabeanus, Helminthosporium oryzae)
2B009	Pseudocercospora ulei (Microcyclus ulei, Dothidella ulei)

2C Parasites, whether natural, enhanced or modified, either in the form of isolated live cultures or as material including living material which has been deliberately inoculated or contaminated with such cultures for the following:

2C001	Entamoeba histolytica
2C002	Babesia microti
2C003	Babesia divergens
2C004	Blastidium coli
2C005	Cryptosporidium spp.
2C006	Leishmania species
2C007	Naegleria australiensis
2C008	Naegleria fowleri
2C009	Plasmodium falciparum
2C010	(Reserved)
2C011	Schistosoma mansoni
2C012	Schistosoma japonicum
2C013	Schistosoma hematobium
2C014	Toxoplasma gondii
2C015	Trichinella spiralis
2C016	Trypanosoma brucei

2D Viruses, whether natural, enhanced or modified, either in the form of isolated live cultures or as material including living material which has been deliberately inoculated or contaminated with such cultures, related technology and software:

2D001	African Horse Sickness virus
2D002	African Swine Fever virus
2D003	Avian influenza virus <i>Note This includes only those Avian influenza viruses of high pathogenicity as defined by the World Organization for Animal Health (OIE), the European Union (EU), or competent national regulatory bodies.</i>
2D004	Blue tongue virus
2D005	Camel pox virus
2D006	Chikungunya virus
2D007	Crimean-Congo hemorrhagic fever virus
2D008	Dengue virus
2D009	Eastern equine encephalitis virus
2D010	Ebolavirus: all members of the Ebolavirus genus
2D011	Encephalomyocarditis virus (EMC)

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2D012	Foot and Mouth Disease virus (all serotypes and subtypes)
2D013	Guanarito virus
2D014	Goatpox virus
2D015	Hantaan virus
2D016	Herpes virus simiae (monkey B virus)
2D017	Herpes ateles, Herpes saimiri
2D018	HIV- 1 & HIV-2 and other strains of SIV
2D019	Classical swine fever virus (Hog cholera virus)
2D020	Human T-cell Leukemia virus
2D021	Junin virus
2D022	Japanese encephalitis virus
2D023	Kyasanur Forest Disease virus
2D024	Korean hemorrhagic fever virus
2D025	Lymphocytic choriomeningitis virus (LCM)
2D026	Lassa virus
2D027	Marburgvirus: all members of the Marburgvirusgenus
2D028	Murray valley encephalitis virus
2D029	Machupo virus
2D030	Mason-pfizer monkey virus
2D031	Middle East respiratory syndrome coronavirus (MERS-CoV)
2D032	Monkey pox virus
2D033	Newcastle disease virus
2D034	Omsk hemorrhagic fever virus
2D035	Peste des petits ruminant virus
2D036	Teschen disease virus (Porcine entero virus type 1)
2D037	Powassan virus
2D038	Rabies virus and other members of the Lyssavirus genus
2D039	Respiratory syncitial virus
2D040	Rift Valley Fever virus
2D041	Rinderpest virus
2D042	Sabia virus
2D043	Sheeppox virus
2D044	Sin Nombre virus
2D045	Smallpox virus
2D046	St.Louis encephalitis virus
2D047	Swine vesicular disease virus
2D048	Tick-borne encephalitis virus (Far Eastern subtype)
2D049	(Reserved)
2D050	Variola virus
2D051	Venezuelan equine encephalitis virus
2D052	Vesicular stomatitis virus
2D053	Western equine encephalitis virus
2D054	Yellow fever virus
2D055	Andes virus
2D056	Chapare virus
2D057	Choclo virus
2D058	Dobrava-Belgrade virus
2D059	Suid herpesvirus 1 (Pseudorabies virus; Aujeszky's disease)

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2D060	Hendra virus (Equine morbillivirus)
2D061	Laguna Negra virus
2D062	Louping ill virus
2D063	Lujo virus
2D064	Lumpy skin disease virus
2D065	(Reserved)
2D066	Nipah virus
2D067	Oropouche virus
2D068	(Reserved)
2D069	Rocio virus
2D070	Seoul virus
2D071	Severe acute respiratory syndrome-related coronavirus (SARS-related coronavirus)
2D072	Reconstructed 1918 influenza virus
2E	[Reserved]
2F	Toxins, related technology and software
2F001	Abrin
2F002	Aflatoxins
2F003	Anatoxins
2F004	Botulinum toxins
	<i>Note Excluding botulinum toxins in product form meeting all of the following criteria:</i>
	<i>a. are pharmaceutical formulations designed for testing and human administration in the treatment of medical conditions;</i>
	<i>b. are pre-packaged for distribution as clinical or medical products; and</i>
	<i>c. are authorised by a state authority to be marketed as clinical or medical products</i>
2F005	Bungarotoxins
2F006	Clostridium perfringens alpha, beta 1, beta 2, epsilon and iota toxins
2F007	Corynebacterium diphtheriae toxins
2F008	Cyanginosins (Microcystins) (Microcystic aeuginosa)
2F009	Staphylococcus aureus enterotoxins, hemolysin alpha toxin, and toxic shock syndrome toxin (formerly known as Staphylococcus enterotoxin F)
2F010	Neurotoxin (Shigella dysenteriae)
2F011	(Reserved)
2F012	Shiga toxins (shiga-like toxins, verotoxins, and verocytotoxins)
2F013	(Reserved)
2F014	Trichothecene mycotoxins
2F015	Tetanus toxin (clostridium tetani)
2F016	Tetrodotoxin (Spheroides rufripes)
2F017	Verrucologen (M. verrucadia)
2F018	Cholera toxin
2F019	Conotoxins
	<i>Note Excluding conotoxins in product form meeting all of the following criteria:</i>
	<i>a. are pharmaceutical formulations designed for testing and human administration in the treatment of medical conditions;</i>
	<i>b. are pre-packaged for distribution as clinical or medical products; and</i>
	<i>are authorised by a state authority to be marketed as clinical or medical products</i>
2F020	Diacetoxyscirpenol toxin
2F021	HT-2 toxin
2F022	Modeccin toxin
2F023	T-2 mycotoxin
2F024	Verotoxin and shiga-like ribosome inactivating proteins
2F025	Viscum Album Lectin 1 (Viscumin)

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2F026	Volkensin toxin
2F027	Brevetoxins
2F028	Gonyautoxins
2F029	Nodularins
2F030	Palytoxin
2G	Plant pathogens, related technology and software
2G001	Bemisia tabaci
2G002	Colletotrichum kahawae (Colletotrichum coffeanum var. virulans)
2G003	Claviceps purpurea
2G004	Dothistroma pini (Scirrhia pini)
2G005	Erwinia amylovora
2G006	Frankliniella occidentalis
2G007	Microcyclus ulei (syn. Dothidella ulei)
2G008	Peronospora hyoscyami f.sp. tabacina
2G009	Phytophthora infestans
2G010	Puccinia graminis ssp. graminis var. graminis / Puccinia graminis ssp. graminis var. stakmanii (Puccinia graminis [syn. Puccinia graminis f. sp. tritici])
2G011	Puccinia erianthi
2G012	Puccinia striiformis f. sp. tritici (Puccinia glumarum)
2G013	Magnaporthe oryzae (Pyricularia oryzae)
2G014	Ralstonia solanacearum
2G015	Peronospora hyscyami de Bary
2G016	(Reserved)
2G017	Sugar cane Fiji disease virus
2G018	Sclerotinia sclerotiorum
2G019	Tilletia indica
2G020	Thrips palmi
2G021	Ustilago Maydis
2G022	Xanthomonas albilineans
2G023	Xanthomonas axonopodis pv. citri (Xanthomonas campestris pv. citri A) [Xanthomonas campestris pv. citri]
2G024	Xanthomonas oryzae pv. oryzae (Pseudomonas campestris pv. oryzae)
2G025	Clavibacter michiganensis subsp. sepedonicus (Corynebacterium michiganensis subsp. sepedonicum or Corynebacterium sepedonicum)
2G026	Cochliobolus miyabeanus (Helminthosporium oryzae)
2G027	Andean potato latent virus (Potato Andean latent tymovirus)
2G028	Potato spindle tuber viroid
2G029	Thecaphora solani
2G030	Synchytrium endobioticum
2G031	Sclerophthora rayssiae var. zeae
2G032	Peronosclerospora philippinensis (Peronosclerospora sacchari)
2G033	Xanthomonas citri pv. citri (Xanthomonas axonopodis pv. citri, Xanthomonas campestris pv. citri)
2H¹⁰	Genetic Elements and Genetically-modified Organisms, related technology and software
2H001¹¹	Any genetically-modified organism which contains, or genetic element that codes for:
(1)	Gene(s) specific to virus, bacterium, fungus, parasites or plant pathogens listed in Categories 2A, 2B, 2C, 2D and 2G and which
a.	in itself or through its transcribed or translated products represents a significant hazard to

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- b. human, animal or plant health, or
could endow or enhance pathogenicity
- (2) Toxin(s) or their sub-unit(s) listed in Category 2F.

Technical note

- (1) Genetically-modified organisms include organisms in which the nucleic acid sequences have been created or altered by deliberate molecular manipulation.
- (2) Genetic elements include, inter alia: chromosomes, genomes, plasmids, transposons, vectors, and inactivated organisms containing recoverable nucleic acid fragments, whether genetically modified or unmodified, or chemically synthesized in whole or in part. For the purposes of the genetic elements control, nucleic acids from an inactivated organism, virus, or sample are considered recoverable if the inactivation and preparation of the material is intended or known to facilitate isolation, purification, amplification, detection, or identification of nucleic acid(s).
- (3) Endow or enhance pathogenicity is defined as when the insertion or integration of the nucleic acid sequence(s) are likely to enable or increase a recipient organism's ability to be used to deliberately cause disease or death. This might include alterations to (not limited to) inter alia: virulence, transmissibility, stability, route of infection, host range, reproducibility, ability to evade or suppress host immunity, resistance to medical countermeasures, or detectability.'

¹⁰ Para 1(A) of Notification No. 29/2015-20 dated 21.09.2017

¹¹ Para 1(C) of Notification No. 29/2015-20 dated 21.09.2017

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Category 3 Materials, Materials Processing Equipment and related technologies

3A Materials

3A1 Special Materials

- 3A101** Zirconium, beryllium, magnesium, and alloys of these in particle size less than 60 µm
- 3A102** Maraging steel in any form in which any linear dimension exceeds 75 mm, or in the form of sheet, plate or tubing with a wall or plate thickness equal or less than 5 mm.
- 3A103** Tungsten, molybdenum, and alloys of those metals in particulate form and a particle size of 50 x10⁻⁶ m (50 µm) or less; ”
- 3A104** Germanium
- 3A105** Gallium
- 3A106** Indium
- 3A107** Titanium-stabilised Duplex Stainless Steel (Ti-DSS) and specially designed components therefor having all of the following:
- a. Having all of the following characteristics:
 1. Containing 17.0 - 23.0% by weight of chromium and 4.5 - 7.0% by weight of nickel;
 2. Having a titanium content of greater than 0.10% by weight; and
 3. A ferritic-austenitic microstructure (also referred to as a two-phase microstructure) of which at least 10% by volume (according to ASTM E-1181-87 or national equivalents) is austenite; and
 - b. Any of the following forms:
 1. Ingots or bars having a size of 100 mm or more in each dimension;
 2. Sheets having a width of 600 mm or more and a thickness of 3 mm or less; or
 3. Tubes having an outer diameter of 600 mm or more and a wall thickness of 3 mm or less
- 3A108** Aluminium alloys in any form ‘capable of acquiring’ an ultimate tensile strength of 460 MPa or more at 293 K (20 degrees C)
- Note The phrase ‘capable of acquiring’ encompasses alloys before or after heat treatment*
- 3A109** Bismuth having a purity of 99.99% or greater by weight and containing less than 10 parts per million by weight of silver
- 3A110** Calcium containing less than 1000 parts per million by weight of metallic impurities other than magnesium and containing less than 10 parts per million by weight of Boron
- 3A111** Chlorine trifluoride (ClF₃)
- 3A112** Magnesium containing less than 200 parts per million by weight of metallic impurities other than calcium and containing less than 10 parts per million by weight of boron
- 3A113** (a) Tungsten, tungsten carbide, and alloys containing more than 90% tungsten by weight in forms with a hollow cylindrical symmetry (including cylinder segments) with an inside diameter between 100 and 300 mm and a mass greater than 20 kg.
- (b) Tungsten materials in the solid form usable for the fabrication of missile components in complete rocket systems of 5A and unmanned aerial vehicles of 5B, having all of the following:

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1. Any of the following material compositions:
 - i. Tungsten and alloys containing 97% by weight or more of tungsten;
 - ii. Copper infiltrated tungsten containing 80% by weight or more of tungsten; or
 - iii. Silver infiltrated tungsten containing 80% by weight or more of tungsten; and
 2. Able to be machined to any of the following products:
 - i. Cylinders having a diameter of 120 mm or greater and a length of 50 mm or greater;
 - ii. Tubes having an inner diameter of 65 mm or greater and a wall thickness of 25 mm or greater and a length of 50 mm or greater; or
 - iii. Blocks having a size of 120 mm x 120 mm x 50 mm or greater.
- 3A114** a. Nickel powder of purity 99.0% or greater by weight; and having a mean particle size of less than 10 µm measured by the ASTM B 330 standard;
b. Porous nickel metal produced from the nickel powder specified above
- 3A115** Natural boron, boron carbide or metal borides having a boron purity of 85% or more.
- 3A116** Fibrous or filamentary materials, and preregs, as follows:
- a. Carbon or aramid fibrous or filamentary materials having ‘specific modulus’ of 12.7×10^6 m or greater; or ‘specific tensile strength’ of 23.5×10^4 m or greater;
 - b. Glass fibrous or filamentary materials having ‘specific modulus’ of 3.18×10^6 m or greater; and ‘specific tensile strength’ of 7.62×10^4 m or greater;
 - c. Thermoset resin impregnated continuous yarns, rovings, tows or tapes with a width of 15 mm or less (preregs), made from carbon or glass fibrous or filamentary materials specified in (a) or (b) above.
- 3A117** Carbon - carbon composites.
- 3A118** Titanium alloys having both of the following characteristics:
- a. Capable of an ultimate tensile strength of 900 MPa or more at 293 K (20 degrees C); and
 - b. In the form of tubes or cylindrical solid forms (including forgings) with an outside diameter of more than 75 mm.
- Technical Note* The phrase ‘capable of’ encompasses titanium alloys before or after heat treatment
- 3A119** Rhenium, and alloys containing 90% by weight or more rhenium; and alloys of rhenium and tungsten containing 90% by weight or more of any combination of rhenium and tungsten, have both of the following characteristics:
- a. In forms with a hollow cylindrical symmetry (including cylinder segments) with an inside diameter between 100 and 300 mm; and
 - b. A mass greater than 20kg
- 3A120** **Technology and Software**
Technology and software for the development, production or use of items specified in 3A1 or 3A4
- 3A2 Structural Materials**
- 3A201** Structural materials such as:
- a. Composite structures, laminates, resin impregnated fibre preregs and metal coated fibre preforms made either with an organic matrix or metal matrix utilizing fibrous or filamentary reinforcements, and manufactures thereof, specially designed for use in rocket systems (including ballistic missile systems, space launch vehicles and sounding rockets), unmanned aerial vehicles and cruise missiles or subsystems thereof;
 - b. Resaturated pyrolyzed (i.e. Carbon-Carbon) materials specially designed for rocket systems (including ballistic missile systems, space launch vehicles and sounding rockets), unmanned aerial vehicles and cruise missiles;
 - c. Fine grain re-crystallised bulk graphites and pyrolytic or fibrous reinforced graphites usable for rocket nozzles and re-entry vehicles nose tips;
 - d. Ceramic composite materials (dielectric constant less than 6 at any frequency from 100 MHz to 100 GHz) for use in missile radomes;
 - e. Materials and coatings for reduced radar reflectivity;

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- f. Bulk machinable silicon-carbide reinforced unfired ceramic usable in re-entry vehicles nose tips.
- g. Reinforced silicon-carbide ceramic composites usable for nose tips, re-entry vehicles, nozzle flaps, usable in complete rocket systems of 5A and complete unmanned aerial vehicles of 5B .
- h. Bulk machinable ceramic composite materials consisting of an 'Ultra High Temperature Ceramic (UHTC)' matrix with a melting point equal to or greater than 3000°C and reinforced with fibres or filaments, usable for missile components (such as nose-tips, re-entry vehicles, leading edges, jet vanes, control surfaces or rocket motor throat inserts) in the systems specified in 5A and 5B.

Note:

Item 3A201.h does not control 'Ultra High Temperature Ceramic (UHTC)' materials in non-composite form.

Technical Note

'Ultra High Temperature Ceramics (UHTC)' includes:

1. *Titanium diboride (TiB₂);*
2. *Zirconium diboride (ZrB₂);*
3. *Niobium diboride (NbB₂);*
4. *Hafnium diboride (HfB₂);*
5. *Tantalum diboride (TaB₂);*
6. *Titanium carbide (TiC);*
7. *Zirconium carbide (ZrC);*
8. *Niobium carbide (NbC);*
9. *Hafnium carbide (HfC);*
10. *Tantalum carbide (TaC). ”*

3A3 Rocket propellants and constituent chemicals:

3A301 Fuel substances as follows:

- a. Hydrazine (CAS 302-01-2) with a concentration of more than 70%
- b. Hydrazine derivatives as follows:
 1. Monomethylhydrazine (MMH) (CAS 60-34-4);
 2. Unsymmetrical dimethylhydrazine (UDMH) (CAS 57-14-7);
 3. Hydrazine mononitrate (CAS 13464-97-6);
 4. Trimethylhydrazine (CAS 1741-01-1);
 5. Tetramethylhydrazine (CAS 6415-12-9);
 6. N, N diallylhydrazine (CAS 5164-11-4);
 7. Allylhydrazine (CAS 7422-78-8);
 8. Ethylene dihydrazine;
 9. Monomethylhydrazine dinitrate;
 10. Unsymmetrical dimethylhydrazine nitrate;
 11. Hydrazinium azide (CAS 14546-44-2);
 12. Dimethylhydrazinium azide;
 13. Hydrazinium dinitrate (CAS 13464-98-7);
 14. Diimido oxalic acid dihydrazine (CAS 3457-37-2);

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15. 2-hydroxyethylhydrazine nitrate (HEHN);
16. Hydrazinium perchlorate (CAS 27978-54-7);
17. Hydrazinium diperchlorate (CAS 13812-39-0);
18. Methylhydrazine nitrate (MHN) (CAS 29674-96-2);
19. Diethylhydrazine nitrate (DEHN);
20. 3, 6-dihydrazino tetrazine nitrate (DHTN);

‘Technical Note 1: 3, 6-dihydrazino tetrazine nitrate is also referred to as 1, 4-dihydrazine nitrate

Technical Note 2 Substance groupings in Item 3A301 and 3A306 (e.g. fuels, oxidisers, etc.) describe typical applications of propellant substances. A substance remains specified by 3A301 and 3A306 even when used in an application other than the typical one indicated by its grouping [e.g. hydrazinium perchlorate (CAS 27978-54-7) is grouped as a fuel but can also be used as an oxidizer]’.

- c. Spherical or spheroidal aluminium powder (CAS 7429-90-5) in particle size of less than 200×10^{-6} m (200 μ m) and an aluminium content of 97% by weight or more, if at least 10% of the total weight is made up of particles of less than 63 μ m, according to ISO 2591-1:1988 or national equivalents;

Technical Note A particle size of 63 μ m (ISO R-565) corresponds to 250 mesh (Tyler) or 230 mesh (ASTM standard E-11).

- d. Hydrazine replacement fuels as follows:

1.2-Dimethylaminoethylazide (DMAZ) (CAS 86147-04-8)”;

3A302 Metal fuels containing any of the following: Zirconium(CAS 7440-67-7), beryllium(CAS 7440-41-7), magnesium, titanium, tungsten, boron and boron alloys, zinc, and alloys of magnesium(CAS 7439-95-4);

3A303 Polymeric substances, as follows:

- a. Carboxy-terminated polybutadiene (including carboxyl – terminated polybutadiene) (CTPB);
- b. Hydroxy Terminated Polybutadiene (including Hydroxyl Terminated polybutadiene) (HTPB) (CAS 69102-90-5)
- c. Glycidyl azide polymer (GAP);
- d. Polybutadiene - Acrylic Acid (PBAA);
- e. Polybutadiene - Acrylic Acid - Acrylonitrile (PBAN);
- f. Polytetrahydrofuran polyethylene glycol (TPEG).
- g. Polyglycidyl nitrate (PGN or poly-GLYN) (CAS 27814-48-8).

Technical Note

Polytetrahydrofuran polyethylene glycol (TPEG) is a block co-polymer of poly 1, 4- Butanediol (CAS 110-63-4) and polyethylene glycol (PEG) (CAS 25322-68-3).”

3A304 Composite propellants and composite modified double base propellants;

3A305 High energy density materials such as boron slurry;

3A306 Oxidizers/fuels - Perchlorates, chlorates or chromates mixed with powdered metals or other high energy fuel components; Dinitrogen trioxide, Nitrogen dioxide / Dinitrogen tetroxide, Mixed Oxides of Nitrogen (MON), Dinitrogen pentoxide, Inhibited red fuming nitric acid (IRFNA) (CAS 8007-58-7), Ammonium perchlorate (CAS 7790-98-9), Ammonium Dinitramide (ADN) (CAS 140456-78-6), Hydrazinium Nitroformate (HNF), 2,4,6,8,10,12- Hexanitrohexaazaisowurtzitane (CL-20) (CAS 135285-90-4), Compounds composed of fluorine and one more of other halogens, oxygen or nitrogen.

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- 3A307** Bonding agents - Tris (1-2 (2-methyl)) aziridinyl phosphine oxide (MAPO)(CAS 57-39- 6), Trimesoyl-1-(2-ethyl) aziridene (HX-868, BITA)(CAS 7722-73-8), Tepanol (HX- 878)(CAS 68412-46-4), Tepan (HX-879) reaction product of tetraethylenepentamine and acrylonitrile (CAS 68412-45-3), and Polyfunctional aziridine amides with isophthalic, trimesic, isocyanuric, or trimethyladipic backbone also having a 2-methyl or 2-ethyl aziridine group including 1,1'-Isophthaloyl-bis(2-methylaziridene (CAS 7652-64-4), (HX-752, HX-874, and HX-877);
- 3A308** Curing agents and reaction catalysts - Triphenyl bismuth (TPB)(CAS 603-33-8);
- 3A309** Burning rate modifiers –
- Carboranes, decaboranes, pentaboranes, and derivatives thereof;
 - Ferrocene derivatives, as follows:1.
Catocene (CAS 37206-42-1);
2. Ethyl ferrocene;
3. n-Propyl ferrocene (CAS 1273-92-3) / iso-propyl ferrocene (CAS 12126-81-7)
4. n-Butyl ferrocene(CAS 31904-29-7);
5. Pentyl ferrocene (CAS 1274-00-6);
6. Dicyclopentyl ferrocene(CAS 20773-28-8);
7. Dicyclohexyl ferrocene;
8. Diethyl ferrocene;
9. Dipropyl ferrocene;
10. Dibutyl ferrocene(CAS 1274-08-4);
11. Dihexyl ferrocene (CAS 93894-59-8);
12. Acetyl ferrocenes;
13. Ferrocene Carboxylic acids;
14. Butacene;
 - Other ferrocene derivatives usable as rocket propellant burning ratemodifiers.
- 3A310** Nitrate esters and nitrated plasticisers as follows:
- Triethylene glycol dinitrate (TEGDN);
 - Trimethylolethane trinitrate (TMETN)(CAS 3032-55-1) ;
 - 1,2,4-butanetriol trinitrate (BTTN)(CAS 6659-60-5) ;
 - Diethylene glycol dinitrate (DEGDN);
 - 4,5 diazidomethyl-2-methyl-1,2,3-triazole (iso-DAMTR);
 - Nitratoethylnitramine (NENA) based plasticisers, as follows:
 - Methyl-NENA (CAS 17096-47-8);
 - Ethyl-NENA (CAS 85068-73-1);
 - Butyl-NENA (CAS 82486-82-6);
 - Dinitropropyl based plasticisers, as follows:
 - Bis (2,2-dinitropropyl) acetal (BDNPA) (CAS 5108-69-0);
 - Bis (2,2-dinitropropyl) formal (BDNPF) (CAS 5917-61-3).
- 3A311** Stabilisers as follows:
- 2-Nitrodiphenylamine (CAS 119-75-5);
 - N-methyl-p-nitroaniline (CAS 100-15-2).

3A4 High explosives

- 3A401** High explosive substances or mixtures, containing more than 2 % by weight of any of the following:
- Cyclotetramethylenetetranitramine (HMX) (CAS 2691-41-0);
 - Cyclotrimethylenetrinitramine (RDX) (CAS 121-82-4);
 - Triaminotrinitrobenzene (TATB) (CAS 3058-38-6);
 - Aminodinitrobenzo-furoxan or 7-amino-4,6 nitrobenzofurazane-1-oxide (ADNBF) (CAS 97096-78-1);
 - 1,1-diamino-2,2-dinitroethylene (DADE or FOX7) (CAS 145250-81-3);
 - 2,4-dinitroimidazole (DNI) (CAS 5213-49-0);
 - Diaminoazoxyfuran (DAAOF or DAAF) (CAS 78644-89-0);
 - Diaminotrinitrobenzene (DATB) (CAS 1630-08-6);

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- i. Dinitroglycoluril (DNGU or DINGU) (CAS 55510-04-8);
- j. 2,6-Bis (picrylamino)-3,5-dinitropyridine (PYX) (CAS 38082-89-2);
- k. 3,3'-diamino-2,2',4,4',6,6'-hexanitrobiphenyl or dipicramide (DIPAM) (CAS 17215-44-0);
- l. Diaminoazofurazan (DAAzF) (CAS 78644-90-3);
- m. 1,4,5,8-tetranitro-pyridazino[4,5-d] pyridazine (TNP) (CAS 229176-04-9);
- n. Hexanitrostilbene (HNS) (CAS 20062-22-0); or
- o. Any explosive with a crystal density greater than 1.8 g/cm³ and having a detonation velocity greater than 8000 m/s.

Note License applications for the export of items at 3A401a and 3A401b will normally be denied.

3A5 Stealth materials

- 3A501** a. Materials for reduced observables such as radar reflectivity, ultraviolet/infrared signatures and acoustic signatures;

b. Devices, including made from non-stealth material, for reduced observables such as radar reflectivity, ultraviolet/infrared signatures and acoustic signatures;
- 3A502** Materials and coatings (including paints) specially designed for reduced or tailored reflectivity or emissivity in the microwave, infrared or ultraviolet spectra other than coatings (including paints) when specially used for thermal control of satellites.
- 3A503** Technology related to the development, production or use of items in 3A.

3B Materials processing and “production equipment”, related “technology” and specially designed components and accessories therefor.

- 3B001** Remote manipulators that provide mechanical translation of human operator actions by electrical, hydraulic or mechanical means and operating arm and terminal fixture that can be used to provide remote actions;
- 3B002** Multidirectional, multidimensional weaving and interlacing machines, including adapters and modification kits for weaving, interlacing or braiding fibres to fabricate composite structures except textile machinery which has not been modified for rocket systems;
- 3B003** Equipment designed or modified for production of fibrous or filamentary materials as follows: converting polymeric substances; vapour deposition on heated filament substrates; wet spinning of refractory ceramics.
- 3B004** Equipment designed or modified for special fibre surface treatment or for producing prepregs and preforms, including rollers, tension stretchers, coating equipment, cutting equipment and clicker dies;
- 3B005** Chemical vapour deposition furnaces designed or modified for the densification of carbon-carbon composites.
- 3B006** Pyrolytic deposition and densification equipment including:
 - a. Technology for producing pyrolytically derived materials formed on a mould, mandrel or other substrate from precursor gases.
 - b. Specially designed nozzles for the above process.
 - c. Equipment and process controls and specially designated software thereof, specially designed or modified for densification and pyrolysis of structural composite rocket nozzles and re-entry vehicle nose tips.
- 3B007** Production equipment usable for or specially designed or modified for production, handling, mixing, curing, casting, pressing, machining, extruding or acceptance testing of the solid or liquid rocket propellants or rocket propellant constituents and related technology.

*Note: Item 3B007 does not control batch mixers, continuous mixers and fluid energy mills.
N.B. For batch mixers, continuous mixers and fluid energy mills, see 3B014 and 3B015.*

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3B008	Refrigeration units and equipment capable of cooling hydrogen or helium to -250 degrees Celsius (23K) or lower.
3B009	Continuous nitrators.
3B010	Dehydration presses.
3B011	Screw extruders usable for or specially designed or modified for high explosive extrusion.
3B012	Cutting machines for the sizing of extruded propellant.
3B013	Sweetie barrels (tumblers) 1.85 m or more in diameter and having over 227 kg product capacity;
3B014	Equipment as follows, and specially designed components therefor: a. Batch mixers having all of the following: 1. Designed or modified for mixing under vacuum in the range of zero to 13.326 kPa; 2. Capable of controlling the temperature of the mixing chamber; 3. A total volumetric capacity of 110 litres or more; <u>and</u> 4. At least one 'mixing/kneading shaft' mounted off centre; <i><u>Note:</u> In Item 3B014.a.4. the term 'mixing/kneading shaft' does not refer to deagglomerators or knife-spindles.</i> b. Continuous mixers having all of the following: 1. Designed or modified for mixing under vacuum in the range of zero to 13.326 kPa; 2. Capable of controlling the temperature of the mixing chamber; <u>and</u> 3. Any of the following: a. Two or more mixing/kneading shafts; <u>or</u> b. All of the following 1. A single rotating and oscillating shaft with kneading teeth/pins, <u>and</u> 2. Kneading teeth/pins inside the casing of the mixing chamber;
3B015	Fluid energy mills usable for grinding or milling any of the items in 3A3.
3B016	Metal powder production equipment usable for the production, in a controlled environment, of spherical, spheroidal or atomised materials specified in 3A301.c. or 3A302 <u>Note</u> This entry includes: a. Plasma generators (high frequency arc-jet) usable for obtaining sputtered or spherical metallic powders with organization of the process in an argon-water environment; b. Electrobust equipment usable for obtaining sputtered or spherical metallic powders with organization of the process in an argon-water environment; c. Equipment usable for the production of spherical aluminium powders by powdering a melt in an inert medium (e.g. nitrogen).
3B017	Sputter ion pumps
3B018	Technical data (including processing conditions) and procedures for the regulation of temperature, pressure or atmosphere in autoclaves or hydroclaves when used for the production of composites or partially processed composites.
3B019	Software specially designed or modified for the use of equipment for the production and handling of materials specified in 3A
3B020	Technology for the development, production or use of items in 3B
3C	[Reserved]
3D	Chemical and biomaterial manufacturing and handling equipment and facilities:

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3D001 (1) Reaction Vessels, Reactors or Agitators

- (i) Reaction vessels or reactors, with or without agitators, with total internal (geometric) volume greater than 0.1 m³ (100 l) and less than 20 m³ (20000 l), where all surfaces that come in direct contact with the chemical(s) being processed or contained are made from the following materials:
 - a. nickel or alloys with more than 40% nickel by weight;
 - b. alloys with more than 25% nickel and 20% chromium by weight;
 - c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
- (ii) glass or glass-lined (including vitrified mentioned reaction vessels or reactors; and impellers, blades or shafts designed for such agitators where all surfaces of the agitator that come in direct contact with the chemical(s) being processed or contained are made from any of the following materials:
 - a. nickel or alloys with more than 40% nickel by weight;
 - b. alloys with more than 25% nickel and 20% chromium by weight;
 - c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
 - d. glass or glass-lined (including vitrified or enamelled coating);
 - e. tantalum or tantalum alloys;
 - f. titanium or titanium alloys;
 - g. zirconium or zirconium alloys; or
 - h. niobium (columbium) or niobium alloys.
- (iii)¹² Prefabricated repair assemblies and their specially designed components, that:
 - d. are designed for mechanical attachment or enamelled coating);
 - e. tantalum or tantalum alloys;
 - f. titanium or titanium alloys;
 - g. zirconium or zirconium alloys; or
 - h. niobium (columbium) or niobium alloys.
- (iii) Agitators designed for use in the above-
 - a. to glass-lined reaction vessels or reactors that meet the parameters above; and,
 - b. have metallic surfaces that come in direct contact with the chemical(s) being processed which are made from tantalum or tantalum alloys’.

(2) Storage Tanks, Containers or Receivers

- (i) Storage tanks, containers or receivers with a total internal (geometric) volume greater than 0.1 m³ (100 l) where all surfaces that come in direct contact with the chemical(s) being processed or contained are made from the following materials:
 - a. nickel or alloys with more than 40% nickel by weight;
 - b. alloys with more than 25% nickel and 20% chromium by weight;
 - c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
 - d. glass or glass-lined (including vitrified or enamelled coating);
 - e. tantalum or tantalum alloys;
 - f. titanium or titanium alloys;
 - g. zirconium or zirconium alloys; or
 - h. niobium (columbium) or niobium alloys.
- (ii)¹³ Prefabricated repair assemblies and their specially designed components, that:

¹² Para 1(D) of Notification No. 29/2015-20 dated 21.09.2017

¹³ Para 1(E) of Notification No. 29/2015-20 dated 21.09.2017

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- a. are designed for mechanical attachment to glass-lined reaction vessels or reactors that meet the parameters above; and,
- b. have metallic surfaces that come in direct contact with the chemical(s) being processed which are made from tantalum or tantalum alloys .

(3) Heat Exchangers or Condensers

Heat exchangers or condensers with a heat transfer surface area of greater than 0.15 m², and less than 20 m²; and tubes, plates, coils or blocks (cores) designed for such heat exchangers or condensers, where all surfaces that come in direct contact with the chemical(s) being processed are made from the following materials:

- a. nickel or alloys with more than 40% nickel by weight;
- b. alloys with more than 25% nickel and 20% chromium by weight;
- c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
- d. glass or glass-lined (including vitrified or enamelled coating);
- e. graphite or carbon-graphite;
- f. tantalum or tantalum alloys;
- g. titanium or titanium alloys;
- h. zirconium or zirconium alloys;
- i. silicon carbide;
- j. titanium carbide; or
- k. niobium (columbium) or niobium alloys.

Technical Note carbon-graphite is a composition consisting of amorphous carbon and graphite, in which the graphite content is eight percent or more by weight.

(4) Distillation or Absorption Columns

Distillation or absorption columns of internal diameter greater than 0.1 m; and liquid distributors, vapour distributors or liquid collectors designed for such distillation or absorption columns, where all surfaces that come in direct contact with the chemical(s) being processed are made from the following materials:

- a. nickel or alloys with more than 40% nickel by weight;
- b. alloys with more than 25% nickel and 20% chromium by weight;
- c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
- d. glass or glass-lined (including vitrified or enamelled coating);
- e. graphite or carbon-graphite;
- f. tantalum or tantalum alloys;
- g. titanium or titanium alloys;
- h. zirconium or zirconium alloys; or
- i. niobium (columbium) or niobium alloys.

Technical Note carbon-graphite is a composition consisting of amorphous carbon and graphite, in which the graphite content is eight percent or more by weight.

(5) Filling Equipment

Remotely operated filling equipment in which all surfaces that come in direct contact with the chemical(s) being processed are made from the following materials:

- a. nickel or alloys with more than 40% nickel by weight; or
- b. alloys with more than 25% nickel and 20% chromium by weight.

(6) Valves

- (i) Valves, having both of the following:
 - a. A nominal size greater than DN 10 or NPS 3/8 or national equivalents, and

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- b. All surfaces that come in direct contact with the chemical(s) being produced, processed, or contained are made from the materials of construction in Technical Note 1 of this entry
- (ii) Valves, not already identified in 3D001(6)(i), having all of the following:
 - a. A nominal size equal to or greater than DN 25 or NPS 1 and equal to or less than DN 100 or NPS 4 or national equivalents.
 - b. Casings (valve bodies) or preformed casing liners,
 - c. A closure element designed to be interchangeable, and
 - d. All surfaces of the casing (valve body) or preformed case liner that come in direct contact with the chemical(s) being produced, processed, or contained are made from the materials of construction in Technical Note 1 of this entry
- (iii) Components, as follows:
 - a. Casings (valve bodies) designed for valves in paragraphs 6.a.or 6.b., in which all surfaces that come in direct contact with the chemical(s) being produced, processed, or contained are made from the materials of construction in Technical Note 1 of this entry;
 - b. Preformed casing liners designed for valves in paragraphs 6.a.or 6.b., in which all surfaces that come in direct contact with the chemical(s) being produced, processed, or contained are made from the materials of construction in Technical Note 1 of this entry.

Technical Note 1. *Materials of construction for valves include any of the following:*

- a. nickel or alloys with more than 40% nickel by weight;
- b. alloys with more than 25% nickel and 20% chromium by weight;
- c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
- d. glass or glass-lined (including vitrified or enamelled coating);
- e. tantalum or tantalum alloys;
- f. titanium or titanium alloys;
- g. zirconium or zirconium alloys;
- h. niobium (columbium) or niobium alloys; or
- i. ceramic materials as follows:
 - 1. silicon carbide with a purity of 80% or more by weight;
 - 2. aluminum oxide (alumina) with a purity of 99.9% or more by weight;
 - 3. zirconium oxide (zirconia).

Technical Note 2. *The 'nominal size' is defined as the smaller of the inlet and outlet port diameters.*

Technical Note 3. *Metric nominal sizes (DN) of valves are in accordance with ISO6708:1995.
National Pipe Sizes (NPS) are in accordance with ASME B36.10 or B36.19, or national equivalents.*

(7) Multi-Walled Piping

Multi-walled piping incorporating a leak detection port, in which all surfaces that come in direct contact with the chemical(s) being processed or contained are made from the following materials:

- a. nickel or alloys with more than 40% nickel by weight;
- b. alloys with more than 25% nickel and 20% chromium by weight;
- c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
- d. glass or glass-lined (including vitrified or enamelled coating);
- e. graphite or carbon-graphite;
- f. tantalum or tantalum alloys;
- g. titanium or titanium alloys;
- h. zirconium or zirconium alloys; or
- i. (columbium) or niobium alloys.

Technical Note *carbon-graphite is a composition consisting of amorphous carbon and graphite, in which the graphite-content is eight percent or more by weight.*

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(8) Pumps

Multiple-seal and seal-less pumps with manufacturer's specified maximum flow-rate greater than 0.6 m³/h, or vacuum pumps with manufacturer's specified maximum flow-rate greater than 5 m³/h (under standard temperature (273 K (0° C)) and pressure (101.3 kPa) conditions), and casings (pump bodies), preformed casing liners, impellers, rotors or jet pump nozzles designed for such pumps, in which all surfaces that come into direct contact with the chemical(s) being processed are made from any of the following materials:

- a. nickel or alloys with more than 40% nickel by weight;
- b. alloys with more than 25% nickel and 20% chromium by weight;
- c. fluoropolymers (polymeric or elastomeric materials with more than 35% fluorine by weight);
- d. glass or glass-lined (including vitrified or enamelled coating);
- e. graphite or carbon-graphite;
- f. tantalum or tantalum alloys;
- g. titanium or titanium alloys;
- h. zirconium or zirconium alloys;
- i. ceramics;
- j. ferrosilicon (high silicon iron alloys); or
- k. niobium (columbium) or niobium alloys.

Technical note 1 carbon-graphite is a composition consisting of amorphous carbon and graphite, in which the graphite content is eight percent or more by weight.

Technical note 2 The seals referred to in this control come into direct contact with the chemical(s) being processed (or are designed to), and provide a sealing function where a rotary or reciprocating drive shaft passes through a pump body.

(9) Incinerators

Incinerators designed to destroy CW agents, Category 1 chemicals or chemical munitions, having specially designed waste supply systems, special handling facilities, and an average combustion chamber temperature greater than 1000° C, in which all surfaces in the waste supply system that come into direct contact with the waste products are made from or lined with the following materials:

- a. nickel or alloys with more than 40% nickel by weight;
- b. alloys with more than 25% nickel and 20% chromium by weight; or ceramics.

Notes to 3D001:

Technical note For the listed materials in 3D001, the term 'alloy' when not accompanied by a specific elemental concentration is understood as identifying those alloys where the identified metal is present in a higher percentage by weight than any other element.

Note 1 The objective of these controls should not be defeated by the transfer of any non-controlled item containing one or more controlled components where the controlled component or components are the principal element of the item and can feasibly be removed or used for other purposes.

N.B. In judging whether the controlled component or components are to be considered the principal element, the licensing authority should weigh the factors of quantity, value, and technological know-how involved and other special circumstances which might establish the controlled component or components as the principal element of the item being procured.

Note 2 The objective of these controls should not be defeated by the transfer of a whole plant, on any scale, which has been designed to produce any CW agent or Category 1 chemical.

Note 3. The materials used for gaskets, packing, seals, screws, washers or other materials performing a sealing function do not determine the status of control of the items listed below, provided that such components are designed to be interchangeable.

Note 4 The controls in 3D001 do not apply to equipment which is specially designed for use in civil applications (for example food processing, pulp and paper processing, or water purification, etc) and

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is, by the nature of its design, inappropriate for use in storing, processing, producing or conducting and controlling the flow of chemical warfare agents or any of the Category 1 chemicals.

Note : Export of chemicals (Excluding Software and Technology) listed in 3D001 below is allowed to the countries specified in Table 1 on the basis of a one time General authorization for export of Chemicals and related equipments (GAEC) issued by DGFT, subject to the following conditions(including those below) and the procedure as prescribed from time to time:

- I. The exporter is required to register and obtain General authorization for export of Chemicals and related equipments only once during the validity period. Subsequent export/re-export is subject to post reporting(s) on quarterly basis to relevant Govt. authorities;
- II. General authorization for export of Chemicals and related equipments issued for export / re-export of SCOMET items under the above Categories / Sub Categories (excluding software and technology) shall be valid for a period of **five years** from the date of issue of General authorization for export of Chemicals and related equipments subject to subsequent post reporting(s) on quarterly basis to be reported within 30 days from the last quarter.

Table 1

Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Republic of Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Republic of Korea, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom, United States.
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Note : Export of items under the 3D001 category to the countries other than those specified in Table 1 and the Category 3D001 related software and technology to any country shall be allowed only against an export authorisation and in compliance to the conditions and documentation specified under the respective Categories of SCOMET List.

3D002 [Reserved]

3D003 [Reserved]

3D004¹⁴ Toxic gas monitors and monitoring systems, and their dedicated detecting components as follows: detectors; sensor devices; replaceable sensor cartridges; and dedicated software for such equipment;

- a. designed for continuous operation and usable for the detection of chemical warfare agents or Category 1 chemicals at concentrations of less than 0.3 mg/m³; or
- b. designed for the detection of cholinesterase-inhibiting activity.

Note : Export of chemicals (Excluding Software and Technology) listed in 3D004 below is allowed to the countries specified in Table 1 on the basis of a one time General authorization for export of Chemicals and related equipments (GAEC) issued by DGFT, subject to the following conditions(including those below) and the procedure as prescribed from time to time:

- I. The exporter is required to register and obtain General authorization and related equipments for export of Chemicals only once during the validity period. Subsequent export/re-export is subject to post reporting(s) on quarterly basis to relevant Govt. authorities;
- II. General authorization for export of Chemicals and related equipments issued for export / re-export of SCOMET items under the above Categories / Sub Categories (excluding software and technology) shall be valid for a period of **five years** from the date of issue of General authorization for export of Chemicals subject to subsequent post reporting(s) on quarterly basis to be reported within 30 days from the last quarter

¹⁴ Para 1(F) of Notification No. 29/2015-20 dated 21.09.2017

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Table 1

Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Republic of Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Republic of Korea, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom, United States.

Note : Export of items under the 3D004 category to the countries other than those specified in Table 1 the Category 3D004 related software and technology to any country shall be allowed only against an export authorisation and in compliance to the conditions and documentation specified under the respective Categories of SCOMET List.

3D005

Containment facilities and related equipment as follows:

- (1) Complete containment facilities that meet the criteria for P3 or P4 (BL3, BL4, L3, L4) containment as specified in the WHO Laboratory Biosafety Manual (3rd edition, Geneva, 2004)
- (2) Equipment designed for fixed installation in containment facilities specified in 3D005a., as follows:
 - (i) Double-door pass-through decontamination autoclaves;
 - (ii) Breathing air suit decontamination showers;
 - (iii) Mechanical-seal or inflatable-seal walkthrough doors.

3D006¹⁵

Fermenters:

- (1) Fermenters capable of cultivation of micro-organisms or of live cells for the production of viruses or toxins, without the propagation of aerosols, having a total internal volume of 20 litres or greater;
- (2) Components designed for such fermenters, as follows:-
 - a. cultivation chambers designed to be sterilized or disinfected in situ;
 - b. cultivation chamber holding devices; or
 - c. process control units capable of simultaneously monitoring and controlling two or more fermentation system parameters (e.g. temperature, pH, nutrients, agitation, dissolved oxygen, air flow, foam control).

Technical Note 1 Fermenters include bioreactors (including single-use (disposable) bioreactors), chemostats and continuous-flow systems.

Technical Note 2 Cultivation chamber holding devices include single-use cultivation chambers with rigid walls.

3D007

Centrifugal separators capable of the continuous separation of pathogenic micro-organisms, without the propagation of aerosols, and having all the following characteristics:

- a. one or more sealing joints within the steam containment area;
- b. a flow rate greater than 100 litres per hour;
- c. components of polished stainless steel or titanium;
- d. capable of in-situ steam sterilisation in a closed state.

Technical Note Centrifugal separators include decanters.

3D008

Cross (tangential) flow filtration equipment

- (1) Cross (tangential) flow filtration equipment capable of separation of micro-organisms, viruses, toxins or cell cultures having all the following characteristics:
 - a. a total filtration area equal to or greater than 1 square metre; and
 - b. having any of the following characteristics:
 - i. capable of being sterilized or disinfected in-situ; or

¹⁵ Para 1(G) of Notification No. 29/2015-20 dated 21.09.2017

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- ii. using disposable or single-use filtration components.

Note This control excludes reverse osmosis and hemodialysis equipment.

- (2) Cross (tangential) flow filtration components (e.g. modules, elements, cassettes, cartridges, units or plates) with filtration area equal to or greater than 0.2 square metres for each component and designed for use in cross (tangential) flow filtration equipment as specified above.

Technical note In this control, 'sterilized' denotes the elimination of all viable microbes from the equipment through the use of either physical (e.g. steam) or chemical agents. 'Disinfected' denotes a process to reduce the number of microorganisms but not usually of bacterial spores, through the use of chemical agents, without necessarily killing or removing all organisms.

3D009 Steam, gas or vapour sterilisable freeze-drying equipment with a condenser capacity of 10 kg of ice or greater in 24 hours and less than 1000 kg of ice in 24 hours.

3D010 Spray drying equipment capable of drying toxins or pathogenic microorganisms having all of the following characteristics:

- a. A water evaporation capacity of ≥ 0.4 kg/h and ≤ 400 kg/h;
- b. The ability to generate a typical mean product particle size of ≤ 10 micrometers with existing fittings or by minimal modification of the spray-dryer with atomization nozzles enabling generation of the required particle size; and
- c. Capable of being sterilized or disinfected in situ.

3D011 Protective and containment equipment as follows:

- a. Protective full or half suits, or hoods dependent upon a tethered external air supply and operating under positive pressure;

Technical Note This does not control suits designed to be worn with self-contained breathing apparatus.

- b. Biocontainment chambers, isolators, or biological safety cabinets having all of the following characteristics, for normal operation:
 - i. fully enclosed workspace where the operator is separated from the work by a physical barrier;
 - ii. able to operate at negative pressure;
 - iii. means to safely manipulate items in the workspace;
 - iv. supply and exhaust air to and from the workspace is HEPA filtered.

Note 1 This control includes class III biosafety cabinets, as described in the latest edition of the WHO Laboratory Biosafety Manual or constructed in accordance with national standards, regulations or guidance.

Note 2 This control includes any isolator meeting all of the above-mentioned characteristics, regardless of its intended use and its designation, except for medical isolators specially designed for barrier nursing or transportation of infected patients.

3D012 Aerosol inhalation equipment designed for aerosol challenge testing with micro-organisms, viruses or toxins as follows:

- a. Whole-body exposure chambers having a capacity of 1 cubic metre or greater.
- b. Nose-only exposure apparatus utilising directed aerosol flow and having capacity for exposure of 12 or more rodents, or 2 or more animals other than rodents; and, closed animal restraint tubes designed for use with such apparatus.

3D013 Spraying or fogging systems and components therefor, as follows:

- a. Complete spraying or fogging systems, specially designed or modified for fitting to aircraft, lighter than air vehicles or UAVs, capable of delivering, from a liquid suspension, an initial droplet "VMD" of less than 50 microns at a flow rate of greater than two litres per minute.
- b. Spray booms or arrays of aerosol generating units, specially designed or modified for fitting to

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aircraft, lighter than air vehicles or UAVs, capable of delivering, from a liquid suspension, an initial droplet “VMD” of less than 50 microns at a flow rate of greater than two litres per minute.

- c. Aerosol generating units specially designed for fitting to systems that fulfil all the criteria specified in 3D011.a and 3D001.b

Technical Notes

- (1) *Aerosol generating units are devices specially designed or modified for fitting to aircraft such as nozzles, rotary drum atomisers and similar devices.*
- (2) *This entry does not control spraying or fogging systems and components as specified in 3D010 that are demonstrated not to be capable of delivering biological agents in the form of infectious aerosols.*
- (3) *Droplet size for spray equipment or nozzles specially designed for use on aircraft or UAVs should be measured using either of the following methods:*
 - (a) *Doppler laser method*
 - (b) *Forward laser diffraction method*

3D014¹⁶ Nucleic acid assemblers and synthesizers, which are partly or entirely automated, and designed to generate continuous nucleic acids greater than 1.5 kilobases in length with error rates less than 5% in a single run.’

3D015 Technology and software for the development, production or use of items in 3D001 to 3D014.

3D016 Combustors or pyrolysers capable of a heat-zone (‘burner’) temperature greater than 1,273 K (1000 Degree Centigrade), and in which any surfaces that come into direct contact with material coming into the containing chamber are made from, or lined with, any of the following materials:

- a. Alloys with more than 25% nickel and 25% chromium by weight; (e.g., ‘Hastelloy’, ‘Illium’, ‘Inconel’, ‘Incoloy’)
- b. Nickel, or alloys with more than 40% nickel by weight;
- c. Titanium; or
- d. Ceramics’

Category 4 Nuclear-related other equipment, assemblies and components; test and production equipment; and related technology not controlled under Category 0

4A Equipment, assemblies, components including test and production equipment

4A001 Flow-forming machines, spin-forming machines capable of flow-forming functions, and mandrels, as follows:

- a. For flow forming machines refer to 5A205
- b. Spin forming machines having both of the following characteristics:
 - 1. Three or more rollers (active or guiding); and
 - 2. Which, according to the manufacturer’s technical specification, can be equipped with ‘numerical control’ units or a computer control.
- c. Rotor-forming mandrels designed to form cylindrical rotors of inside diameter between 75 and 650 mm.

Note Item 4A001a and 4A001b include machines which have only a single roller designed to deform metal plus two auxiliary rollers which support the mandrel, but do not participate directly in the deformation process.

¹⁶ Para 1(H) of Notification No. 29/2015-20 dated 21.09.2017

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4A002

Machine tools, as follows, and any combination thereof, for removing or cutting metals, ceramics, or composites, which, according to the manufacturer's technical specifications, can be equipped with electronic devices for simultaneous "contouring control" in two or more axes;

N.B.: For "numerical control" units controlled by their associated "software", see Item 4C

- a. Machine tools for turning, that have "positioning accuracies" with all compensations available better (less) than 6 μm according to ISO 230/2 (1988) along any linear axis (overall positioning) for machines capable of machining diameters greater than 35 mm;

Note Item 4A002.a. does not control bar machines (Swissturn), limited to machining only bar feed thru, if maximum bar diameter is equal to or less than 42 mm and there is no capability of mounting chucks. Machines may have drilling and/or milling capabilities for machining parts with diameters less than 42 mm.

- b. Machine tools for milling, having any of the following characteristics:

1. "Positioning accuracies" with all compensations available better (less) than 6 μm according to ISO 230/2 (1988) along any linear axis (overall positioning);
2. Two or more contouring rotary axes; or
3. Five or more axes which can be coordinated simultaneously for "contouring control".

Note Item 4A002.b. does not control milling machines having both of the following characteristics:

1. X-axis travel greater than 2 m; and
2. Overall "positioning accuracy" on the x-axis worse (more) than 30 μm according to ISO 230/2 (1988).

- c. Machine tools for grinding, having any of the following characteristics:

1. "Positioning accuracies" with all compensations available better (less) than 4 μm according to ISO 230/2 (1988) along any linear axis (overall positioning);
2. Two or more contouring rotary axes; or
3. Five or more axes which can be coordinated simultaneously for "contouring control".

Note Item 4A002.c. does not control grinding machines as follows:

1. Cylindrical external, internal, and external-internal grinding machines having all the following characteristics:

- a. Limited to a maximum workpiece capacity of 150 mm outside diameter or length; and

- b. Axes limited to x, z and c.

2. Jig grinders that do not have a z-axis or a w-axis with an overall "positioning accuracy" less (better) than 4 μm according to ISO 230/2 (1988).'

- d. Non-wire type Electrical Discharge Machines (EDM) that have two or more contouring rotary axes and that can be coordinated simultaneously for "contouring control".

Notes 1 Stated "positioning accuracy" levels derived under the following procedures from measurements made according to ISO 230/2 (1988) or national equivalents may be used for each machine tool model if provided to, and accepted by, national authorities instead of individual machine tests.

Stated "positioning accuracy" levels are to be derived as follows:

- a. Select five machines of a model to be evaluated;
- b. Measure the linear axis accuracies according to ISO 230/2 (1988);
- c. Determine the accuracy values (A) for each axis of each machine. The method of calculating the accuracy value is described in the ISO 230/2 (1988) standard;

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- d. Determine the average accuracy value of each axis. This average value becomes the stated “positioning accuracy” of each axis for the model ($\hat{A}_x, \hat{A}_y...$);
 - e. Since Item 4A002. refers to each linear axis, there will be as many stated “positioning accuracy” values as there are linear axes;
 - f. If any axis of a machine tool not controlled by Items 4A002.a., 4A002.b., or 4A002.c. has a stated “positioning accuracy” of 6 μm or better (less) for grinding machines, and 8 μm or better (less) for milling and turning machines, both according to ISO 230/2 (1988), then the builder should be required to reaffirm the accuracy level once every eighteen months.
2. Item 4A002. does not control special purpose machine tools limited to the manufacture of any of the following parts:
 - a. Gears;
 - b. Crankshafts or cam shafts;
 - c. Tools or cutters;
 - d. Extruder worms.

Technical Notes

1. Axis nomenclature shall be in accordance with ISO 841 (2001), “Numerical Control Machines - Axis and Motion Nomenclature”.
2. Not counted in the total number of contouring axes are secondary parallel contouring axes (e.g., the w-axis on horizontal boring mills or a secondary rotary axis the centreline of which is parallel to the primary rotary axis).
3. Rotary axes do not necessarily have to rotate over 360°. A rotary axis can be driven by a linear device, e.g., a screw or a rack-and-pinion.
4. For the purposes of 4A002. the number of axes which can be coordinated simultaneously for “contouring control” is the number of axes along or around which, during processing of the workpiece, simultaneous and interrelated motions are performed between the workpiece and a tool. This does not include any additional axes along or around which other relative motions within the machine are performed, such as:
 - a. Wheel-dressing systems in grinding machines;
 - b. Parallel rotary axes designed for mounting of separate workpieces;
 - c. Co-linear rotary axes designed for manipulating the same workpiece by holding it in a chuck from different ends.
5. A machine tool having at least 2 of the 3 turning, milling or grinding capabilities (e.g., a turning machine with milling capability) must be evaluated against each applicable entry, 4A002.a., 4A002.b. and 4A002.c.
6. Items 4A002.b.3. and 4A002.c.3. include machines based on a parallel linear kinematic design (e.g., hexapods) that have 5 or more axes none of which is a rotary axis.’

4A003

Dimensional inspection machines, instruments, or systems, as follows:

- a. Computer controlled or numerically controlled coordinate measuring machines (CMM) having either of the following characteristics:
 1. Having only two axes and having a maximum permissible error of length measurement along any axis (one dimensional), identified as any combination of E0x MPE, E0y MPE or E0z MPE, equal to or less(better) than $(1.25 + L/1000) \mu\text{m}$ (where L is the measured

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length in mm) at any point within the operating range of the machine (i.e., within the length of the axis), according to ISO 10360-2(2009); or

2. Three or more axes and having a three dimensional (volumetric) maximum permissible error of length measurement (E_0 , MPE equal to or less (better) than $(1.7 + L/800) \mu\text{m}$ (where L is the measured length in mm) at any point within the operating range of the machine (i.e., within the length of the axis), according to ISO 10360-2(2009).

Technical Note The E_0 , MPE of the most accurate configuration of the CMM specified according to ISO 10360-2(2009) by the manufacturer (e.g., best of the following: probe stylus length, motion parameters, environment) and with all compensations available shall be compared to the $1.7 + L/800 \mu\text{m}$ threshold.

b. Linear displacement measuring instruments, as follows:

1. Non-contact type measuring systems with a “resolution” equal to or better (less) than $0.2 \mu\text{m}$ within a measuring range up to 0.2 mm ;
2. Linear variable differential transformer (LVDT) systems having both of the following characteristics:
 - a. 1. “Linearity” equal to or less (better) than 0.1% measured from 0 to the full operating range, for LVDTs with an operating range up to 5 mm ; or
 2. “Linearity” equal to or less (better) than 0.1% measured from 0 to 5 mm for LVDTs with an operating range greater than 5 mm ; and
 - b. Drift equal to or better (less) than 0.1% per day at a standard ambient test room temperature $\pm 1 \text{ K}$ ($\pm 1^\circ\text{C}$);
3. Measuring systems having both of the following characteristics:
 - a. Containing a laser; and
 - b. Capable of maintaining for at least 12 hours, over a temperature range of $\pm 1 \text{ K}$ ($\pm 1^\circ\text{C}$) around a standard temperature and a standard pressure:
 1. A “resolution” over their full scale of $0.1 \mu\text{m}$ or better; and
 2. With a “measurement uncertainty” equal to or better (less) than $(0.2 + L/2000) \mu\text{m}$ (L is the measured length in mm);

Note Item 4A003.b.3. does not control measuring interferometer systems, without closed or open loop feedback, containing a laser to measure slide movement errors of machine tools, dimensional inspection machines, or similar equipment

*Technical Note In Item 4A003.b.3. ‘linear displacement’ means the change of distance between the measuring **probe** and the measured object.*

- c. Angular displacement measuring instruments having an “angular position deviation” equal to or better (less) than 0.00025° ;

Note Item 4A003.c does not control optical instruments, such as autocollimators, using collimated light (e.g., laser light) to detect angular displacement of a mirror.

- d. Systems for simultaneous linear-angular inspection of hemishells, having both of the following characteristics:
 1. “Measurement uncertainty” along any linear axis equal to or better (less) than $3.5 \mu\text{m}$ per 5 mm ; and
 2. “Angular position deviation” equal to or less than 0.02° .

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Notes

1. “Item 4A003.d includes machine tools other than those specified by 4A002, that can be used as measuring machines if they meet or exceed the criteria specified for the measuring machine function.
2. Machines described in Item 4A003.d. are controlled if they exceed the threshold specified anywhere within their operating range”.

Technical Note All parameters of measurement values in this item represent plus/minus, i.e., not total band.

4A004

Controlled atmosphere (vacuum or inert gas) induction furnaces, and power supplies therefor, as follows:

- a. Furnaces having all of the following characteristics:
 1. Capable of operation at temperatures above 1123 K (850 °C);
 2. Induction coils 600 mm or less in diameter; and
 3. Designed for power inputs of 5 kW or more;

Note Item 4A004.a. does not control furnaces designed for the processing of semiconductor wafers.

- b. Power supplies, with a specified output power of 5 kW or more, specially designed for furnaces specified in Item 4A004.a.

4A005

Isostatic presses', and related equipment, as follows:

- a. 'Isostatic presses' as specified in 5A208;
- b. Dies, moulds, and controls specially designed for the 'isostatic presses' specified in Item 4A005.a.

Technical Notes

1. In Item 4A005 'Isostatic presses' means equipment capable of pressurizing a closed cavity through various media (gas, liquid, solid particles, etc.) to create equal pressure in all directions within the cavity upon a work piece or material.
2. In Item 4A005 the inside chamber dimension is that of the chamber in which both the working temperature and the working pressure are achieved and does not include fixtures. That dimension will be the smaller of either the inside diameter of the pressure chamber or the inside diameter of the insulated furnace chamber, depending on which of the two chambers is located inside the other.

4A006

Vibration test systems, equipment, and components as follows:

- a. Electrodynamic vibration test systems, having all of the following characteristics:
 1. Employing feedback or closed loop control techniques and incorporating a digital control unit;
 2. Capable of vibrating at 10 g RMS or more between 20 and 2000 Hz; and
 3. Capable of imparting forces of 50 kN or greater measured 'bare table';
- b. Digital control units, combined with 'software' specially designed for vibration testing, with a real-time bandwidth greater than 5 kHz and being designed for a system specified in Item 4A006.a.;
- c. Vibration thrusters (shaker units), with or without associated amplifiers, capable of imparting a force of 50 kN or greater measured 'bare table', which are usable for the systems specified in Item 4A006.a.;

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d. Test piece support structures and electronic units designed to combine multiple shaker units into a complete shaker system capable of providing an effective combined force of 50 kN or greater, measured 'bare table,' which are usable for the systems specified in Item 4A006.a..

Technical Note In Item 4A006 'bare table' means a flat table, or surface, with no fixtures or fittings.

4A007¹⁷

Vacuum or other controlled atmosphere metallurgical melting and casting furnaces and related equipment, as follows:-

a. Arc remelt furnaces, arc melt furnaces and arc melt and casting furnaces having both of the following characteristics:-

1. Consumable electrode capacities between 1000 and 20000 cm³; and
2. Capable of operating with melting temperatures above 1973 K (1700 °C);

b. Electron beam melting furnaces, plasma atomisation furnaces and plasma melting furnaces, having both of the following characteristics:

1. A power of 50 kW or greater; and
2. Capable of operating with melting temperatures above 1473 K (1200 °C);

c. Computer control and monitoring systems specially configured for any of the furnaces specified in 4A007.a. or 4A007.b;

d. Plasma torches specially designed for the furnaces specified in 4A007.b. having both of the following characteristics:-

1. Operating at a power greater than 50kW; and
2. Capable of operating above 1473 K (1200°C);

e. Electron beam guns specially designed for the furnaces specified in 4A007.b. operating at a power greater than 50kW. '

4A008

Crucibles made of materials resistant to liquid actinide metals, as follows:

a. Crucibles having both of the following characteristics:

1. A volume of between 150 cm³ (150 ml) and 8000 cm³ (8 litres); and
2. Made of or coated with any of the following materials, or combination of the following materials, having an overall impurity level of 2% or less by weight:
 - a. Calcium fluoride (CaF₂);
 - b. Calcium zirconate (metazirconate) (CaZrO₃);
 - c. Cerium sulphide (Ce₂S₃);
 - d. Erbium oxide (erbia) (Er₂O₃);
 - e. Hafnium oxide (hafnia) (HfO₂);
 - f. Magnesium oxide (MgO);
 - g. Nitrided niobium-titanium-tungsten alloy (approximately 50% Nb, 30% Ti, 20% W);
 - h. Yttrium oxide (yttria) (Y₂O₃); or
 - i. Zirconium oxide (zirconia) (ZrO₂);

¹⁷ Para 1(I) of Notification No. 29/2015-20 dated 21.09.2017

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- b. Crucibles having both of the following characteristics:
 - 1. A volume of between 50 cm³ (50 ml) and 2000 cm³ (2 litres); and
 - 2. Made of or lined with tantalum, having a purity of 99.9% or greater by weight;
 - c. Crucibles having all of the following characteristics:
 - 1. A volume of between 50 cm³ (50 ml) and 2000 cm³ (2 litres);
 - 2. Made of or lined with tantalum, having a purity of 98% or greater by weight; and
 - 3. Coated with tantalum carbide, nitride, boride, or any combination thereof.
- 4A009** Platinized catalysts specially designed or prepared for promoting the hydrogen isotope exchange reaction between hydrogen and water for the recovery of tritium from heavy water or for the production of heavy water.
- 4A010** Composite structures in the form of tubes having both of the following characteristics:
- a. An inside diameter of between 75 and 400 mm; and
 - b. Made with any of the materials specified in Item 3A116.
- 4A011** Frequency changers or generators, usable as a variable frequency or fixed frequency motor drive, having all of the following characteristics:
- N.B.1: Frequency changers and generators specially designed or prepared for the gas centrifuge process are controlled under Prescribe Equipment (0B)
- N.B.2: “Software” specially designed to enhance or release the performance of frequency changers or generators to meet the characteristics below is controlled (see Item 4C).
- a. Multiphase output providing a power of 40 VA or greater;
 - b. Operating at a frequency of 600 Hz or more; and
 - c. Frequency control better (less) than 0.2%.
- Notes 1. Item 4A011 only controls frequency changers intended for specific industrial machinery and/or consumer goods (machine tools, vehicles, etc.) if the frequency changers can meet the characteristics above when removed,
2. For the purpose of export control, the Government will determine whether or not a particular frequency changer meets the characteristics above, taking into account hardware and software constraints.
- Technical Notes 1. Frequency changers in Item 4A011. are also known as converters or inverters.
2. The characteristics specified in item 4A011 may be met by certain equipment marketed such as: Generators, Electronic Test Equipment, AC Power Supplies, Variable Speed Motor Drives, Variable Speed Drives (VSDs), Variable Frequency Drives (VFDs), Adjustable Frequency Drives (AFDs), or Adjustable Speed Drives (ASDs).
- 4A012** Lasers, laser amplifiers and oscillators as follows:
- a. Copper vapour lasers having both of the following characteristics:
 - 1. Operating at wavelengths between 500 and 600 nm; and
 - 2. An average output power equal to or greater than 40 W;
 - b. Argon ion lasers having both of the following characteristics:
 - 1. Operating at wavelengths between 400 and 515 nm; and
 - 2. An average output power greater than 40 W;

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- c. Neodymium-doped (other than glass) lasers with an output wavelength between 1000 and 1100 nm having either of the following:
 - 1. Pulse-excited and Q-switched with a pulse duration equal to or greater than 1 ns, and having either of the following:
 - a. A single-transverse mode output with an average output power greater than 40 W; or
 - b. A multiple-transverse mode output with an average output power greater than 50 W;
 - or
 - 2. Incorporating frequency doubling to give an output wavelength between 500 and 550 nm with an average output power of greater than 40 W;
 - d. Tuneable pulsed single-mode dye laser oscillators having all of the following characteristics:
 - 1. Operating at wavelengths between 300 and 800 nm;
 - 2. An average output power greater than 1 W;
 - 3. A repetition rate greater than 1 kHz; and
 - 4. Pulse width less than 100 ns;
 - e. Tuneable pulsed dye laser amplifiers and oscillators having all of the following characteristics:
 - 1. Operating at wavelengths between 300 and 800 nm;
 - 2. An average output power greater than 30 W;
 - 3. A repetition rate greater than 1 kHz; and
 - 4. Pulse width less than 100 ns;
- Note Item 4A012e does not control single mode oscillators.*
- f. Alexandrite lasers having all of the following characteristics:
 - 1. Operating at wavelengths between 720 and 800 nm;
 - 2. A bandwidth of 0.005 nm or less;
 - 3. A repetition rate greater than 125 Hz; and
 - 4. An average output power greater than 30 W;
 - g. Pulsed carbon dioxide lasers having all of the following characteristics:
 - 1. Operating at wavelengths between 9000 and 11000 nm;
 - 2. A repetition rate greater than 250 Hz;
 - 3. An average output power greater than 500 W; and
 - 4. Pulse width of less than 200 ns;
- Note Item 4A012g does not control the higher power (typically 1 to 5 kW) industrial CO₂ lasers used in applications such as cutting and welding, as these latter lasers are either continuous wave or are pulsed with a pulse width greater than 200 ns.*
- h. Pulsed excimer lasers (XeF, XeCl, KrF) having all of the following characteristics:
 - 1. Operating at wavelengths between 240 and 360 nm;
 - 2. A repetition rate greater than 250 Hz; and
 - 3. An average output power greater than 500 W;
 - i. Para-hydrogen Raman shifters designed to operate at 16 µm output wavelength and at a repetition rate greater than 250 Hz.
 - j. Pulsed carbon monoxide lasers having all of the following characteristics:

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1. Operating at wavelengths between 5000 and 6000 nm;
2. A repetition rate greater than 250 Hz;
3. An average output power greater than 200 W; and
4. Pulse width of less than 200 ns.

Note Item 4A012.j. does not control the higher power (typically 1 to 5 kW) industrial CO lasers used in applications such as cutting and welding, as these latter lasers are either continuous wave or are pulsed with a pulse width greater than 200 ns.

4A013

Valves having all of the following characteristics:

- a. A nominal size of 5 mm or greater;
- b. Having a bellows seal; and
- c. Wholly made of or lined with aluminium, aluminium alloy, nickel, or nickel alloy containing more than 60% nickel by weight.

Technical Note For valves with different inlet and outlet diameter, the nominal size parameter in Item 4A013a refers to the smallest diameter.

4A014

Superconducting solenoidal electromagnets having all of the following characteristics:

- a. Capable of creating magnetic fields greater than 2 T;
- b. A ratio of length to inner diameter greater than 2;
- c. Inner diameter greater than 300 mm; and
- d. Magnetic field uniform to better than 1% over the central 50% of the inner volume.

Note: Item 4A014 does not control magnets specially designed for and exported as part of medical nuclear magnetic resonance (NMR) imaging systems. ('As part of' does not necessarily mean physical part in the same shipment. Separate shipments from different sources are allowed, provided the related export documents clearly specify the 'as part of' relationship.)

4A015

High-power direct current power supplies having both of the following characteristics:

- a. Capable of continuously producing, over a time period of 8 hours, 100 V or greater with current output of 500 A or greater; and
- b. Current or voltage stability better than 0.1% over a time period of 8 hours

4A016

High-voltage direct current power supplies having both of the following characteristics:

- a. Capable of continuously producing, over a time period of 8 hours, 20 kV or greater with current output of 1 A or greater; and
- b. Current or voltage stability better than 0.1% over a time period of 8 hours.

4A017

All types of pressure transducers capable of measuring absolute pressures and having all of the following characteristics:

- a. Pressure sensing elements made of or protected by aluminium, aluminium alloy, aluminium oxide (alumina or sapphire), nickel, nickel alloy with more than 60% nickel by weight, or fully fluorinated hydrocarbon polymers;
- b. Seals, if any, essential for sealing the pressure sensing element, and in direct contact with the process medium, made of or protected by aluminium, aluminium alloy, aluminium oxide (alumina or sapphire), nickel, nickel alloy with more than 60% nickel by weight, or fully fluorinated hydrocarbon polymers; and
- c. Having either of the following characteristics:
 1. A full scale of less than 13 kPa and an "accuracy" of better than 1% of full scale; or

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2. A full scale of 13 kPa or greater and an “accuracy” of better than 130 Pa when measuring at 13 kPa”.

Technical Notes

1. In Item 4A017, pressure transducers are devices that convert pressure measurements into a signal.
2. In Item 4A017, “accuracy” includes non-linearity, hysteresis and repeatability at ambient temperature.

4A018

Vacuum pumps having all of the following characteristics:

- a. Input throat size equal to or greater than 380 mm;
- b. Pumping speed equal to or greater than 15 m³/s; and
- c. Capable of producing an ultimate vacuum better than 13.3 mPa.

Technical Notes

1. The pumping speed is determined at the measurement point with nitrogen gas or air.
2. The ultimate vacuum is determined at the input of the pump with the input of the pump blocked off.

4A019

Electrolytic cells for fluorine production with an output capacity greater than 250 g of fluorine per hour.

4A020

Rotor fabrication or assembly equipment, rotor straightening equipment, bellows-forming mandrels and dies, as follows:

- a. Rotor assembly equipment for assembly of gas centrifuge rotor tube sections, baffles, and end caps;
Note Item 4A020a includes precision mandrels, clamps, and shrink fit machines.
- b. Rotor straightening equipment for alignment of gas centrifuge rotor tube sections to a common axis;

Technical Note In Item 4A020b such equipment normally consists of precision measuring probes linked to a computer that subsequently controls the action of, for example, pneumatic rams used for aligning the rotor tube sections.

- c. Bellows-forming mandrels and dies for producing single -convolution bellows.

Technical Note The bellows Deposition Techniques referred to in Item 4A020c have all of the following characteristics:

1. Inside diameter between 75 and 650mm;
2. Length equal to or greater than 12.7 mm;
3. Single convolution depth greater than 2 mm; and
4. Made of high-strength aluminium alloys, maraging steel, or high strength fibrous or filamentary materials.

4A021

Centrifugal multi-plane balancing machines, fixed or portable, horizontal or vertical, as follows:

- a. Centrifugal balancing machines designed for balancing flexible rotors having a length of 600 mm or more and having all of the following characteristics:
 1. Swing or journal diameter greater than 75 mm;
 2. Mass capability of from 0.9 to 23 kg; and
 3. Capable of balancing speed of revolution greater than 5000 rpm;
- b. Centrifugal balancing machines designed for balancing hollow cylindrical rotor components and having all of the following characteristics:

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1. Journal diameter greater than 75 mm;
2. Mass capability of from 0.9 to 23 kg;
3. A minimum achievable residual specific unbalance equal to or less than 10 g-mm/kg per plane; and
4. Belt drive type.

4A022

Filament winding machines and related equipment, as follows:

- a. Filament winding machines as specified in 5A206; and having all of the following characteristics:
 1. Having motions for positioning, wrapping, and winding fibers coordinated and programmed in two or more axes;
 2. Specially designed to fabricate composite structures or laminates from “fibrous or filamentary materials”; and
 3. Capable of winding cylindrical tubes with an internal diameter between 75 and 650 mm and lengths of 300 mm or greater;
- b. Coordinating and programming controls for the filament winding machines specified in Item 4A022a;
- c. Precision mandrels for the filament winding machines specified in Item 4A022a.

4A023

Electromagnetic isotope separators designed for, or equipped with, single or multiple ion sources capable of providing a total ion beam current of 50 mA or greater.

Notes

1. Item 4A023 includes separators capable of enriching stable isotopes as well as those for uranium. (A separator capable of separating the isotopes of lead with a one-mass unit difference is inherently capable of enriching the isotopes of uranium with a three-unit mass difference.)
2. Item 4A023 includes separators with the ion sources and collectors both in the magnetic field and those configurations in which they are external to the field.

Technical Note A single 50 mA ion source cannot produce more than 3 g of separated highly enriched uranium (HEU) per year from natural abundance feed.

4A024 Mass spectrometers capable of measuring ions of 230 u or greater and having a resolution of better than 2 parts in 230, as follows, and ion sources therefor:

N.B.: Mass spectrometers especially designed or prepared for analysing on-line samples of uranium hexafluoride (UF₆) are controlled under Prescribed Equipment (OB Category).

- a. Inductively coupled plasma mass spectrometers (ICP/MS);
- b. Glow discharge mass spectrometers (GDMS);
- c. Thermal ionisation mass spectrometers (TIMS);
- d. Electron bombardment mass spectrometers having both of the following features:
 1. A molecular beam inlet system that injects a collimated beam of analyte molecules into a region of the ion source where the molecules are ionised by an electron beam; and
 2. One or more cold traps that can be cooled to a temperature of 193 K (-80 °C) or less in order to trap analyte molecules that are not ionised by the electron beam;
- e. Mass spectrometers equipped with a microfluorination ion source designed for actinides or actinide fluorides.

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Note: 'u' stands for unified atomic mass unit or dalton and is defined as one twelfth of the mass of an unbound neutral atom of Carbon-12 in its nuclear and electronic ground state and at rest, and has a value of $1.660539040 \times 10^{-27}$ kg. 'u' is a non-SI unit accepted for use with the SI units.

Technical Notes

1. 4A024.d describes mass spectrometers that are typically used for isotopic analysis of UF₆ gas samples.
2. Electron bombardment mass spectrometers in Item 4A024.d. are also known as electron impact mass spectrometers or electron ionization mass spectrometers.
3. In 4A024.d.2, a 'cold trap' is a device that traps gas molecules by condensing or freezing them on cold surfaces. For the purposes of this entry, a closed-loop gaseous helium cryogenic vacuum pump is not a cold trap.

4A025 Specialized packings which may be used in separating heavy water from ordinary water, having both of the following characteristics:

- a. Made of phosphor bronze mesh chemically treated to improve wettability; and
- b. Designed to be used in vacuum distillation towers.

4A026 Pumps capable of circulating solutions of concentrated or dilute potassium amide catalyst in liquid ammonia (KNH₂/NH₃), having all of the following characteristics:

- a. Airtight (i.e., hermetically sealed);
- b. A capacity greater than 8.5 m³/h; and
- c. Either of the following characteristics:
 1. For concentrated potassium amide solutions (1% or greater), an operating pressure of 1.5 to 60 MPa; or
 2. For dilute potassium amide solutions (less than 1%), an operating pressure of 20 to 60 MPa.

4A027 Turboexpanders or turboexpander-compressor sets having both of the following characteristics:

- a. Designed for operation with an outlet temperature of 35 K (- 238 °C) or less; and
- b. Designed for a throughput of hydrogen gas of 1000 kg/h or greater.

4A028 Water-hydrogen sulphide exchange tray columns and internal contactors, as follows:

N.B.: For columns which are specially designed or prepared for the production of heavy water, see Prescribed Equipment (0B002).

- a. Water-hydrogen sulphide exchange tray columns, having all of the following characteristics:
 1. Can operate at pressures of 2 MPa or greater;
 2. Constructed of carbon steel having an austenitic ASTM (or equivalent standard) grain size number of 5 or greater; and
 3. With a diameter of 1.8 m or greater;
- b. Internal contactors for the water-hydrogen sulphide exchange tray columns specified in Item 4A028a.

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Technical Note Internal contactors of the columns are segmented trays which have an effective assembled diameter of 1.8 m or greater; are designed to facilitate counter current contacting and are constructed of stainless steels with a carbon content of 0.03% or less. These may be sieve trays, valve trays, bubble cap trays or turbo grid trays.

4A029

Hydrogen-cryogenic distillation columns having all of the following characteristics:

- a. Designed for operation at internal temperatures of 35 K (-238 °C) or less;
- b. Designed for operation at internal pressures of 0.5 to 5 MPa;
- c. Constructed of either:
 1. Stainless steel of the 300 series with low sulfur content and with an austenitic ASTM (or equivalent standard) grain size number of 5 or greater; or
 2. Equivalent materials which are both cryogenic and H₂-compatible; and
- d. With internal diameters of 30 cm or greater and 'effective lengths' of 4 m or greater.

Technical Note The term 'effective length' means the active height of packing material in a packed-type column, or the active height of internal contactor plates in a plate-type column.

4A030

Bellows-sealed scroll-type compressors and bellows-sealed scroll-type vacuum pumps having all of the following characteristics:

- a. Capable of an inlet volume flow rate of 50 m³/h or greater;
- b. Capable of a pressure ratio of 2:1 or greater; and
- c. Having all surfaces that come in contact with the process gas made from any of the following materials:
 1. Aluminium or aluminium alloy;
 2. Aluminium oxide;
 3. Stainless steel;
 4. Nickel or nickel alloy;
 5. Phosphor bronze; or
 6. Fluoropolymers.

Technical Notes

1. In a scroll compressor or vacuum pump, crescent-shaped pockets of gas are trapped between one or more pairs of intermeshed spiral vanes, or scrolls, one of which moves while the other remains stationary. The moving scroll orbits the stationary scroll; it does not rotate. As the moving scroll orbits the stationary scroll, the gas pockets diminish in size (i.e., they are compressed) as they move toward the outlet port of the machine.

2. In a bellows-sealed scroll compressor or vacuum pump, the process gas is totally isolated from the lubricated parts of the pump and from the external atmosphere by a metal bellows. One end of the bellows is attached to the moving scroll and the other end is attached to the stationary housing of the pump.

3. Fluoropolymers include, but are not limited to, the following materials:

- a. Polytetrafluoroethylene (PTFE),
- b. Fluorinated Ethylene Propylene (FEP),
- c. Perfluoroalkoxy (PFA),
- d. Polychlorotrifluoroethylene (PCTFE); and
- e. Vinylidene fluoride-hexafluoropropylene copolymer.

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4A031

Industrial equipment including assemblies and components as follows:

- a. High-density (lead glass or other) radiation shielding windows, having all of the following characteristics, and specially designed frames therefor:

1. A ‘cold area’ greater than 0.09 m²;
2. A density greater than 3 g/cm³; and
3. A thickness of 100 mm or greater.

Technical Note In Item 4A031.a.1. the term ‘cold area’ means the viewing area of the window exposed to the lowest level of radiation in the design application.

- b. Radiation-hardened TV cameras, or lenses therefor, specially designed or rated as radiation hardened to withstand a total radiation dose greater than 5×10^4 Gy (silicon) without operational degradation.

Technical Note The term Gy (silicon) refers to the energy in Joules per kilogram absorbed by an unshielded silicon sample when exposed to ionizing radiation.

- c. ‘Robots’, ‘end-effectors’ and control units as follows:

1. ‘Robots’ or ‘end-effectors’ having either of the following characteristics:
 - (a) Specially designed to comply with national safety standards applicable to handling high explosives (for example, meeting electrical code ratings for high explosives); or
 - (b) Specially designed or rated as radiation hardened to withstand a total radiation dose greater than 5×10^4 Gy (silicon) without operational degradation;
2. Control units specially designed for any of the ‘robots’ or ‘end-effectors’ specified in Item 4A031.c.1.

Note: Item 1.A.3. does not control ‘robots’ specially designed for non-nuclear industrial applications such as automobile paint-spraying booths.

Technical Notes

1. ‘Robots’ In Item 4A031.c. ‘robot’ means a manipulation mechanism, which may be of the continuous path or of the point-to-point variety, may use “sensors”, and has all of the following characteristics:

- (a) is multifunctional;
- (b) is capable of positioning or orienting material, parts, tools, or special devices through variable movements in three-dimensional space;
- (c) incorporates three or more closed or open loop servo-devices which may include stepping motors; and
- (d) has “user-accessible programmability” by means of teach/playback method or by means of an electronic computer which may be a programmable logic controller, i.e., without mechanical intervention.

N.B.1:

In the above definition “sensors” means detectors of a physical phenomenon, the output of which (after conversion into a signal that can be interpreted by a control unit) is able to generate “programs” or modify programmed instructions or numerical “program” data. This includes “sensors” with machine vision, infrared imaging, acoustical imaging, tactile

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feel, inertial position measuring, optical or acoustic ranging or force or torque measuring capabilities.

N.B.2:

In the above definition “user-accessible programmability” means the facility allowing a user to insert, modify or replace “programs” by means other than:

- (a) a physical change in wiring or interconnections; or*
- (b) the setting of function controls including entry of parameters.*

N.B.3:

The above definition does not include the following devices:

- (a) Manipulation mechanisms which are only manually/teleoperator controllable;*
- (b) Fixed sequence manipulation mechanisms which are automated moving devices operating according to mechanically fixed programmed motions. The “program” is mechanically limited by fixed stops, such as pins or cams. The sequence of motions and the selection of paths or angles are not variable or changeable by mechanical, electronic, or electrical means;*
- (c) Mechanically controlled variable sequence manipulation mechanisms which are automated moving devices operating according to mechanically fixed programmed motions. The “program” is mechanically limited by fixed, but adjustable, stops such as pins or cams. The sequence of motions and the selection of paths or angles are variable within the fixed “program” pattern. Variations or modifications of the “program” pattern (e.g., changes of pins or exchanges of cams) in one or more motion axes are accomplished only through mechanical operations;*
- (d) Non-servo-controlled variable sequence manipulation mechanisms which are automated moving devices, operating according to mechanically fixed programmed motions. The “program” is variable but the sequence proceeds only by the binary signal from mechanically fixed electrical binary devices or adjustable stops;*
- (e) Stacker cranes defined as Cartesian coordinate manipulator systems manufactured as an integral part of a vertical array of storage bins and designed to access the contents of those bins for storage or retrieval.*

2. ‘End-effectors’

In Item 4A031.c. ‘end-effectors’ are grippers, ‘active tooling units’, and any other tooling that is attached to the baseplate on the end of a ‘robot’ manipulator arm.

N.B.:

In the above definition ‘active tooling units’ is a device for applying motive power, process energy or sensing to the workpiece.

d. Remote manipulators that can be used to provide remote actions in radiochemical separation operations or hot cells, having either of the following characteristics:

- 1. A capability of penetrating 0.6 m or more of hot cell wall (through-the-wall operation); or*
- 2. A capability of bridging over the top of a hot cell wall with a thickness of 0.6 m or more (over-the-wall operation).*

Technical Note Remote manipulators provide translation of human operator actions to a remote operating arm and terminal fixture. They may be of a master/slave type or operated by joystick or keypad.

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4A032¹⁸

Target assemblies and components for the production of tritium as follows:-

- a. Target assemblies made of or containing lithium enriched in the lithium-6 isotope specially designed for the production of tritium through irradiation, including insertion in a nuclear reactor;
- b. Components specially designed for the target assemblies specified in 4A032.a.

Technical Note

Components specially designed for target assemblies for the production of tritium may include lithium pellets, tritium getters, and specially-coated cladding.’

4B

Equipment, assemblies and components, including test and measurement equipment usable in development of nuclear explosive devices

4B001

Photomultiplier tubes having both of the following characteristics:

- a. Photocathode area of greater than 20 cm²; and
- b. Anode pulse rise time of less than 1 ns.

4B002

Flash X-ray generators or pulsed electron accelerators having either of the following sets of characteristics:

- a. 1. An accelerator peak electron energy of 500 keV or greater but less than 25 MeV; and
2. With a figure of merit (K) of 0.25 or greater; or
- b. 1. An accelerator peak electron energy of 25 MeV or greater; and
2. A peak power greater than 50 MW.

Note Item 4B002 does not control accelerators that are component parts of devices designed for purposes other than electron beam or X-ray radiation (electron microscopy, for example) nor those designed for medical purposes.

Technical Notes

1. The figure of merit K is defined as: $K = 1.7 \times 10^3 V^{2.65} Q$. V is the peak electron energy in million electron volts. If the accelerator beam pulse duration is less than or equal to 1 μs, then Q is the total accelerated charge in Coulombs. If the accelerator beam pulse duration is greater than 1 μs, then Q is the maximum accelerated charge in 1 μs. Q equals the integral of i with respect to t, over the lesser of 1 μs or the time duration of the beam pulse ($Q = \int i dt$) where i is beam current in amperes and t is the time in seconds.

2. Peak power = (peak potential in volts) x (peak beam current in amperes).

3. In machines based on microwave accelerating cavities, the time duration of the beam pulse is the lesser of 1 μs or the duration of the bunched beam packet resulting from one microwave modulator pulse.

4. In machines based on microwave accelerating cavities, the peak beam current is the average current in the time duration of a bunched beam packet.

4B003

High-velocity gun systems (propellant, gas, coil, electromagnetic, and electrothermal types, and other advanced systems) capable of accelerating projectiles to 1.5 km/s or greater.

Note: This item does not control guns specially designed for high velocity weapon systems.

¹⁸ Para 1(J) of Notification No. 29/2015-20 dated 21.09.2017

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4B004 High-speed cameras and imaging devices and components therefor, as follows:

N.B. “Software” specially designed to enhance or release the performance of cameras or imaging devices to meet the characteristics below is controlled (See Item 4C).

- a. Streak cameras, and specially designed components therefor, as follows:
 1. Streak cameras with writing speeds greater than 0.5 mm/μs;
 2. Electronic streak cameras capable of 50 ns or less time resolution;
 3. Streak tubes for cameras specified in 4B004.a.2.;
 4. Plug-ins specially designed for use with streak cameras which have modular structures and that enable the performance specifications in 4B004.a.1 or 4B004.a.2.;
 5. Synchronizing electronics units, rotor assemblies consisting of turbines, mirrors and bearings specially designed for cameras specified in 4B004.a.1.
- b. Framing cameras and specially designed components therefor as follows:
 1. Framing cameras with recording rates greater than 225,000 frames per second;
 2. Framing cameras capable of 50 ns or less frame exposure time;
 3. Framing tubes and solid-state imaging devices having a fast image gating (shutter) time of 50ns or less specially designed for cameras specified in 4B004.b.1 or 4B004.b.2.;
 4. Plug-ins specially designed for use with framing cameras which have modular structures and that enable the performance specifications in 4B004.b.1 or 4B004.b.2.;
 5. Synchronizing electronics units, rotor assemblies consisting of turbines, mirrors and bearings specially designed for cameras specified in 4B004.b.1 or 4B004.b.2.
- c. Solid state or electron tube cameras and specially designed components therefor as follows:
 1. Solid-state cameras or electron tube cameras with a fast image gating (shutter) time of 50 ns or less;
 2. Solid-state imaging devices and image intensifiers tubes having a fast image gating (shutter) time of 50 ns or less specially designed for cameras specified in 4B004.c.1.;
 3. Electro-optical shuttering devices (Kerr or Pockels cells) with a fast image gating (shutter) time of 50 ns or less;
 4. Plug-ins specially designed for use with cameras which have modular structures and that enable the performance specifications in 4B004.c.1.

Technical Note High speed single frame cameras can be used alone to produce a single image of a dynamic event, or several such cameras can be combined in a sequentially-triggered system to produce multiple images of an event.

4B005 High explosive containment vessels, chambers, containers and other similar containment devices designed for the testing of high explosives or explosive devices and having both of the following characteristics:

- a. Designed to fully contain an explosion equivalent to 2 kg of TNT or greater; and
- b. Having design elements or features enabling real time or delayed transfer of diagnostic or measurement information.

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4B006 Specialized instrumentation for hydrodynamic experiments, as follows:

- a. Velocity interferometers for measuring velocities exceeding 1 km/s during time intervals of less than 10 μ s;
- b. Shock pressure gauges capable of measuring pressures greater than 10 GPa, including gauges made with manganin, ytterbium, and polyvinylidene fluoride (PVDF) / polyvinyl difluoride (PVF2).
- c. Quartz pressure transducers for pressures greater than 10 GPa.

Note Item 4B006.a. includes velocity interferometers such as VISARs (Velocity Interferometer Systems for Any Reflector), DLIs (Doppler Laser Interferometers) and PDV (Photonic Doppler Velocimeters) also known as Het-V (Heterodyne Velocimeters).

4B007 High-speed pulse generators, and pulse heads therefor, having both of the following characteristics:

- a. Output voltage greater than 6 V into a resistive load of less than 55 ohms; and
- b. 'Pulse transition time' less than 500 ps.

Technical Notes

1. In Item 4B007.b. 'pulse transition time' is defined as the time interval between 10% and 90% voltage amplitude.
2. Pulse heads are impulse forming networks designed to accept a voltage step function and shape it into a variety of pulse forms that can include rectangular, triangular, step, impulse, exponential, or monocycle types. Pulse heads can be an integral part of the pulse generator, they can be a plug-in module to the device or they can be an externally connected device.

4B008 Detonators and multipoint initiation systems, as follows:

- a. Electrically driven explosive detonators, as follows:
 1. Exploding bridge (EB);
 2. Exploding bridge wire (EBW);
 3. Slapper;
 4. Exploding foil initiators (EFI);
- b. Arrangements using single or multiple detonators designed to nearly simultaneously initiate an explosive surface over an area greater than 5000 mm² from a single firing signal with an initiation timing spread over the surface of less than 2.5 μ s.

Note Item 4B008. does not control detonators using only primary explosives, such as lead azide.

Technical Note

In Item 4B008. the detonators of concern all utilize a small electrical conductor (bridge, bridge wire, or foil) that explosively vaporizes when a fast, high-current electrical pulse is passed through it. In nonslapper types, the exploding conductor starts a chemical detonation in a contacting high-explosive material such as PETN (pentaerythritol tetranitrate). In slapper detonators, the explosive vaporization of the electrical conductor drives a flyer or slapper across a gap, and the impact of the slapper on an explosive starts a chemical detonation. The slapper in some designs is driven by magnetic force. The term exploding foil detonator may refer to either an EB or a slapper-type detonator. Also, the word initiator is sometimes used in place of the word detonator.

4B009 Firing sets and equivalent high-current pulse generators, as follows:

- a. Detonator firing sets (initiation systems, firesets), including electronically-charged, explosively-driven and optically-driven firing sets designed to drive multiple controlled detonators specified by Item 4B008 above;

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- b. Modular electrical pulse generators (pulsers) having all of the following characteristics:
 - 1. Designed for portable, mobile, or ruggedized-use;
 - 2. Capable of delivering their energy in less than 15 μ s into loads of less than 40 ohms;
 - 3. Having an output greater than 100 A;
 - 4. No dimension greater than 30 cm;
 - 5. Weight less than 30 kg ; and
 - 6. Specified to operate over an extended temperature range of 223 to 373 K (-50 °C to 100 °C) or specified as suitable for aerospace applications.
- c. Micro-firing units having all of the following characteristics:
 - 1. No dimension greater than 35 mm;
 - 2. Voltage rating of equal to or greater than 1 kV; and
 - 3. Capacitance of equal to or greater than 100 nF.

Note Optically driven firing sets include both those employing laser initiation and laser charging. Explosively-driven firing sets include both explosive ferroelectric and explosive ferromagnetic firing set types. Item 4B009.b. includes xenon flashlamp drivers.

4B010

Switching devices as follows:

- a. Cold-cathode tubes, whether gas filled or not, operating similarly to a spark gap, having all of the following characteristics:
 - 1. Containing three or more electrodes;
 - 2. Anode peak voltage rating of 2.5 kV or more;
 - 3. Anode peak current rating of 100 A or more; and
 - 4. Anode delay time of 10 μ s or less;

Note Item 4B010.a. includes gas krytron tubes and vacuum spraytron tubes.

- b. Triggered spark-gaps having both of the following characteristics:
 - 1. Anode delay time of 15 μ s or less; and
 - 2. Rated for a peak current of 500 A or more;
- c. Modules or assemblies with a fast switching function having all of the following characteristics:
 - 1. Anode peak voltage rating greater than 2 kV;
 - 2. Anode peak current rating of 500 A or more; and
 - 3. Turn-on time of 1 μ s or less.

4B011

Pulse discharge capacitors having either of the following sets of characteristics:

- a. 1. Voltage rating greater than 1.4 kV;

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- 2. Energy storage greater than 10 J;
 - 3. Capacitance greater than 0.5 μF ; and
 - 4. Series inductance less than 50 nH; or
 - b.
 - 1. Voltage rating greater than 750 V;
 - 2. Capacitance greater than 0.25 μF ; and
 - 3. Series inductance less than 10 nH.
- 4B012** Neutron generator systems, including tubes, having both of the following characteristics:
- a. Designed for operation without an external vacuum system; and
 - b.
 - 1. Utilizing electrostatic acceleration to induce a tritium-deuterium nuclear reaction; or
 - 2. Utilizing electrostatic acceleration to induce a deuterium-deuterium nuclear reaction and capable of an Output of 3×10^9 neutrons/s or greater.
- 4B013** Striplines to provide low inductance path to detonators with the following characteristics:
- a. Voltage rating greater than 2 kV; and
 - b. Inductance of less than 20 nH
- 4C** **Technology and Software**
Technology and software for the development, production or use of items specified in 4A or 4B.

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5 Aerospace systems, equipment including production and test equipment, related technology, and specially designed components and accessories therefor.

5A Rocket Systems (including ballistic missiles, space launch vehicles and sounding rockets)

Note: Ballistic missiles (regardless of their end use) and their related systems will be classified under sub-category 6A004 (Munitions List)

5A1 Systems

5A101 Systems for missiles and rockets, including:

- a. complete rocket systems (including ballistic missile systems, space launch vehicles and sounding rockets)
- b. complete rocket stages with engines
- c. solid or liquid fuel rocket engines and their control systems including liquid propellant apogee engines designed or modified for satellites

Note 1: 5A101 does not control JATO units, propulsion units for flares, ejection seats, emergency escape equipment and rockets for display fireworks;

Note 2: As per the extant Guidelines for Sensitive Missile-Relevant Transfers of the Missile Technology Control Regime (MTCR)¹⁹ license applications for the export of items in 5A101 shall be subject to a presumption of denial.

5A102 Subsystems and components usable in missiles and rockets including:

- a. rocket motor cases, interior lining, insulation and nozzles;
- b. rocket staging mechanisms, separation mechanisms and inter-stages;
- c. Liquid, slurry and gel propellant (including oxidisers) control systems, and specially designed components therefor, usable in missiles and rockets, designed or modified to operate in vibration environments greater than 10 g rms between 20 Hz and 2 kHz.

Notes

1. *The only servo valves, pumps and gas turbines specified in 5A102.c are the following:*
 - a. *Servo valves designed for flow rates equal to or greater than 24 litres per minute, at an absolute pressure equal to or greater than 7 MPa, that have an actuator response time of less than 100ms.*
 - b. *Pumps, for liquid propellants, with shaft speeds equal to or greater than 8,000 rpm at the maximum operating mode or with discharge pressures equal to or greater than 7 MPa*
 - c. *Gas turbines, for liquid propellant turbopumps, with shaft speeds equal to or greater than 8,000 rpm at the maximum operating mode.*
2. *Systems and components specified in this clause may be exported as part of a satellite;*
- d. re-entry vehicles and equipment including
 1. Heat-shields, and components thereof, fabricated of ceramic or ablative materials;
 2. Heat sinks and components thereof, fabricated of light weight, high heat capacity materials;
 3. Electronic equipment specially designed for re-entry vehicles.
- e. guidance systems and their components such as gyros and inertial reference units;
- f. thrust-vector control subsystems including methods of achieving thrust vector control such as flexible nozzle, fluid or secondary gas injection, movable engine or nozzle, deflection of exhaust gas stream (jet vanes or probes) and use of thrust tabs;
- g. hybrid rocket motors and components thereof;
- h. safing, arming, fusing and firing mechanisms for weapons or warhead.
- i. software specially designed for reduced observables such as radar reflectivity, ultraviolet/infrared signatures and acoustic signatures.
- j. Combustion chambers and nozzles for liquid propellant rocket engines or gel propellant rocket motors

¹⁹ <http://mtcr.info/guidelines-for-sensitive-missile-relevant-transfers/>

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- k. Turbojet and turbofan engines (including turbo compound engines).
- l. Ramjet, scramjet, pulse jet, detonation, or 'combined cycle' engines, including devices to regulate combustion, and specially designed components.

5A2 Production and Test Equipment

- 5A201** Transonic, supersonic, hypersonic wind tunnels; shock tunnels; gun tunnels; aeroballistic ranges.
- 5A202** Test and production equipment and facilities designed to handle systems in 5A1.
Note: As per the extant Guidelines for Sensitive Missile-Relevant Transfers of the Missile Technology Control Regime (MTCR) license applications for transfer of items in 5A202 shall not be authorised.
- 5A203** Test benches or test stands, usable for complete rocket systems and subsystems (including ballistic missile systems, space launch vehicles and sounding rockets) which have the capacity to handle solid or liquid propellant rockets, motors or engines, or which are capable of simultaneously measuring the three axial thrust components.
- 5A204** Vibration test equipment (vibration test systems and vibration thrusters) and components using digital control techniques and feedback or closed loop test equipment and software thereof (Refer 4A006).
- 5A205** Flow-forming machines, usable in the "production" of propulsion components and equipment (e.g. motor cases and interstages) and specially designed components therefor having all of the following:
- a. Equipped with, or according to the manufacturer's technical specification are capable of being equipped with, numerical control units or computer control; and
 - b. More than two axes which can be co-ordinated simultaneously for contouring control.

Technical Note

Machines combining the function of spin-forming and flow-forming are, for the purpose of this item, regarded as flow-forming machines.

- 5A206** Filament winding machines or fibre placement machines for which the motion for positioning wrapping and winding fibres can be coordinated and programmed in two or more axes; precision mandrels thereof, and coordinating and programming controls.
- 5A207** Tape-laying machines of which the motions for positioning and laying tape and sheets can be coordinated and programmed in two or more axes;
- 5A208** Isostatic presses having all of the characteristics of maximum working pressure equal to or greater than 69 MPa or greater; designed to achieve and maintain a controlled thermal environment of 600°C or greater; and possessing a chamber cavity with an inside diameter of 152 mm or greater.
- 5A209** Environmental chambers simulating vibration environments, with altitudes equal to or greater than 15 km, or temperature ranging between minus 50 and plus 125 degrees centigrade.
- 5A210** Environmental chambers simulating acoustic pressure level of 140 dB or greater or rated acoustic power output of 4 KW or greater, with altitudes equal to or greater than 15 km, or temperature ranging between minus 50 and plus 125 degrees centigrade.
- 5A211** Accelerators delivering electro-magnetic radiation produced by Bremsstrahlung from accelerated electrons.
- 5A212** Pulsed electron accelerators
- 5A213** Radial ball bearings having all tolerances specified in accordance with ISO 492 Tolerance Class 2 or better and having all the following characteristics:
- a. An inner ring bore diameter between 12 and 50 mm;

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	b. An outer ring outside diameter between 25 and 100 mm; and c. A width between 10 and 20 mm.	
5A214	Liquid propellant tanks specially designed for the propellants controlled in liquid propellants used in the systems specified in 5A and 5B.	Item 3A3 or other
5A215	Production facilities and production equipment specially designed for equipment or materials for 5A101 and 5A102.	
5A216	Production equipment and specially designed components thereof, for the production, handling or acceptance testing of liquid propellants or propellant constituents as referred in 3A3;	
5A217	Launch and ground support equipment and facilities usable for rocket systems (including ballistic missile systems, space launch vehicles and sounding rockets), unmanned airborne system and cruise missiles as follows:- a. Apparatus, devices and vehicles, designed or modified for the transport, handling, control, activation and launching of the systems. <i>Note: Apparatus and devices specified in 5A217.a include those installed on a manned aircraft or an unmanned aerial vehicle.</i> b. gravity meters (gravimeters), gravity gradiometers, and specially designed components thereof, designed or modified for airborne or marine use usable for complete rocket systems and for complete unmanned aerial vehicle systems (including cruise missile systems target drones and reconnaissance drones). c. telemetry and tele-command equipment, including ground equipment, designed or modified for complete rocket systems and complete unmanned aerial vehicle systems and cruise missiles, excluding equipment designed or modified for manned aircraft or satellites, ground based equipment designed or modified for terrestrial or marine application, and equipment designed for commercial, civil or 'safety of life' (e.g. data integrity, flight safety) GNSS services. d. radomes designed to withstand a combined thermal and pressure shock usable in protecting rocket systems, unmanned aerial vehicles and cruise missiles against nuclear effects (eg. electro-magnetic pulse (EMP), X-rays, combined blast and thermal effects). e. Software which processes post-flight, recorded data, enabling determination of vehicle position throughout its flight path f. Thermal batteries designed or modified for complete rocket systems of 5A or complete unmanned aerial vehicles of 5B <i>Note 'Thermal batteries' are single use batteries that contain a solid non- conducting inorganic salt as the electrolyte. These batteries incorporate a pyrolytic material that, when ignited, melts the electrolyte and activates the battery.</i>	
5A218	Systems, specially designed for radar cross section measurement, usable for rocket systems (including ballistic missile systems, space launch vehicles and sounding rockets), unmanned airborne system and cruise missiles and their subsystems.	
5A3	Technology	
5A301	Technology related to the development, production, testing and use of items in 5A1 and 5A2.	
5A302	Software for the development, production, and testing and use of items in 5A1 and 5A2.	
5A303	Software which coordinates the function of more than one subsystem, specially designed or modified for use in the systems specified in 5A1 and 5A2.	

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5B Unmanned aerial vehicle systems including cruise missiles, drones, remotely piloted air vehicles, autonomous programmable vehicles, related equipment, technology or software and specially designed components therefor:

- a. i. Complete unmanned aerial vehicle systems (including cruise missile systems, target drones, exclusive delivery drones, drones with encrypted/satellite communication, drones with explosive/warhead/electronic warfare payload capability) or any related technology or software.

Note: Cruise missiles (regardless of their end use) and their related systems will be classified under sub-category 6A004 (Munitions List)

- ii. Unmanned aerial vehicle systems including reconnaissance and other drones, remotely piloted air vehicles and autonomous programmable vehicles [not specified above at (a)]
- b. Complete unmanned aerial vehicle systems having an autonomous flight control and navigation capability or capability of controlled flight out of the direct vision range involving a human operator, designed or modified to incorporate an aerosol dispensing system/mechanism, with a capacity greater than 20 liters.

Technical Note: For the purposes of this entry, an aerosol consists of particulate or liquids other than fuel components, by-products or additives, as part of the payload to be dispersed in the atmosphere.

Note: This entry does not control unpowered airborne vehicles such as gliders, hot air balloons, etc.

- c. Associated launchers and ground support equipment;
 - d. Related equipment for command and control.
 - e. Light weight Turbojet and turbofan engines (including turbo compound engines).
 - f. Ramjet, Scramjet, pulse jet, combined cycle engines, including devices to regulate combustion, and specially designed components.
 - g. Safing, arming, fusing and firing mechanisms for weapons or warhead.
 - h. Production facilities and Production equipment specially designed for equipment or materials for 5B.
 - i. Technology, for the development, production or use of equipment, materials or software specified for 5B.
 - j. Software, for the development, production or use of equipment or materials specified for 5B.
 - k. Software which coordinates the function of more than one subsystem, specially designed or modified for use in the systems specified in 5B.
 - l. Turboprop engine systems' specially designed for the systems in 5B.a, and specially designed components therefor, having a maximum power greater than 10 kW (achieved uninstalled at sea level standard conditions), excluding civil certified engines.
- Technical Note: For the purposes of this entry, a turboprop engine system" incorporates all of the following:*
- i. Turboshift engine; and
 - ii. Power transmission system to transfer the power to a propeller

Note: Unmanned aerial vehicle systems including drones, remotely piloted air vehicles and autonomous programmable vehicles specified at 5B(a)(ii), and not covered under SCOMET Categories/sub-categories 5B(a)(i) & 5B(b), 6A010, 8A912, and capable of range equal to or less than 25 km and delivering a payload of not more than 25 kgs (excluding the software and technology of these items), will be subject to the General Licensing procedure under General Authorisation for Export of Drones (GAED) policy.

5C Avionics and navigation systems designed or modified for use in, or usable in rocket systems (including ballistic missile systems, space launch vehicles and sounding rockets), unmanned aerial vehicles and cruise missiles

- 5C001** Guidance systems and their components such as gyros and inertial reference units, and specially designed components therefor;
- 5C002** Integrated flight instrument systems which include gyrostabilizers or automatic pilots, and specially designed components therefor;
- 5C003** Compasses (including gyro-astro compasses), gyroscopes, accelerometers and inertial equipment and specially designed software thereof and specially designed components therefor.

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- | | |
|--------------|---|
| 5C004 | Inertial or other equipment using accelerometers or systems incorporating such equipment, and specially designed integration software therefor; |
| 5C005 | Encrypted telemetry systems, equipment and software thereof. |
| 5C006 | <p>Flight control system (including servo valves) designed or modified for the systems as follows:</p> <ol style="list-style-type: none"> a. Hydraulic, mechanical, electro-optical or electro-mechanical flight control systems (including fly-by-wire systems); b. Attitude control equipment; c. Design technology for integration of flight control, guidance, and propulsion data into a flight management system for optimisation of rocket system trajectory. d. Specially designed test, calibration, and alignment thereof. <p><i><u>Note1</u> Systems, equipment or valves specified in 5C006 may be exported as part of a manned aircraft or satellite or in quantities appropriate for replacement parts for manned aircraft.</i></p> <p><i><u>Note 2</u> For conversion of manned aircraft to operate as unmanned aerial vehicles specified in 5A101, entry 5C006 includes the systems, equipment and valves designed or modified to enable operation of manned aircraft as unmanned aerial vehicles.</i></p> |
| 5C007 | <ol style="list-style-type: none"> 1. Integrated navigation system incorporating an inertial measurement device (example: an attitude and heading reference system, inertial reference unit, or inertial navigation system); one or more external sensor used to update the position and/or velocity, either periodically or continuously throughout the flight (example: satellite navigation receiver, radar altimeter and/or Doppler radar); integration hardware and software. 2. Three axis magnetic heading sensors having all of the following characteristics, and specially designed components therefor: <ol style="list-style-type: none"> a. Internal tilt compensation in pitch (+/- 90 degrees) and having roll (+/- 180 degrees) axes; b. Capable of providing azimuthal accuracy better (less) than 0.5 degrees rms at latitudes of +/- 80 degrees, referenced to local magnetic field; and c. Designed or modified to be integrated with flight control and navigation systems. <p><i><u>Note:</u></i>
 <i>Flight control and navigation systems in this item include gyrostabilizers, automatic pilots and inertial navigation systems.</i></p> |
| 5C008 | Production equipment and other test, calibration and alignment equipment, designed or modified to be used with equipment specified in 5C001 – 5C004 and 5C007. |
| 5C009 | Equipment used to characterize mirrors for laser gyros such as scatterometer, reflectometer and profilometer and for other inertial equipments such as Inertial measurement unit (IMU Module) tester, IMU Platform tester, IMU stable element handling fixture, Gyro tuning test station, Gyro dynamic balance station, Gyro run-in/motor test station, Gyro evacuation and filling station, Centrifuge fixture for gyro bearings, Accelerometer axis align station, Accelerometer test station and Fiber Optic Gyro Coil Winding Machines. |
| 5C010 | <p>Avionics equipment and embedded or specially designed software and components thereof, including but not limited to:</p> <ol style="list-style-type: none"> a. Radar and laser radar system including altimeter; b. Electronic assemblies and components including umbilical and interstage electrical connectors <p><i><u>Technical Note</u> Interstage connectors also include electrical connectors installed between systems and their payload.</i></p> <ol style="list-style-type: none"> c. Design technology for protection of avionics and electrical subsystems against electromagnetic pulse (EMP) and electromagnetic interference (EMI) hazards from external sources. d. Passive sensors for determining bearings to electromagnetic sources (direction finding devices) or terrain characteristics |

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- e. Receiving equipment for 'navigation satellite systems' and specially designed components therefor, usable in missiles and rockets or for airborne applications and having any of the following characteristics:

1. Capable of providing navigation information at speeds in excess of 600 m/s;
2. Employing decryption, designed or modified for military or governmental services, to gain access to a 'navigation satellite system' secure signal/data; or
3. Being specially designed to employ anti-jam features (e.g. null steering antenna or electronically steerable antenna) to function in an environment of active or passive countermeasures.

Note

5C010.e.2 and e.3 do not control equipment designed for commercial, civil or 'Safety of Life' (e.g. data integrity, flight safety) 'navigation satellite system' services.

Technical Note

In entry 5C010.e, 'navigation satellite system' includes Global Navigation Satellite Systems (GNSS; e.g. GPS, GLONASS, Galileo or BeiDou) and Regional Navigation Satellite Systems (RNSS; e.g. NavIC, QZSS).

- f. Terrain contour mapping equipment, Scene mapping and correlation (both digital and analogue) equipment, Doppler navigation radar equipment, Passive interferometer equipment and Imaging sensor equipment (both active and passive)
- g. Design technology for electromagnetic shielding systems, the configuration of hardened electrical circuits and subsystems and for the determination of hardening criteria.

5C011 On-board electronic equipment, devices and their design and manufacturing know-how (except warhead fuses, timers and sequencers), and embedded or specially designed software thereof.

5C012 Detectors designed or modified, in protecting rocket systems, unmanned aerial vehicles and cruise missiles against nuclear effects (eg. electro-magnetic pulse (EMP), X-rays, combined blast and thermal effects).

5C013²⁰ a. Radiation Hardened microcircuits usable in protecting rocket systems, unmanned aerial vehicles and cruise missiles against nuclear effects (e.g. electro-magnetic pulse (EMP), X-rays, combined blast and thermal effects);

- b. Analogue-to-digital converters, usable in the systems specified in 5A, having any of the following characteristics:

1. Designed to meet military specifications for ruggedised equipment; or
2. Designed or modified for military use and being any of the following types:
 - (i) Analogue-to-digital converter "microcircuits", which are "radiation hardened" or have all of the following characteristics:

a. Rated for operation in the temperature range from below -54°C to above +125°C; and

b. Hermetically sealed;

or

- (ii) Electrical input type analogue-to-digital converter printed circuit boards or modules, having all of the following characteristics:

a. Rated for operation in the temperature range from below -45°C to above +80°C; and

b. Incorporating "microcircuits" specified in 5C013.b.2.(i)'

²⁰ Para 1(K) of Notification No. 29/2015-20 dated 21.09.2017

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- 5C014** Precision tracking systems using a code translator installed on the rocket or unmanned aerial vehicle in conjunction with either surface or airborne references or navigation satellite systems to provide real-time measurement of inflight position and velocity; Range instrumentation radars including associated optical/infrared trackers and related software.
- 5C015** Balancing machines capable of balancing rotors/assemblies and correcting unbalance in two planes or more.
- 5C016** Indicator heads or balancing instrumentation designed or modified for use with balancing machines
- 5C017** Motion simulators/rate tables (equipment capable of simulating motion) having all of the following characteristics:
1. Two or more axes;
 2. Designed or modified to incorporate sliprings or integrated non-contact devices capable of transferring electrical power, signal information, or both; and
 3. Having any of the following characteristics:
 - a. For any single axis having all of the following:
 1. Capable of rates of 400 degrees/s or more, or 30 degrees/s or less; and
 2. A rate resolution equal to or less than 6 degrees/s and an accuracy equal to or less than 0.6 degrees/s;
 - b. Having a worst-case rate stability equal to or better (less) than plus or minus 0.05% averaged over 10 degrees or more; or
 - c. A positioning "accuracy" equal to or less (better) than 5 arc second;

Note:

5C017 applies whether or not sliprings or integrated non-contact devices are fitted at the time of export.

- 5C018** Positioning tables (equipment capable of precise rotary positioning in any axes) having all of the following characteristics:
1. Two or more axes; and
 2. A positioning "accuracy" equal to or less (better) than 5 arc second;

Note 1 *5C017 and 5C018 do not control rotary tables designed or modified for machine tools or for medical equipment.*

Note 2 *Rate tables not controlled by 5C017 and providing the characteristics of a positioning table are to be evaluated according to 5C018.*

Note 3 *Equipment that has the characteristics specified in 5C018 which also meets the characteristics of 5C017 will be treated as equipment specified in 5C017.*

- 5C019** Centrifuges capable of imparting accelerations greater than 100 g and designed or modified to incorporate sliprings or integrated non-contact devices capable of transferring electrical power, signal information, or both

Note *5C019 applies whether or not sliprings or integrated non-contact devices are fitted at the time of export.*

- 5C020** Design technology for integration of air vehicle fuselage, propulsion system and lifting control surfaces designed or modified for the unmanned aerial vehicle systems to optimize aerodynamic performance throughout the flight regime of an unmanned aerial vehicle system

- 5C021** Design technology for integration of the flight control, guidance, and propulsion data into a flight management system, designed or modified for the complete rocket systems, unmanned aerial vehicles and cruise missiles for optimization of the trajectory.

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5C022	Technology for the development, production, or use of items in 5C.
5C023	Software for the development, production and use of items in 5C.
5C024	Software specially designed or modified for use in the systems specified in 5C.
5D	[Reserved]
5E	Microlight aircraft and powered ‘hang-gliders’

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Category 6 Munitions List

Note 1 Terms in "quotations" are defined terms. Refer to 'Glossary'

Note 2 In some instances chemicals are listed by name and CAS number. The list applies to chemicals of the same structural formula (including hydrates) regardless of name or CAS number. CAS numbers are shown to assist in identifying a particular chemical or mixture, irrespective of nomenclature. CAS numbers cannot be used as unique identifiers because some forms of the listed chemical have different CAS numbers, and mixtures containing a listed chemical may also have different CAS numbers.

Note 3 Subject to Notes 4-6 below, an authorization from Department of Defence Production, Ministry of Defence would be required for export of items in the Munitions list. This is as per the Standard Operating Procedures issued by Department of Defence Production. However, for export of items under sub-categories 6A007 and 6A008, an export authorization would be required from Directorate General of Foreign Trade (DGFT) as per the SCOMET policy and procedures.

Note 4 Notwithstanding anything contained in Note 3 above, the following items will be classified under the relevant description in category 0 and would be subject to authorisation by Department of Atomic Energy(refer a) to d) of Commodity Identification Note 2 of SCOMET):-

- a) Radioactive materials covered under 6A007;
- b) 'Reactive material' powders and shapes and any material containing Beryllium or "Zirconium with Hafnium content less than 2000 ppm" as the major constituent covered under 6A008;
- c) Nuclear power generating equipment or propulsion equipment, including "nuclear reactors", and specially designed for military use and components therefor specially designed or 'modified for military use' covered under 6A017;
- d) Simulators specially designed for military "nuclear reactors" covered under 6A017;

Note 5 Notwithstanding anything contained in Note 3 above, items corresponding to Schedule I of the Chemical Weapons Convention as specified in Category 6A007.b are prohibited for exports.

Note 6 Notwithstanding anything contained in Note 3 above, licence applications for Items in 6A008 a.13 and 6A008.a 21 would normally be denied.

Note 7 Notwithstanding anything contained in Note 3 above, export of item 6A010 will be subject to 'No Objection' from Defence Research and Development Organisation.

Note 8 Notwithstanding anything contained in Note 3 above, export of item 6A011.c would be subject to 'No Objection' from ISRO, Department of Space and Defence Research and Development Organisation.

6A001 Smooth-bore weapons with a calibre of less than 20 mm, other arms and automatic weapons with a calibre of 12.7 mm (calibre 0.50 inches) or less and accessories, as follows, and specially designed components therefor:

Note 6A001 does not apply to:

- a. Firearms specially designed for dummy ammunition and which are incapable of discharging a projectile;
- b. Firearms specially designed to launch tethered projectiles having no high explosive charge or communications link, to a range of less than or equal to 500 m;
- c. Weapons using non-centre fire cased ammunition and which are not of the fully automatic firing type;
- d. "Deactivated firearms".

Technical Note A 'deactivated firearm' is a firearm that has been made incapable of firing any projectile by processes defined by the Wassenaar Arrangement Participating State's national authority. These processes irreversibly modify the essential elements of the firearm. According to national laws and regulations, deactivation of the firearm may be attested by a certificate delivered by a competent authority and may be marked on the firearm by a stamp on an essential part.

6A001 a. Rifles and combination guns, handguns, machine, sub-machine and volley guns;

Note 6A001.a does not apply to the following:

- a. Rifles and combination guns, manufactured earlier than 1938;
- b. Reproductions of rifles and combination guns, the originals of which were manufactured earlier than 1890;

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- c. *Handguns, volley guns and machine guns, manufactured earlier than 1890, and their reproductions;*
- d. *Rifles or handguns, specially designed to discharge an inert projectile by compressed air or CO₂.*
- e. *Handguns specially designed for any of the following:*
1. *Slaughtering of domestic animals; or*
 2. *Tranquilising of animals.*
- 6A001 b. Smooth-bore weapons as follows:
1. Smooth-bore weapons specially designed for military use;
 2. Other smooth-bore weapons as follows:
 - a. Fully automatic type weapons;
 - b. Semi-automatic or pump-action type weapons;
- Note 6A001.b.2 does not apply to weapons specially designed to discharge an inert projectile by compressed air or CO₂.
- Note 6A001.b. does not apply to the following:
- a. Smooth-bore weapons manufactured earlier than 1938;
 - b. Reproductions of smooth-bore weapons, the originals of which were manufactured earlier than 1890.
 - c. Smooth-bore weapons used for hunting or sporting purposes. These weapons must not be specially designed for military use or of the fully automatic firing type;
 - d. Smooth-bore weapons specially designed for any of the following:
 1. Slaughtering of domestic animals;
 2. Tranquilizing of animals;
 3. Seismic testing;
 4. Firing of industrial projectiles; or
 5. Disrupting Improvised Explosive Devices (IEDs).
- N.B. For disruptors, see 6A004 and 8A106.
- 6A001 c. Weapons using caseless ammunition;
- 6A001 d. Accessories designed for arms specified by 6A001.a 6A001.b or 6A001.c, as follows:
1. Detachable cartridge magazines;
 2. Sound suppressors or moderators;
 3. **Gun-mountings;**
 4. Flash suppressors;
 5. Optical weapon-sights with electronic image processing;
 6. Optical weapon-sights specially designed for military use.

Technical Note For the purposes of 6A001.d.3., a 'gun-mounting' is a fixture designed to mount a gun onto a ground vehicle, "aircraft", vessel or structure.

6A002 Smooth-bore weapons with a calibre of 20 mm or more, other weapons or armament with a calibre greater than 12.7 mm (calibre 0.50 inches), projectors specially designed or modified for military use and accessories, as follows, and specially designed components therefor:

- a. Guns, howitzers, cannon, mortars, anti-tank weapons, projectile launchers, military flame throwers, rifles, recoilless rifles and smooth-bore weapons;

Note 1 6A002.a includes injectors, metering devices, storage tanks and other specially designed components for use with liquid propelling charges for any of the equipment specified by 6A002.a.

Note 2 6A002.a does not apply to weapons as follows:

- a. Rifles, smooth-bore weapons and combination guns, manufactured earlier than 1938;
- b. Reproductions of rifles, smooth-bore weapons and combination guns, the originals of which were manufactured earlier than 1890;
- c. Guns, howitzers, cannons, mortars, manufactured earlier than 1890;
- d. Smooth-bore weapons used for hunting or sporting purposes. These weapons must not be specially designed for military use or of the fully automatic firing type;
- e. Smooth-bore weapons specially designed for any of the following:

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1. *Slaughtering of domestic animals;*
2. *Tranquilizing of animals;*
3. *Seismic testing;*
4. *Firing of industrial projectiles; or*
5. *Disrupting Improvised Explosive Devices (IEDs);*

N.B. For disruptors, see 6A004 and 8A106

- f. *Hand-held projectile launchers specially designed to launch tethered projectiles having no high explosive charge or communications link, to a range of less than or equal to 500 m.*

6A002 b. Projectors, specially designed or modified for military use, as follows:

1. Smoke canister projectors;
2. Gas canister projectors;
3. Pyrotechnics projectors;

Note 6A002.b does not apply to signal pistols.

6A002 c. Accessories specially designed for the weapons specified in 6A002.a., as follows:

1. Weapon sights and weapon sight mounts, specially designed for military use;
2. Signature reduction devices;
3. Mountings;
4. Detachable cartridge magazines;

6A002 d. Reserved

6A003 a. Ammunition and fuze setting devices, as follows, and specially designed components therefor:

- b. Ammunition for weapons specified by 6A001, 6A002, or 6A012;
- c. Fuze setting devices specially designed for ammunition specified by 6A003.a.

Note 1 Specially designed components specified by 6A003 include:

- a. *Metal or plastic fabrications such as primer anvils, bullet cups, cartridge links, rotating bands and munitions metal parts;*
- b. *Safing and arming devices, fuzes, sensors and initiation devices ;*
- c. *Power supplies with high one-time operational output;*
- d. *Combustible cases for charges;*
- e. *Submunitions including bomblets, minelets and terminally guided projectiles.*

Note 2 6A003.a does not apply to any of the following:

- a. *Ammunition crimped without a projectile (blank star);*
- b. *Dummy ammunition with a pierced powder chamber;*
- c. *Other blank and dummy ammunition, not incorporating components designed for live ammunition; or*
- d. *Components specially designed for blank or dummy ammunition, specified in this Note 2.a, b. or c.*

Note 3 6A003.a does not apply to cartridges specially designed for any of the following purposes:

- a. *Signalling;*
- b. *Bird scaring; or*
- c. *Lighting of gas flares at oil wells.*

6A004 Bombs, torpedoes, rockets, missiles, other explosive devices and charges and related equipment and accessories, as follows, and specially designed components therefor:

N.B.1. For guidance and navigation equipment, see 6A011.

N.B.2. For Aircraft Missile Protection Systems (AMPS), see 6A004.c.

- a. Bombs, torpedoes, grenades, smoke canisters, rockets, mines, missiles, depth charges, demolition-charges, demolition-devices, demolition-kits, "pyrotechnic" devices, cartridges, submunitions therefor and simulators (i.e., equipment simulating the characteristics of any of these items), specially designed for military use;

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Note 6A004.a. includes:

- a. Smoke grenades, fire bombs, incendiary bombs and explosive devices;
- b. Missile or rocket nozzles and re-entry vehicle nosetips
- c.

N.B. For grenade or canister ammunition for weapons or projectors specified in 6A001. or 6A002. and submunitions specially designed for ammunition, see 6A003.

6A004

- b. Equipment having all of the following:
 1. Specially designed for military use; and
 2. Specially designed for 'activities' relating to any of the following:
 - a. Items specified by 6A004.a; or
 - b. Improvised Explosive Devices (IEDs).

Technical Note

For the purpose of 6A004.b.2 'activities' applies to handling, launching, laying, controlling, discharging, detonating, activating, powering with one-time operational output, decoying, jamming, sweeping, detecting, disrupting or disposing.

Note 1 6A004.b includes:

- a. Mobile gas liquefying equipment;
- b. Buoyant electric conducting cable suitable for sweeping magnetic mines.

Note 2 6A004.b does not apply to hand-held devices limited by design solely to the detection of metal objects and incapable of distinguishing between mines and other metal objects.

6A004. c. Aircraft Missile Protection Systems (AMPS).

Note 6A004.c does not apply to AMPS having all of the following:

- a. Any of the following missile warning sensors:
 1. Passive sensors having peak response between 100-400 nm; or
 2. Active pulsed Doppler missile warning sensors;
- b. Countermeasures dispensing systems;
- c. Flares, which exhibit both a visible signature and an infrared signature, for decoying surface-to-air missiles; and
- d. Installed on "civil aircraft" and having all of the following:
 1. The AMPS is only operable in a specific "civil aircraft" in which the specific AMPS is installed and for which any of the following has been issued:
 - a. A civil Type Certificate issued by civil aviation authority of India; or
 - b. An equivalent document recognised by the International Civil Aviation Organisation (ICAO);
 2. The AMPS employs protection to prevent unauthorised access to "software"; and
 3. The AMPS incorporates an active mechanism that forces the system not to function when it is removed from the "civil aircraft" in which it was installed.

6A005

Fire control, surveillance and warning equipment, and related systems, test and alignment and countermeasure equipment, as follows, specially designed for military use, and specially designed components and accessories therefor:

- a. Weapon sights, bombing computers, gun laying equipment and weapon control systems;
- b. Other fire control, surveillance and warning equipment, and related systems, as follows:
 1. Target acquisition, designation, range-finding, surveillance or tracking systems;
 2. Detection, recognition or identification equipment;
 3. Data fusion or sensor integration equipment;
- c. Countermeasure equipment for items specified by 6A005.a. or 6A005.b.;
Note For the purposes of 6A005.c., countermeasure equipment includes detection equipment.
- d. Field test or alignment equipment, specially designed for items specified by 6A005.a., 6A005.b. or 6A005.c.

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6A006 Ground vehicles and components, as follows:

N.B. For guidance and navigation equipment, see 6A011.

- a. Ground vehicles and components therefor, specially designed or modified for military use;

Note 1 6A006.a. includes:

- a. Tanks and other military armed vehicles and military vehicles fitted with mountings for arms or equipment for mine laying or the launching of munitions specified by 6A004.;
- b. Armoured vehicles;
- c. Amphibious and deep water fording vehicles;
- d. Recovery vehicles and vehicles for towing or transporting ammunition or weapon systems and associated load handling equipment;
- e. Trailers.

Note 2 Modification of a ground vehicle for military use specified by 6A006.a. entails a structural, electrical or mechanical change involving one or more components that are specially designed for military use. Such components include:

- a. Pneumatic tyre casings of a kind specially designed to be bullet-proof;
- b. Armoured protection of vital parts (e.g., fuel tanks or vehicle cabs);
- c. Special reinforcements or mountings for weapons;
- d. Black-out lighting.

6A006. b. Other ground vehicles and components, as follows:

1. Vehicles having all of the following:

- a. Manufactured or fitted with materials or components to provide ballistic protection equal to or better than level III (NIJ 0108.01, September 1985), or "equivalent standards";
- b. A transmission to provide drive to both front and rear wheels simultaneously, including those for vehicles having additional wheels for load bearing purposes whether driven or not;
- c. Gross Vehicle Weight Rating (GVWR) greater than 4,500 kg; and
- d. Designed or modified for off-road use;
- e. Mine-Protected vehicle

2. Components having all of the following:

- a. Specially designed for vehicles specified in 6A006.b.1.; and
- b. Providing ballistic protection equal to or better than level III (NIJ 0108.01, September 1985), or "equivalent standards".

N.B. See also 6A013.a.

Note 1 6A006. does not apply to civil vehicles designed or modified for transporting money or valuables.

Note 2 6A006. does not apply to vehicles that meet all of the following;

- a. Were manufactured before 1946;
- b. Do not have items specified by the Munitions List and manufactured after 1945, except for reproductions of original components or accessories for the vehicle; and
- c. Do not incorporate weapons specified in 6A001., 6A002. or 6A004. unless they are inoperable and incapable of discharging a projectile.

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6A007 Chemical agents, "biological agents", "riot control agents", radioactive materials, related equipment, components and materials, as follows:

N.B (See Commodity Identification Note of SCOMET list)

- a. "Biological agents" or radioactive materials selected or modified to increase their effectiveness in producing casualties in humans or animals, degrading equipment or damaging crops or the environment;
- b. Chemical warfare (CW) agents including:
 1. CW nerve agents:
 - a. O-Alkyl (equal to or less than C₁₀, including cycloalkyl) alkyl (Methyl, Ethyl, n-Propyl or Isopropyl) - phosphonofluoridates, such as:
Sarin (GB):O-Isopropyl methylphosphonofluoridate (CAS 107-44-8); and
Soman (GD):O-Pinacolyl methylphosphonofluoridate (CAS 96-64-0);
 - b. O-Alkyl (equal to or less than C₁₀, including cycloalkyl)
N,N-dialkyl (Methyl, Ethyl, n-Propyl or Isopropyl) phosphoramidocyanidates, such as: Tabun (GA):O-Ethyl
N,N-dimethylphosphoramidocyanidate (CAS 77-81-6);
 - c. O-Alkyl (H or equal to or less than C₁₀, including cycloalkyl)
S-2-dialkyl (Methyl, Ethyl, n-Propyl or Isopropyl)-aminoethyl alkyl (Methyl, Ethyl, n-Propyl or Isopropyl) phosphonothiolates and corresponding alkylated and protonated salts, such as:
VX: O-Ethyl S-2-diisopropylaminoethyl methyl phosphonothiolate (CAS 50782-69-9);
 2. CW vesicant agents:
 - a. Sulphur mustards, such as:
 1. 2-Chloroethylchloromethylsulphide (CAS 2625-76-5);
 2. Bis(2-chloroethyl) sulphide (CAS 505-60-2);
 3. Bis(2-chloroethylthio) methane (CAS 63869-13-6);
 4. 1,2-bis (2-chloroethylthio) ethane (CAS 3563-36-8);
 5. 1,3-bis (2-chloroethylthio) -n-propane (CAS 63905-10-2);
 6. 1,4-bis (2-chloroethylthio) -n-butane (CAS 142868-93-7) ;
 7. 1,5-bis (2-chloroethylthio) -n-pentane (CAS 142868-94-8);
 8. Bis (2-chloroethylthiomethyl) ether (CAS 63918-90-1);
 9. Bis (2-chloroethylthioethyl) ether (CAS 63918-89-8);
 - b. Lewisites, such as:
 1. 2-chlorovinylchloroarsine (CAS 541-25-3);
 2. Tris (2-chlorovinyl) arsine (CAS 40334-70-1);
 3. Bis (2-chlorovinyl) chloroarsine (CAS 40334-69-8);
 - c. Nitrogen mustards, such as:
 1. HN1: bis (2-chloroethyl) ethylamine (CAS 538-07-8);
 2. HN2: bis (2-chloroethyl) methylamine (CAS 51-75-2);
 3. HN3: tris (2-chloroethyl) amine (CAS 555-77-1);
 3. CW incapacitating agents, such as:
 - a. 3-Quinuclidinyl benzilate (BZ) (CAS 6581-06-2);
 4. CW defoliants, such as:
 - a. Butyl 2-chloro-4-fluorophenoxyacetate (LNF);
 - b. 2,4,5-trichlorophenoxyacetic acid (CAS 93-76-5) mixed with
2,4-dichlorophenoxyacetic acid (CAS 94-75-7)
(Agent Orange (CAS 39277-47-9));
 5. CW binary precursors and key precursors, as follows:
 1. Alkyl (Methyl, Ethyl, n-Propyl or Isopropyl) Phosphonyl Difluorides, such as: DF: Methyl Phosphonyldifluoride (CAS 676-99-3);
 2. O-Alkyl (H or equal to or less than C₁₀, including cycloalkyl) O-2-dialkyl (Methyl, Ethyl, n-Propyl or Isopropyl)-aminoethyl alkyl (Methyl, Ethyl, n-Propyl or Isopropyl) phosphonites and corresponding alkylated and protonated salts, such as:

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QL: O-Ethyl O-2-di-isopropylaminoethyl methylphosphonite
(CAS 57856-11-8);

3. Chlorosarin: O-Isopropyl methylphosphonochloridate (CAS 1445-76-7);
4. Chlorosoman: O-Pinacolyl methylphosphonochloridate (CAS 7040-57-5);

- 6A007 d. "Riot control agents", active constituent chemicals and combinations thereof, including:
1. α -Bromobenzeneacetonitrile, (Bromobenzyl cyanide) (CA) (CAS 5798-79-8);
 2. [(2-chlorophenyl) methylene] propanedinitrile,
(o-Chlorobenzylidenemalononitrile) (CS) (CAS 2698-41-1);
 3. 2-Chloro-1-phenylethanone, Phenylacetyl chloride
(ω -chloroacetophenone) (CN) (CAS 532-27-4);
 4. Dibenz-(b,f)-1,4-oxazephine, (CR) (CAS 257-07-8);
 5. 10-Chloro-5,10-dihydrophenarsazine, (Phenarsazine chloride), (Adamsite), (DM) (CAS 578-94-9);
 6. N-Nonanoylmorpholine, (MPA) (CAS 5299-64-9);

Note 1 6A007.d does not apply to "riot control agents" individually packaged for personal self-defence purposes.

Note 2 6A007.d does not apply to active constituent chemicals, and combinations thereof, identified and packaged for food production or medical purposes.

- 6A007 e. Equipment, specially designed or modified for military use, designed or modified for the dissemination of any of the following, and specially designed components thereof:
1. Materials or agents specified by 6A007.a, 6A007.b, or 6A007.d;or
 2. CW agents made up of precursors specified by 6A007.c;

- 6A007 f. Protective and decontamination equipment, specially designed or modified for military use, components and chemical mixtures, as follows:
1. Equipment designed or modified for defence against materials specified by 6A007.a, 6A007.b, or 6A007.d, and specially designed components thereof;
 2. Equipment designed or modified for decontamination of objects contaminated with materials specified by 6A007.a, or 6A007.b, and specially designed components thereof;
 3. Chemical mixtures specially developed or formulated for the decontamination of objects contaminated with materials specified by 6A007.a or 6A007.b;

Note 6A007.f.1 includes:

- a. Air conditioning units specially designed or modified for nuclear, biological or chemical filtration;
- b. Protective clothing.

N.B. For civil gas masks, protective and decontamination equipment, see also 8A104.

- 6A007 g. Equipment, specially designed or modified for military use designed or modified for the detection or identification of materials specified by 6A007.a, 6A007.b, or 6A007.d, and specially designed components thereof;

Note 6A007.g does not apply to personal radiation monitoring dosimeters.

N.B. See also 8A104.

- 6A007 h. "Biopolymers" specially designed or processed for the detection or identification of CW agents specified by 6A007.b, and the cultures of specific cells used to produce them;

- 6A007 i. "Biocatalysts" for the decontamination or degradation of CW agents, and biological systems therefor, as follows:
1. "Biocatalysts" specially designed for the decontamination or degradation of CW agents specified by 6A007.b, and resulting from directed laboratory selection or genetic manipulation of biological systems;
 2. Biological systems containing the genetic information specific to the production of

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"biocatalysts" specified by 6A007.i.1, as follows:

- a. "Expression vectors";
- b. Viruses;
- c. Cultures of cells.

Note 1 6A007.b and 6A007.d do not apply to the following:

- a. Cyanogen chloride (CAS 506-77-4);
- b. Hydrocyanic acid (CAS 74-90-8);
- c. Chlorine (CAS 7782-50-5);
- d. Carbonyl chloride (phosgene) (CAS 75-44-5);
- e. Diphosgene (trichloromethyl-chloroformate) (CAS 503-38-8);
- f. ((Reserved))
- g. Xylyl bromide, ortho: (CAS 89-92-9), meta: (CAS 620-13-3), para: (CAS 104-81-4);
- h. Benzyl bromide (CAS 100-39-0);
- i. Benzyl iodide (CAS 620-05-3);
- j. Bromo acetone (CAS 598-31-2);
- k. Cyanogen bromide (CAS 506-68-3);
- l. Bromo methylethylketone (CAS 816-40-0);
- m. Chloro acetone (CAS 78-95-5);
- n. Ethyl iodoacetate (CAS 623-48-3);
- o. Iodo acetone (CAS 3019-04-3);
- p. Chloropicrin (CAS 76-06-2).

Note 2 The cultures of cells and biological systems specified by 6A007.h and 6A007.i.2 are exclusive and these sub-items do not apply to cells or biological systems for civil purposes, such as agricultural, pharmaceutical, medical, veterinary, environmental, waste management, or in the food industry.

6A008 "Energetic materials" and related substances, as follows:

N.B.1. See also 8C111.

N.B.2. For charges and devices, see 6A004 and

8A108. Technical Notes

1. For the purposes of 6A008, excluding 6A008.c.11, or 6A008.c.12, 'mixture' refers to a composition of two or more substances with at least one substance being listed in the 6A008 sub-items.
2. Any substance listed in the 6A008 sub-items is subject to this list, even when utilised in an application other than that indicated. (e.g. TAGN is predominantly used as an explosive but can also be used either as a fuel or an oxidizer.)
3. For the purposes of 6A008, particle size is the mean particle diameter on a weight or volume basis. International standards will be used in sampling and determining particle size.

6A008 a. "Explosives" as follows, and 'mixtures' thereof:

1. ADNBF (aminodinitrobenzofuroxan or 7-amino-4,6-dinitrobenzofurazane-1-oxide) (CAS 97096-78-1);
2. BNCP (cis-bis (5-nitrotetrazolato) tetra amine-cobalt (III) perchlorate) (CAS 117412-28-9);
3. CL-14 (diamino dinitrobenzofuroxan or 5,7-diamino-4,6-dinitrobenzofurazane-1-oxide) (CAS 117907-74-1);
4. CL-20 (HNIW or Hexanitrohexaazaisowurtzitane) (CAS 135285-90-4); chlathrates of CL-20 (see also 6A008.g.3. and g.4. for its "precursors");
5. CP (2-(5-cyanotetrazolato) penta amine-cobalt (III) perchlorate) (CAS 70247-32-4);
6. DADE (1,1-diamino-2,2-dinitroethylene, FOX-7) (CAS 145250-81-3);
7. DATB (diaminotrinitrobenzene) (CAS 1630-08-6);
8. DDFP (1,4-dinitrodifurazanopiperazine);
9. DDPO (2,6-diamino-3,5-dinitropyrazine-1-oxide, PZO) (CAS 194486-77-6);
10. DIPAM (3,3'-diamino-2,2',4,4',6,6'-hexanitrobiphenyl or dipicramide) (CAS 17215-44-0);

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11. DNGU (DINGU or dinitroglycoluril) (CAS 55510-04-8);
 12. Furazans as follows:
 - a. DAAOF (DAAF, DAAFox, or diaminoazoxyfurazan);
 - b. DAAzF (diaminoazofurazan) (CAS 78644-90-3);
 13. HMX and derivatives (see also 6A008.g.5. for its "precursors"), as follows:
 - a. HMX (Cyclotetramethylenetetranitramine, octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazine, 1,3,5,7-tetranitro-1,3,5,7-tetraza-cyclooctane, octogen or octogene) (CAS 2691-41-0);
 - b. difluoroaminated analogs of HMX;
 - c. K-55 (2,4,6,8-tetranitro-2,4,6,8-tetraazabicyclo [3,3,0]-octanone-3, tetranitrosemiglycouril or keto-bicyclic HMX) (CAS 130256-72-3);
 14. HNAD (hexanitroadamantane) (CAS 143850-71-9);
 15. HNS (hexanitrostilbene) (CAS 20062-22-0);
 16. Imidazoles as follows:
 - a. BNNII (Octahydro-2,5-bis(nitroimino)imidazo [4,5-d]imidazole);
 - b. DNI (2,4-dinitroimidazole) (CAS 5213-49-0);
 - c. FDIA (1-fluoro-2,4-dinitroimidazole);
 - d. NTDNIA (N-(2-nitrotriazolo)-2,4-dinitroimidazole);
 - e. PTIA (1-picryl-2,4,5-trinitroimidazole);
- 6A008 **a.**
17. NTNMH (1-(2-nitrotriazolo)-2-dinitromethylene hydrazine);
 18. NTO (ONTA or 3-nitro-1,2,4-triazol-5-one) (CAS 932-64-9);
 19. Polynitrocubanes with more than four nitro groups;
 20. PYX (2,6-Bis(picrylamino)-3,5-dinitropyridine) (CAS 38082-89-2);
 21. RDX and derivatives, as follows:
 - a. RDX (cyclotrimethylenetrinitramine, cyclonite, T4, hexahydro-1,3,5-trinitro-1,3,5-triazine, 1,3,5-trinitro-1,3,5-triaza-cyclohexane, hexogen or hexogene) (CAS 121-82-4);
 - b. Keto-RDX (K-6 or 2,4,6-trinitro-2,4,6-triazacyclohexanone) (CAS 115029-35-1);
 22. TAGN (triaminoguanidinenitrate) (CAS 4000-16-2);
 23. TATB (triaminotrinitrobenzene) (CAS 3058-38-6) (see also 6A008.g.7 for its "precursors");
 24. TEDDZ (3,3,7,7-tetrakis(difluoroamine) octahydro-1,5-dinitro-1,5-diazocine);
 25. Tetrazoles as follows:
 - a. NTAT (nitrotriazol aminotetrazole);
 - b. NTNT (1-N-(2-nitrotriazolo)-4-nitrotetrazole);
 26. Tetryl (trinitrophenylmethylnitramine) (CAS 479-45-8);
 27. TNAD (1,4,5,8-tetranitro-1,4,5,8-tetraazadecalin) (CAS 135877-16-6) (see also 6A008.g.6. for its "precursors");
 28. TNAZ (1,3,3-trinitroazetidine) (CAS 97645-24-4) (see also 6A008.g.2. for its "precursors");
 29. TNGU (SORGUYL or tetranitroglycoluril) (CAS 55510-03-7);
 30. TNP (1,4,5,8-tetranitro-pyridazino[4,5-d]pyridazine) (CAS 229176-04-9);
 31. Triazines as follows:
 - a. DNAM (2-oxy-4,6-dinitroamino-s-triazine) (CAS 19899-80-0);
 - b. NNHT (2-nitroimino-5-nitro-hexahydro-1,3,5-triazine) (CAS 130400-13-4);
 32. Triazoles as follows:
 - a. 5-azido-2-nitrotriazole;
 - b. ADHTDN (4-amino-3,5-dihydrazino-1,2,4-triazole dinitramide) (CAS 1614-08-0);
 - c. ADNT (1-amino-3,5-dinitro-1,2,4-triazole);
 - d. BDNTA ((bis-dinitrotriazole)amine);
 - e. DBT (3,3'-dinitro-5,5-bi-1,2,4-triazole) (CAS 30003-46-4);
 - f. DNBT (dinitrobistriazole) (CAS 70890-46-9);
 - g. (Reserved)
 - h. NTDNT (1-N-(2-nitrotriazolo) 3,5-dinitrotriazole);

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- i. PDNT (1-picryl-3,5-dinitrotriazole);
- j. TACOT (tetranitrobenzotriazolobenzotriazole) (CAS 25243-36-1);
- 33. "Explosives" not listed elsewhere in 6A008.a and having any of the following:
 - a. Detonation velocity exceeding 8,700 m/s, at maximum density, or
 - b. Detonation pressure exceeding 34 GPa (340 kbar);
- 34. (Reserved)
- 35. DNAN (2,4-dinitroanisole) (CAS 119-27-7);
- 36. TEX (4,10-Dinitro-2,6,8,12-tetraoxa-4,10-diazaisowurtzitane);
- 37. GUDN (Guanylurea dinitramide) FOX-12 (CAS 217464-38-5);
- 38. Tetrazines as follows:
 - a. BTAT (Bis(2,2,2-trinitroethyl)-3,6-diaminotetrazine);
 - b. LAX-112 (3,6-diamino-1,2,4,5-tetrazine-1,4-dioxide);
- 39. Energetic ionic materials melting between 343 K (70°C) and 373 K (100°C) and with detonation velocity exceeding 6,800 m/s or detonation pressure exceeding 18 GPa (180 kbar);
- 40. BTNEN (Bis(2,2,2-trinitroethyl)-nitramine) (CAS 19836-28-3);
- 41. FTDO (5,6-(3',4'-furazano)- 1,2,3,4-tetrazine-1,3-dioxide);
- 42. EDNA (Ethylenedinitramine) (CAS 505-71-5)
- 43. TKX-50 (Dihydroxylammonium 5,5'-bistetrazole-1,1'-diolate);

Note 6A008.a includes 'explosive co-crystals'.

Technical Note

An 'explosive co-crystal' is a solid material consisting of an ordered three dimensional arrangement of two or more explosive molecules, where at least one is specified in 6A008.a.

- 6A008 **b.** "Propellants" as follows:
 - 1. Any solid "propellant" with a theoretical specific impulse (under standard conditions) of more than:
 - a. 240 seconds for non-metallized, non-halogenized "propellant";
 - b. 250 seconds for non-metallized, halogenized "propellant"; or
 - c. 260 seconds for metallized "propellant";
 - 2. (Reserved)
 - 3. "Propellants" having a force constant of more than 1,200 kJ/kg;
 - 4. "Propellants" that can sustain a steady-state linear burning rate of more than 38 mm/s under standard conditions (as measured in the form of an inhibited single strand) of 6.89 MPa (68.9 bar) pressure and 294K (21°C);
 - 5. Elastomer Modified Cast Double Base (EMCDB) "propellants" with extensibility at maximum stress of more than 5% at 233K (-40°C);
 - 6. Any "propellant" containing substances specified by 6A008.a;
 - 7. "Propellants", not specified elsewhere in Category 6, specially designed for military use;
- 6A008. **c.** "Pyrotechnics", fuels and related substances, as follows, and 'mixtures' thereof:
 - 1. "Aircraft" fuels specially formulated for military purposes;
 - Note. "Aircraft" fuels specified by 6A008.c.1 are finished products, not their constituents.
 - Note 1: 6A008.c.1 does not apply to the following "aircraft" fuels: JP-4, JP-5, and JP-8.
 - 2. Alane (aluminium hydride) (CAS 7784-21-6);
 - 3. Boranes, as follows, and their derivatives:
 - a. Carboranes;
 - b. Borane homologues, as follows:
 - 1. Decaborane (14) (CAS 17702-41-9);
 - 2. Pentaborane (9) (CAS 19624-22-7);
 - 3. Pentaborane (11) (CAS 18433-84-6);
 - 4. Hydrazine and derivatives, as follows (see also 6A008.d.8. and d.9. for oxidising hydrazine derivatives):
 - a. Hydrazine (CAS 302-01-2) in concentrations of 70% or more;

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- b. Monomethyl hydrazine (CAS 60-34-4);
 - c. Symmetrical dimethyl hydrazine (CAS 540-73-8);
 - d. Unsymmetrical dimethyl hydrazine (CAS 57-14-7);
- Note 6A008.c.4.a does not apply to hydrazine 'mixtures' specially formulated for corrosion control.
- 6A008 c. 5. Metal fuels, fuel 'mixtures' or "pyrotechnic" 'mixtures', in particle form whether spherical, atomized, spheroidal, flaked or ground, manufactured from material consisting of 99 % or more of any of the following:
- a. Metals as follows and 'mixtures' thereof:
 - 1. Beryllium (CAS 7440-41-7) in particle sizes of less than 60 µm;
 - 2. Iron powder (CAS 7439-89-6) with particle size of 3 µm or less produced by reduction of iron oxide with hydrogen;
 - b. 'Mixtures' containing any of the following:
 - 1. Zirconium (CAS 7440-67-7), magnesium (CAS 7439-95-4) or alloys of these in particle sizes of less than 60 µm; or
 - 2. Boron (CAS 7440-42-8) or boron carbide (CAS 12069-32-8) fuels of 85% purity or higher and particle sizes of less than 60 µm;
- Note 1 6A008.c.5 applies to "explosives" and fuels, whether or not the metals or alloys are encapsulated in aluminium, magnesium, zirconium, or beryllium.
- Note 2 6A008.c.5.b only applies to metal fuels in particle form when they are mixed with other substances to form a 'mixture' formulated for military purposes such as liquid "propellant" slurries, solid "propellants", or "pyrotechnic" 'mixtures'.
- Note 3 6A008.c.5.b.2 does not apply to boron and boron carbide enriched with boron-10 (20% or more of total boron-10 content.)
- 6A008 c. 6. Military materials, containing thickeners for hydrocarbon fuels, specially formulated for use in flame throwers or incendiary munitions, such as metal stearates (e.g. octal (CAS 637-12-7)) or palmitates;
7. Perchlorates, chlorates and chromates, composited with powdered metal or other high energy fuel components;
8. Spherical or spheroidal aluminium powder (CAS 7429-90-5) with a particle size of 60 µm or less and manufactured from material with an aluminium content of 99% or more;
9. Titanium subhydride (TiH_n) of stoichiometry equivalent to n= 0.65-1.68;
10. Liquid high energy density fuels not specified in 6A008.c.1, as follows:
- a. Mixed fuels, that incorporate both solid and liquid fuels (e.g. boron slurry), having a mass-based energy density of 40 MJ/kg or greater;
 - b. Other high energy density fuels and fuel additives (e.g. cubane, ionic solutions, JP-7, JP-10), having a volume-based energy density of 37.5 GJ per cubic meter or greater, measured at 293 K (20°C) and one atmosphere (101.325 kPa) pressure;
- Note 6A008.c.10.b does not apply to fossil refined fuels or biofuels, or fuels for engines certified for use in civil aviation.
- 6A008 c. 11. "Pyrotechnic" and pyrophoric materials as follows:
- a. "Pyrotechnic" or pyrophoric materials specifically formulated to enhance or control the production of radiated energy in any part of the IR spectrum;
 - b. Mixtures of magnesium, polytetrafluoroethylene (PTFE) and a vinylidene difluoride-hexafluoropropylene copolymer (e.g. MTV);
12. Fuel mixtures, "pyrotechnic" mixtures or "energetic materials", not specified elsewhere in 6A008, having all of the following:
- a. Containing greater than 0.5% of particles of any of the following:
 - 1. Aluminium;
 - 2. Beryllium;
 - 3. Boron;
 - 4. Zirconium;
 - 5. Magnesium; or
 - 6. Titanium;
 - b. Particles specified by 6A008.c.12. with a size less than 200 nm in any direction; and

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c. Particles specified by 6A008.c.12.a. with a metal content of 60% or greater;

Note 6A008.c.12. includes termites.

- 6A008 d. Oxidizers as follows, and 'mixtures' thereof:
1. ADN (ammonium dinitramide or SR 12) (CAS 140456-78-6);
 2. AP (ammonium perchlorate) (CAS 7790-98-9);
 3. Compounds composed of fluorine and any of the following:
 - a. Other halogens;
 - b. Oxygen; or
 - c. Nitrogen;
- Note 1 6A008.d.3 does not apply to chlorine trifluoride (CAS 7790-91-2).*
- Note 2 6A008.d.3 does not apply to nitrogen trifluoride (CAS 7783-54-2) in its gaseous state.*
4. DNAD (1,3-dinitro-1,3-diazetidine) (CAS 78246-06-7);
 5. HAN (hydroxylammonium nitrate) (CAS 13465-08-2);
 6. HAP (hydroxylammonium perchlorate) (CAS 15588-62-2);
 7. HNF (hydrazinium nitroformate) (CAS 20773-28-8);
 8. Hydrazine nitrate (CAS 37836-27-4);
 9. Hydrazine perchlorate (CAS 27978-54-7);
 10. Liquid oxidisers comprised of or containing inhibited red fuming nitric acid (IRFNA) (CAS 8007-58-7);

Note 6A008.d.10 does not apply to non-inhibited fuming nitric acid.

- 6A008 e. Binders, plasticizers, monomers and polymers, as follows:
1. AMMO (azidomethylmethyloxetane and its polymers) (CAS 90683-29-7)
(see also 6A008.g.1. for its "precursors");
 2. BAMO (3,3-bis(azidomethyl)oxetane and its polymers)
(CAS 17607-20-4) (see also 6A008.g.1. for its "precursors");
 3. BDNPA (bis (2,2-dinitropropyl)acetal) (CAS 5108-69-0);
 4. BDNPF (bis (2,2-dinitropropyl)formal) (CAS 5917-61-3);
 5. BTTN (butanetrioltrinitrate) (CAS 6659-60-5)
(see also 6A008.g.8 for its "precursors");
 6. Energetic monomers, plasticizers or polymers, specially formulated for military use and containing any of the following:
 - a. Nitro groups;
 - b. Azido groups;
 - c. Nitrate groups;
 - d. Nitro groups; or
 - e. Difluoroamino groups;
 7. FAMAO (3-difluoroaminomethyl-3-azidomethyl oxetane) and its polymers;
 8. FEFO (bis-(2-fluoro-2,2-dinitroethyl) formal) (CAS 17003-79-1);
 9. FPF-1 (poly-2,2,3,3,4,4-hexafluoropentane-1,5-diol formal) (CAS 376-90-9);
 10. FPF-3 (poly-2,4,4,5,5,6,6-heptafluoro-2-tri-fluoromethyl-3-oxaheptane-1,7-diol formal);
 11. GAP (glycidylazide polymer) (CAS 143178-24-9) and its derivatives;
 12. HTPB (hydroxyl terminated polybutadiene) with a hydroxyl functionality equal to or greater than 2.2 and less than or equal to 2.4, a hydroxyl value of less than 0.77 meq/g, and a viscosity at 30°C of less than 47 poise
(CAS 69102-90-5);
 13. Alcohol functionalised poly(epichlorohydrin) with a molecular weight less than 10,000, as follows:
 - a. Poly(epichlorohydrindiol);
 - b. Poly(epichlorohydrintriol).
 14. NENAs (nitroethylnitramine compounds) (CAS 17096-47-8, 85068-73-1, 82486-83-7, 82486-82-6 And 85954-06-9);
 15. PGN (poly-GLYN, polyglycidyl nitrate or poly(nitratomethyl oxirane)) (CAS 27814-48-8);
 16. Poly-NIMMO (poly (nitratomethylmethyloxetane), poly-NMMO or poly (3-Nitratomethyl-3-methyloxetane)) (CAS 84051-81-0);
 17. Polynitroorthocarbonates;

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18. TVOPA (1,2,3-tris[1,2-bis(difluoroamino)ethoxy] propane or tris vinoxy propane adduct) (CAS 53159-39-0);
 19. 4,5 diazidomethyl-2-methyl-1,2,3-triazole (iso- DAMTR);
 20. PNO (Poly(3-nitrato oxetane));
 21. TMETN (Trimethylolethane trinitrate) (CAS 3032-55-1)
- 6A008
- f. "Additives" as follows:
 1. Basic copper salicylate (CAS 62320-94-9);
 2. BHEGA (bis-(2-hydroxyethyl) glycolamide) (CAS 17409-41-5);
 3. BNO (butadienenitrileoxide);
 4. Ferrocene derivatives as follows:
 - a. Butacene (CAS 125856-62-4);
 - b. Catocene (2,2-bis-ethylferrocenyl propane) (CAS 37206-42-1);
 - c. Ferrocene carboxylic acids and ferrocene carboxylic acid esters;
 - d. n-butyl-ferrocene (CAS 31904-29-7);
 - e. Other adducted polymer ferrocene derivatives not specified elsewhere in 6A008.f.4;
 - f. Ethyl ferrocene (CAS 1273-89-8);
 - g. Propyl ferrocene;
 - h. Pentyl ferrocene (CAS 1274-00-6);
 - i. Dicyclopentyl ferrocene;
 - j. Dicyclohexyl ferrocene;
 - k. Diethyl ferrocene (CAS 1273-97-8);
 - l. Dipropyl ferrocene;
 - m. Dibutyl ferrocene (CAS 1274-08-4);
 - n. Dihexyl ferrocene (CAS 93894-59-8);
 - o. Acetyl ferrocene (CAS 1271-55-2)/1,1'-diacetyl ferrocene (CAS 1273-94-5);
 5. Lead beta-resorcyate (CAS 20936-32-7) or copper beta-resorcyate (CAS 70983-44-7);
 6. Lead citrate (CAS 14450-60-3);
 7. Lead-copper chelates of beta-resorcyate or salicylates (CAS 68411-07-4);
 8. Lead maleate (CAS 19136-34-6);
 9. Lead salicylate (CAS 15748-73-9);
 10. Lead stannate (CAS 12036-31-6);
 11. MAPO (tris-1-(2-methyl)aziridinyl phosphine oxide) (CAS 57-39-6); BOBBA 8 (bis(2-methyl aziridinyl) 2-(2-hydroxypropanoxy) propylamino phosphine oxide); and other MAPO derivatives;
 12. Methyl BAPO (bis(2-methyl aziridinyl) methylamino phosphine oxide) (CAS 85068-72-0);
 13. N-methyl-p-nitroaniline (CAS 100-15-2);
 14. 3-Nitrazo-1,5-pentane diisocyanate (CAS 7406-61-9);
 15. Organo-metallic coupling agents as follows:
 - a. Neopentyl[diallyl]oxy, tri[dioctyl]phosphato-titanate (CAS 103850-22-2); also known as titanium IV, 2,2[bis 2-propenolato-methyl, butanolato, tris (dioctyl) phosphato] (CAS 110438-25-0); or LICA 12 (CAS 103850-22-2);
 - b. Titanium IV, [(2-propenolato-1) methyl, n-propanolatomethyl] butanolato-1, tris[dioctyl] pyrophosphate or KR3538;
 - c. Titanium IV, [(2-propenolato-1)methyl, n-propanolatomethyl] butanolato-1, tris(dioctyl)phosphate;
 16. Polycyanodifluoroaminoethyleneoxide;
 17. Bonding agents as follows:
 - a. 1,1R,1S-trimesoyl-tris(2-ethylaziridine) (HX-868, BITA) (CAS 7722-73-8);
 - b. Polyfunctional aziridine amides with isophthalic, trimesic, isocyanuric or trimethyladipic backbone also having a 2-methyl or 2-ethyl aziridine group;

Note Item 6A008.f.17.b. includes:

 - a. 1,1H-Isophthaloyl-bis(2-methylaziridine)(HX-752) (CAS 7652-64-4);
 - b. 2,4,6-tris(2-ethyl-1-aziridinyl)-1,3,5-triazine (HX-874) (CAS 18924-91-9);
 - c. 1,1'-trimethyladipoyl-bis(2-ethylaziridine) (HX-877) (CAS 71463-62-2).

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18. Propyleneimine (2-methylaziridine) (CAS 75-55-8);
19. Superfine iron oxide (Fe₂O₃) (CAS 1317-60-8) with a specific surface area more than 250 m²/g and an average particle size of 3.0 nm or less;
20. TEPAN (tetraethylenepentaamineacrylonitrile) (CAS 68412-45-3); cyanoethylated polyamines and their salts;
21. TEPANOL (tetraethylenepentaamineacrylonitrileglycidol) (CAS 68412-46-4); cyanoethylated polyamines adducted with glycidol and their salts;
22. TPB (triphenyl bismuth) (CAS 603-33-8);
23. TEPB (Tris (ethoxyphenyl) bismuth) (CAS 90591-48-3);

6A008 g. "Precursors" as follows:

N.B. In 6A008.g the references are to specified "Energetic Materials" manufactured from these substances.

1. BCMO (3,3-bis(chloromethyl)oxetane) (CAS 78-71-7) (see also 6A008.e.1 and e.2);
2. Dinitroazetidide-t-butyl salt (CAS 125735-38-8) (see also 6A008.a.28);
3. Hexaazaisowurtzitane derivatives including HBIW (hexabenzylhexaazaisowurtzitane) (CAS 124782-15-6) (see also 6A008.a.4) and TAIW (tetraacetyldibenzylhexaazaisowurtzitane) (CAS 182763-60-6) (see also 6A008.a.4);
4. (Reserved)
5. TAT (1,3,5,7 tetraacetyl-1,3,5,7-tetraaza cyclo-octane) (CAS 41378-98-7) (see also 6A008.a.13);
6. 1,4,5,8-tetraazadecalin (CAS 5409-42-7) (see also 6A008.a.27);
7. 1,3,5-trichlorobenzene (CAS 108-70-3) (see also 6A008.a.23);
8. 1,2,4-trihydroxybutane (1,2,4-butanetriol) (CAS 3068-00-6) (see also 6A008.e.5);
9. DADN (1,5-diacetyl-3,7-dinitro-1, 3, 5, 7-tetraaza-cyclooctane) (see also 6A008.a.13).

6A008 h. 'Reactive material' powders and shapes, as follows:

1. Powders of any of the following materials, with a particle size less than 250 µm in any direction and not specified elsewhere by 6A008:
 - a. Aluminium;
 - b. Niobium;
 - c. Boron;
 - d. Zirconium;
 - e. Magnesium;
 - f. Titanium;
 - g. Tantalum;
 - h. Tungsten;
 - i. Molybdenum; or
 - j. Hafnium;
2. Shapes, not specified by 6A003, 6A004, 6A012 or 6A016, fabricated from powders specified by 6A008.h.1.

Technical Notes

1. 'Reactive materials' are designed to produce an exothermic reaction only at high shear rates and for use as liners or casings in warheads.
2. 'Reactive material' powders are produced by, for example, a high energy ball milling process.
3. 'Reactive material' shapes are produced by, for example, selective laser sintering.

Note 1 6A008 does not apply to the following substances unless they are compounded or mixed with the "energetic material" specified by 6A008.a. or powdered metals specified by 6A008.c:

- a. Ammonium picrate (CAS 131-74-8);
- b. Black powder;
- c. Hexanitrodiphenylamine (CAS 131-73-7);
- d. Difluoroamine (CAS 10405-27-3);
- e. Nitrostarch (CAS 9056-38-6);
- f. Potassium nitrate (CAS 7757-79-1);
- g. Tetranitronaphthalene;
- h. Trinitroanisole;

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- i. Trinitronaphthalene;
- j. Trinitroxylene;
- k. N-pyrrolidinone; 1-methyl-2-pyrrolidinone (CAS 872-50-4);
- l. Dioctylmaleate (CAS 142-16-5);
- m. Ethylhexylacrylate (CAS 103-11-7);
- n. Triethylaluminium (TEA) (CAS 97-93-8), trimethylaluminium (TMA) (CAS 75-24-1), and other pyrophoric metal alkyls and aryls of lithium, sodium, magnesium, zinc or boron;
- o. Nitrocellulose (CAS 9004-70-0);
- p. Nitroglycerin (or glyceroltrinitrate, trinitroglycerine) (NG) (CAS 55-63-0);
- q. 2,4,6-trinitrotoluene (TNT) (CAS 118-96-7);
- r. Ethylenediaminedinitrate (EDDN) (CAS 20829-66-7);
- s. Pentaerythritoltetranitrate (PETN) (CAS 78-11-5);
- t. Lead azide (CAS 13424-46-9), normal lead styphnate (CAS 15245-44-0) and basic lead styphnate (CAS 12403-82-6), and primary explosives or priming compositions containing azides or azide complexes;
- u. Triethyleneglycoldinitrate (TEGDN) (CAS 111-22-8);
- v. 2,4,6-trinitroresorcinol (styphnic acid) (CAS 82-71-3);
- w. Diethyldiphenylurea; (CAS 85-98-3); dimethyldiphenylurea; (CAS 611-92-7), methylethyldiphenylurea; [Centralites]
- x. N,N-diphenylurea (unsymmetrical diphenylurea) (CAS 603-54-3);
- y. Methyl-N,N-diphenylurea (methyl unsymmetrical diphenylurea) (CAS 13114-72-2);
- z. Ethyl-N,N-diphenylurea (ethyl unsymmetrical diphenylurea) (CAS 64544-71-4);
- aa. 2-Nitrodiphenylamine (2-NDPA) (CAS 119-75-5);
- bb. 4-Nitrodiphenylamine (4-NDPA) (CAS 836-30-6);
- cc. 2,2-dinitropropanol (CAS 918-52-5);
- dd. Nitroguanidine (CAS 556-88-7) (see 8C111.d.).

Note 2 6A008 does not apply to ammonium perchlorate (6A008.d.2), NTO (6A008.a.18) or catocene (6A008.f.4.b), and meeting all of the following:

- a. Specially shaped and formulated for civil-use gas generation devices;
- b. Compounded or mixed, with non-active thermoset binders or plasticizers, and having a mass of less than 250 g;
- c. Having a maximum of 80% ammonium perchlorate (6A008.d.2) in mass of active material;
- d. Having less than or equal to 4 g of NTO (6A008.a.18); and
- e. Having less than or equal to 1 g of catocene (6A008.f.4.b).

6A009 Vessels of war (surface or underwater), special naval equipment, accessories, components and other surface vessels, as follows:

N.B. For guidance and navigation equipment, see 6A011.

a. Vessels and components, as follows:

- 1. Vessels (surface or underwater) specially designed or modified for military use, regardless of current state of repair or operating condition, and whether or not they contain weapon delivery systems or armour, and hulls or parts of hulls for such vessels, and components therefor specially designed for military use;

Note 6A009.a.1. includes vehicles specially designed or modified for the delivery of divers.

- 2. Surface vessels, not specified in 6A009.a.1., having any of the following, fixed or integrated into the vessel:

- a. Automatic weapons specified in 6A001., or weapons specified in 6A002., 6A004., 6A012. Or 6A019., or 'mountings' or hard points for weapons having a calibre of 12.7 mm or greater;

Technical Note

'Mountings' refers to weapon mounts or structural strengthening for the purpose of installing weapons.

- b. Fire control systems specified in 6A005;

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- c. Having all of the following:
 - 1. 'Chemical, Biological, Radiological and Nuclear (CBRN) protection'; and
 - 2. 'Pre-wet or wash down system' designed for decontamination purposes; or

Technical Notes

 - 1. 'CBRN protection' is a self-contained interior space containing features such as over-pressurization, isolation of ventilation systems, limited ventilation openings with CBRN filters and limited personnel access points incorporating air-locks.
 - 2. 'Pre-wet or wash down system' is a seawater spray system capable of simultaneously wetting the exterior superstructure and decks of a vessel.
 - d. Active weapon countermeasure systems specified in 6A004.b, 6A005.c or 6A011.a and having any of the following:
 - 1. 'CBRN protection';
 - 2. Hull and superstructure, specially designed to reduce the radar cross section;
 - 3. Thermal signature reduction devices, (e.g. an exhaust gas cooling system), excluding those specially designed to increase overall power plant efficiency or to reduce the environmental impact; or
 - 4. A degaussing system designed to reduce the magnetic signature of the whole vessel;
- 6A009
- b. Engines and propulsion systems, as follows, specially designed for military use and components therefor specially designed for military use:
 - 1. Diesel engines specially designed for submarines:
 - a. Power output of 1.12 MW (1,500 hp) or more; and
 - b. Rotary speed of 700 rpm or more;
 - 2. Electric motors specially designed for submarines and having all of the following:
 - a. Power output of more than 0.75 MW (1,000 hp);
 - b. Quick reversing;
 - c. Liquid cooled; and
 - d. Totally enclosed;
 - 3. Diesel engines having all of the following:
 - a. Power output of 37.3 kW (50 hp) or more; and
 - b. 'Non-magnetic' content in excess of 75% of total mass;

Technical Note

For the purposes of 6A009.b.3., 'non-magnetic' means the relative permeability is less than 2.

 - 4. 'Air Independent Propulsion' (AIP) systems specially designed for submarines;
- Technical Note
- 'Air Independent Propulsion' (AIP) allows a submerged submarine to operate its propulsion system, without access to atmospheric oxygen, for a longer time than the batteries would have otherwise allowed. For the purposes of 6A009.b.4, AIP does not include nuclear power.*
- 6A009
- c. Underwater detection devices, specially designed for military use, controls therefor and components therefor specially designed for military use;
 - d. Anti-submarine nets and anti-torpedo nets, specially designed for military use;
 - e. (Reserved)
 - f. Hull penetrators and connectors, specially designed for military use, that enable interaction with equipment external to a vessel, and components therefor specially designed for military use;
- Note 6A009.f includes connectors for vessels which are of the single-conductor, multi-conductor, coaxial or waveguide type, and hull penetrators for vessels, both of which are capable of remaining impervious to leakage from without and of retaining required characteristics at marine depths exceeding 100 m; and fibre-optic connectors and optical hull penetrators, specially designed for "laser" beam transmission, regardless of depth. 6A009.f. does not apply to ordinary propulsive shaft and hydrodynamic control-rod hull penetrators.
- 6A009
- g. Silent bearings having any of the following, components therefor and equipment containing those bearings, specially designed for military use:

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1. Gas or magnetic suspension;
2. Active signature controls; or
3. Vibration suppression controls.

6A009 h. Nuclear power generating equipment or propulsion equipment, specially designed for vessels specified in 6A009.a. and components therefor specially designed or 'modified' for military use.

Technical Note

For the purpose of 6A009.h., 'modified' means any structural, electrical, mechanical, or other change that provides a non-military item with military capabilities equivalent to an item which is specially designed for military use.

Note

6A009.h. includes "nuclear reactors".

N.B.

See Commodity Identification Note 2 to SCOMET.

6A010 "Aircraft", "lighter-than-air vehicles", "Unmanned Aerial Vehicles" ("UAVs"), aero-engines and "aircraft" equipment, related equipment, and components, as follows, specially designed or modified for military use:
N.B. For guidance and navigation equipment, see 6A011.

- a. Manned "aircraft" and "lighter-than-air vehicles", and specially designed components therefor;
- b. (Reserved)
- c. Unmanned "aircraft" and "lighter-than-air vehicles", and related equipment, as follows, and specially designed components therefor:
 1. "UAVs", Remotely Piloted Air Vehicles (RPVs), autonomous programmable vehicles and unmanned "lighter-than-air vehicles";
 2. Launchers, recovery equipment and ground support equipment;
 3. Equipment designed for command or control;
- d. Propulsion aero-engines and specially designed components therefor;
- e. Airborne refuelling equipment specially designed or modified for any of the following, and specially designed components therefor:
 1. "Aircraft" specified by 6A010.a; or
 2. Unmanned "aircraft" specified by 6A010.c;
- f. 'Ground equipment' specially designed for "aircraft" specified by 6A010.a or aero-engines specified by 6A010.d;

Technical Note

Note 1 6A010.f. includes pressure refuelling equipment and equipment designed to facilitate operations in confined areas, including equipment located on board a ship.

Note 2 6A010.f. does not apply to:

1. Towbars;
2. Protective mats and covers;
3. Ladders, steps and platforms;
4. Chocks, lashings and tie-down equipment

- g. Aircrew life support equipment, aircrew safety equipment and other devices for emergency escape, not specified in 6A010.a, designed for "aircraft" specified by 6A010.a;

Note

6A010.g does not control aircrew helmets that do not incorporate, or have mountings or fittings for, equipment specified in Category 6.

N.B.

For helmets see also 6A013.c.

- h. Parachutes, paragliders and related equipment, as follows, and specially designed components therefor:
 1. Parachutes not specified elsewhere in Category 6;
 2. Paragliders;
 3. Equipment specially designed for high altitude parachutists (e.g. suits, special helmets, breathing systems, navigation equipment);
- i. Controlled opening equipment or automatic piloting systems, designed for parachuted loads.

Note 1

6A010.a does not apply to "aircraft" and "lighter-than-air vehicles" or variants of those "aircraft", specially designed for military use and which are all of the following:

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- a. Not a combat "aircraft";
- b. Not configured for military use and not fitted with equipment or attachments specially designed or modified for military use; and
- c. Certified for civil use by civil aviation authority of India

Note 2 6A010.d does not apply to:

- a. Aero-engines designed or modified for military use which have been certified by civil aviation authority of India, for use in "civil aircraft" or specially designed components therefor;
- b. Reciprocating engines or specially designed components therefor, except those specially designed for "UAVs".

Note 3 For the purposes of 6A010.a, and 6A010.d, specially designed components and related equipment for non-military "aircraft" or aero-engines modified for military use applies only to those military components and to military related equipment required for the modification to military use.

Note 4 For the purposes of 6A010.a, military use includes: combat, military reconnaissance, assault, military training, logistics support, and transporting and airdropping troops or military equipment.

Note 5 6A010.a. does not apply to "aircraft" or "lighter-than-air vehicles" that meet all of the following:

- a. Were first manufactured before 1946;
- b. Do not incorporate items specified by the unless the items are required to meet safety or airworthiness standards of civil aviation authorities of one or more Wassenaar Arrangement Participating States; and
- c. Do not incorporate weapons specified by the Munitions List, unless inoperable and incapable of being returned to operation.

Note 6 6A010.d. does not apply to propulsion aero-engines that were first manufactured before 1946.

6A011 Electronic equipment, "spacecraft" and components, not specified elsewhere on Category 6, as follows:

- a. Electronic equipment specially designed for military use and specially designed components therefor;

Note 6A011.a includes:

- a. Electronic countermeasure and electronic counter-countermeasure equipment (i.e, equipment designed to introduce extraneous or erroneous signals into radar or radio communication receivers or otherwise hinder the reception, operation or effectiveness of adversary electronic receivers including their countermeasure equipment), including jamming and counter-jamming equipment;
- b. Frequency agile tubes;
- c. Electronic systems or equipment, designed either for surveillance and monitoring of the electromagnetic spectrum for military intelligence or security purposes or for counteracting such surveillance and monitoring;
- d. Underwater countermeasures, including acoustic and magnetic jamming and decoy, equipment designed to introduce extraneous or erroneous signals into sonar receivers;
- e. Data processing security equipment, data security equipment and transmission and signalling line security equipment, using **cryptographic functionality**;
- f. Identification, authentication and keyloader equipment and key management, manufacturing and distribution equipment;
- g. Guidance and navigation equipment;
- h. Digital troposcatter-radio communications transmission equipment;
- i. Digital demodulators specially designed for signals intelligence;
- j. "Automated Command and Control Systems".

N.B. For "software" associated with military "Software" Defined Radio (SDR), see 6A021.

6A011 b. **Jamming equipment designed or modified to hinder the reception, operation or effectiveness of positioning, navigation or timing services provided by "satellite navigation systems", and specially designed components therefor;**

6A011 c. "Spacecraft" specially designed or modified for military use, and "spacecraft" components specially designed for military use.

6A012 High velocity kinetic energy weapon systems and related equipment, as follows, and specially designed components therefor:

- a. Kinetic energy weapon systems specially designed for destruction or effecting mission-abort of a target;

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- b. Specially designed test and evaluation facilities and test models, including diagnostic instrumentation and targets, for dynamic testing of kinetic energy projectiles and systems.

N.B. For weapon systems using sub-calibre ammunition or employing solely chemical propulsion, and ammunition therefor, see 6A001 to 6A004.

Note 1 6A012 includes the following when specially designed for kinetic energy weapon systems:

- a. Launch propulsion systems capable of accelerating masses larger than 0.1 g to velocities in excess of 1.6 km/s, in single or rapid fire modes;
- b. Prime power generation, electric armour, energy storage (e.g. high energy storage capacitors), thermal management, conditioning, switching or fuel-handling equipment; and electrical interfaces between power supply, gun and other turret electric drive functions;

N.B. See also 8A301.e.2 for high energy storage capacitors.

- c. Target acquisition, tracking, fire control or damage assessment systems;
- d. Homing seeker, guidance or divert propulsion (lateral acceleration) systems for projectiles.

Note 2 6A012 applies to weapon systems using any of the following methods of propulsion:

- a. Electromagnetic;
- b. Electrothermal;
- c. Plasma;
- d. Light gas; or
- e. Chemical (when used in combination with any of the above).

6A013 Armoured or protective equipment, constructions, components and **accessories** as follows:

- a. Metallic or non-metallic armoured plate, having any of the following:
 1. Manufactured to comply with a military standard or specification; or
 2. Suitable for military use;
- N.B. For body armour plates, see 6A013.d.2.
- b. Constructions of metallic or non-metallic materials, or combinations thereof, specially designed to provide ballistic protection for military systems, and specially designed components therefor;
 - c. **Helmets and specially designed components and accessories therefor, as follows:**
 1. **Helmets manufactured according to military standards or specifications, or comparable national standards;**
 2. **Shells, liners, or comfort pads, specially designed for helmets specified in 6A013.c.1.;**
 3. **Add-on ballistic protection elements, specially designed for helmets specified in 6A013.c.1.**

N.B. For other military helmet components or accessories, see the relevant Category 6 entry.

- d. Body armour or protective garments, and components therefor, as follows:
 1. Soft body armour or protective garments, manufactured to military standards or specifications, or to their equivalents, and specially designed components therefor;
Note For the purposes of 6A013.d.1, military standards or specifications include, at a minimum, specifications for fragmentation protection.
 2. Hard body armour plates providing ballistic protection equal to or greater than level III (NIJ 0101.06, July 2008), or "equivalent standards".

Note 1 6A013.b includes materials specially designed to form explosive reactive armour or to construct military shelters.

Note 2 **6A013.c. does not apply to helmets that meet all of the following:**

- a. Were first manufactured before 1970; and**
- b. Are neither designed or modified to accept, nor equipped with items specified by Category 6.**

Note 3 6A013.c and d. do not apply to helmets, body armour or protective garments, when accompanying their user for the user's own personal protection.

Note 4 The only helmets specially designed for bomb disposal personnel that are specified by 6A013.c. are those specially designed for military use.

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Note 5 6A013.d.1. does not apply to protective eyewear.

N.B. For laser protective eyewear, see 6A017.o.

N.B. 1 See also entry 8A105 on the Dual-Use List.

N.B. 2 For "fibrous or filamentary materials" used in the manufacture of body armour and helmets, see entry 8C110 on the Dual-Use List

N.B. 1 See also 8A105.

N.B. 2 For "fibrous or filamentary materials" used in the manufacture of body armour and helmets, see 8C110.

6A014 'Specialised equipment for military training' or for simulating military scenarios, simulators specially designed for training in the use of any firearm or weapon specified by 6A001 or 6A002 and specially designed components and accessories therefor.

Technical Note

The term 'specialised equipment for military training' includes military types of attack trainers, operational flight trainers, radar target trainers, radar target generators, gunnery training devices, anti-submarine warfare trainers, flight simulators (including human-rated centrifuges for pilot/astronaut training), radar trainers, instrument flight trainers, navigation trainers, missile launch trainers, target equipment, drone "aircraft", armament trainers, pilotless "aircraft" trainers, mobile training units and training equipment for ground military operations.

Note 1 6A014 includes image generating and interactive environment systems for simulators, when specially designed or modified for military use.

Note 2 6A014 does not apply to equipment specially designed for training in the use of hunting or sporting weapons.

6A015 Imaging or countermeasure equipment, as follows, specially designed for military use, and specially designed components and accessories therefor:

- a. Recorders and image processing equipment;
- b. Cameras, photographic equipment and film processing equipment;
- c. Image intensifier equipment;
- d. Infrared or thermal imaging equipment;
- e. Imaging radar sensor equipment;
- f. Countermeasure or counter-countermeasure equipment, for the equipment specified by 6A015.a to 6A015.e.

Note 6A015.f includes equipment designed to degrade the operation or effectiveness of military imaging systems or to minimize such degrading effects.

Note 1 In 6A015 the term specially designed components includes the following, when specially designed for military use:

- a. Infrared image converter tubes;
- b. Image intensifier tubes (other than first generation);
- c. Microchannel plates;
- d. Low-light-level television camera tubes;
- e. Detector arrays (including electronic interconnection or read out systems);
- f. Pyroelectric television camera tubes;
- g. Cooling systems for imaging systems;
- h. Electrically triggered shutters of the photochromic or electro-optical type having a shutter speed of less than 100 μ s, except in the case of shutters which are an essential part of a high speed camera;
- i. Fibre optic image inverters;
- j. Compound semiconductor photocathodes.

Note 2 6A015 does not apply to "first generation image intensifier tubes" or equipment specially designed to incorporate "first generation image intensifier tubes".

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N.B. For the classification of weapons sights incorporating "first generation image intensifier tubes" see 6A001, 6A002, and 6A005.a.

N.B. See also 8A602.a.2 and 8A602.b. *on the Dual-Use List.*

6A016 Forgings, castings and other unfinished products, specially designed for items specified by 6A001, to 6A004, 6A006, 6A009, 6A010, 6A012, or 6A019.

Note 6A016 applies to unfinished products when they are identifiable by material composition, geometry or function.

6A017 Miscellaneous equipment, materials and "libraries", as follows, and specially designed components therefor:

- a. Diving and underwater swimming apparatus, specially designed or modified for military use, as follows:
 1. ISO intermodal containers or demountable vehicle bodies (i.e., swap bodies), specially designed or 'modified' for military use;
 2. Underwater swimming apparatus specially designed for use with the diving apparatus specified in 6A017.a.1;

N.B. See also 8A802.q.

- b. Construction equipment specially designed for military use;
- c. Fittings, coatings and treatments, for signature suppression, specially designed for military use;
- d. Field engineer equipment specially designed for use in a combat zone;
- e. "Robots", "robot" controllers and "robot" "end-effectors", having any of the following characteristics:
 1. Specially designed for military use;
 2. Incorporating means of protecting hydraulic lines against externally induced punctures caused by ballistic fragments (e.g. incorporating self-sealing lines) and designed to use hydraulic fluids with flash points higher than 839 K (566°C); or
 3. Specially designed or rated for operating in an electro-magnetic pulse (EMP) environment;

Technical Note

Electro-magnetic pulse does not refer to unintentional interference caused by electromagnetic radiation from nearby equipment (e.g. machinery, appliances or electronics) or lightning.

- f. "Libraries" specially designed or modified for military use with systems, equipment or components, specified by Category 6;
- g. Nuclear power generating equipment or propulsion equipment, not specified elsewhere, specially designed for military use and components therefor specially designed or 'modified' for military use;

Note 6A017.g. includes "nuclear reactors".

N.B. See Commodity Identification Note of SCOMET List.

- h. Equipment and material, coated or treated for signature suppression, specially designed for military use, not specified elsewhere in the Munitions List;
- i. Simulators specially designed for military "nuclear reactors";
- N.B. (See Commodity Identification Note of SCOMET list)
- j. Mobile repair shops specially designed or 'modified' to service military equipment;
- k. Field generators specially designed or 'modified' for military use;

- l. Containers specially designed or 'modified' for military use;
- m. Ferries, not specified elsewhere in the Munitions List, bridges and pontoons, specially designed for military use;
- n. Test models specially designed for the "development" of items specified by 6A004, 6A006, 6A009, or 6A010;
- o. "Laser" protection equipment (e.g., eye or sensor protection) specially designed for military use;
- p. "Fuel cells", not specified elsewhere in the Munitions List, specially designed or 'modified' for military use.

Technical Note

1. (Reserved)
2. For the purpose of 6A017, 'modified' means any structural, electrical, mechanical, or other change that provides a non-military item with military capabilities equivalent to an item which is specially designed for military use.

6A018 'Production' equipment, environmental test facilities and components, as follows:

- a. Specially designed or modified 'production' equipment for the 'production' of products specified by Category 6, and specially designed components therefor;

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- b. Specially designed environmental test facilities and specially designed equipment **therefor, not specified elsewhere**, for the certification, qualification or testing of products specified by Category 6.

Technical Note

For the purposes of 6A018 the term 'production' includes design, examination, manufacture, testing and checking.

Note

6A018.a and 6A018.b include the following equipment:

- a. Continuous nitrators;
- b. Centrifugal testing apparatus or equipment, having any of the following:
 1. Driven by a motor or motors having a total rated horsepower of more than 298 kW (400 hp);
 2. Capable of carrying a payload of 113 kg or more; or
 3. Capable of exerting a centrifugal acceleration of 8 g or more on a payload of 91 kg or more;
- c. Dehydration presses;
- d. Screw extruders specially designed or modified for military "explosive" extrusion;
- e. Cutting machines for the sizing of extruded "propellants";
- f. Sweetie barrels (tumblers) 1.85 m or more in diameter and having over 227 kg product capacity;
- g. Continuous mixers for solid "propellants";
- h. Fluid energy mills for grinding or milling the ingredients of military "explosives";
- i. Equipment to achieve both sphericity and uniform particle size in metal powder listed in 6A008.c.8;
- j. Convection current converters for the conversion of materials listed in 6A008.c.3.

6A019 Directed Energy Weapon (DEW) systems, related or countermeasure equipment and test models, as follows, and specially designed components therefor:

- a. "Laser" systems specially designed for destruction or effecting mission-abort of a target;
- b. Particle beam systems capable of destruction or effecting mission-abort of a target;
- c. High power Radio-Frequency (RF) systems capable of destruction or effecting mission-abort of a target;
- d. Equipment specially designed for the detection or identification of, or defence against, systems specified by 6A019.a to 6A019.c;
- e. Physical test models for the systems, equipment and components, specified by 6A019.
- f. "Laser" systems specially designed to cause permanent blindness to unenhanced vision, i.e, to the naked eye or to the eye with corrective eyesight devices.

Note 1

DEW systems specified by 6A019 include systems whose capability is derived from the controlled application of:

- a. "Lasers" of sufficient power to effect destruction similar to the manner of conventional ammunition;
- b. Particle accelerators which project a charged or neutral particle beam with destructive power;
- c. High pulsed power or high average power radio frequency beam transmitters, which produce fields sufficiently intense to disable electronic circuitry at a distant target.

Note 2

6A019 includes the following when specially designed for DEW systems:

- a. Prime power generation, energy storage, switching, power conditioning or fuel-handling equipment;
- b. Target acquisition or tracking systems;
- c. Systems capable of assessing target damage, destruction or mission-abort;
- d. Beam-handling, propagation or pointing equipment;
- e. Equipment with rapid beam slew capability for rapid multiple target operations;
- f. Adaptive optics and phase conjugators;
- g. Current injectors for negative hydrogen ion beams;
- h. "Space-qualified" accelerator components;
- i. Negative ion beam funnelling equipment;
- j. Equipment for controlling and slewing a high energy ion beam;
- k. "Space-qualified" foils for neutralising negative hydrogen isotope beams.

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6A020 Cryogenic and "superconductive" equipment, as follows, and specially designed components and accessories therefor:

- a. Equipment specially designed or configured to be installed in a vehicle for military ground, marine, airborne or space applications, capable of operating while in motion and of producing or maintaining temperatures below 103 K (- 170°C);

Note 6A020.a includes mobile systems incorporating or employing accessories or components manufactured from non-metallic or non-electrical conductive materials, such as plastics or epoxy-impregnated materials.

- b. "Superconductive" electrical equipment (rotating machinery or transformers) specially designed or configured to be installed in a vehicle for military ground, marine, airborne or space applications and capable of operating while in motion".

Note 6A020.b does not apply to direct-current hybrid homopolar generators that have single-pole normal metal armatures which rotate in a magnetic field produced by superconducting windings, provided those windings are the only superconducting components in the generator.

6A021 "Software" as follows:

- a. "Software" specially designed or modified for any of the following:
 1. "Development", "production", operation or maintenance of equipment specified by Category 6;
 2. "Development" or "production" of materials specified by Category 6; or
 3. "Development", "production", operation or maintenance of "software" specified by Category 6.
- b. Specific "software", other than that specified by 6A021.a as follows:
 1. "Software" specially designed for military use and specially designed for modelling, simulating or evaluating military weapon systems;
 2. "Software" specially designed for military use and specially designed for modelling or simulating military operational scenarios;
 3. "Software" for determining the effects of conventional, nuclear, chemical or biological weapons;
 4. "Software" specially designed for military use and specially designed for Command, Communications, Control and Intelligence (C³I) or Command, Communications, Control, Computer and Intelligence (C⁴I) applications;
 5. "Software" specially designed or modified for the conduct of military offensive cyber operations;
Note 1 6A021.b.5. includes "software" designed to destroy, damage, degrade or disrupt systems, equipment or "software", specified by Category 6, cyber reconnaissance and cyber command and control "software", therefor.
Note 2 6A021.b.5. does not apply to "vulnerability disclosure" or to "cyber incident response", limited to non-military defensive cybersecurity readiness or response.
- c. "Software", not specified by 6A021.a. or 6A021.b., specially designed or modified to enable equipment not specified by Category 6 to perform the military functions of equipment specified by Category 6.

N.B. See systems, equipment or components specified by Category 6 for general purpose "digital computers" with installed "software" specified by 6A021.c.

6A022 "Technology" as follows:

- a. "Technology", other than specified in 6A022.b which is "required" for the "development", "production", operation, installation, maintenance (checking), repair, overhaul or refurbishing of items specified in Category 6;
- b. "Technology" as follows:
 1. "Technology" "required" for the design of, the assembly of components into, and the operation, maintenance and repair of, complete production installations for items specified by Category 6, even if the components of such production installations are not specified;
 2. "Technology" "required" for the "development" and "production" of small arms, even if used to produce reproductions of antique small arms;
 3. (Reserved)
 4. (Reserved)

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5. "Technology" "required" exclusively for the incorporation of "biocatalysts", specified by 6A007.i.1 into military carrier substances or military material.

Note 1 *"Technology" "required" for the "development", "production", operation, installation, maintenance (checking), repair, overhaul or refurbishing of items specified by Category 6 remains under control even when applicable to any item not specified by Category 6.*

Note 2 6A022 does not apply to:

- a. *"Technology" that is the minimum necessary for the installation, operation, maintenance (checking) or repair, of those items which are not controlled or whose export has been authorised;*
- b. *"Technology" that is "in the public domain", "basic scientific research" or the minimum necessary information for patent applications.*
- c. *"Technology" for magnetic induction for continuous propulsion of civil transport devices.*

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Category 7²¹:

7A	[Reserved]
7B	[Reserved]
7C	[Reserved]
7D	[Reserved]
7E	[Reserved]

²¹ Para 1(L) of Notification No. 29/2015-20 dated 21.09.2017

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CATEGORY 8 Special Materials and Related Equipment, Material Processing, Electronics, Computers, Telecommunications, Information Security, Sensors and Lasers, Navigation and Avionics, Marine, Aerospace and Propulsion

Note: Terms in "quotations" are defined terms. Refer to 'Glossary'.

8A1 SPECIAL MATERIALS AND RELATED EQUIPMENT (SYSTEMS, EQUIPMENT AND COMPONENTS)

8A101 Components made from fluorinated compounds, as follows:

- a. Seals, gaskets, sealants or fuel bladders, specially designed for "aircraft" or aerospace use, made from more than 50 % by weight of any of the materials specified by 8C109.b or 8C109.c;

8A102 "Composite" structures or laminates, as follows:

- a. Made from any of the following:
 1. An organic "matrix" and "fibrous or filamentary materials" specified by 8C110.c, 8C110.d or 8C110.e;
 2. Prepregs or preforms specified by 8C110.e;
- b. Made from a metal or carbon "matrix", and any of the following:
 1. Carbon "fibrous or filamentary materials" having all of the following:
 - a. A "specific modulus" exceeding 10.15×10^6 m; and
 - b. A "specific tensile strength" exceeding 17.7×10^4 m; or
 2. Materials specified by 8C110.c.

Note 1 8A102 does not apply to "composite" structures or laminates, made from epoxy resin impregnated carbon "fibrous or filamentary materials", for the repair of "civil aircraft" structures or laminates, having all of the following:

- a. An area not exceeding 1 m²;
- b. A length not exceeding 2.5 m; and
- c. A width exceeding 15 mm.

Note 2 8A102 does not apply to semi-finished items, specially designed for purely civilian applications as follows:

- a. Sporting goods;
- b. Automotive industry;
- c. Machine tool industry;
- d. Medical applications.

Note 3 8A102.b.1. does not apply to semi-finished items containing a maximum of two dimensions of interwoven filaments and specially designed for applications as follows:

- a. Metal heat-treatment furnaces for tempering metals;
- b. Silicon boule production equipment.

Note 4 8A102 does not apply to finished items specially designed for a specific application.

Note 5 8A102.b.1. does not apply to mechanically chopped, milled, or cut carbon "fibrous or filamentary materials" 25.0 mm or less in length.

8A103 Manufactures of non-"fusible" aromatic polyimides in film, sheet, tape or ribbon form having any of the following:

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- a. A thickness exceeding 0.254 mm; or
- b. Coated or laminated with carbon, graphite, metals or magnetic substances.

Note 8A103 does not apply to manufactures when coated or laminated with copper and designed for the production of electronic printed circuit boards.

N.B. For "fusible" aromatic polyimides in any form, see 8C108.a.3.

8A104 Protective and detection equipment and components, not specially designed for military use, as follows:

- a. Full face masks, filter canisters and decontamination equipment therefor, designed or modified for defence against any of the following, and specially designed components therefor:

Note 8A104.a includes Powered Air Purifying Respirators (PAPR) that are designed or modified for defence against agents or materials, listed in 8A104.a.

Technical Notes

For the purposes of 8A104.a:

- 1. Full face masks are also known as gas masks.
- 2. Filter canisters include filter cartridges.

- 8A104 a.
 - 1. "Biological agents";
 - 2. 'Radioactive materials'
 - 3. Chemical warfare (CW) agents; or
 - 4. "Riot control agents", including:
 - a. α -Bromobenzeneacetonitrile, (Bromobenzyl cyanide) (CA)(CAS 5798-79-8);
 - b. [(2-chlorophenyl) methylene] propanedinitrile, (o-Chlorobenzylidenemalononitrile) (CS) (CAS 2698-41-1);
 - c. 2-Chloro-1-phenylethanone, Phenylacetyl chloride(ω -chloroacetophenone) (CN) (CAS 532-27-4);
 - d. Dibenz-(b,f)-1,4-oxazepine, (CR) (CAS 257-07-8);
 - e. 10-Chloro-5,10-dihydrophenarsazine, (Phenarsazine chloride), (Adamsite), (DM) (CAS 578-94-9);
 - f. N-Nonanoylmorpholine, (MPA) (CAS 5299-64-9);
 - b. Protective suits, gloves and shoes, specially designed or modified for defence against any of the following:
 - 1. "Biological agents";
 - 2. 'Radioactive materials' ; or
 - 3. Chemical warfare (CW) agents;
 - c. Detection systems, specially designed or modified for detection or identification of any of the following, and specially designed components therefor:
 - 1. "Biological agents";
 - 2. 'Radioactive materials'; or
 - 3. Chemical warfare (CW) agents.
- 8A104 d. Electronic equipment designed for automatically detecting or identifying the presence of "explosives" residues and utilising 'trace detection' techniques (e.g., surface acoustic wave, ion mobility spectrometry, differential mobility spectrometry, mass spectrometry).

Technical Note

'Trace detection' is defined as the capability to detect less than 1 ppm vapour, or 1 mg solid or liquid.

Note 1 8A104.d does not apply to equipment specially designed for laboratory use.

Note 2 8A104.d does not apply to non-contact walk-through security portals.

Note 8A104 does not apply to:

- a. Personal radiation monitoring dosimeters;
- b. Occupational health or safety equipment limited by design or function to protect against hazards specific to residential safety or civil industries, including:
 - 1. mining;
 - 2. quarrying;

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3. agriculture;
4. pharmaceutical;
5. medical;
6. veterinary;
7. environmental;
8. waste management;
9. food industry.

Technical Notes

1. 8A104 includes equipment and components that have been identified, successfully tested to standards or otherwise proven effective, for the detection of or defence against 'radioactive materials', "biological agents", chemical warfare agents, 'simulants' or "riot control agents", even if such equipment or components are used in civil industries such as mining, quarrying, agriculture, pharmaceuticals, medical, veterinary, environmental, waste management, or the food industry.
2. 'Simulant': A substance or material that is used in place of toxic agent (chemical or biological) in training, research, testing or evaluation.
3. For the purposes of 8A104, 'radioactive materials' are those selected or modified to increase their effectiveness in producing casualties in humans or animals, degrading equipment or damaging crops or the environment.

8A105 Body armour and components therefor, as follows:

- a. Soft body armour not manufactured to military standards or specifications, or to their equivalents, and specially designed components therefor;
- b. Hard body armour plates providing ballistic protection equal to or less than level IIIA (NIJ 0101.06, July 2008), or "equivalent standards".

N.B.1. For "fibrous or filamentary materials" used in the manufacture of body armour, see 8C110.

N.B.2. For body armour manufactured to military standards or specifications, see 6A013.d.

Note 1 8A105 does not apply to body armour when accompanying its user for the user's own personal protection.

Note 2 8A105 does not apply to body armour designed to provide frontal protection only from both fragment and blast from non-military explosive devices.

Note 3 8A105 does not apply to body armour designed to provide protection only from knife, spike, needle or blunt trauma.

8A106 Equipment, specially designed or modified for the disposal of Improvised Explosive Devices (IEDs), as follows, and specially designed components and accessories therefor:

- a. Remotely operated vehicles;
- b. 'Disruptors';

Technical Note

For the purpose of 8A106.b., 'disruptors' are devices specially designed for the purpose of preventing the operation of an explosive device by projecting a liquid, solid or frangible projectile.

N.B. For equipment specially designed for military use for the disposal of IEDs, see also 6A004.

Note 8A106 does not apply to equipment when accompanying its operator.

8A107 Equipment and devices, specially designed to initiate charges and devices containing "energetic materials", by electrical means, as follows:

- a. Explosive detonator firing sets designed to drive explosive detonators specified in 8A107.b
- b. Electrically driven explosive detonators as follows:

1. Exploding bridge (EB);
2. Exploding bridge wire (EBW);
3. Slapper;
4. Exploding foil initiators (EFI).

Technical Notes

1. The word initiator or igniter is sometimes used in place of the word detonator.

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2. For the purpose of 8A107.b the detonators of concern all utilise a small electrical conductor (bridge, bridge wire, or foil) that explosively vaporises when a fast, high-current electrical pulse is passed through it. In non-slapper types, the exploding conductor starts a chemical detonation in a contacting high explosive

material such as PETN (pentaerythritoltetranitrate). In slapper detonators, the explosive vaporisation of the electrical conductor drives a flyer or slapper across a gap, and the impact of the slapper on an explosive starts a chemical detonation. The slapper in some designs is driven by magnetic force. The term exploding foil detonator may refer to either an EB or a slapper-type detonator.

N.B. For equipment and devices specially designed for military use see Category 6.

8A108 Charges, devices and components, as follows:

- a. 'Shaped charges' having all of the following:
 - 1. Net Explosive Quantity (NEQ) greater than 90 g; and
 - 2. Outer casing diameter equal to or greater than 75 mm;
- b. Linear shaped cutting charges having all of the following, and specially designed components therefor:
 - 1. An explosive load greater than 40 g/m; and
 - 2. A width of 10 mm or more;
- c. Detonating cord with explosive core load greater than 64 g/m;
- d. Cutters, other than those specified by 8A108.b and severing tools, having a NEQ greater than 3.5 kg.

NoteThe only charges and devices specified in 8A108 are those containing "explosives" listed in the Annex-A to Category 8 and mixtures thereof. See also 6A008.

Technical Note

'Shaped charges' are explosive charges shaped to focus the effects of the explosive blast.

8B1SPECIAL MATERIALS AND RELATED EQUIPMENT(TEST, INSPECTION AND PRODUCTION EQUIPMENT)

8B101 Equipment for the production or inspection of "composite" structures or laminates specified by 8A102 or "fibrous or filamentary materials" specified by 8C110, as follows, and specially designed components and accessories therefor:

- a. Filament winding machines, of which the motions for positioning, wrapping and winding fibres are coordinated and programmed in three or more 'primary servo positioning' axes, specially designed for the manufacture of "composite" structures or laminates, from "fibrous or filamentary materials";
- b. 'Tape-laying machines', of which the motions for positioning and laying tape are coordinated and programmed in five or more 'primary servo positioning' axes, specially designed for the manufacture of "composite" airframe or missile structures;

Technical Note

For the purposes of 8B101.b, 'tape-laying machines' have the ability to lay one or more 'filament bands' limited to widths greater than 25.4 mm and less than or equal to 304.8 mm, and to cut and restart individual 'filament band' courses during the laying process.

- c. Multidirectional, multidimensional weaving machines or interlacing machines, including adapters and modification kits, specially designed or modified for weaving, interlacing or braiding fibres for "composite" structures;

Technical Note

For the purposes of 8B101.c, the technique of interlacing includes knitting.

- d. Equipment specially designed or adapted for the production of reinforcement fibres, as follows:
 - 1. Equipment for converting polymeric fibres (such as polyacrylonitrile, rayon, pitch or polycarbosilane) into carbon fibres or silicon carbide fibres, including special equipment to strain the fibre during heating;
 - 2. Equipment for the chemical vapour deposition of elements or compounds, on heated filamentary substrates, to manufacture silicon carbide fibres;
 - 3. Equipment for the wet-spinning of refractory ceramics (such as aluminium oxide);

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4. Equipment for converting aluminium containing precursor fibres into alumina fibres by heat treatment;
- 8B101 e. Equipment for producing preregs specified by 8C110.e by the hot melt method;
- f. Non-destructive inspection equipment specially designed for "composite" materials, as follows:
1. X-ray tomography systems for three dimensional defect inspection;
 2. Numerically controlled ultrasonic testing machines of which the motions for positioning transmitters or receivers are simultaneously coordinated and programmed in four or more axes to follow the three dimensional contours of the component under inspection;
- g. 'Tow-placement machines', of which the motions for positioning and laying tows are coordinated and programmed in two or more 'primary servo positioning' axes, specially designed for the manufacture of "composite" airframe or missile structures.
- Technical Note
For the purposes of 8B101.g, 'tow-placement machines' have the ability to place one or more 'filament bands' having widths less than or equal to 25.4 mm, and to cut and restart individual 'filament band' courses during the placement process.
- Technical Notes
1. For the purposes of 8B101, 'primary servo positioning' axes control, under computer program direction, the position of the end effector (i.e., head) in space relative to the work piece at the correct orientation and direction to achieve the desired process.
 2. For the purposes of 8B101, a 'filament band' is a single continuous width of fully or partially resin-impregnated tape, tow or fibre. Fully or partially resin-impregnated 'filament bands' include those coated with dry powder that tacks upon heating.
- 8B102 Equipment designed to produce metal alloy powder or particulate materials and having all of the following:
- a. Specially designed to avoid contamination; and
 - b. Specially designed for use in one of the processes specified by 8C102.c.2.
- 8B103 Tools, dies, moulds or fixtures, for "superplastic forming" or "diffusion bonding" titanium, aluminium or their alloys, specially designed for the manufacture of any of the following:
- a. Airframe or aerospace structures;
 - b. "Aircraft" or aerospace engines; or
 - c. Specially designed components for structures specified by 8B103.a or for engines specified by 8B103.b.

8C1SPECIAL MATERIALS AND RELATED EQUIPMENT(MATERIALS)

Technical Note

Metals and alloys

Unless provision to the contrary is made, the words 'metals' and 'alloys' cover crude and semi-fabricated forms, as follows:

Crude forms

Anodes, balls, bars (including notched bars and wire bars), billets, blocks, blooms, brickets, cakes, cathodes, crystals, cubes, dice, grains, granules, ingots, lumps, pellets, pigs, powder, rondelles, shot, slabs, slugs, sponge, sticks;

Semi-fabricated forms (whether or not coated, plated, drilled or punched):

- a. Wrought or worked materials fabricated by rolling, drawing, extruding, forging, impact extruding, pressing, graining, atomising, and grinding, i.e.: angles, channels, circles, discs, dust, flakes, foils and leaf, forging, plate, powder, pressings and stampings, ribbons, rings, rods (including bare welding rods, wire rods, and rolled wire), sections, shapes, sheets, strip, pipe and tubes (including tube rounds, squares, and hollows), drawn or extruded wire;
- b. Cast material produced by casting in sand, die, metal, plaster or other types of moulds, including high pressure castings, sintered forms, and forms made by powder metallurgy.
The object of the control should not be defeated by the export of non-listed forms alleged to be finished products but representing in reality crude forms or semi-fabricated forms.

- 8C101 Materials specially designed for absorbing electromagnetic radiation, or intrinsically conductive polymers, as follows:

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a. Materials for absorbing frequencies exceeding 2×10^8 Hz but less than 3×10^{12} Hz;

Note 1: 8C101.a does not apply to:

- a. Hair type absorbers, constructed of natural or synthetic fibres, with non-magnetic loading to provide absorption;
- b. Absorbers having no magnetic loss and whose incident surface is non-planar in shape, including pyramids, cones, wedges and convoluted surfaces;
- c. Planar absorbers, having all of the following:
 1. Made from any of the following:
 - a. Plastic foam materials (flexible or non-flexible) with carbon-loading, or organic materials, including binders, providing more than 5% echo compared with metal over a bandwidth exceeding $\pm 15\%$ of the centre frequency of the incident energy, and not capable of withstanding temperatures exceeding 450 K (177° C); or
 - b. Ceramic materials providing more than 20% echo compared with metal over a bandwidth exceeding $\pm 15\%$ of the centre frequency of the incident energy, and not capable of withstanding temperatures exceeding 800 K (527°C);
 2. Tensile strength less than $7 \times 10^6 \text{ N/m}^2$; and
 3. Compressive strength less than $14 \times 10^6 \text{ N/m}^2$;
- d. Planar absorbers made of sintered ferrite, having all of the following:
 1. A specific gravity exceeding 4.4; and
 2. A maximum operating temperature of 548 K (275°C) or less;
- e. Planar absorbers having no magnetic loss and fabricated from 'open-cell foam' plastic material with a density of 0.15 grams/cm³ or less.

Technical Note

'Open-cell foams' are flexible and porous materials, having an inner structure open to the atmosphere. 'Open-cell foams' are also known as reticulated foams.

Note 2: Nothing in Note 1 releases magnetic materials to provide absorption when contained in paint.

8C101.b Materials not transparent to visible light and specially designed for absorbing near-infrared radiation having a wavelength exceeding 810 nm but less than 2000 nm (frequencies exceeding 150 THz but less than 370 THz);

Note: 8C101.b does not apply to materials, specially designed or formulated for any of the following applications:

- a. "Laser" marking of polymers; or
- b. "Laser" welding of polymers.

8C101.c Intrinsically conductive polymeric materials with a 'bulk electrical conductivity' exceeding 10,000 S/m (Siemens per metre) or a 'sheet (surface) resistivity' of less than 100 ohms/square, based on any of the following polymers:

1. Polyaniline;
2. Polypyrrole;
3. Polythiophene;
4. Poly phenylene-vinylene; or
5. Poly thienylene-vinylene.

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Technical Note

'Bulk electrical conductivity' and 'sheet (surface) resistivity' should be determined using ASTM D-257 or national equivalents.

Note: 8C101.c does not apply to materials in a liquid form

8C102 Metal alloys, metal alloy powder and alloyed materials, as follows:

(See Commodity Identification Note 1 of SCOMET list)

Note 8C102 does not apply to metal alloys, metal alloy powder and alloyed materials, specially formulated for coating purposes.

Technical Notes

1. The metal alloys in 8C102 are those containing a higher percentage by weight of the stated metal than of any other element.
2. 'Stress-rupture life' should be measured in accordance with ASTM standard E-139.
3. 'Low cycle fatigue life' should be measured in accordance with ASTM Standard E-606 'Recommended Practice for Constant-Amplitude Low-Cycle Fatigue Testing' or national equivalents. Testing should be axial with an average stress ratio equal to 1 and a stress-concentration factor (Kt) equal to 1. The average stress ratio is defined as maximum stress minus minimum stress divided by maximum stress.

8C102 a. Aluminides, as follows:

1. Nickel aluminides containing a minimum of 15% by weight aluminium, a maximum of 38% by weight aluminium and at least one additional alloying element;
2. Titanium aluminides containing 10% by weight or more aluminium and at least one additional alloying element;

8C102 b. Metal alloys, as follows, made from the powder or particulate material specified by 8C102.c:

1. Nickel alloys having any of the following:
 - a. A 'stress-rupture life' of 10,000 hours or longer at 923 K (650°C) at a stress of 676 MPa; or
 - b. A 'low cycle fatigue life' of 10,000 cycles or more at 823 K (550°C) at a maximum stress of 1,095 MPa;

8C102 b. 2. Niobium alloys having any of the following:

- a. A 'stress-rupture life' of 10,000 hours or longer at 1,073 K (800°C) at a stress of 400 MPa; or
- b. A 'low cycle fatigue life' of 10,000 cycles or more at 973 K (700°C) at a maximum stress of 700 MPa;

3. Titanium alloys having any of the following:

- a. A 'stress-rupture life' of 10,000 hours or longer at 723 K (450°C) at a stress of 200 MPa; or
- b. A 'low cycle fatigue life' of 10,000 cycles or more at 723 K (450°C) at a maximum stress of 400 MPa;

4. Aluminium alloys having any of the following:

- a. A tensile strength of 240 MPa or more at 473 K (200°C); or
- b. A tensile strength of 415 MPa or more at 298 K (25°C);

5. Magnesium alloys having all of the following:

- a. A tensile strength of 345 MPa or more; and
- b. A corrosion rate of less than 1 mm/year in 3% sodium chloride aqueous solution measured in accordance with ASTM standard G-31;

8C102 c. Metal alloy powder or particulate material, having all of the following:

1. Made from any of the following composition systems:
Technical Note
X in the following equals one or more alloying elements.

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- | | | |
|-------|-------|--|
| | a. | Nickel alloys (Ni-Al-X, Ni-X-Al) qualified for turbine engine parts or components, i.e. with less than 3 non-metallic particles (introduced during the manufacturing process) larger than 100 µm in 10 ⁹ alloy particles; |
| | b. | Niobium alloys (Nb-Al-X or Nb-X-Al, Nb-Si-X or Nb-X-Si, Nb-Ti-X or Nb-X-Ti); |
| | c. | Titanium alloys (Ti-Al-X or Ti-X-Al); |
| | d. | Aluminium alloys (Al-Mg-X or Al-X-Mg, Al-Zn-X or Al-X-Zn, Al-Fe-X or Al-X-Fe); <u>or</u> |
| | e. | Magnesium alloys (Mg-Al-X or Mg-X-Al); |
| 8C102 | c. 2. | Made in a controlled environment by any of the following processes:
a. 'Vacuum atomisation';
b. 'Gas atomisation';
c. 'Rotary atomisation';
d. 'Splat quenching';
e. 'Melt spinning' and 'comminution';
f. 'Melt extraction' and 'comminution';
g. 'Mechanical alloying'; <u>or</u>
h. 'Plasma atomisation'; <u>and</u> |
| 8C102 | c. 3. | Capable of forming materials specified by 8C102.a or 8C102.b; |
| 8C102 | d. | Alloyed materials having all of the following:
1. Made from any of the composition systems specified by 8C102.c.1;
2. In the form of uncomminuted flakes, ribbons or thin rods; <u>and</u>

3. Produced in a controlled environment by any of the following:
a. 'Splat quenching';
b. 'Melt spinning'; <u>or</u>
c. 'Melt extraction'.
<u>Technical Notes</u>
1. 'Vacuum atomisation' is a process to reduce a molten stream of metal to droplets of a diameter of 500 µm or less by the rapid evolution of a dissolved gas upon exposure to a vacuum.
2. 'Gas atomisation' is a process to reduce a molten stream of metal alloy to droplets of 500 µm diameter or less by a high pressure gas stream.
3. 'Rotary atomisation' is a process to reduce a stream or pool of molten metal to droplets to a diameter of 500 µm or less by centrifugal force.
4. 'Splat quenching' is a process to 'solidify rapidly' a molten metal stream impinging upon a chilled block, forming a flake-like product.
5. 'Melt spinning' is a process to 'solidify rapidly' a molten metal stream impinging upon a rotating chilled block, forming a flake, ribbon or rod-like product.
6. 'Comminution' is a process to reduce a material to particles by crushing or grinding.
7. 'Melt extraction' is a process to 'solidify rapidly' and extract a ribbon-like alloy product by the insertion of a short segment of a rotating chilled block into a bath of a molten metal alloy.
8. 'Mechanical alloying' is an alloying process resulting from the bonding, fracturing and rebonding of elemental and master alloy powders by mechanical impact. Non-metallic particles may be incorporated in the alloy by addition of the appropriate powders.
9. 'Plasma atomisation' is a process to reduce a molten stream or solid metal to droplets of 500 µm diameter or less, using plasma torches in an inert gas environment.
10. Solidify rapidly' is a process involving the solidification of molten material at cooling rates exceeding 1000 K/sec. |
| 8C104 | | Uranium titanium alloys or tungsten alloys with a "matrix" based on iron, nickel or copper, having all of the following: (See Commodity Identification Note 1 of SCOMET list)

a. A density exceeding 17.5 g/cm ³ ;
b. An elastic limit exceeding 880 MPa;
c. An ultimate tensile strength exceeding 1,270 MPa; <u>and</u>
d. An elongation exceeding 8%. |

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8C105 "Superconductive" "composite" conductors in lengths exceeding 100 m or with a mass exceeding 100 g, as follows:

- a. "Superconductive" "composite" conductors containing one or more niobium-titanium 'filaments', having all of the following:
 1. Embedded in a "matrix" other than a copper or copper-based mixed "matrix"; and
 2. Having a cross-section area less than $0.28 \times 10^{-4} \text{ mm}^2$ (6 μm in diameter for circular 'filaments');
- b. "Superconductive" "composite" conductors consisting of one or more "superconductive" 'filaments' other than niobium-titanium, having all of the following:
 1. A "critical temperature" at zero magnetic induction exceeding 9.85 K (-263.31°C); and
 2. Remaining in the "superconductive" state at a temperature of 4.2 K (-268.96°C) when exposed to a magnetic field oriented in any direction perpendicular to the longitudinal axis of conductor and corresponding to a magnetic induction of 12 T with critical current density exceeding 1,750 A/mm² on overall cross-section of the conductor.

8C105 c. "Superconductive" "composite" conductors consisting of one or more "superconductive" 'filaments', which remain "superconductive" above 115 K (-158.16°C).

Technical Note

For the purpose of 8C105, 'filaments' may be in wire, cylinder, film, tape or ribbon form.

8C106 Fluids and lubricating materials, as follows:

- a. (Reserved)
- b. Lubricating materials containing, as their principal ingredients, phenylene or alkylphenylene ethers or thioethers, or their mixtures, containing more than two ether or thio-ether functions or mixtures thereof;
- c. Damping or flotation fluids having all of the following:
 1. Purity exceeding 99.8%;
 2. Containing less than 25 particles of 200 μm or larger in size per 100 ml; and
 3. Made from at least 85% of any of the following:
 - a. Dibromotetrafluoroethane (CAS 25497-30-7, 124-73-2, 27336-23-8);
 - b. Polychlorotrifluoroethylene (oily and waxy modifications only); or
 - c. Polybromotrifluoroethylene;
- d. Fluorocarbon fluids designed for electronic cooling and having all of the following:
 1. Containing 85% by weight or more of any of the following, or mixtures thereof:
 - a. Monomeric forms of perfluoropolyalkylether-triazines or perfluoroaliphatic-ethers;
 - b. Perfluoroalkylamines;
 - c. Perfluorocycloalkanes; or
 - d. Perfluoroalkanes;
 2. Density at 298 K (25°C) of 1.5 g/ml or more;
 3. In a liquid state at 273 K (0°C); and
 4. Containing 60% or more by weight of fluorine.

Note 8C106.d. does not apply to materials specified and packaged as medical products.

8C107 Ceramic powders, ceramic-"matrix" "composite" materials and 'precursor materials', as follows:

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- a. Ceramic powders of titanium diboride (TiB₂) (CAS 12045-63-5) having total metallic impurities, excluding intentional additions, of less than 5,000 ppm, an average particle size equal to or less than 5 µm and no more than 10% of the particles larger than 10 µm;
 - b. (Reserved)
- 8C107
- c. Ceramic-"matrix" "composite" materials as follows:
 1. Ceramic-ceramic "composite" materials with a glass or oxide-"matrix" and reinforced with any of the following:
 - a. Continuous fibres made from any of the following materials:
 1. Al₂O₃ (CAS 1344-28-1); or
 2. Si-C-N; or
- Note* 8C107.c.1.a. does not apply to "composites" containing fibres with a tensile strength of less than 700 MPa at 1,273 K (1,000°C) or tensile creep resistance of more than 1% creep strain at 100 MPa load and 1,273 K (1,000°C) for 100 hours.
- b. Fibres being all of the following:
 1. Made from any of the following materials:
 - a. Si-N;
 - b. Si-C;
 - c. Si-Al-O-N; or
 - d. Si-O-N; and
 2. Having a "specific tensile strength" exceeding 12.7×10^3 m;
- 8C107
- c. 2. Ceramic-"matrix" "composite" materials with a "matrix" formed of carbides or nitrides of silicon, zirconium or boron;
- N.B.* For items previously specified by 8C107.c see 8C107.c.1.b.
- d. (Reserved)
- N.B.* For items previously specified by 8C107.d see 8C107.c.2
- e. 'Precursor materials' specially designed for the "production" of materials specified by 8C107.c., as follows:
 1. Polydiorganosilanes;
 2. Polysilazanes;
 3. Polycarbosilazanes;
- Technical Note*
- For the purposes of 8C107, 'precursor materials' are special purpose polymeric or metallo-organic materials used for the "production" of silicon carbide, silicon nitride, or ceramics with silicon, carbon and nitrogen
- f. (Reserved)
- N.B.* For items previously specified by 8C107.f see 8C107.c.1.a
- 8C108
- Non-fluorinated polymeric substances as follows:
- a. Imides as follows:
 1. Bismaleimides;
 2. Aromatic polyamide-imides (PAI) having a 'glass transition temperature (T_g)' exceeding 563 K (290°C);
 3. Aromatic polyimides having a 'glass transition temperature (T_g)' exceeding 505 K (232°C);
 4. Aromatic polyetherimides having a 'glass transition temperature (T_g)' exceeding 563 K (290°C);
- Note* 8C108.a applies to the substances in liquid or solid "fusible" form, including resin, powder, pellet, film, sheet, tape, or ribbon.
- N.B.* For non-"fusible" aromatic polyimides in film, sheet, tape, or ribbon form, see 8A103.
- 8C108
- b. (Reserved)
 - c. (Reserved)

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- d. Polyarylene ketones;
- e. Polyarylene sulphides, where the arylene group is biphenylene, triphenylene or combinations thereof;
- f. Polybiphenylenethersulphone having a 'glass transition temperature (Tg)' exceeding 563 K (290° C).

Technical Notes

1. The 'glass transition temperature (Tg)' for 8C108.a.2 thermoplastic materials, 8C108.a.4 materials and 8C108.f materials is determined using the method described in ISO 11357-2 (1999).
2. The 'glass transition temperature (Tg)' for 8C108.a.2 thermosetting materials and 8C108.a.3 materials is determined using the 3-point bend method described in ASTM D 7028-07. The test is to be performed using a dry test specimen which has attained a minimum of 90% degree of cure as specified by ASTM E 2160-04, and was cured using the combination of standard- and post-cure processes that yield the highest Tg.

8C109 Unprocessed fluorinated compounds as follows:

- a. (Reserved)
- b. Fluorinated polyimides containing 10% by weight or more of combined fluorine;
- c. Fluorinated phosphazene elastomers containing 30% by weight or more of combined fluorine.

8C110 "Fibrous or filamentary materials" as follows:

Technical Notes

1. For the purpose of calculating "specific tensile strength", "specific modulus" or specific weight of "fibrous or filamentary materials" in 8C110.a, 8C110.b, 8C110.c or 8C110.e.1.b, the tensile strength and modulus should be determined by using Method A described in ISO 10618 (2004).
2. Assessing the "specific tensile strength", "specific modulus" or specific weight of non-unidirectional "fibrous or filamentary materials" (e.g., fabrics, random mats or braids) in 8C110 is to be based on the mechanical properties of the constituent unidirectional monofilaments (e.g., monofilaments, yarns, rovings or tows) prior to processing into the non-unidirectional "fibrous or filamentary materials".

8C110 a. Organic "fibrous or filamentary materials", having all of the following:

1. "Specific modulus" exceeding 12.7×10^6 m; and
2. "Specific tensile strength" exceeding 23.5×10^4 m;

Note 8C110.a does not apply to polyethylene.

b. Carbon "fibrous or filamentary materials", having all of the following:

1. "Specific modulus" exceeding 14.65×10^6 m; and
2. "Specific tensile strength" exceeding 26.82×10^4 m;

Note 8C110.b does not apply to:

- a. "Fibrous or filamentary materials", for the repair of "civil aircraft" structures or laminates, having all of the following:
 1. An area not exceeding 1 m^2 ;
 2. A length not exceeding 2.5 m; and
 3. A width exceeding 15 mm.
- b. Mechanically chopped, milled or cut carbon "fibrous or filamentary materials" 25.0 mm or less in length.

8C110 c. Inorganic "fibrous or filamentary materials", having all of the following:

1. Having any of the following:
 - a. Composed of 50% or more by weight silicon dioxide and having a "specific modulus" exceeding 2.54×10^6 m; or
 - b. Not specified in 8C110.c.1.a. and having a "specific modulus" exceeding 5.6×10^6 m; and
2. Melting, softening, decomposition or sublimation point exceeding 1,922 K (1,649°C) in an inert environment;

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Note 8C110.c does not apply to:

- a. Discontinuous, multiphase, polycrystalline alumina fibres in chopped fibre or random mat form, containing 3% by weight or more silica, with a "specific modulus" of less than 10×10^6 m;
- b. Molybdenum and molybdenum alloy fibres;
- c. Boron fibres;
- d. Discontinuous ceramic fibres with a melting, softening, decomposition or sublimation point lower than 2,043 K (1,770°C) in an inert environment.

8C110 d. "Fibrous or filamentary materials", having any of the following:

1. Composed of any of the following:
 - a. Polyetherimides specified by 8C108.a; or
 - b. Materials specified by 8C108.d to 8C108.f; or
2. Composed of materials specified by 8C110.d.1.a or 8C110.d.1.b and 'commingled' with other fibres specified by 8C110.a, 8C110.b or 8C110.c';

Technical Note 'Commingled' is filament to filament blending of thermoplastic fibres and reinforcement fibres in order to produce a fibre reinforcement "matrix" mix in total fibre form.

8C110 e. Fully or partially resin-impregnated or pitch-impregnated "fibrous or filamentary materials" (prepregs), metal or carbon-coated "fibrous or filamentary materials" (preforms) or 'carbon fibre preforms', having all of the following:

1. Having any of the following:
 - a. Inorganic "fibrous or filamentary materials" specified by 8C110.c; or
 - b. Organic or carbon "fibrous or filamentary materials", having all of the following:
 1. "Specific modulus" exceeding 10.15×10^6 m; and
 2. "Specific tensile strength" exceeding 17.7×10^4 m; and
2. Having any of the following:
 - a. Resin or pitch, specified by 8C108 or 8C109.b;
 - b. 'Dynamic Mechanical Analysis glass transition temperature (DMA T_g)' equal to or exceeding 453 K (180°C) and having a phenolic resin; or
 - c. 'Dynamic Mechanical Analysis glass transition temperature (DMA T_g)' equal to or exceeding 505 K (232°C) and having a resin or pitch, not specified by 8C108 or 8C109.b, and not being a phenolic resin;

Note 1 Metal or carbon-coated "fibrous or filamentary materials" (preforms) or 'carbon fibre preforms', not impregnated with resin or pitch, are specified by "fibrous or filamentary materials" in 8C110.a, 8C110.b or 8C110.c.

Note 2 8C110.e does not apply to:

- a. Epoxy resin "matrix" impregnated carbon "fibrous or filamentary materials" (prepregs) for the repair of "civil aircraft" structures or laminates, having all of the following:
 1. An area not exceeding 1 m^2 ;
 2. A length not exceeding 2.5 m; and
 3. A width exceeding 15 mm;
- b. Fully or partially resin-impregnated or pitch-impregnated mechanically chopped, milled or cut carbon "fibrous or filamentary materials" 25.0 mm or less in length when using a resin or pitch other than those specified by 8C108 or 8C109.b.

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Technical Notes

1. 'Carbon fibre preforms' are an ordered arrangement of uncoated or coated fibres intended to constitute a framework of a part before the "matrix" is introduced to form a "composite".
2. The 'Dynamic Mechanical Analysis glass transition temperature (DMA T)_g' for materials specified by 1.C.10.e. is determined using the method described in ASTM D 7028-07, or equivalent national standard, on a dry test specimen. In the case of thermoset materials, degree of cure of a dry test specimen shall be a minimum of 90% as defined by ASTM E 2160-04 or equivalent national standard.

8C111

Metals and compounds, as follows:

- a. Metals in particle sizes of less than 60 µm whether spherical, atomised, spheroidal, flaked or ground, manufactured from material consisting of 99% or more of zirconium, magnesium and alloys thereof;

Note The metals or alloys specified by 8C111.a also refer to metals or alloys encapsulated in aluminium, magnesium, zirconium or beryllium.

Technical Note

The natural content of hafnium in the zirconium (typically 2% to 7%) is counted with the zirconium.

- b. Boron or boron alloys, with a particle size of 60 µm or less, as follows:

1. Boron with a purity of 85% by weight or more;
2. Boron alloys with a boron content of 85% by weight or more;

Note The metals or alloys specified by 8C111.b also refer to metals or alloys encapsulated in aluminium, magnesium, zirconium or beryllium.

- c. Guanidine nitrate (CAS 506-93-4);
- d. Nitroguanidine (NQ) (CAS 556-88-7).

N.B. See 6A008.c.5.b for metal powders mixed with other substances to form a mixture formulated for military purposes

8C112

Materials as follows:

Technical Note

These materials are typically used for nuclear heat sources.

- a. Plutonium in any form with a plutonium isotopic assay of plutonium-238 of more than 50% by weight;

Note 8C112.a does not apply to:

- a. Shipments with a plutonium content of 1 g or less;

Technical Note 'Previously separated' is the application of any process intended to increase the concentration of the controlled isotope.

- b. Shipments of 3 'effective grams' or less when contained in a sensing component in instruments.

Technical Note 'Effective grams' for plutonium isotope is defined as the isotope weight in grams

8C112

- b. Previously separated' neptunium-237 in any form.

8D1

SPECIAL MATERIALS AND RELATED EQUIPMENT(SOFTWARE)

8D101 "Software" specially designed or modified for the "development", "production" or "use" of equipment specified by 8B1.

8D102 "Software" for the "development" of organic "matrix", metal "matrix" or carbon "matrix" laminates or "composites".

8D103 "Software" specially designed or modified to enable equipment to perform the functions of equipment specified by 8A104.c or 8A104.d.

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8E1SPECIAL MATERIALS AND RELATED EQUIPMENT(TECHNOLOGY)

8E101 "Technology" according to the General Technology Note for the "development" or "production" of equipment or materials specified by 8A102 to 8A105, 8A106.b, 8A107, 8B1 or 8C.

8E102 Other "technology" as follows:

- a. "Technology" for the "development" or "production" of polybenzothiazoles or polybenzoxazoles;
- b. "Technology" for the "development" or "production" of fluoroelastomer compounds containing at least one vinyl ether monomer;
- c. "Technology" for the design or "production" of the following ceramic powders or non-"composite" ceramic materials:
 1. Ceramic powders having all of the following:
 - a. Any of the following compositions:
 1. *Single or complex oxides of zirconium and complex oxides of silicon or aluminium;*
 2. *Single nitrides of boron (cubic crystalline forms);*
 3. *Single or complex carbides of silicon or boron; or*
 4. *Single or complex nitrides of silicon;*
 - b. Any of the following total metallic impurities (excluding intentional additions):
 1. *Less than 1,000 ppm for single oxides or carbides; or*
 2. *Less than 5,000 ppm for complex compounds or single nitrides; and*
 - c. Being any of the following:
 1. *Zirconia (CAS 1314-23-4) with an average particle size equal to or less than 1 µm and no more than 10% of the particles larger than 5 µm; or*
 2. *Other ceramic powders with an average particle size equal to or less than 5 µm and no more than 10% of the particles larger than 10 µm;*

8E102 c. 2. Non-"composite" ceramic materials composed of the materials specified by 8E102.c.1;
Note 8E102.c.2 does not apply to "technology" for abrasives.

8E102 d. (Reserved)

- e. "Technology" for the installation, maintenance or repair of materials specified by 8C101;
- f. "Technology" for the repair of "composite" structures, laminates or materials specified by 8A102 or 8C107.c;
Note 8E102.f does not apply to "technology" for the repair of "civil aircraft" structures using carbon "fibrous or filamentary materials" and epoxy resins, contained in "aircraft" manufacturers' manuals.

8E102 g. "Libraries" specially designed or modified to enable equipment to perform the functions of equipment specified by 8A104.c or 8A104.d.

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ANNEX-A (Refer 8A108)

LIST - "EXPLOSIVES"

1. ADNBF (aminodinitrobenzofuroxan or 7-amino-4,6-dinitrobenzofurazane-1-oxide) (CAS 97096-78-1);
2. BNCP (cis-bis (5-nitrotetrazolato) tetra amine-cobalt (III) perchlorate) (CAS 117412-28-9);
3. CL-14 (diamino dinitrobenzofuroxan or 5,7-diamino-4,6-dinitrobenzofurazane-1-oxide) (CAS 117907-74-1);
4. CL-20 (HNIW or Hexanitrohexaazaisowurtzitane) (CAS 135285-90-4);
chlathrates of CL-20;
5. CP (2-(5-cyanotetrazolato) penta amine-cobalt (III) perchlorate) (CAS 70247-32-4);
6. DADE (1,1-diamino-2,2-dinitroethylene, FOX7) (CAS 145250-81-3);
7. DATB (diaminotrinitrobenzene) (CAS 1630-08-6);
8. DDFP (1,4-dinitrodifurazanopiperazine);
9. DDPO (2,6-diamino-3,5-dinitropyrazine-1-oxide, PZO) (CAS 194486-77-6);
10. DIPAM (3,3'-diamino-2,2',4,4',6,6'-hexanitrobiphenyl or dipicramide) (CAS 17215-44-0);
11. DNGU (DINGU or dinitroglycoluril) (CAS 55510-04-8);
12. Furazans as follows:
 - a. DAAOF (diaminoazoxyfurazan);
 - b. DAAzF (diaminoazofurazan) (CAS 78644-90-3);
13. HMX and derivatives, as follows:
 - a. HMX (Cyclotetramethylenetetranitramine, octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazine, 1,3,5,7-tetranitro-1,3,5,7-tetraza-cyclooctane, octogen or octogene) (CAS 2691-41-0);
 - b. difluoroaminated analogs of HMX;
 - c. K-55 (2,4,6,8-tetranitro-2,4,6,8-tetraazabicyclo [3,3,0]-octanone-3, tetranitrosemiglycouril or keto-bicyclic HMX) (CAS 130256-72-3);
14. HNAD (hexanitroadamantane) (CAS 143850-71-9);
15. HNS (hexanitrostilbene) (CAS 20062-22-0);
16. Imidazoles as follows:
 - a. BNNII (Octahydro-2,5-bis(nitroimino)imidazo [4,5-d]imidazole);
 - b. DNI (2,4-dinitroimidazole) (CAS 5213-49-0);
 - c. FDIA (1-fluoro-2,4-dinitroimidazole);
 - d. NTDNIA (N-(2-nitrotriazolo)-2,4-dinitroimidazole);
 - e. PTIA (1-picryl-2,4,5-trinitroimidazole);
17. NTNMH (1-(2-nitrotriazolo)-2-dinitromethylene hydrazine);
18. NTO (ONTA or 3-nitro-1,2,4-triazol-5-one) (CAS 932-64-9);
19. Polynitrocubanes with more than four nitro groups;
20. PYX (2,6-Bis(picrylamino)-3,5-dinitropyridine) (CAS 38082-89-2);
21. RDX and derivatives, as follows:
 - a. RDX (cyclotrimethylenetrinitramine, cyclonite, T4, hexahydro-1,3,5-trinitro-1,3,5-triazine, 1,3,5-trinitro-1,3,5-triaza-cyclohexane, hexogen or hexogene) (CAS 121-82-4);
 - b. Keto-RDX (K-6 or 2,4,6-trinitro-2,4,6-triazacyclohexanone) (CAS 115029-35-1);
22. TAGN (triaminoguanidinenitrate) (CAS 4000-16-2);
23. TATB (triaminotrinitrobenzene) (CAS 3058-38-6);
24. TEDDZ (3,3,7,7-tetrakis(difluoroamine) octahydro-1,5-dinitro-1,5-diazocine);
25. Tetrazoles as follows:
 - a. NTAT (nitrotriazol aminotetrazole);
 - b. NTNT (1-N-(2-nitrotriazolo)-4-nitrotetrazole);
26. Tetryl (trinitrophenylmethylnitramine) (CAS 479-45-8);
27. TNAD (1,4,5,8-tetranitro-1,4,5,8-tetraazadecalin) (CAS 135877-16-6);
28. TNAZ (1,3,3-trinitroazetidine) (CAS 97645-24-4);
29. TNGU (SORGUYL or tetranitroglycoluril) (CAS 55510-03-7);
30. TNP (1,4,5,8-tetranitro-pyridazino[4,5-d]pyridazine) (CAS 229176-04-9);
31. Triazines as follows:
 - a. DNAM (2-oxy-4,6-dinitroamino-s-triazine) (CAS 19899-80-0);

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- b. NNHT (2-nitroimino-5-nitro-hexahydro-1,3,5-triazine) (CAS 130400-13-4);
- 32. Triazoles as follows:
 - a. 5-azido-2-nitrotriazole;
 - b. ADHTDN (4-amino-3,5-dihydrazino-1,2,4-triazole dinitramide) (CAS 1614-08-0);
 - c. ADNT (1-amino-3,5-dinitro-1,2,4-triazole);
 - d. BDNTA ((bis-dinitrotriazole)amine);
 - e. DBT (3,3'-dinitro-5,5-bi-1,2,4-triazole) (CAS 30003-46-4);
 - f. DNBT (dinitrobistriazole) (CAS 70890-46-9);
 - g. (Reserved)
 - h. NTDNT (1-N-(2-nitrotriazolo) 3,5-dinitrotriazole);
 - i. PDNT (1-picryl-3,5-dinitrotriazole);
 - j. TACOT (tetranitrobenzotriazolobenzotriazole) (CAS 25243-36-1);
- 33. "Explosives" not listed elsewhere in this list having a detonation velocity exceeding 8,700 m/s, at maximum density, or a detonation pressure exceeding 34 GPa (340 kbar);
- 34. (Reserved)
- 35. Nitrocellulose (containing more than 12.5% nitrogen) (CAS 9004-70-0);
- 36. Nitroglycol (CAS 628-96-6);
- 37. Pentaerythritol tetranitrate (PETN) (CAS 78-11-5);
- 38. Picryl chloride (CAS 88-88-0);
- 39. 2,4,6-Trinitrotoluene (TNT) (CAS 118-96-7);
- 40. Nitroglycerine (NG) (CAS 55-63-0);
- 41. Triacetone Triperoxide (TATP) (CAS 17088-37-8);
- 42. Guanidine nitrate (CAS 506-93-4);
- 43. Nitroguanidine (NQ) (CAS 556-88-7);
- 44. DNAN (2,4-dinitroanisole) (CAS 119-27-7);
- 45. TEX (4,10-Dinitro-2,6,8,12-tetraoxa-4,10-diazaisowurtzitane);
- 46. GUDN (Guanylurea dinitramide) FOX-12 (CAS 217464-38-5);
- 47. Tetrazines as follows:
 - a. BTAT (Bis(2,2,2-trinitroethyl)-3,6-diaminotetrazine);
 - b. LAX-112 (3,6-diamino-1,2,4,5-tetrazine-1,4-dioxide);
- 48. Energetic ionic materials melting between 343 K (70°C) and 373 K (100°C) and with detonation velocity exceeding 6,800 m/s or detonation pressure exceeding 18 GPa (180 kbar);
- 49. BTNEN (Bis(2,2,2-trinitroethyl)-nitramine) (CAS 19836-28-3);
- 50. FTDO (5,6-(3',4'-furazano)- 1,2,3,4-tetrazine-1,3-dioxide).
- 51. EDNA (Ethylenedinitramine) (CAS 505-71-5);
- 52. TKX-50 (Dihydroxylammonium 5,5'-bistetrazole-1,1'-diolate).

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8A2 MATERIAL PROCESSING (SYSTEMS, EQUIPMENT AND COMPONENTS)

N.B. For quiet running bearings, see 6A009

- 8A201 1. Anti-friction bearings, bearing systems and components, as follows:
- a. Ball bearings and solid roller bearings, having all tolerances specified by the manufacturer in accordance with ISO 492 Tolerance Class 4 or Class 2 (or national equivalents), or better, and having both 'rings' and 'rolling elements', made from monel or beryllium;
Note 8A201.a does not apply to tapered roller bearings.

Technical Notes

1. 'Ring' - annular part of a radial rolling bearing incorporating one or more raceways (ISO 5593:1997).
2. 'Rolling element' - ball or roller which rolls between raceways (ISO 5593:1997).

- b. (Reserved)
- c. Active magnetic bearing systems using any of the following, and specially designed components therefor:
 1. Materials with flux densities of 2.0 T or greater and yield strengths greater than 414 MPa;
 2. All-electromagnetic 3D homopolar bias designs for actuators; or
 3. High temperature (450 K (177°C) and above) position sensors.

8B2 MATERIAL PROCESSING (TEST, INSPECTION AND PRODUCTION EQUIPMENT)

Technical Notes

1. Secondary parallel contouring axes, (e.g., the w-axis on horizontal boring mills or a secondary rotary axis the centre line of which is parallel to the primary rotary axis) are not counted in the total number of contouring axes. Rotary axes need not rotate over 360°. A rotary axis can be driven by a linear device (e.g., a screw or a rack-and-pinion).
2. For the purposes of 8B201, the number of axes which can be co-ordinated simultaneously for "contouring control" is the number of axes along or around which, during processing of the workpiece, simultaneous and interrelated motions are performed between the workpiece and a tool. This does not include any additional axes along or around which other relative motions within the machine are performed, such as:
 - a. Wheel-dressing systems in grinding machines;
 - b. Parallel rotary axes designed for mounting of separate workpieces;
 - c. Co-linear rotary axes designed for manipulating the same workpiece by holding it in a chuck from different ends.
3. Axis nomenclature shall be in accordance with International Standard ISO 841:2001, Industrial automation systems and integration - Numerical control of machines - Coordinate system and motion nomenclature.
4. For the purposes of 8A2, 8B2, 8C2, 8D2 and 8E2 "tilting spindle" is counted as a rotary axis.
5. 'Stated "unidirectional positioning repeatability"' may be used for each machine tool model as an alternative to individual machine tests, and is determined as follows:
 - a. Select five machines of a model to be evaluated;
 - b. Measure the linear axis repeatability ($R_{\uparrow}, R_{\downarrow}$) according to ISO 230-2:2014 and evaluate "unidirectional positioning repeatability" for each axis of each of the five machines;
 - c. Determine the arithmetic mean value of the "unidirectional positioning repeatability"-values for each axis of all five machines together. These arithmetic mean values of "unidirectional positioning repeatability" ($\overline{UPR}$) become the stated value of each axis for the model ($\overline{UPR}_x, \overline{UPR}_y, \dots$)
 - d. Since 8A2 refers to each linear axis there will be as many 'stated "unidirectional positioning repeatability"'-values as there are linear axes;
 - e. If any axis of a machine model not specified by 8B201.a to 8B201.c has a 'stated "unidirectional positioning repeatability"' equal to or less than the specified "unidirectional positioning

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repeatability" of each machine tool model plus 0.7 μm , the builder should be required to reaffirm the accuracy level once every eighteen months.

6. For the purposes of 8B2, measurement uncertainty for the "unidirectional positioning repeatability" of machine tools, as defined in the International Standard ISO 230-2:2014, shall not be considered.
7. For the purpose of 8B2, the measurement of axes shall be made according to test procedures in 5.3.2. of ISO 230-2:2014. Tests for axes longer than 2 meters shall be made over 2 m segments. Axes longer than 4 m require multiple tests (e.g., two tests for axes longer than 4 m and up to 8 m, three tests for axes longer than 8 m and up to 12 m), each over 2 m segments and distributed in equal intervals over the axis length. Test segments are equally spaced along the full axis length, with any excess length equally divided at the beginning, in between, and at the end of the test segments. The smallest "unidirectional positioning repeatability"-value of all test segments is to be reported.

8B201 1. Machine tools and any combination thereof, for removing (or cutting) metals, ceramics or "composites", which, according to the manufacturer's technical specification, can be equipped with electronic devices for "numerical control", as follows:

Note 1 8B201 does not apply to special purpose machine tools limited to the manufacture of gears. For such machines, see 8B203.

Note 2 8B201 does not apply to special purpose machine tools limited to the manufacture of any of the following:

- a. Crank shafts or cam shafts;
- b. Tools or cutters;
- c. Extruder worms;
- d. Engraved or faceted jewellery parts; or
- e. Dental prostheses;

Note 3 A machine tool having at least two of the three turning, milling or grinding capabilities (e.g., a turning machine with milling capability), must be evaluated against each applicable entry 8B201.a, b. or c.

Note 4 A machine tool having an additive manufacturing capability in addition to a turning, milling or grinding capability must be evaluated against each applicable entry 8B201.a, b. or c.

N.B. For optical finishing machines, see 8B202.

- 8B201 a. Machine tools for turning having two or more axes which can be coordinated simultaneously for "contouring control" having any of the following:
1. "Unidirectional positioning repeatability" equal to or less (better) than 0.9 μm along one or more linear axis with a travel length less than 1.0 m; or
 2. "Unidirectional positioning repeatability" equal to or less (better) than 1.1 μm along one or more linear axis with a travel length equal to or greater than 1.0 m;

Note 1 8B201.a does not apply to turning machines specially designed for producing contact lenses, having all of the following:

- a. Machine controller limited to using ophthalmic based "software" for part programming data input; and
- b. No vacuum chucking.

Note 2 8B201.a does not apply to bar machines (Swissturn), limited to machining only bar feed thru, if maximum bar diameter is equal to or less than 42 mm and there is no capability of mounting chucks. Machines may have drilling or milling capabilities for machining parts with diameters less than 42 mm.

- 8B201 b. Machine tools for milling having any of the following:
1. Three linear axes plus one rotary axis which can be coordinated simultaneously for "contouring control" having any of the following:
 - a. "Unidirectional positioning repeatability" equal to or less (better) than 0.9 μm along one or more linear axis with a travel length less than 1.0 m; or
 - b. "Unidirectional positioning repeatability" equal to or less (better) than 1.1 μm along one or more linear axis with a travel length equal to or greater than 1.0 m;

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2. Five or more axes which can be coordinated simultaneously for "contouring control" having any of the following:
 - a. "Unidirectional positioning repeatability" equal to or less (better) than 0.9 μm along one or more linear axis with a travel length less than 1.0 m;
 - b. "Unidirectional positioning repeatability" equal to or less (better) than 1.4 μm along one or more linear axis with a travel length equal to or greater than 1 m and less than 4 m;
 - c. "Unidirectional positioning repeatability" equal to or less (better) than 6.0 μm along one or more linear axis with a travel length equal to or greater than 4 m; or

- 8B201 b. 3. A "unidirectional positioning repeatability" for jig boring machines, equal to or less (better) than 1.1 μm along one or more linear axis; or

- 8B201 b. 4. Fly cutting machines having all of the following:
 - a. Spindle "run-out" and "camming" less (better) than 0.0004 mm TIR; and
 - b. Angular deviation of slide movement (yaw, pitch and roll) less (better) than 2 seconds of arc, TIR, over 300 mm of travel;

- 8B201 c. Machine tools for grinding having any of the following:
 1. Having all of the following:
 - a. "Unidirectional positioning repeatability" equal to or less (better) than 1.1 μm along one or more linear axis; and
 - b. Three or more axes which can be coordinated simultaneously for "contouring control"; or
 2. Five or more axes which can be coordinated simultaneously for "contouring control" having any of the following:
 - a. "Unidirectional positioning repeatability" equal to or less (better) than 1.1 μm along one or more linear axis with a travel length less than 1 m;
 - b. "Unidirectional positioning repeatability" equal to or less (better) than 1.4 μm along one or more linear axis with a travel length equal to or greater than 1 m and less than 4 m; or
 - c. "Unidirectional positioning repeatability" equal to or less (better) than 6.0 μm along one or more linear axis with a travel length equal to or greater than 4 m.

Note 8B201.c does not apply to grinding machines as follows:

 - a. Cylindrical external, internal, and external-internal grinding machines, having all of the following:
 1. Limited to cylindrical grinding; and
 2. Limited to a maximum workpiece capacity of 150 mm outside diameter or length.
 - b. Machines designed specifically as jig grinders that do not have a z-axis or a w-axis, with a "unidirectional positioning repeatability" less (better) than 1.1 μm .
 - c. Surface grinders.

- 8B201 d. Electrical discharge machines (EDM) of the non-wire type which have two or more rotary axes which can be coordinated simultaneously for "contouring control";

- 8B201 e. Machine tools for removing metals, ceramics or "composites", having all of the following:
 1. Removing material by means of any of the following:
 - a. Water or other liquid jets, including those employing abrasive additives;
 - b. Electron beam; or
 - c. "Laser" beam; and
 2. At least two rotary axes having all of the following:
 - a. Can be coordinated simultaneously for "contouring control"; and
 - b. A positioning "accuracy" of less (better) than 0.003°;

- 8B201 f. Deep-hole-drilling machines and turning machines modified for deep-hole-drilling, having a maximum depth-of-bore capability exceeding 5 m.

- 8B202 Numerically controlled optical finishing machine tools equipped for selective material removal to produce non-spherical optical surfaces having all of the following characteristics:
 - a. Finishing the form to less (better) than 1.0 μm ;
 - b. Finishing to a roughness less (better) than 100 nm rms;

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- c. Four or more axes which can be coordinated simultaneously for "contouring control"; and
- d. Using any of the following processes:
 - 1. 'Magnetorheological finishing (MRF)';
 - 2. 'Electrorheological finishing (ERF)';
 - 3. 'Energetic particle beam finishing';
 - 4. 'Inflatable membrane tool finishing'; or
 - 5. 'Fluid jet finishing'.

Technical Notes

For the purposes of 8B202:

- 1. 'MRF' is a material removal process using an abrasive magnetic fluid whose viscosity is controlled by a magnetic field.
- 2. 'ERF' is a removal process using an abrasive fluid whose viscosity is controlled by an electric field.
- 3. 'Energetic particle beam finishing' uses Reactive Atom Plasmas (RAP) or ion-beams to selectively remove material.
- 4. 'Inflatable membrane tool finishing' is a process that uses a pressurized membrane that deforms to contact the workpiece over a small area.
- 5. 'Fluid jet finishing' makes use of a fluid stream for material removal.

8B203 "Numerically controlled" or manual machine tools, and specially designed components, controls and accessories therefor, specially designed for the shaving, finishing, grinding or honing of hardened ($R_c = 40$ or more) spur, helical and double-helical gears with a pitch diameter exceeding 1,250 mm and a face width of 15% of pitch diameter or larger finished to a quality of AGMA 14 or better (equivalent to ISO 1328 class 3).

8B204 Hot "isostatic presses" having all of the following, and specially designed components and accessories therefor:

- a. A controlled thermal environment within the closed cavity and a chamber cavity with an inside diameter of 406 mm or more; and
- b. Having any of the following:
 - 1. A maximum working pressure exceeding 207 MPa;
 - 2. A controlled thermal environment exceeding 1,773 K (1,500°C); or
 - 3. A facility for hydrocarbon impregnation and removal of resultant gaseous degradation products.

Technical Note

The inside chamber dimension is that of the chamber in which both the working temperature and the working pressure are achieved and does not include fixtures. That dimension will be the smaller of either the inside diameter of the pressure chamber or the inside diameter of the insulated furnace chamber, depending on which of the two chambers is located inside the other.

N.B. For specially designed dies, moulds and tooling see 8B103, 8B909 and 6A018

8B205 Equipment specially designed for the deposition, processing and in-process control of inorganic overlays, coatings and surface modifications, as follows, for substrates specified in column 2, by processes shown in column 1 in the Table following 8E203.f., and specially designed automated handling, positioning, manipulation and control components therefor:

- a. Chemical vapour deposition (CVD) production equipment having all of the following:
 - 1. A process modified for one of the following:
 - a. Pulsating CVD;
 - b. Controlled nucleation thermal deposition (CNTD); or
 - c. Plasma enhanced or plasma assisted CVD; and
 - 2. Having any of the following:
 - a. Incorporating high vacuum (equal to or less than 0.01 Pa) rotating seals; or
 - b. Incorporating *in situ* coating thickness control;

8B205 b. Ion implantation production equipment having beam currents of 5 mA or more;

- c. Electron beam physical vapour deposition (EB-PVD) production equipment incorporating power systems rated for over 80 kW and having any of the following:
 - 1. A liquid pool level "laser" control system which regulates precisely the ingots feed rate; or

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2. A computer controlled rate monitor operating on the principle of photo-luminescence of the ionised atoms in the evaporant stream to control the deposition rate of a coating containing two or more elements;
- d. Plasma spraying production equipment having any of the following:
 1. Operating at reduced pressure controlled atmosphere (equal to or less than 10 kPa measured above and within 300 mm of the gun nozzle exit) in a vacuum chamber capable of evacuation down to 0.01 Pa prior to the spraying process; or
 2. Incorporating *in situ* coating thickness control;
- e. Sputter deposition production equipment capable of current densities of 0.1 mA/mm² or higher at a deposition rate of 15 µm/h or more;
- f. Cathodic arc deposition production equipment incorporating a grid of electromagnets for steering control of the arc spot on the cathode;
- g. Ion plating production equipment capable of *in situ* measurement of any of the following:
 1. Coating thickness on the substrate and rate control; or
 2. Optical characteristics.

Note 8B205.a, 8B205.b, 8B205.e, 8B205.f and 8B205.g do not apply to chemical vapour deposition, cathodic arc, sputter deposition, ion plating or ion implantation equipment, specially designed for cutting or machining tools.

8B206 Dimensional inspection or measuring systems, equipment, position feedback units and "electronic assemblies", as follows:

- a. Computer controlled or "numerical controlled" Coordinate Measuring Machines (CMM), having a three dimensional (volumetric) maximum permissible error of length measurement ($E_{0,MPE}$) at any point within the operating range of the machine (i.e., within the length of axes) equal to or less (better) than $1.7 + L/1,000$ µm (L is the measured length in mm), according to ISO 10360-2 (2009);

Technical Note

The $E_{0,MPE}$ of the most accurate configuration of the CMM specified by the manufacturer (e.g., best of the following: probe, stylus length, motion parameters, environment) and with "all compensations available" shall be compared to the $1.7 + L/1,000$ µm threshold.

- b. Linear displacement measuring instruments or systems, linear position feedback units, and "electronic assemblies", as follows:

Note: Interferometer and optical-encoder measuring systems containing a "laser" are only specified in 8B206.b.3.

1. 'Non-contact type measuring systems' with a 'resolution' equal to or less (better) than 0.2 µm within a measuring range up to 0.2 mm;

Technical Note

For the purposes of 8B206.b.1, 'non-contact type measuring systems' are designed to measure the distance between the probe and measured object along a single vector, where the probe or measured object is in motion.

2. Linear position feedback units specially designed for machine tools and having an overall "accuracy" less (better) than $(800 + (600 \times L/1,000))$ nm (L equals effective length in mm);

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3. Measuring systems having all of the following:

- a. Containing a "laser";
- b. A 'resolution' over their full scale of 0.200 nm or less (better); and
- c. Capable of achieving a "measurement uncertainty" equal to or less (better) than $(1.6 + L/2,000)$ nm (L is the measured length in mm) at any point within a measuring range, when compensated for the refractive index of air and measured over a period of 30 seconds at a temperature of $20 \pm 0.01^{\circ}\text{C}$; or

Technical Note

For the purposes of 8B206.b, 'resolution' is the least increment of a measuring device; on digital instruments, the least significant bit.

8B206 b. 4 "Electronic assemblies" specially designed to provide feedback capability in systems specified by 8B206.b.3;

8B206 c. Rotary position feedback units specially designed for machine tools or angular displacement measuring instruments, having an angular position "accuracy" equal to or less (better) than 0.9 second of arc;

Note: 8B206.c does not apply to optical instruments, such as autocollimators, using collimated light (e.g., laser light) to detect angular displacement of a mirror.

8B206 d. Equipment for measuring surface roughness (including surface defects), by measuring optical scatter with a sensitivity of 0.5 nm or less (better).

Note: 8B206 includes machine tools, other than those specified by 8B201 that can be used as measuring machines if they meet or exceed the criteria specified for the measuring machine function.

8B207 "Robots" having any of the following characteristics and specially designed controllers and "end-effectors" therefor:

- a. (Reserved);
- b. Specially designed to comply with Indian safety standards applicable to potentially explosive munitions environments;

Note 8B207.b does not apply to "robots" specially designed for paint-spraying booths.

- c. Specially designed or rated as radiation-hardened to withstand greater than 5×10^3 Gy (Si) without operational degradation; or
- d. Specially designed to operate at altitudes exceeding 30,000 m.

8B208 'Compound rotary tables' and "tilting spindles", specially designed for machine tools, as follows:

- a. (Reserved)
- b. (Reserved)
- c. 'Compound rotary tables' having all of the following:
 1. Designed for machine tools for turning, milling or grinding; and
 2. Two rotary axes designed to be coordinated simultaneously for "contouring control";

Technical Note

A 'compound rotary table' is a table allowing the workpiece to rotate and tilt about two non-parallel axes.

- d. "Tilting spindles" having all of the following:
 1. Designed for machine tools for turning, milling or grinding; and
2. Designed to be coordinated simultaneously for "contouring control".

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- 8B209 Spin-forming machines and flow-forming machines, which, according to the manufacturer's technical specification, can be equipped with "numerical control" units or a computer control and having all of the following:
- Three or more axes which can be coordinated simultaneously for "contouring control"; and
 - A roller force more than 60 kN.

Technical Note

For the purpose of 8B209, machines combining the function of spin-forming and flow-forming are regarded as flow-forming machines.

8C2. MATERIAL PROCESSING (MATERIALS) (Reserved)

8D2. MATERIAL PROCESSING (SOFTWARE)

8D201 "Software", other than that specified by 8D202 as follows:

- "Software" specially designed or modified for the "development" or "production" of equipment specified by 8A2 or 82B;
- "Software" specially designed or modified for the "use" of equipment specified by 8A201.c, 8B201, or 8B203 to 8B209.

Note 8D201 does not apply to part programming "software" that generates "numerical control" codes for machining various parts.

8D202 "Software" for electronic devices, even when residing in an electronic device or system, enabling such devices or systems to function as a "numerical control" unit, capable of co-ordinating simultaneously more than 4 axes for "contouring control".

Note1 8D202 does not apply to "software" specially designed or modified for the operation of items not specified by 8A2.

Note 2 8D202 does not apply to "software" for items specified by 8B202 See 8D201 and 8D203 for "software" for items specified by 8B202.

Note 3 8D202 does not apply to "software" that is exported with, and the minimum necessary for the operation of, items not specified by Categories 8A2, 8B2, 8C2, 8D2 and 8E2.

8D203 "Software", designed or modified for the operation of equipment specified by 8B202 that converts optical design, workpiece measurements and material removal functions into "numerical control" commands to achieve the desired workpiece form.

8E2MATERIAL PROCESSING (TECHNOLOGY)

8E201 "Technology" according to the General Technology Note for the "development" of equipment or "software" specified by 8A2, 8B2 or 8D2.

Note 8E201 includes "technology" for the integration of probe systems into coordinate measurement machines specified by 8B206.a.

8E202 "Technology" according to the General Technology Note for the "production" of equipment specified by 8A2 or 8B2

8E203 Other "technology", as follows:

- (Reserved);
- "Technology" for metal-working manufacturing processes, as follows:
 - "Technology" for the design of tools, dies or fixtures specially designed for any of the following processes:
 - "Superplastic forming";
 - "Diffusion bonding"; or
 - "Direct-acting hydraulic pressing";

2. **Reserved**

N.B. For "technology" for metal-working manufacturing processes for gas turbine engines and components, see 8E903 and 6A022.

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- | | |
|-------|---|
| 8E203 | c. "Technology" for the "development" or "production" of hydraulic stretch-forming machines and dies therefor, for the manufacture of airframe structures; |
| 8E203 | d. (Reserved); |
| 8E203 | e. "Technology" for the "development" of integration "software" for incorporation of expert systems for advanced decision support of shop floor operations into "numerical control" units; |
| 8E203 | f. "Technology" for the application of inorganic overlay coatings or inorganic surface modification coatings (specified in column 3 of the following table) to non-electronic substrates (specified in column 2 of the following table), by processes specified in column 1 of the following table and defined in the Technical Note. |

N.B. This Table should be read to specify the "technology" of a particular 'Coating Process' only when the Resultant Coating in column 3 is in a paragraph directly across from the relevant 'Substrate' under column 2. For example, Chemical Vapour Deposition (CVD) 'coating process' technical data are included for the application of 'silicides' to 'Carbon-carbon, Ceramic and Metal "matrix" "composites" substrates, but are not included for the application of 'silicides' to 'Cemented tungsten carbide (16), Silicon carbide (18)' substrates. In the second case, the resultant coating is not listed in the paragraph under column 3 directly across from the paragraph under column 2 listing 'Cemented tungsten carbide (16), Silicon carbide (18)'.

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TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)*	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
A. Chemical Vapour Deposition (CVD)	"Superalloys"	Aluminides for internal passages
	Ceramics (19) and Low-expansion glasses (14)	Silicides Carbides Dielectric layers (15) Diamond Diamond-like carbon (17)
	Carbon-carbon, Ceramic and Metal "matrix" "composites"	Silicides Carbides Refractory metals Mixtures thereof (4) Dielectric layers (15) Aluminides Alloyed aluminides (2) Boron nitride
	Cemented tungsten carbide (16), Silicon carbide (18)	Carbides Tungsten Mixtures thereof (4) Dielectric layers (15)
	Molybdenum and Molybdenum alloys	Dielectric layers (15)
	Beryllium and Beryllium alloys	Dielectric layers (15) Diamond Diamond-like carbon (17)
	Sensor window materials (9)	Dielectric layers (15) Diamond Diamond-like carbon (17)

* The numbers in parenthesis refer to the Notes following this Table.

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TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
B. Thermal-Evaporation Physical Vapour Deposition (TE-PVD)		
B.1. Physical Vapour Deposition (PVD): Electron-Beam (EB-PVD)	"Superalloys"	Alloyed silicides Alloyed aluminides (2) MCrAlX (5) Modified zirconia (12) Silicides Aluminides Mixtures thereof (4)
	Ceramics (19) and Low- expansion glasses (14)	Dielectric layers (15)
	Corrosion resistant steel (7)	MCrAlX (5) Modified zirconia (12) Mixtures thereof (4)
	Carbon-carbon, Ceramic and Metal "matrix" "composites"	Silicides Carbides Refractory metals Mixtures thereof (4) Dielectric layers (15) Boron nitride
	Cemented tungsten carbide (16), Silicon carbide (18)	Carbides Tungsten Mixtures thereof (4) Dielectric layers (15)
	Molybdenum and Molybdenum alloys	Dielectric layers (15)
	Beryllium and Beryllium alloys	Dielectric layers (15) Borides Beryllium
	Sensor window materials (9)	Dielectric layers (15)
	Titanium alloys (13)	Borides Nitrides

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TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
B.2. Ion assisted resistive heating Physical Vapour Deposition (PVD) (Ion Plating)	Ceramics (19) and Low-expansion glasses (14)	Dielectric layers (15) Diamond-like carbon (17)
	Carbon-carbon, Ceramic and Metal "matrix" "composites"	Dielectric layers (15)
	Cemented tungsten carbide (16), Silicon carbide	Dielectric layers (15)
	Molybdenum and Molybdenum alloys	Dielectric layers (15)
	Beryllium and Beryllium alloys	Dielectric layers (15)
	Sensor window materials (9)	Dielectric layers (15) Diamond-like carbon (17)
B.3. Physical Vapour Deposition (PVD): "Laser" Vaporization	Ceramics (19) and Low-expansion glasses (14)	Silicides Dielectric layers (15) Diamond-like carbon (17)
	Carbon-carbon, Ceramic and Metal "matrix" "composites"	Dielectric layers (15)
	Cemented tungsten carbide (16), Silicon carbide	Dielectric layers (15)
	Molybdenum and Molybdenum alloys	Dielectric layers (15)
	Beryllium and Beryllium alloys	Dielectric layers (15)
	Sensor window materials (9)	Dielectric layers (15) Diamond-like carbon

TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
B.4. Physical Vapour Deposition (PVD): Cathodic Arc Discharge	"Superalloys"	Alloyed silicides Alloyed aluminides (2) MCrAlX (5)
	Polymers (11) and Organic "matrix" "composites"	Borides Carbides Nitrides Diamond-like carbon (17)

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C.	Pack cementation (see A above for out-of-pack cementation) (10)	Carbon-carbon, Ceramic and Metal "matrix" "composites"	Silicides Carbides Mixtures thereof (4)
		Titanium alloys (13)	Silicides Aluminides Alloyed aluminides (2)
		Refractory metals and alloys (8)	Silicides Oxides
D.	Plasma spraying	"Superalloys"	MCrAlX (5) Modified zirconia (12) Mixtures thereof (4) Abradable Nickel-Graphite Abradable materials containing Ni-Cr-Al Abradable Al-Si-Polyester Alloyed aluminides (2)
		Aluminium alloys (6)	MCrAlX (5) Modified zirconia (12) Silicides Mixtures thereof (4)
		Refractory metals and alloys (8)	Aluminides Silicides Carbides

TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
D. (continued)	Corrosion resistant steel (7)	MCrAlX (5) Modified zirconia (12) Mixtures thereof (4)
	Titanium alloys (13)	Carbides Aluminides Silicides Alloyed aluminides (2) Abradable Nickel-Graphite Abradable materials containing Ni-Cr-Al Abradable Al-Si-Polyester
E.	Refractory metals and alloys (8)	Fused silicides Fused aluminides except for resistance heating elements
	Carbon-carbon,	Silicides

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	Ceramic and Metal "matrix" "composites"	Carbides Mixtures thereof (4)
F. Sputter Deposition	"Superalloys"	Alloyed silicides Alloyed aluminides (2) Noble metal modified aluminides (3) MCrAlX (5) Modified zirconia (12) Platinum Mixtures thereof (4)
	Ceramics and Low-expansion glasses (14)	Silicides Platinum Mixtures thereof (4) Dielectric layers (15) Diamond-like carbon (17)

TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
F. (continued)	Titanium alloys (13)	Borides Nitrides Oxides Silicides Aluminides Alloyed aluminides (2) Carbides
	Carbon-carbon, Ceramic and Metal "matrix" "composites"	Silicides Carbides Refractory metals Mixtures thereof (4) Dielectric layers (15) Boron nitride
	Cemented tungsten carbide (16), Silicon carbide (18)	Carbides Tungsten Mixtures thereof (4) Dielectric layers (15) Boron nitride
	Molybdenum and Molybdenum alloys	Dielectric layers (15)
	Beryllium and Beryllium alloys	Borides Dielectric layers (15) Beryllium
	Sensor window materials (9)	Dielectric layers (15) Diamond-like carbon (17)
	Refractory metals and alloys (8)	Aluminides Silicides Oxides Carbides

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TABLE - DEPOSITION TECHNIQUES

1. <u>Coating Process</u> (1)	2. <u>Substrate</u>	3. <u>Resultant Coating</u>
G. Ion Implantation	High temperature bearing steels	Additions of Chromium Tantalum or Niobium (Columbium)
	Titanium alloys (13)	Borides Nitrides
	Beryllium and Beryllium alloys	Borides
	Cemented tungsten carbide (16)	Carbides Nitrides

TABLE - DEPOSITION TECHNIQUES - NOTES

- The term 'coating process' includes coating repair and refurbishing as well as original coating.
- The term 'alloyed aluminide coating' includes single or multiple-step coatings in which an element or elements are deposited prior to or during application of the aluminide coating, even if these elements are deposited by another coating process. It does not, however, include the multiple use of single-step pack cementation processes to achieve alloyed aluminides.
- The term 'noble metal modified aluminide' coating includes multiple-step coatings in which the noble metal or noble metals are laid down by some other coating process prior to application of the aluminide coating.
- The term 'mixtures thereof' includes infiltrated material, graded compositions, co-deposits and multilayer deposits and are obtained by one or more of the coating processes specified in the Table.
- 'MCrAlX' refers to a coating alloy where M equals cobalt, iron, nickel or combinations thereof and X equals hafnium, yttrium, silicon, tantalum in any amount or other intentional additions over 0.01% by weight in various proportions and combinations, except:
 - CoCrAlY coatings which contain less than 22% by weight of chromium, less than 7% by weight of aluminium and less than 2 weight percent of yttrium;
 - CoCrAlY coatings which contain 22 to 24% by weight of chromium, 10 to 12% by weight of aluminium and 0.5 to 0.7% by weight of yttrium; or
 - NiCrAlY coatings which contain 21 to 23% by weight of chromium, 10 to 12% by weight of aluminium and 0.9 to 1.1% by weight of yttrium.
- The term 'aluminium alloys' refers to alloys having an ultimate tensile strength of 190 MPa or more measured at 293 K (20°C).
- The term 'corrosion resistant steel' refers to AISI (American Iron and Steel Institute) 300 series or equivalent Indian standard steels.
- 'Refractory metals and alloys' include the following metals and their alloys: niobium (columbium), molybdenum, tungsten and tantalum.

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9. 'Sensor window materials', as follows: alumina, silicon, germanium, zinc sulphide, zinc selenide, gallium arsenide, diamond, gallium phosphide, sapphire and the following metal halides: sensor window materials of more than 40 mm diameter for zirconium fluoride and hafnium fluoride.
10. 8A2, 8B2, 8C2, 8D2, 8E2 do not include "technology" for single-step pack cementation of solid **aerofoils**.
11. 'Polymers', as follows: polyimide, polyester, polysulphide, polycarbonates and polyurethanes.
12. 'Modified zirconia' refers to additions of other metal oxides (e.g., calcia, magnesia, yttria, hafnia, rare earth oxides) to zirconia in order to stabilise certain crystallographic phases and phase compositions. Thermal barrier coatings made of zirconia, modified with calcia or magnesia by mixing or fusion, are not included.
13. 'Titanium alloys' refers only to aerospace alloys having an ultimate tensile strength of 900 MPa or more measured at 293 K (20°C).
14. 'Low-expansion glasses' refers to glasses which have a coefficient of thermal expansion of $1 \times 10^{-7} \text{ K}^{-1}$ or less measured at 293 K (20°C).
15. 'Dielectric layers' are coatings constructed of multi-layers of insulator materials in which the interference properties of a design composed of materials of various refractive indices are used to reflect, transmit or absorb various wavelength bands. Dielectric layers refers to more than four dielectric layers or dielectric/metal "composite" layers.
16. 'Cemented tungsten carbide' does not include cutting and forming tool materials consisting of tungsten carbide/(cobalt, nickel), titanium carbide/(cobalt, nickel), chromium carbide/nickel-chromium and chromium carbide/nickel.
17. "Technology" for depositing diamond-like carbon on any of the following is not included: magnetic disk drives and heads, equipment for the manufacture of disposables, valves for faucets, acoustic diaphragms for speakers, engine parts for automobiles, cutting tools, punching-pressing dies, office automation equipment, microphones, medical devices or moulds, for casting or moulding of plastics, manufactured from alloys containing less than 5% beryllium.
18. 'Silicon carbide' does not include cutting and forming tool materials.
19. Ceramic substrates, as used in this entry, does not include ceramic materials containing 5% by weight, or greater, clay or cement content, either as separate constituents or in combination.

TABLE - DEPOSITION TECHNIQUES - TECHNICAL NOTE

Processes specified in Column 1 of the Table are defined as follows:

- a. Chemical Vapour Deposition (CVD) is an overlay coating or surface modification coating process wherein a metal, alloy, "composite", dielectric or ceramic is deposited upon a heated substrate. Gaseous reactants are decomposed or combined in the vicinity of a substrate resulting in the deposition of the desired elemental, alloy or compound material on the substrate. Energy for this decomposition or chemical reaction process may be provided by the heat of the substrate, a glow discharge plasma, or "laser" irradiation.
 - N.B.1 CVD includes the following processes: directed gas flow out-of-pack deposition, pulsating CVD, controlled nucleation thermal deposition (CNTD), plasma enhanced or plasma assisted CVD processes.
 - N.B.2 Pack denotes a substrate immersed in a powder mixture.
 - N.B.3 The gaseous reactants used in the out-of-pack process are produced using the same basic reactions and parameters as the pack cementation process, except that the substrate to be coated is not in contact with the powder mixture.
- b. Thermal Evaporation-Physical Vapour Deposition (TE-PVD) is an overlay coating process conducted in a vacuum with a pressure less than 0.1 Pa wherein a source of thermal energy is used to vaporize the coating

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material. This process results in the condensation, or deposition, of the evaporated species onto appropriately positioned substrates.

The addition of gases to the vacuum chamber during the coating process to synthesize compound coatings is an ordinary modification of the process.

The use of ion or electron beams, or plasma, to activate or assist the coating's deposition is also a common modification in this technique. The use of monitors to provide in-process measurement of optical characteristics and thickness of coatings can be a feature of these processes.

Specific TE-PVD processes are as follows:

1. Electron Beam PVD uses an electron beam to heat and evaporate the material which forms the coating;
2. Ion Assisted Resistive Heating PVD employs electrically resistive heating sources in combination with impinging ion beam(s) to produce a controlled and uniform flux of evaporated coating species;
3. "Laser" Vaporization uses either pulsed or continuous wave "laser" beams to vaporize the material which forms the coating;
4. Cathodic Arc Deposition employs a consumable cathode of the material which forms the coating and has an arc discharge established on the surface by a momentary contact of a ground trigger. Controlled motion of arcing erodes the cathode surface creating a highly ionized plasma. The anode can be either a cone attached to the periphery of the cathode, through an insulator, or the chamber. Substrate biasing is used for non line-of-sight deposition.

N.B. This definition does not include random cathodic arc deposition with non-biased substrates.

5. Ion Plating is a special modification of a general TE-PVD process in which a plasma or an ion source is used to ionize the species to be deposited, and a negative bias is applied to the substrate in order to facilitate the extraction of the species from the plasma. The introduction of reactive species, evaporation of solids within the process chamber, and the use of monitors to provide in-process measurement of optical characteristics and thicknesses of coatings are ordinary modifications of the process.
- c. Pack Cementation is a surface modification coating or overlay coating process wherein a substrate is immersed in a powder mixture (a pack), that consists of:
1. The metallic powders that are to be deposited (usually aluminium, chromium, silicon or combinations thereof);
 2. An activator (normally a halide salt); and
 3. An inert powder, most frequently alumina.

The substrate and powder mixture is contained within a retort which is heated to between 1,030 K (757°C) and 1,375 K (1,102°C) for sufficient time to deposit the coating.

- d. Plasma Spraying is an overlay coating process wherein a gun (spray torch) which produces and controls a plasma accepts powder or wire coating materials, melts them and propels them towards a substrate, whereon an integrally bonded coating is formed. Plasma spraying constitutes either low pressure plasma spraying or high velocity plasma spraying.

N.B.1 Low pressure means less than ambient atmospheric pressure.

N.B.2 High velocity refers to nozzle-exit gas velocity exceeding 750 m/s calculated at 293 K (20°C) at 0.1 MPa.

- e. Slurry Deposition is a surface modification coating or overlay coating process wherein a metallic or ceramic powder with an organic binder is suspended in a liquid and is applied to a substrate by either spraying, dipping or painting, subsequent air or oven drying, and heat treatment to obtain the desired coating.
- f. Sputter Deposition is an overlay coating process based on a momentum transfer phenomenon, wherein positive ions are accelerated by an electric field towards the surface of a target (coating material). The kinetic energy of the impacting ions is sufficient to cause target surface atoms to be released and deposited on an appropriately positioned substrate.

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N.B.1 The Table refers only to triode, magnetron or reactive sputter deposition which is used to increase adhesion of the coating and rate of deposition and to radio frequency (RF) augmented sputter deposition used to permit vaporisation of non-metallic coating materials.

N.B.2 Low-energy ion beams (less than 5 keV) can be used to activate the deposition.

- g. Ion Implantation is a surface modification coating process in which the element to be alloyed is ionized, accelerated through a potential gradient and implanted into the surface region of the substrate. This includes processes in which ion implantation is performed simultaneously with electron beam physical vapour deposition or sputter deposition.

TABLE - DEPOSITION TECHNIQUES - STATEMENT OF UNDERSTANDING

It is understood that the following technical information, accompanying the table of deposition techniques, is for use as appropriate.

1. Technical information for pretreatments of the substrates listed in the Table, as follows:
 - a. Chemical stripping and cleaning bath cycle parameters, as follows:
 1. Bath composition
 - a. For the removal of old or defective coatings, corrosion product or foreign deposits;
 - b. For preparation of virgin substrates;
 2. Time in bath;
 3. Temperature of bath;
 4. Number and sequences of wash cycles;
 - b. Visual and macroscopic criteria for acceptance of the cleaned part;
 - c. Heat treatment cycle parameters, as follows:
 1. Atmosphere parameters, as follows:
 - a. Composition of the atmosphere;
 - b. Pressure of the atmosphere;
 2. Temperature for heat treatment;
 3. Time of heat treatment;
 - d. Substrate surface preparation parameters, as follows:
 1. Grit blasting parameters, as follows:
 - a. Grit composition;
 - b. Grit size and shape;
 - c. Grit velocity;
 2. Time and sequence of cleaning cycle after grit blast;
 3. Surface finish parameters;
 4. Application of binders to promote adhesion;
 - e. Masking technique parameters, as follows:
 1. Material of mask;
 2. Location of mask;
2. Technical information for in situ quality assurance techniques for evaluation of the coating processes listed in the Table, as follows:
 - a. Atmosphere parameters, as follows:
 1. Composition of the atmosphere;
 2. Pressure of the atmosphere;
 - b. Time parameters;
 - c. Temperature parameters;
 - d. Thickness parameters;
 - e. Index of refraction parameters;
 - f. Control of composition;
3. Technical information for post deposition treatments of the coated substrates listed in the Table, as follows:
 - a. Shot peening parameters, as follows:
 1. Shot composition;
 2. Shot size;

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3. Shot velocity
- b. Post shot peening cleaning parameters;
- c. Heat treatment cycle parameters, as follows:
 1. Atmosphere parameters, as follows:
 - a. Composition of the atmosphere;
 - b. Pressure of the atmosphere;
 2. Time-temperature cycles;
 - d. Post heat treatment visual and macroscopic criteria for acceptance of the coated substrates;
4. Technical information for quality assurance techniques for the evaluation of the coated substrates listed in the Table, as follows:
 - a. Statistical sampling criteria;
 - b. Microscopic criteria for:
 1. Magnification;
 2. Coating thickness uniformity;
 3. Coating integrity;
 4. Coating composition;
 5. Coating and substrates bonding;
 6. Microstructural uniformity;
 - c. Criteria for optical properties assessment (measured as a function of wavelength):
 1. Reflectance;
 2. Transmission;
 3. Absorption;
 4. Scatter;
5. Technical information and parameters related to specific coating and surface modification processes listed in the Table, as follows:
 - a. For Chemical Vapour Deposition (CVD):
 1. Coating source composition and formulation;
 2. Carrier gas composition;
 3. Substrate temperature;
 4. Time-temperature-pressure cycles;
 5. Gas control and part manipulation;
 - b. For Thermal Evaporation - Physical Vapour Deposition (PVD):
 1. Ingot or coating material source composition;
 2. Substrate temperature;
 3. Reactive gas composition;
 4. Ingot feed rate or material vaporisation rate;
 5. Time-temperature-pressure cycles;
 6. Beam and part manipulation;
 7. "Laser" parameters, as follows:
 - a. Wave length;
 - b. Power density;
 - c. Pulse length;
 - d. Repetition ratio;
 - e. Source;
 - c. For Pack Cementation:
 1. Pack composition and formulation;
 2. Carrier gas composition;
 3. Time-temperature-pressure cycles;
 - d. For Plasma Spraying:
 1. Powder composition, preparation and size distributions;
 2. Feed gas composition and parameters;
 3. Substrate temperature;
 4. Gun power parameters;
 5. Spray distance;
 6. Spray angle;
 7. Cover gas composition, pressure and flow rates;
 8. Gun control and part manipulation;

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- e. For Sputter Deposition:
 - 1. Target composition and fabrication;
 - 2. Geometrical positioning of part and target;
 - 3. Reactive gas composition;
 - 4. Electrical bias;
 - 5. Time-temperature-pressure cycles;
 - 6. Triode power;
 - 7. Part manipulation;

- f. For Ion Implantation:
 - 1. Beam control and part manipulation;
 - 2. Ion source design details;
 - 3. Control techniques for ion beam and deposition rate parameters;
 - 4. Time-temperature-pressure cycles;

- g. For Ion Plating:
 - 1. Beam control and part manipulation;
 - 2. Ion source design details;
 - 3. Control techniques for ion beam and deposition rate parameters;
 - 4. Time-temperature-pressure cycles;
 - 5. Coating material feed rate and vaporisation rate;
 - 6. Substrate temperature;
 - 7. Substrate bias parameters.

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8A3 ELECTRONICS (SYSTEMS, EQUIPMENT AND COMPONENTS)

Note 1: The status of equipment and components described in 8A3, other than those described in 8A301.a.3 to 8A301.a.10, or 8A301.a.12 to 8A301.a.14 or 8A301.b.12, which are specially designed for or which have the same functional characteristics as other equipment is determined by the status of the other equipment.

Note 2 The status of integrated circuits described in 8A301.a.3. to 8A301.a.9., 8A301.a.12. or 8A301.a.14., which are unalterably programmed or designed for a specific function for another equipment is determined by the status of the other equipment
N.B. When the manufacturer or applicant cannot determine the status of the other equipment, the status of the integrated circuits is determined in 8A301.a.3. to 8A301.a.9., 8A301.a.12. and 8A301.a.13.

Note 3 The status of wafers (finished or unfinished), in which the function has been determined, is to be evaluated against the parameters of 8A301.a, 8A301.b., 8A301.d., 8A301.e., 8A301.g., 8A301.h., or 8A301.i.

8A301 Electronic items as follows:

a. General purpose integrated circuits as follows:

Note Integrated circuits include the following types:

- "Monolithic integrated circuits";
- "Hybrid integrated circuits";
- "Multichip integrated circuits";
- "Film type integrated circuits", including silicon-on-sapphire integrated circuits;
- "Optical integrated circuits";
- "Three dimensional integrated circuits";
- "Monolithic Microwave Integrated Circuits" ("MMICs").

8A301 a. 1. Integrated circuits designed or rated as radiation hardened to withstand any of the following:

- a. A total dose of 5×10^3 Gy (Si) or higher;
- b. A dose rate upset of 5×10^6 Gy (Si)/s or higher; or
- c. A fluence (integrated flux) of neutrons (1 MeV equivalent) of 5×10^{13} n/cm² or higher on silicon, or its equivalent for other materials;

Note 8A301.a.1.c does not apply to Metal Insulator Semiconductors (MIS).

8A301 a. 2. Microprocessor microcircuits", "microcomputer microcircuits", microcontroller microcircuits, storage integrated circuits manufactured from a compound semiconductor, analogue-to-digital converters, integrated circuits that contain analogue-to-digital converters and store or process the digitized data, digital-to-analogue converters, electro-optical or "optical integrated circuits" designed for "signal processing", field programmable logic devices, custom integrated circuits for which either the function is unknown or the status of the equipment in which the integrated circuit will be used is unknown, Fast Fourier Transform (FFT) processors, Static Random-Access Memories (SRAMs) or 'non-volatile memories', having any of the following:

Technical Note

'Non-volatile memories' are memories with data retention over a period of time after a power shutdown.

- 8A301.a.2 a. Rated for operation at an ambient temperature above 398 K (+125°C);
- b. Rated for operation at an ambient temperature below 218 K (- 55°C); or

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- c. Rated for operation over the entire ambient temperature range from 218 K (-55°C) to 398 K (+125°C);

Note: 8A301.a.2 does not apply to integrated circuits for civil automobile or railway train applications'.

- 8A301 a. 3. "Microprocessor microcircuits", "microcomputer microcircuits" and microcontroller microcircuits, manufactured from a compound semiconductor and operating at a clock frequency exceeding 40 MHz;

Note 8A301.a.3 includes digital signal processors, digital array processors and digital coprocessors.

- 8A301 a. 4. (Reserved)

- 8A301 a. 5. Analogue-to-Digital Converter (ADC) and Digital-to-Analogue Converter (DAC) integrated circuits, as follows:

a. ADCs having any of the following:

1. A resolution of 8 bit or more, but less than 10 bit, with a "sample rate" greater than 1.3 Giga Samples Per Second (GSPS);
2. A resolution of 10 bit or more, but less than 12 bit, with a "sample rate" greater than 600 Mega Samples Per Second (MSPS);
3. A resolution of 12 bit or more, but less than 14 bit, with a "sample rate" greater than 400 MSPS;
4. A resolution of 14 bit or more, but less than 16 bit, with a "sample rate" greater than 250 MSPS; or
5. A resolution of 16 bit or more with a "sample rate" greater than 65 MSPS;

N.B. For integrated circuits that contain analogue-to-digital converters and store or process the digitized data, see 8A301.a.14.

Technical Notes

1. A resolution of n bit corresponds to a quantisation of 2^n levels.
2. The resolution of the ADC is the number of bits of the digital output that represents the measured analogue input. Effective Number of Bits (ENOB) is not used to determine the resolution of the ADC.
3. For "multiple channel ADCs", the "sample rate" is not aggregated and the "sample rate" is the maximum rate of any single channel.
4. For "interleaved ADCs" or for "multiple channel ADCs" that are specified to have an interleaved mode of operation, the "sample rates" are aggregated and the "sample rate" is the maximum combined total rate of all of the interleaved channels.

- 8A301 a. 5. b. Digital-to-Analogue Converters (DAC) having any of the following:
1. A resolution of 10 bit or more but less than 12 bit, with an 'adjusted update rate' exceeding 3,500 MSPS; or
 2. A resolution of 12 bit or more and having any of the following:
 - a. An 'adjusted update rate' exceeding 1,250 MSPS but not exceeding 3,500 MSPS, and having any of the following:
 1. A settling time less than 9 ns to arrive at or within 0.024% of full scale from a full scale step; or
 2. A 'Spurious Free Dynamic Range' (SFDR) greater than 68 dBc (carrier) when synthesising a full scale analogue signal of 100 MHz or the highest full scale analogue signal frequency specified below 100 MHz; or
 - b. An 'adjusted update rate' exceeding 3,500 MSPS;

Technical Notes

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1. 'Spurious Free Dynamic Range' (SFDR) is defined as the ratio of the RMS value of the carrier frequency (maximum signal component) at the input of the DAC to the RMS value of the next largest noise or harmonic distortion component at its output.
 2. SFDR is determined directly from the specification table or from the characterisation plots of SFDR versus frequency.
 3. A signal is defined to be full scale when its amplitude is greater than -3 dBfs (full scale).
 4. 'Adjusted update rate' for DACs:
 - a. For conventional (non-interpolating) DACs, the 'adjusted update rate' is the rate at which the digital signal is converted to an analogue signal and the output analogue values are changed by the DAC. For DACs where the interpolation mode may be bypassed (interpolation factor of one), the DAC should be considered as a conventional (non-interpolating) DAC.
 - b. For interpolating DACs (oversampling DACs), the 'adjusted update rate' is defined as the DAC update rate divided by the smallest interpolating factor. For interpolating DACs, the 'adjusted update rate' may be referred to by different terms including:
 - input data rate
 - input word rate
 - input sample rate
 - maximum total input bus rate
 - maximum DAC clock rate for DAC clock input.
- 8A301 a. 6. Electro-optical and "optical integrated circuits", designed for "signal processing" and having all of the following:
- a. One or more than one internal "laser" diode;
 - b. One or more than one internal light detecting element; and
 - c. Optical waveguides;
- 8A301 a. 7. Field programmable logic devices having any of the following:
- a. A maximum number of single-ended digital input/outputs of greater than 700; or
 - b. An 'aggregate one-way peak serial transceiver data rate' of 500 Gb/s or greater;
- Note 8A301.a.7 includes:
- Complex Programmable Logic Devices (CPLDs)
 - Field Programmable Gate Arrays (FPGAs)
 - Field Programmable Logic Arrays (FPLAs)
 - Field Programmable Interconnects (FPICs)
- N.B. For integrated circuits having field programmable logic devices that are combined with an analogue-to-digital converter, see 8A301.a.14.
- Technical Notes
1. Maximum number of digital input/outputs in 8A301.a.7.a is also referred to as maximum user input/outputs or maximum available input/ outputs, whether the integrated circuit is packaged or bare die.
 2. 'Aggregate one-way peak serial transceiver data rate' is the product of the peak serial one-way transceiver data rate times the number of transceivers on the FPGA.
- 8A301 a. 8. (Reserved)
9. Neural network integrated circuits;
10. Custom integrated circuits for which the function is unknown, or the status of the equipment in which the integrated circuits will be used is unknown to the manufacturer, having any of the following:
- a. More than 1,500 terminals;
 - b. A typical "basic gate propagation delay time" of less than 0.02 ns; or
 - c. An operating frequency exceeding 3 GHz;

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11. Digital integrated circuits, other than those described in 8A301.a.3 to 8A301.a.10 and 8A301.a.12, based upon any compound semiconductor and having any of the following:
 - a. An equivalent gate count of more than 3,000 (2 input gates); or
 - b. A toggle frequency exceeding 1.2 GHz;
12. Fast Fourier Transform (FFT) processors having a rated execution time for an N-point complex FFT of less than $(N \log_2 N)/20,480$ ms, where N is the number of points;

Technical Note

When N is equal to 1,024 points, the formula in 8A301.a.12 gives an execution time of 500 μ s.

- 8A301 a.13. Direct Digital Synthesizer (DDS) integrated circuits having any of the following:
- a. A Digital-to-Analogue Converter (DAC) clock frequency of 3.5 GHz or more and a DAC resolution of 10 bit or more, but less than 12 bit; or
 - b. A DAC clock frequency of 1.25 GHz or more and a DAC resolution of 12 bit or more;

Technical Note

The DAC clock frequency may be specified as the master clock frequency or the input clock frequency.

- a.14. Integrated circuits that perform or are programmable to perform all of the following:
- a. Analogue-to-digital conversions meeting any of the following:
 1. A resolution of 8 bit or more, but less than 10 bit, with a "sample rate" greater than 1.3 Giga Samples Per Second (GSPS);
 2. A resolution of 10 bit or more, but less than 12 bit, with a "sample rate" greater than 1.0 GSPS;
 3. A resolution of 12 bit or more, but less than 14 bit, with a "sample rate" greater than 1.0 GSPS;
 4. A resolution of 14 bit or more, but less than 16 bit, with a "sample rate" greater than 400 Mega Samples Per Second (MSPS); or
 5. A resolution of 16 bit or more with a "sample rate" greater than 180 MSPS; and
 - b. Any of the following:
 1. Storage of digitized data; or
 2. Processing of digitized data.

N.B.1 For analogue-to-digital converter integrated circuits see 8A301.a.5.a.

N.B.2 For field programmable logic devices see 8A301.a.7.

Technical Notes

1. A resolution of n bit corresponds to a quantisation of 2^n levels.
2. The resolution of the ADC is the number of bits of the digital output of the ADC that represents the measured analogue input. Effective Number of Bits (ENOB) is not used to determine the resolution of the ADC.
3. For integrated circuits with non-interleaving "multiple channel ADCs", the "sample rate" is not aggregated and the "sample rate" is the maximum rate of any single channel.
4. For integrated circuits with "interleaved ADCs" or with "multiple channel ADCs" that are specified to have an interleaved mode of operation, the "sample rates" are aggregated and the "sample rate" is the maximum combined total rate of all of the interleaved channels.

- 8A301 b.²² Microwave or millimetre wave items, as follows:

Technical Note

²² Para 1(M) of Notification No. 29/2015-20 dated 21.09.2017

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For purposes of 8A301.b., the parameter peak saturated power output may also be referred to on product data sheets as output power, saturated power output, maximum power output, peak power output, or peak envelope power output.

1. "Vacuum electronic devices" and cathodes, as follows:

Note 1 8A301.b.1. does not apply to "vacuum electronic devices" designed or rated for operation in any frequency bands and having all of the following:

- a. Does not exceed 31.8 GHz; and
- b. Is "allocated by the ITU" for radio-communications services, but not for radio-determination.

Note 2 8A301.b 1. does not apply to non-"space-qualified" "vacuum electronic devices" having all of the following:

- a. An average output power equal to or less than 50 W; and
- b. Designed or rated for operation in any frequency band and having all of the following:
 1. Exceeds 31.8 GHz but does not exceed 43.5 GHz; and
 2. Is "allocated by the ITU" for radio-communications services, but not for radio-determination.

- 8A301 b. 1. a. Travelling-wave "vacuum electronic devices", pulsed or continuous wave, as follows:
1. Devices operating at frequencies exceeding 31.8 GHz;
 2. Devices having a cathode heater with a turn on time to rated RF power of less than 3 seconds;
 3. Coupled cavity devices, or derivatives thereof, with a "fractional bandwidth" of more than 7% or a peak power exceeding 2.5 kW;
 4. Devices based on helix, folded waveguide, or serpentine waveguide circuits, or derivatives thereof, having any of the following:
 - a. An "instantaneous bandwidth" of more than one octave, and average power (expressed in kW) times frequency (expressed in GHz) of more than 0.5;
 - b. An "instantaneous bandwidth" of one octave or less, and average power (expressed in kW) times frequency (expressed in GHz) of more than 1;
 - c. Being "space-qualified"; or
 - d. Having a gridded electron gun;
 5. Devices with a "fractional bandwidth" greater than or equal to 10%, with any of the following:
 - a. An annular electron beam;
 - b. A non-axisymmetric electron beam; or
 - c. Multiple electron beams;

- 8A301 b. 1. b. Crossed-field amplifier "vacuum electronic devices" with a gain of more than 17 dB;
- c. Thermionic cathodes designed for "vacuum electronic devices" producing an emission current density at rated operating conditions exceeding 5 A/cm² or a pulsed (non-continuous) current density at rated operating conditions exceeding 10 A/cm²;
 - d. "Vacuum electronic devices" with the capability to operate in a 'dual mode'.

Technical Note

'Dual mode' means the "vacuum electronic device" beam current can be intentionally changed between continuous-wave and pulsed mode operation by use of a grid and produces a peak pulse output power greater than the continuous-wave output power.

- 8A301 b. 1. b. Crossed-field amplifier tubes with a gain of more than 17 dB;
- c. Thermionic cathodes designed for 'vacuum electronic devices' producing an emission current density at rated operating conditions exceeding 5 A/cm² or a pulsed (non-continuous) current density at rated operating conditions exceeding 10 A/cm²;
 - d. 'Vacuum electronic devices' with the capability to operate in a 'dual mode'.

Technical Note

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'Dual mode' means the 'vacuum electronic device' beam current can be intentionally changed between continuous-wave and pulsed mode operation by use of a grid and produces a peak pulse output power greater than the continuous-wave output power.

- 8A301 b. 2. "Monolithic Microwave Integrated Circuit" ("MMIC") amplifiers that are any of the following:
- N.B. For "MMIC" amplifiers that have an integrated phase shifter see 8A301.b.12.*
- a. Rated for operation at frequencies exceeding 2.7 GHz up to and including 6.8 GHz with a "fractional bandwidth" greater than 15%, and having any of the following:
1. A peak saturated power output greater than 75 W (48.75 dBm) at any frequency exceeding 2.7 GHz up to and including 2.9 GHz;
 2. A peak saturated power output greater than 55 W (47.4 dBm) at any frequency exceeding 2.9 GHz up to and including 3.2 GHz;
 3. A peak saturated power output greater than 40 W (46 dBm) at any frequency exceeding 3.2 GHz up to and including 3.7 GHz; or
 4. A peak saturated power output greater than 20 W (43 dBm) at any frequency exceeding 3.7 GHz up to and including 6.8 GHz;
- b. Rated for operation at frequencies exceeding 6.8 GHz up to and including 16 GHz with a "fractional bandwidth" greater than 10%, and having any of the following:
1. A peak saturated power output greater than 10W (40 dBm) at any frequency exceeding 6.8 GHz up to and including 8.5 GHz; or
 2. A peak saturated power output greater than 5W (37 dBm) at any frequency exceeding 8.5 GHz up to and including 16 GHz;
- c. Rated for operation with a peak saturated power output greater than 3 W (34.77 dBm) at any frequency exceeding 16 GHz up to and including 31.8 GHz, and with a "fractional bandwidth" of greater than 10%;
- d. Rated for operation with a peak saturated power output greater than 0.1n W (-70 dBm) at any frequency exceeding 31.8 GHz up to and including 37 GHz;
- e. Rated for operation with a peak saturated power output greater than 1 W (30 dBm) at any frequency exceeding 37 GHz up to and including 43.5 GHz, and with a "fractional bandwidth" of greater than 10%;
- f. Rated for operation with a peak saturated power output greater than 31.62 mW (15 dBm) at any frequency exceeding 43.5 GHz up to and including 75 GHz, and with a "fractional bandwidth" of greater than 10%;
- g. Rated for operation with a peak saturated power output greater than 10 mW (10 dBm) at any frequency exceeding 75 GHz up to and including 90 GHz, and with a "fractional bandwidth" of greater than 5%; or
- h. Rated for operation with a peak saturated power output greater than 0.1 nW (-70 dBm) at any frequency exceeding 90 GHz;

Note 1 (Reserved)

Note 2 The status of the "MMIC" whose rated operating frequency includes frequencies listed in more than one frequency range, as defined by 8A301.b.2.a through 8A301.b.2.h, is determined by the lowest peak saturated power output threshold.

Note 3 Notes 1 and 2 in 8A3 mean that 8A301.b.2 does not apply to "MMICs" if they are specially designed for other applications, e.g., telecommunications, radar, automobiles.

- 8A301 b. 3. Discrete microwave transistors that are any of the following:
- a. Rated for operation at frequencies exceeding 2.7 GHz up to and including 6.8 GHz and having any of the following:
1. A peak saturated power output greater than 400 W (56 dBm) at any frequency exceeding 2.7 GHz up to and including 2.9 GHz;
 2. A peak saturated power output greater than 205 W (53.12 dBm) at any frequency exceeding 2.9 GHz up to and including 3.2 GHz;
 3. A peak saturated power output greater than 115 W (50.61 dBm) at any frequency exceeding 3.2 GHz up to and including 3.7 GHz; or
 4. A peak saturated power output greater than 60 W (47.78 dBm) at any frequency

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exceeding 3.7 GHz up to and including 6.8 GHz;

- b. Rated for operation at frequencies exceeding 6.8 GHz up to and including 31.8 GHz and having any of the following:
 - 1. A peak saturated power output greater than 50 W (47 dBm) at any frequency exceeding 6.8 GHz up to and including 8.5 GHz;
 - 2. A peak saturated power output greater than 15 W (41.76 dBm) at any frequency exceeding 8.5 GHz up to and including 12 GHz;
 - 3. A peak saturated power output greater than 40 W (46 dBm) at any frequency exceeding 12 GHz up to and including 16 GHz; or
 - 4. A peak saturated power output greater than 7 W (38.45 dBm) at any frequency exceeding 16 GHz up to and including 31.8 GHz;
- c. Rated for operation with a peak saturated power output greater than 0.5 W (27 dBm) at any frequency exceeding 31.8 GHz up to and including 37 GHz;
- d. Rated for operation with a peak saturated power output greater than 1 W (30 dBm) at any frequency exceeding 37 GHz up to and including 43.5 GHz;
- e. Rated for operation with a peak saturated power output greater than 0.1 nW (-70 dBm) at any frequency exceeding 43.5 GHz; or
- f. Other than those specified by 8A301.b.3.a. to 8A301.b.3.e. and rated for operation with a peak saturated power output greater than 5 W (37.0 dBm) at all frequencies exceeding 8.5 GHz up to and including 31.8 GHz;

Note 1 *The status of a transistor in 8A301.b.3.a. through 8A301.b.3.e. whose rated operating frequency includes frequencies listed in more than one frequency range, as defined by 8A301.b.3.a. through 8A301.b.3.e., is determined by the lowest peak saturated power output threshold.*

Note 2 *8A301.b.3. includes bare dice, dice mounted on carriers, or dice mounted in packages. Some discrete transistors may also be referred to as power amplifiers, but the status of these discrete transistors is determined by 8A301.b.3.*

- | | |
|-------|---|
| 8A301 | <ul style="list-style-type: none"> b. 4. Microwave solid state amplifiers and microwave assemblies/modules containing microwave solid state amplifiers, that are any of the following: <ul style="list-style-type: none"> a. Rated for operation at frequencies exceeding 2.7 GHz up to and including 6.8 GHz with a "fractional bandwidth" greater than 15%, and having any of the following: <ul style="list-style-type: none"> 1. A peak saturated power output greater than 500 W (57 dBm) at any frequency exceeding 2.7 GHz up to and including 2.9 GHz; 2. A peak saturated power output greater than 270 W (54.3 dBm) at any frequency exceeding 2.9 GHz up to and including 3.2 GHz; 3. A peak saturated power output greater than 200 W (53 dBm) at any frequency exceeding 3.2 GHz up to and including 3.7 GHz; <u>or</u> 4. A peak saturated power output greater than 90 W (49.54 dBm) at any frequency exceeding 3.7 GHz up to and including 6.8 GHz; b. Rated for operation at frequencies greater than 6.8 GHz up to and including 31.8 GHz with a "fractional bandwidth" greater than 10%, and having any of the following: <ul style="list-style-type: none"> 1. A peak saturated power output greater than 70 W (48.45 dBm) at any frequency exceeding 6.8 GHz up to and including 8.5 GHz; 2. A peak saturated power output greater than 50 W (47 dBm) at any frequency exceeding 8.5 GHz up to and including 12 GHz; 3. A peak saturated power output greater than 30 W (44.77 dBm) at any frequency exceeding 12 GHz up to and including 16 GHz; <u>or</u> 4. A peak saturated power output greater than 20 W (43 dBm) at any frequency exceeding 16 GHz up to and including 31.8 GHz; c. Rated for operation with a peak saturated power output greater than 0.5 W (27 dBm) at any frequency exceeding 31.8 GHz up to and including 37 GHz; d. Rated for operation with a peak saturated power output greater than 2 W (33 dBm) at any frequency exceeding 37 GHz up to and including 43.5 GHz, and with a "fractional |
|-------|---|

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- bandwidth" of greater than 10%;
- e. Rated for operation at frequencies exceeding 43.5 GHz and having any of the following:
1. A peak saturated power output greater than 0.2 W (23 dBm) at any frequency exceeding 43.5 GHz up to and including 75 GHz, and with a "fractional bandwidth" of greater than 10%;
 2. A peak saturated power output greater than 20 mW (13 dBm) at any frequency exceeding 75 GHz up to and including 90 GHz, and with a "fractional bandwidth" of greater than 5%; or
 3. A peak saturated power output greater than 0.1 nW (-70 dBm) at any frequency exceeding 90 GHz; or
- 8A301 b. 4. f. (Reserved)
- N.B.1. For "MMIC" amplifiers see 8A301.b.2.
- N.B.2. For 'transmit/receive modules' and 'transmit modules' see 8A301.b.12.
- N.B.3. for converters and harmonic mixers, designed to extend the operating or frequency range of signal analysers, signal generators, network analysers or microwave test receivers, see 8A301.b.7
- Note 1 (Reserved)
- Note 2 The status of an item whose rated operating frequency includes frequencies listed in more than one frequency range, as defined by 8A301.b.4.a through 8A301.b.4.e, is determined by the lowest peak saturated power output threshold.
- 8A301 b. 5. Electronically or magnetically tunable band-pass or band-stop filters, having more than 5 tunable resonators capable of tuning across a 1.5:1 frequency band ($f_{\max}/f_{\min}$) in less than 10 μ s and having any of the following:
- a. A band-pass bandwidth of more than 0.5% of centre frequency; or
 - b. A band-stop bandwidth of less than 0.5% of centre frequency;
6. (Reserved)
7. Converters and harmonic mixers, that are any of the following:
- a. Designed to extend the frequency range of "signal analysers" beyond 90 GHz;
 - b. Designed to extend the operating range of signal generators as follows:
 1. Beyond 90 GHz;
 2. To an output power greater than 100 mW (20 dBm) anywhere within the frequency range exceeding 43.5 GHz but not exceeding 90 GHz;
 - c. Designed to extend the operating range of network analysers as follows:
 1. Beyond 110 GHz;
 2. To an output power greater than 31.62 mW (15 dBm) anywhere within the frequency range exceeding 43.5 GHz but not exceeding 90 GHz;
 3. To an output power greater than 1 mW (0 dBm) anywhere within the frequency range exceeding 90 GHz but not exceeding 110 GHz; or
- 8A301 b. 7. d. Designed to extend the frequency range of microwave test receivers beyond 110 GHz.
- 8A301 b. 8. Microwave power amplifiers containing "vacuum electronic devices" specified by 8A301.b.1. and having all of the following:
- a. Operating frequencies above 3 GHz;
 - b. An average output power to mass ratio exceeding 80 W/kg; and
 - c. A volume of less than 400 cm³;
- Note 8A301.b.8. does not apply to equipment designed or rated for operation in any frequency band which is "allocated by the ITU" for radio-communications services, but not for radio-determination.

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- 8A301 b. 9. Microwave Power Modules (MPMs) consisting of, at least, a travelling-wave "vacuum electronic device", a "Monolithic Microwave Integrated Circuit" ("MMIC") and an integrated electronic power conditioner and having all of the following:
- a. A 'turn-on time' from off to fully operational in less than 10 seconds;
 - b. A volume less than the maximum rated power in Watts multiplied by $10 \text{ cm}^3/\text{W}$; and
 - c. An "instantaneous bandwidth" greater than 1 octave ($f_{\text{max.}} > 2f_{\text{min.}}$) and having any of the following:
 1. For frequencies equal to or less than 18 GHz, an RF output power greater than 100 W; or
 2. A frequency greater than 18 GHz;

Technical Notes

1. To calculate the volume in 8A301.b.9.b., the following example is provided: for a maximum rated power of 20 W, the volume would be: $20 \text{ W} \times 10 \text{ cm}^3/\text{W} = 200 \text{ cm}^3$.
2. The 'turn-on time' in 8A301.b.9.a. refers to the time from fully-off to fully operational, i.e., it includes the warm-up time of the MPM.

- 8A301 b. 10. Oscillators or oscillator assemblies, specified to operate with a single sideband (SSB) phase noise, in dBc/Hz, less (better) than $-(126 + 20\log_{10}F - 20\log_{10}f)$ anywhere within the range of $10 \text{ Hz} \leq F \leq 10 \text{ kHz}$;

Technical Note

In 8A301.b.10, F is the offset from the operating frequency in Hz and f is the operating frequency in MHz.

- 8A301 b. 11. "Frequency synthesiser" "electronic assemblies" having a "frequency switching time" as specified by any of the following:

- a. Less than 143 ps;
- b. Less than 100 μs for any frequency change exceeding 2.2 GHz within the synthesised frequency range exceeding 4.8 GHz but not exceeding 31.8 GHz;
- c. (Reserved)
- d. Less than 500 μs for any frequency change exceeding 550 MHz within the synthesised frequency range exceeding 31.8 GHz but not exceeding 37 GHz;
- e. Less than 100 μs for any frequency change exceeding 2.2 GHz within the synthesised frequency range exceeding 37 GHz but not exceeding 75 GHz; or
- f. Less than 100 μs for any frequency change exceeding 5.0 GHz within the synthesised frequency range exceeding 75 GHz but not exceeding 90 GHz; or
- g. Less than 1 ms within the synthesised frequency range exceeding 90 GHz;

Technical Note

A 'frequency synthesiser' is any kind of frequency source, regardless of the actual technique used, providing a multiplicity of simultaneous or alternative output frequencies, from one or more outputs, controlled by, derived from or disciplined by a lesser number of standard (or master) frequencies.

N.B.

For general purpose "signal analysers", signal generators, network analysers and microwave test receivers, see 8A302.c, 8A302.d, 8A302.e and 8A302.f respectively.

- 8A301 b. 12. 'Transmit/receive modules', 'transmit/receive MMICs', 'transmit modules', and 'transmit MMICs', rated for operation at frequencies above 2.7 GHz and having all of the following:
- a. A peak saturated power output (in watts), P_{sat} , greater than 505.62 divided by the maximum operating frequency (in GHz) squared $[P_{\text{sat}} > 505.62 \text{ W} \cdot \text{GHz}^2 / f_{\text{GHz}}^2]$ for any channel;
 - b. A "fractional bandwidth" of 5% or greater for any channel;
 - c. Any planar side with length d (in cm) equal to or less than 15 divided by the lowest operating frequency in GHz $[d \leq 15 \text{ cm} \cdot \text{GHz} \cdot N / f_{\text{GHz}}]$ where N is the number of transmit or transmit/receive channels; and
 - d. An electronically variable phase shifter per channel.

Technical Notes

1. A 'transmit/receive module': is a multifunction "electronic assembly" that provides bi-directional amplitude and phase control for transmission and reception of signals.
2. A 'transmit module': is an "electronic assembly" that provides amplitude and phase control for transmission of signals.
3. A 'transmit/receive MMIC': is a multifunction "MMIC" that provides bi-directional

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amplitude and phase control for transmission and reception of signals.

4. A 'transmit MMIC': is a "MMIC" that provides amplitude and phase control for transmission of signals.
5. 2.7 GHz should be used as the lowest operating frequency (fGHz) in the formula in 8A301.b.12.c. for transmit/receive or transmit modules that have a rated operation range extending downward to 2.7 GHz and below $[d \leq 15 \text{ cm} * \text{GHz} * N / 2.7 \text{ GHz}]$.
6. 8A301.b.12. applies to 'transmit/receive modules' or 'transmit modules' with or without a heat sink. The value of d in 8A301.b.12.c. does not include any portion of the 'transmit/receive module' or 'transmit module' that functions as a heat sink.
7. 'Transmit/receive modules', or 'transmit modules', or 'transmit/receive MMICs' or 'transmit MMICs' may or may not have N integrated radiating antenna elements where N is the number of transmit or transmit/receive channels.

- 8A301 c. Acoustic wave devices as follows and specially designed components therefor:
1. Surface acoustic wave and surface skimming (shallow bulk) acoustic wave devices, having any of the following:
 - a. A carrier frequency exceeding 6 GHz;
 - b. A carrier frequency exceeding 1 GHz, but not exceeding 6 GHz and having any of the following:
 1. A 'frequency side-lobe rejection' exceeding 65 dB;
 2. A product of the maximum delay time and the bandwidth (time in μs and bandwidth in MHz) of more than 100;
 3. A bandwidth greater than 250 MHz; or
 4. A dispersive delay of more than 10 μs ; or
 - c. A carrier frequency of 1 GHz or less and having any of the following:
 1. A product of the maximum delay time and the bandwidth (time in μs and bandwidth in MHz) of more than 100;
 2. A dispersive delay of more than 10 μs ; or
 3. A 'frequency side-lobe rejection' exceeding 65 dB and a bandwidth greater than 100 MHz;

Technical Note

'Frequency side-lobe rejection' is the maximum rejection value specified in data sheet.

- 8A301 c. 2. Bulk (volume) acoustic wave devices which permit the direct processing of signals at frequencies exceeding 6 GHz;
3. Acoustic-optic "signal processing" devices employing interaction between acoustic waves (bulk wave or surface wave) and light waves which permit the direct processing of signals or images, including spectral analysis, correlation or convolution;

Note 8A301.c does not apply to acoustic wave devices that are limited to a single band pass, low pass, high pass or notch filtering, or resonating function.

- 8A301 d. Electronic devices and circuits containing components, manufactured from "superconductive" materials, specially designed for operation at temperatures below the "critical temperature" of at least one of the "superconductive" constituents and having any of the following:
1. Current switching for digital circuits using "superconductive" gates with a product of delay time per gate (in seconds) and power dissipation per gate (in watts) of less than 10^{-14} J; or
 2. Frequency selection at all frequencies using resonant circuits with Q-values exceeding 10,000;

- 8A301 e. High energy devices as follows:
1. 'Cells' as follows:
 - a. 'Primary cells' having any of the following at 20°C:
 1. 'Energy density' exceeding 550 Wh/kg and a 'continuous power density' exceeding 50 W/kg; or
 2. 'Energy density' exceeding 50 Wh/kg and a 'continuous power density' exceeding 350 W/kg;
 - b. 'Secondary cells' having an 'energy density' exceeding 350 Wh/kg at 20°C;

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Technical Notes

1. For the purpose of 8A301.e.1, 'energy density' (Wh/kg) is calculated from the nominal voltage multiplied by the nominal capacity in ampere-hours (Ah) divided by the mass in kilograms. If the nominal capacity is not stated, energy density is calculated from the nominal voltage squared then multiplied by the discharge duration in hours divided by the discharge load in Ohms and the mass in kilograms.
2. For the purpose of 8A301.e.1., a 'cell' is defined as an electrochemical device, which has positive and negative electrodes, an electrolyte, and is a source of electrical energy. It is the basic building block of a battery.
3. For the purpose of 8A301.e.1.a, a 'primary cell' is a 'cell' that is not designed to be charged by any other source.
4. For the purpose of 8A301.e.1.b, a 'secondary cell' is a 'cell' that is designed to be charged by an external electrical source.
5. *For the purpose of 3.A.1.e.1.a., 'continuous power density (W/kg) is calculated from the nominal voltage multiplied by the specified maximum continuous discharge current in ampere (A) divided by the mass in kilograms. 'Continuous power density' is also referred to as specific power*

Note 8A301.e does not apply to batteries, including single-cell batteries.

- 8A301 e. 2. High energy storage capacitors as follows:
- a. Capacitors with a repetition rate of less than 10 Hz (single shot capacitors) and having all of the following:
 1. A voltage rating equal to or more than 5 kV;
 2. An energy density equal to or more than 250 J/kg; and
 3. A total energy equal to or more than 25 kJ;
 - b. Capacitors with a repetition rate of 10 Hz or more (repetition rated capacitors) and having all of the following:
 1. A voltage rating equal to or more than 5 kV;
 2. An energy density equal to or more than 50 J/kg;
 3. A total energy equal to or more than 100 J; and
 4. A charge/discharge cycle life equal to or more than 10,000;
- N.B. See also 6A012
- 8A301 e. 3. "Superconductive" electromagnets and solenoids, specially designed to be fully charged or discharged in less than one second and having all of the following:
- Note 8A301.e.3 does not apply to "superconductive" electromagnets or solenoids specially designed for Magnetic Resonance Imaging (MRI) medical equipment.
- a. Energy delivered during the discharge exceeding 10 kJ in the first second;
 - b. Inner diameter of the current carrying windings of more than 250 mm; and
 - c. Rated for a magnetic induction of more than 8 T or "overall current density" in the winding of more than 300 A/mm²;
- 8A301 e. 4. Solar cells, cell-interconnect-coverglass (CIC) assemblies, solar panels, and solar arrays, which are "space-qualified", having a minimum average efficiency exceeding 20% at an operating temperature of 301 K (28°C) under simulated 'AM0' illumination with an irradiance of 1,367 Watts per square meter (W/m²);
- Technical Note
'AM0', or 'Air Mass Zero', refers to the spectral irradiance of sun light in the earth's outer atmosphere when the distance between the earth and sun is one astronomical unit (AU).
- 8A301 f. Rotary input type absolute position encoders having an "accuracy" equal to or less (better) than 1.0 second of arc and specially designed encoder rings, discs or scales therefor;
- 8A301 g. Solid-state pulsed power switching thyristor devices and 'thyristor modules', using either electrically, optically, or electron radiation controlled switch methods and having any of the following:
1. A maximum turn-on current rate of rise (di/dt) greater than 30,000 A/μs and off-state voltage greater than 1,100 V; or

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2. A maximum turn-on current rate of rise (di/dt) greater than 2,000 A/ μ s and having all of the following:
 - a. An off-state peak voltage equal to or greater than 3,000 V; and
 - b. A peak (surge) current equal to or greater than 3,000 A;

Note 1 8A301.g includes:

- Silicon Controlled Rectifiers (SCRs)
- Electrical Triggering Thyristors (ETTs)
- Light Triggering Thyristors (LTTs)
- Integrated Gate Commutated Thyristors (IGCTs)
- Gate Turn-off Thyristors (GTOs)
- MOS Controlled Thyristors (MCTs)
- Solidtrons

Note 2 8A301.g does not apply to thyristor devices and 'thyristor modules' incorporated into equipment designed for civil railway or "civil aircraft" applications.

Technical Note

For the purposes of 8A301.g, a 'thyristor module' contains one or more thyristor devices.

- 8A301 h. Solid-state power semiconductor switches, diodes, or 'modules', having all of the following:
1. Rated for a maximum operating junction temperature greater than 488 K (215°C);
 2. Repetitive peak off-state voltage (blocking voltage) exceeding 300 V; and
 3. Continuous current greater than 1 A;

Note 1 Repetitive peak off-state voltage in 8A301.h includes drain to source voltage, collector to emitter voltage, repetitive peak reverse voltage and peak repetitive off-state blocking voltage.

Note 2 8A301.h includes:

- Junction Field Effect Transistors (JFETs)
- Vertical Junction Field Effect Transistors (VJFETs)
- Metal Oxide Semiconductor Field Effect Transistors (MOSFETs)
- Double Diffused Metal Oxide Semiconductor Field Effect Transistor (DMOSFET)
- Insulated Gate Bipolar Transistor (IGBT)
- High Electron Mobility Transistors (HEMTs)
- Bipolar Junction Transistors (BJTs)
- Thyristors and Silicon Controlled Rectifiers (SCRs)
- Gate Turn-Off Thyristors (GTOs)
- Emitter Turn-Off Thyristors (ETOs)
- PiN Diodes
- Schottky Diodes

Note 3 8A301.h does not apply to switches, diodes, or 'modules', incorporated into equipment designed for civil automobile, civil railway, or "civil aircraft" applications.

Technical Note

For the purposes of 8A301.h, 'modules' contain one or more solid-state power semiconductor switches or diodes.

- 8A301.i Intensity, amplitude, or phase electro-optic modulators, designed for analogue signals and having any of the following:

1. A maximum operating frequency of more than 10 GHz but less than 20 GHz, an optical insertion loss equal to or less than 3 dB and having any of the following:
 - a. A 'half-wave voltage' (V_{π}) less than 2.7 V when measured at a frequency of 1 GHz or below; or
 - b. A V_{π} of less than 4 V when measured at a frequency of more than 1 GHz; or
2. A maximum operating frequency equal to or greater than 20 GHz, an optical insertion loss equal to or less than 3 dB and having any of the following:
 - a. A V_{π} less than 3.3 V when measured at a frequency of 1 GHz or below; or
 - b. A V_{π} less than 5 V when measured at a frequency of more than 1 GHz.

Note: 8A301.i includes electro-optic modulators having optical input and output connectors (e.g., fibre-optic pigtails).

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Technical Note

For the purposes of 8A301.i, a 'half-wave voltage' ($V\pi$) is the applied voltage necessary to make a phase change of 180 degrees in the wavelength of light propagating through the optical modulator.

8A302 General purpose "electronic assemblies", modules and equipment, as follows:

a. Recording equipment and oscilloscopes, as follows:

1. (Reserved);
2. (Reserved);
3. (Reserved);
4. (Reserved);
5. (Reserved)

N.B. For waveform digitizers and transient recorders, see 8A302.h.

8A302 a. 6. 'Digital data recorders having all of the following:

- a. A sustained 'continuous throughput' of more than 6.4 Gbit/s to disk or solid-state drive memory; and
- b. "Signal processing" of the radio frequency signal data while it is being recorded."

Technical Notes

1. For recorders with a parallel bus architecture, the 'continuous throughput' rate is the highest word rate multiplied by the number of bits in a word.
2. 'Continuous throughput' is the fastest data rate the instrument can record to disk or solid-state drive memory without the loss of any information while sustaining the input digital data rate or digitizer conversion rate.

8A302 a. 7. Real-time oscilloscopes having a vertical root-mean-square (rms) noise voltage of less than 2% of full-scale at the vertical scale setting that provides the lowest noise value for any input 3dB bandwidth of 60 GHz or greater per channel;

Note 8A302.a.7 does not apply to equivalent-time sampling oscilloscopes.

8A302 b. (Reserved)

8A302 c. "Signal analysers" as follows:

1. "Signal analysers" having a 3 dB resolution bandwidth (RBW) exceeding 40 MHz anywhere within the frequency range exceeding 31.8 GHz but not exceeding 37 GHz;
2. "Signal analysers" having a Displayed Average Noise Level (DANL) less (better) than "-150 dBm/Hz anywhere within the frequency range exceeding 43.5 GHz but not exceeding 90 GHz;
3. "Signal analysers" having a frequency exceeding 90 GHz;
4. "Signal analysers" having all of the following:
 - a. 'Real-time bandwidth' exceeding 170 MHz; and
 - b. Having any of the following:
 1. 100% probability of discovery, with less than a 3 dB reduction from full amplitude due to gaps or windowing effects, of signals having a duration of 15 μ s or less; or
2. A 'frequency mask trigger' function with 100% probability of trigger (capture) for signals having a duration of 15 μ s or less;

Technical Notes

1. 'Real-time bandwidth' is the widest frequency range for which the analyser can continuously transform time-domain data entirely into frequency-domain results, using a Fourier or other discrete time transform that processes every incoming time point, without a reduction of measured amplitude of more than 3 dB below the actual signal amplitude caused by gaps or windowing effects, while outputting or displaying the transformed data.
2. Probability of discovery in 8A302.c.4.b.1 is also referred to as probability of intercept or probability of capture.
3. For the purposes of 8A302.c.4.b.1, the duration for 100% probability of discovery is equivalent to the minimum signal duration necessary for the specified level measurement uncertainty.

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4. A 'frequency mask trigger' is a mechanism where the trigger function is able to select a frequency range to be triggered on as a subset of the acquisition bandwidth while ignoring other signals that may also be present within the same acquisition bandwidth. A 'frequency mask trigger' may contain more than one independent set of limits.

	<u>Note</u>	8A302.c.4 does not apply to those "signal analysers" using only constant percentage bandwidth filters (also known as octave or fractional octave filters)
8A302	c. 5.	(Reserved)
8A302	d.	Signal generators having any of the following: 1. Specified to generate pulse-modulated signals having all of the following, anywhere within the frequency range exceeding 31.8 GHz but not exceeding 37 GHz: a. 'Pulse duration' of less than 25 ns; <u>and</u> b. On/off ratio equal to or exceeding 65 dB; 2. An output power exceeding 100 mW (20 dBm) anywhere within the frequency range exceeding 43.5 GHz but not exceeding 90 GHz; 3. A "frequency switching time" as specified by any of the following: a. (Reserved); b. Less than 100 µs for any frequency change exceeding 2.2 GHz within the frequency range exceeding 4.8 GHz but not exceeding 31.8 GHz; c. (Reserved); d. Less than 500 µs for any frequency change exceeding 550 MHz within the frequency range exceeding 31.8 GHz but not exceeding 37 GHz; <u>or</u> e. Less than 100 µs for any frequency change exceeding 2.2 GHz within the frequency range exceeding 37 GHz but not exceeding 75 GHz; f. (Reserved) g. Less than 100 µs for any frequency change exceeding 5.0 GHz within the frequency range exceeding 75 GHz but not exceeding 90 GHz.
8A302	d. 4.	4. A Single sideband (SSB) phase noise, in dBc/Hz, specified as being any of the following: a. Less (better) than $-(126+20 \log_{10} F-20 \log_{10} f)$ anywhere within the range of $10 \text{ Hz} \leq F \leq 10 \text{ kHz}$ anywhere within the frequency range exceeding 3.2 GHz but not exceeding 90 GHz; <u>or</u> b. Less (better) than $-(206 - 20 \log_{10} f)$ anywhere within the range of $10 \text{ kHz} < F \leq 100 \text{ kHz}$ anywhere within the frequency range exceeding 3.2 GHz but not exceeding 90 GHz; <u>Technical Note</u> In 8A302.d.4., F is the offset from the operating frequency in Hz and f is the operating frequency in MHz.
8A302	d. 5.	5. An 'RF modulation bandwidth' of digital baseband signals as specified by any of the following: a. Exceeding 2.2 GHz within the frequency range exceeding 4.8 GHz but not exceeding 31.8 GHz; b. Exceeding 550 MHz within the frequency range exceeding 31.8 GHz but not exceeding 37 GHz; <u>or</u> c. Exceeding 2.2 GHz within the frequency range exceeding 37 GHz but not exceeding 75 GHz; <u>or</u> d. Exceeding 5.0 GHz within the frequency range exceeding 75 GHz but not exceeding 90 GHz; <u>or</u> <u>Technical Note</u> 'RF modulation bandwidth' is the Radio Frequency (RF) bandwidth occupied by a digitally encoded baseband signal modulated onto an RF signal. It is also referred to as information bandwidth or vector modulation bandwidth. I/Q digital modulation is the technical method for producing a vector-modulated RF output signal, and that output signal is typically specified as having an 'RF modulation bandwidth'.
8A302	d. 6.	6. A maximum frequency exceeding 90 GHz; <u>Note 1</u> For the purpose of 8A302.d., signal generators include arbitrary waveform and function generators. <u>Note 2</u> 8A302.d. does not apply to equipment in which the output frequency is either

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produced by the addition or subtraction of two or more crystal oscillator frequencies, or by an addition or subtraction followed by a multiplication of the result.

Technical Notes

1. *The maximum frequency of an arbitrary waveform or function generator is calculated by dividing the sample rate, in samples/second, by a factor of 2.5.*

2. *For the purposes of 8A302.d.1.a., 'pulse duration' is defined as the time interval from the point on the leading edge that is 50% of the pulse amplitude to the point on the trailing edge that is 50 % of the pulse amplitude.*

- 8A302 e. Network analysers having any of the following:
1. An output power exceeding 31.62 mW (15 dBm) anywhere within the operating frequency range exceeding 43.5 GHz but not exceeding 90 GHz;
 2. An output power exceeding 1 mW (0 dBm) anywhere within the operating frequency range exceeding 90 GHz but not exceeding 110 GHz;
 3. 'Nonlinear vector measurement functionality' at frequencies exceeding 50 GHz but not exceeding 110 GHz; or
- Technical Note
'Nonlinear vector measurement functionality' is an instrument's ability to analyse the test results of devices driven into the large-signal domain or the non-linear distortion range.
4. A maximum operating frequency exceeding 110 GHz;
- 8A302 f. Microwave test receivers having all of the following:
1. A maximum operating frequency exceeding 110 GHz; and
 2. Being capable of measuring amplitude and phase simultaneously;
- 8A302 g. Atomic frequency standards being any of the following:
1. "Space-qualified";
 2. Non-rubidium and having a long-term stability less (better) than 1×10^{-11} /month; or
 3. Non-"space-qualified" and having all of the following:
 - a. Being a rubidium standard;
 - b. Long-term stability less (better) than 1×10^{-11} /month; and
 - c. Total power consumption of less than 1 Watt;
- 8A302 h. "Electronic assemblies", modules or equipment, specified to perform all of the following:
1. Analogue-to-digital conversions meeting any of the following:
 - a. A resolution of 8 bit or more, but less than 10 bit, with a "sample rate" greater than 1.3 Giga Samples Per Second (GSPS);
 - b. A resolution of 10 bit or more, but less than 12 bit, with a "sample rate" greater than 1.0 GSPS;
 - c. A resolution of 12 bit or more, but less than 14 bit, with a "sample rate" greater than 1.0 GSPS;
 - d. A resolution of 14 bit or more but less than 16 bit, with a "sample rate" greater than 400 Mega Samples Per Second (MSPS); or
 - e. A resolution of 16 bit or more with a "sample rate" greater than 180 MSPS; and'
 2. Any of the following:
 - a. Output of digitized data;
 - b. Storage of digitized data; or
 - c. Processing of digitized data;
- N.B. *Digital data recorders, oscilloscopes, "signal analysers", signal generators, network analysers and microwave test receivers, are specified by 8A302.a.6, 8A302.a.7, 8A302.c, 8A302.d, 8A302.e and 8A302.f, respectively.*

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Technical Note

1. A resolution of n bit corresponds to a quantisation of 2^n levels.
2. The resolution of the ADC is the number of bits of the digital output of the ADC that represents the measured analogue input. Effective Number of Bits (ENOB) is not used to determine the resolution of the ADC.
3. For non-interleaved multiple-channel "electronic assemblies", modules, or equipment, the "sample rate" is not aggregated and the "sample rate" is the maximum rate of any single channel.
4. For interleaved channels on multiple-channel "electronic assemblies", modules, or equipment, the "sample rates" are aggregated and the "sample rate" is the maximum combined total rate of all the interleaved channels.

8A303 Spray cooling thermal management systems employing closed loop fluid handling and reconditioning equipment in a sealed enclosure where a dielectric fluid is sprayed onto electronic components using specially designed spray nozzles that are designed to maintain electronic components within their operating temperature range, and specially designed components therefor.

8B3 ELECTRONICS (TEST, INSPECTION AND PRODUCTION EQUIPMENT)

8B301 Equipment for the manufacturing of semiconductor devices or materials, as follows and specially designed components and accessories therefor:

- a. Equipment designed for epitaxial growth as follows:
 1. Equipment designed or modified to produce a layer of any material other than silicon with a thickness uniform to less than $\pm 2.5\%$ across a distance of 75 mm or more;
Note 8B301.a.1 includes atomic layer epitaxy (ALE) equipment.
 2. Metal Organic Chemical Vapour Deposition (MOCVD) reactors designed for compound semiconductor epitaxial growth of material having two or more of the following elements: aluminium, gallium, indium, arsenic, phosphorus, antimony, or nitrogen;
 3. Molecular beam epitaxial growth equipment using gas or solid sources;

8B301 b. Equipment designed for ion implantation and having any of the following:

1. (Reserved);
2. Being designed and optimized to operate at a beam energy of 20 keV or more and a beam current of 10 mA or more for hydrogen, deuterium or helium implant;
3. Direct write capability;
4. A beam energy of 65 keV or more and a beam current of 45 mA or more for high energy oxygen implant into a heated semiconductor material "substrate"; or
5. Being designed and optimized to operate at a beam energy of 20 keV or more and a beam current of 10 mA or more for silicon implant into a semiconductor material "substrate" heated to 600°C or greater;

8B301 c. (Reserved);

8B301 d. (Reserved);

8B301 e. Automatic loading multi-chamber central wafer handling systems having all of the following:

1. Interfaces for wafer input and output, to which more than two functionally different 'semiconductor process tools' specified by 8B301.a.1, 8B301.a.2, 8B301.a.6.3 or 8B301.b are designed to be connected; and
2. Designed to form an integrated system in a vacuum environment for 'sequential multiple wafer processing';

Note 8B301.e does not apply to automatic robotic wafer handling systems specially designed for parallel wafer processing.

Technical Notes

1. For the purpose of 8B301.e, 'semiconductor process tools' refers to modular tools that provide physical processes for semiconductor production that are functionally different, such as deposition, implant or thermal processing.
2. For the purpose of 8B301.e, 'sequential multiple wafer processing' means the capability to process each wafer in different 'semiconductor process tools', such as by transferring each wafer from one tool to a second tool and on to a third tool with the automatic loading multi-chamber central wafer handling systems.

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- 8B301 f. Lithography equipment as follows:
1. Align and expose step and repeat (direct step on wafer) or step and scan (scanner) equipment for wafer processing using photo-optical or X-ray methods and having any of the following:
 - a. A light source wavelength shorter than 193 nm; or
 - b. Capable of producing a pattern with a 'Minimum Resolvable Feature size' (MRF) of 45 nm or less;

Technical Note
The 'Minimum Resolvable Feature size' (MRF) is calculated by the following formula:

$$MRF = \frac{(\text{an exposure light source wavelength in nm}) \times (K \text{ factor})}{\text{numerical aperture}}$$
where the K factor = 0.35
- 8B301 f. 2. Imprint lithography equipment capable of producing features of 45 nm or less;
Note 8B301.f.2 includes:
- Micro contact printing tools
 - Hot embossing tools
 - Nano-imprint lithography tools
 - Step and flash imprint lithography (S-FIL) tools
- 8B301 f. 3. Equipment specially designed for mask making having all of the following:
- a. A deflected focused electron beam, ion beam or "laser" beam; and
 - b. Having any of the following:
 1. A full-width half-maximum (FWHM) spot size smaller than 65 nm and an image placement less than 17 nm (mean + 3 sigma); or
 2. (Reserved)
 3. A second-layer overlay error of less than 23 nm (mean + 3 sigma) on the mask;
- 8B301 f. 4. Equipment designed for device processing using direct writing methods, having all of the following:
- a. A deflected focused electron beam; and
 - b. Having any of the following:
 1. A minimum beam size equal to or smaller than 15 nm; or
 2. An overlay error less than 27 nm (mean + 3 sigma);
- 8B301 g. Masks and reticles, designed for integrated circuits specified by 8A301;
- 8B301 h. Multi-layer masks with a phase shift layer not specified by 8B301.g. and designed to be used by lithography equipment having a light source wavelength less than 245 nm;
Note 8B301.h. does not apply to multi-layer masks with a phase shift layer designed for the fabrication of memory devices not specified by 8A301.
N.B. For masks and reticles, specially designed for optical sensors, see 8B602.
- 8B301 i. Imprint lithography templates designed for integrated circuits specified by 8A301.
- 8B301 j. Mask "substrate blanks" with multilayer reflector structure consisting of molybdenum and silicon, and having all of the following:
1. Specially designed for 'Extreme Ultraviolet' ('EUV') lithography; and
 2. Compliant with SEMI Standard P37.
- Technical Note '*
Extreme Ultraviolet' ('EUV') refers to electromagnetic spectrum wavelengths greater than 5 nm and less than 124 nm.
- 8B302 Test equipment specially designed for testing finished or unfinished semiconductor devices as follows and specially designed components and accessories therefor:
- a. For testing S-parameters of items specified by 8A301.b.3;
 - b. (Reserved)
 - c. For testing items specified by 8A301.b.2.

8C3ELECTRONICS (MATERIALS)

- 8C301 Hetero-epitaxial materials consisting of a "substrate" having stacked epitaxially grown multiple layers of

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any of the following:

- a. Silicon (Si);
- b. Germanium (Ge);
- c. Silicon Carbide (SiC); or
- d. "III/V compounds" of gallium or indium.
- e. Gallium Oxide (Ga₂O₃); or
- f. Diamond.

Note 8C301.d. does not apply to a "substrate" having one or more P-type epitaxial layers of GaN, InGaN, AlGaN, InAlN, InAlGaN, GaP, GaAs, AlGaAs, InP, InGaP, AlInP or InGaIP, independent of the sequence of the elements, except if the P-type epitaxial layer is between N-type layers.

- 8C302 Resist materials as follows and "substrates" coated with the following resists:
- a. Resists designed for semiconductor lithography as follows:
 1. Positive resists adjusted (optimised) for use at wavelengths less than 193 nm but equal to or greater than 15 nm;
 2. Resists adjusted (optimised) for use at wavelengths less than 15 nm but greater than 1 nm;
 - b. All resists designed for use with electron beams or ion beams, with a sensitivity of 0.01 µcoulomb/mm² or better;
 - c. (Reserved);
 - d. All resists optimised for surface imaging technologies;
 - e. All resists designed or optimised for use with imprint lithography equipment specified by 8B301.f.2 that use either a thermal or photo-curable process.
- 8C303 Organo-inorganic compounds as follows:
- a. Organo-metallic compounds of aluminium, gallium or indium, having a purity (metal basis) better than 99.999%;
 - b. Organo-arsenic, organo-antimony and organo-phosphorus compounds, having a purity (inorganic element basis) better than 99.999%.
- Note 8C303 only applies to compounds whose metallic, partly metallic or non-metallic element is directly linked to carbon in the organic part of the molecule.*
- 8C304 Hydrides of phosphorus, arsenic or antimony, having a purity better than 99.999%, even diluted in inert gases or hydrogen.
- Note 8C304 does not apply to hydrides containing 20% molar or more of inert gases or hydrogen.*
- 8C305 High resistivity materials as follows:
- a. Silicon carbide (SiC), gallium nitride (GaN), aluminium nitride (AlN) or aluminium gallium nitride (AlGaIn) gallium oxide (Ga₂O₃) or diamond semiconductor "substrates", or ingots, boules, or other preforms of those materials, having resistivities greater than 10,000 ohm-cm at 20°C;
 - b. Polycrystalline "substrates" or polycrystalline ceramic "substrates", having resistivities greater than 10,000 ohm-cm at 20°C and having at least one non-epitaxial single-crystal layer of silicon (Si), silicon carbide (SiC), gallium nitride (GaN), aluminium nitride (AlN), or aluminium gallium nitride (AlGaIn) gallium oxide (Ga₂O₃) or diamond on the surface of the "substrate".
- 8C306 Materials, not specified by 8C301, consisting of a "substrate" specified by 8C305 with at least one epitaxial layer of silicon carbide, gallium nitride, aluminium nitride or aluminium gallium nitride, gallium oxide (Ga₂O₃) or diamond.

8D3 ELECTRONICS (SOFTWARE)

- 8D301 "Software" specially designed for the "development" or "production" of equipment specified by 8A301.b to 8A302.h or 8B3.
- 8D302 "Software" specially designed for the "use" of equipment specified by 8B301.a to f or 8B302.
- 8D303 'Computational lithography' "software" specially designed for the "development" of patterns on EUV-lithography masks or reticles.

Technical Note

'Computational lithography' is the use of computer modelling to predict, correct, optimise and verify

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imaging performance of the lithography process over a range of patterns, processes, and system conditions.

8D304 "Software" specially designed for the "development" of equipment specified by 8A303.

8D305 "Software" specially designed to restore normal operation of a microcomputer, "microprocessor microcircuit" or "microcomputer microcircuit" within 1 ms after an Electromagnetic Pulse (EMP) or Electrostatic Discharge (ESD) disruption, without loss of continuation of operation.

8D306 'Electronic Computer-Aided Design' ('ECAD') "software" specially designed for the "development" of integrated circuits having any "Gate-All-Around Field-Effect Transistor" ("GAAFET") structure, and having any of the following:

a. Specially designed for implementing 'Register Transfer Level' ('RTL') to 'Geometrical Database Standard II' ('GDSII') or equivalent standard; or

b. Specially designed for optimisation of power or timing rules.

Technical Notes

1. 'Electronic Computer-Aided Design' ('ECAD') is a category of "software" tools used for designing, analysing, optimising, and validating the performance of integrated circuit or printed circuit board.

2. 'Register Transfer Level' ('RTL') is a design abstraction which models a synchronous digital circuit in terms of the flow of digital signals between hardware registers, and the logical operations performed on those signals.

3. 'Geometrical Database Standard II' ('GDSII') is a database file format for data exchange of integrated circuit or integrated circuit layout artwork.

8E3 ELECTRONICS (TECHNOLOGY)

8E301 "Technology" according to the General Technology Note for the "development" or "production" of equipment or materials specified by 8A3, 8B3 or 8C3;

Note 1: 8E301 does not apply to "technology" for equipment or components specified by 8A303.

Note 2: 8E301 does not apply to "technology" for integrated circuits specified by 8A301.a.3 to 8A301.a.12, having all of the following:

a. Using "technology" at or above 0.130 μm ; and

b. Incorporating multi-layer structures with three or fewer metal layers.

Note 3 8E301 does not apply to 'Process Design Kits' ('PDKs') unless they include libraries implementing functions or technologies for items specified by 8A301.

Technical Note

A 'Process Design Kit' ('PDK') is a software tool provided by a semiconductor manufacturer to ensure that the required design practices and rules are taken into account in order to successfully produce a specific integrated circuit design in a specific semiconductor process, in accordance with technological and manufacturing constraints (each semiconductor manufacturing process has its particular 'PDK').

8E302 "Technology" according to the General Technology Note other than that specified by 8E301 for the "development" or "production" of a "microprocessor microcircuit", "microcomputer microcircuit" or microcontroller microcircuit core, having an arithmetic logic unit with an access width of 32 bits or more and any of the following features or characteristics:

8E302 a. A 'vector processor unit' designed to perform more than two calculations on 'floating-point' vectors (one-dimensional arrays of 32-bit or larger numbers) simultaneously;

Technical Note

A 'vector processor unit' is a processor element with built-in instructions that perform multiple calculations on 'floating-point' vectors (one-dimensional arrays of 32-bit or larger numbers) simultaneously, having at least one vector arithmetic logic unit and vector registers of at least 32 elements each.

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- b. Designed to perform more than four 64-bit or larger 'floating-point' operation results per cycle; or
- c. Designed to perform more than eight 16-bit 'fixed-point' multiply-accumulate results per cycle (e.g., digital manipulation of analogue information that has been previously converted into digital form, also known as digital "signal processing").

Note 1 8E302 does not apply to "technology" for multimedia extensions.

Note 2 8E302 does not apply to "technology" for micro-processor cores, having all of the following:

- a. Using "technology" at or above 0.130 μm ; and
- b. Incorporating multi-layer structures with five or fewer metal layers.

Note 3 8E302 includes "technology" for the "development" or "production" of digital signal processors and digital array processors.

Technical Notes

1. For the purpose of 8E302.a. and 8E302.b., 'floating-point' is defined by IEEE-754.
2. For the purpose of 8E302.c., 'fixed-point' refers to a fixed-width real number with both an integer component and a fractional component, and which does not include integer-only formats.

- 8E303 Other "technology" for the "development" or "production" of the following:
- a. Vacuum microelectronic devices;
 - b. Hetero-structure semiconductor electronic devices such as high electron mobility transistors (HEMT), hetero-bipolar transistors (HBT), quantum well and super lattice devices;
Note 8E303.b does not apply to "technology" for high electron mobility transistors (HEMT) operating at frequencies lower than 31.8 GHz and hetero-junction bipolar transistors (HBT) operating at frequencies lower than 31.8 GHz.
 - c. "Superconductive" electronic devices;
 - d. Substrates of diamond for electronic components
 - e. Substrates of silicon-on-insulator (SOI) for integrated circuits in which the insulator is silicon dioxide;
 - f. Substrates of silicon carbide for electronic components;
 - g. "Vacuum electronic devices" operating at frequencies of 31.8 GHz or higher.
 - h. Substrates of gallium oxide for electronic components

8E304 "Technology" "required" for the slicing, grinding and polishing of 300 mm diameter silicon wafers to achieve a 'Site Front least sQuares Range' ('SFQR') less than or equal to 20 nm at any site of 26 mm x 8 mm on the front surface of the wafer and an edge exclusion less than or equal to 2 mm.

Technical Note

For the purpose of 8E304, 'SFQR' is the range of maximum deviation and minimum deviation from front reference plane, calculated by least square method with all front surface data including site boundary within a site.

8A4 COMPUTERS (SYSTEMS, EQUIPMENT AND COMPONENTS)

Note 1 Computers, related equipment and "software" performing telecommunications or "local area network" functions must also be evaluated against the performance characteristics of (8A501,8B501,8C5,8D501,8E5)-Part 1 (Telecommunications).

Note 2 Control units which directly interconnect the buses or channels of central processing units, 'main storage' or disk controllers are not regarded as telecommunications equipment described in (8A501,8B501,8C5,8D501,8E5)- Part 1 (Telecommunications).

N.B. For the status of "software" specially designed for packet switching, see 8D501 (Telecommunications).

Technical Note

'Main storage' is the primary storage for data or instructions for rapid access by a central processing unit. It consists of the internal storage of a "digital computer" and any hierarchical extension thereto, such as cache storage or non-sequentially accessed extended storage.

Note 3 (Reserved)

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- 8A401 Electronic computers and related equipment, having any of the following and "electronic assemblies" and specially designed components therefor:
- a. Specially designed to have any of the following:
 1. Rated for operation at an ambient temperature below 228 K (-45°C) or above 358 K (85°C); or
Note 8A401.a.1 does not apply to computers specially designed for civil automobile, railway train or "civil aircraft" applications.
 2. Radiation hardened to exceed any of the following specifications:
 - a. Total Dose 5×10^3 Gy (Si);
 - b. Dose Rate Upset 5×10^6 Gy (Si)/s; or
 - c. Single Event Upset 1×10^{-8} Error/bit/day.*Note* 8A401.a.2 does not apply to computers specially designed for "civil aircraft" applications.
- 8A401 b. (Reserved)
- 8A402 (Reserved)
- 8A403 "Digital computers", "electronic assemblies", and related equipment therefor, as follows and specially designed components therefor:
- Note 1* 8A403 includes the following:
- 'Vector processors';
 - Array processors;
 - Digital signal processors;
 - Logic processors;
 - Equipment designed for "image enhancement".
- Note 2* The status of the "digital computers" and related equipment described in 8A403 is determined by the status of other equipment or systems provided:
- a. The "digital computers" or related equipment are essential for the operation of the other equipment or systems;
 - b. The "digital computers" or related equipment are not a "principal element" of the other equipment or systems; and
N.B.1 The status of "signal processing" or "image enhancement" equipment specially designed for other equipment with functions limited to those required for the other equipment is determined by the status of the other equipment even if it exceeds the "principal element" criterion.
N.B.2 For the status of "digital computers" or related equipment for telecommunications equipment, see (8A501, 8B501, 8C5, 8D5, 8E5) - Part 1 (Telecommunications).
 - c. The "technology" for the "digital computers" and related equipment is determined by 8E4.
- 8A403 a. (Reserved)
- b. "Digital computers" having an 'Adjusted Peak Performance' ('APP') exceeding 70.0 Weighted TeraFLOPS (WT);
- c. "Electronic assemblies" specially designed or modified for enhancing performance by aggregation of processors so that the 'APP' of the aggregation exceeds the limit specified by 8A403.b;
- Note 1* 8A403.c applies only to "electronic assemblies" and programmable interconnections not exceeding the limit specified by 8A403.b when shipped as unintegrated "electronic assemblies".
- Note 2* 8A403.c does not apply to "electronic assemblies" specially designed for a product or family of products whose maximum configuration does not exceed the limit specified by 8A403.b.
- 8A403 d. (Reserved)
- e. (Reserved)

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N.B. For "electronic assemblies", modules or equipment, performing analogue-to-digital conversions, see 8A302.h.

- f. (Reserved)
- g. Equipment specially designed for aggregating the performance of "digital computers" by providing external interconnections which allow communications at unidirectional data rates exceeding 2.0 Gbyte/s per link.

Note 8A403.g does not apply to internal interconnection equipment (e.g, backplanes, buses), passive interconnection equipment, "network access controllers" or "communications channel controllers".

8A404 Computers as follows and specially designed related equipment, "electronic assemblies" and components therefor:

- a. "Systolic array computers";
- b. "Neural computers";
- c. "Optical computers".

Technical Notes

1. 'Systolic array computers' are computers where the flow and modification of the data is dynamically controllable at the logic gate level by the user.
2. 'Neural computers' are computational devices designed or modified to mimic the behaviour of a neuron or a collection of neurons, i.e., computational devices which are distinguished by their hardware capability to modulate the weights and numbers of the interconnections of a multiplicity of computational components based on previous data.
3. 'Optical computers' are computers designed or modified to use light to represent data and whose computational logic elements are based on directly coupled optical devices'.

8A405 Systems, equipment, and components therefor, specially designed or modified for the generation, command and control, or delivery of "intrusion software".

8B4 COMPUTERS (TEST, INSPECTION AND PRODUCTION EQUIPMENT) – (Reserved)

8C4 COMPUTERS (MATERIALS) – (Reserved)

8D4 COMPUTERS (SOFTWARE)

Note The status of "software" for equipment described in other Categories is dealt with in the appropriate Category.

8D401 "Software" as follows:

- a. "Software" specially designed or modified for the "development" or "production" of equipment or "software" specified by 8A4 or 8D4
- b. "Software", other than that specified by 8D401.a., specially designed or modified for the "development" or "production" of equipment as follows:
 1. "Digital computers" having an 'Adjusted Peak Performance' ('APP') exceeding 24 Weighted TeraFLOPS (WT);
 2. "Electronic assemblies" specially designed or modified for enhancing performance by aggregation of processors so that the 'APP' of the aggregation exceeds the limit in 8D401.b.1.

8D402 (Reserved)

8D403 (Reserved)

8D404 "Software" specially designed or modified for the generation, command and control, or delivery of "intrusion software".

Note: 8D404 does not apply to "software" specially designed and limited to provide "software" updates or upgrades meeting all the following:

- a. The update or upgrade operates only with the authorisation of the owner or administrator of the system receiving it; and

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- b. After the update or upgrade, the “software” updated or upgraded is not any of the following:
1. “Software” specified by 8D404; or
 2. “Intrusion software”.

8E4 COMPUTERS (TECHNOLOGY)

8E401 “Technology” as follows:

- a. “Technology” according to the General Technology Note, for the “development”, “production” or “use” of equipment or “software” specified by 8A4 or 8D4
- b. “Technology” according to the General Technology Note, other than that specified by 8E401.a., for the “development” or “production” of equipment as follows:
 1. “Digital computers” having an ‘Adjusted Peak Performance’ (‘APP’) exceeding 24 Weighted TeraFLOPS (WT);
 2. “Electronic assemblies” specially designed or modified for enhancing performance by aggregation of processors so that the ‘APP’ of the aggregation exceeds the limit in 8E401.b.1.
- c. “Technology” for the “development” of “intrusion software”.

Note 1 8E401.a. and 8E401.c. do not apply to “vulnerability disclosure” or “cyber incident response”.

Note 2 Note 1 does not diminish national authorities’ rights to ascertain compliance with 8E401.a. and 8E401.c.

TECHNICAL NOTE ON ‘ADJUSTED PEAK PERFORMANCE’ (‘APP’)

‘APP’ is an adjusted peak rate at which “digital computers” perform 64-bit or larger floating point additions and multiplications.

Abbreviations used in this Technical Note

n	number of processors in the “digital computer”
i	processor number (i,...n)
t _i	processor cycle time (t _i = 1/F _i)
F _i	processor frequency
R _i	peak floating point calculating rate
W _i	architecture adjustment factor

‘APP’ is expressed in Weighted TeraFLOPS (WT), in units of 10¹² adjusted floating point operations per second.

Outline of ‘APP’ calculation method

1. For each processor i, determine the peak number of 64-bit or larger floating point operations, FPO_i, performed per cycle for each processor in the “digital computer”.

Note In determining FPO, include only 64-bit or larger floating point additions or multiplications. All floating point operations must be expressed in operations per processor cycle; operations requiring multiple cycles may be expressed in fractional results per cycle. For processors not capable of performing calculations on floating point operands of 64-bit or more, the effective calculating rate R is zero.

2. Calculate the floating point rate R for each processor $R_i = FPO_i / t_i$
3. Calculate ‘APP’ as $‘APP’ = W_1 \times R_1 + W_2 \times R_2 + \dots + W_n \times R_n$.
4. For ‘vector processors’, W_i = 0.9. For non-‘vector processors’, W_i = 0.3.

Note 1 For processors that perform compound operations in a cycle, such as addition and multiplication, each operation is counted.

Note 2 For a pipelined processor the effective calculating rate R is the faster of the pipelined rate, once the pipeline is full, or the non-pipelined rate.

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Note 3 The calculating rate R of each contributing processor is to be calculated at its maximum value theoretically possible before the 'APP' of the combination is derived. Simultaneous operations are assumed to exist when the computer manufacturer claims concurrent, parallel, or simultaneous operation or execution in a manual or brochure for the computer.

Note 4 Do not include processors that are limited to input/output and peripheral functions (e.g, disk drive, communication and video display) when calculating 'APP'.

Note 5 'APP' values are not to be calculated for processor combinations (inter)connected by "Local Area Networks", Wide Area Networks, I/O shared connections/devices, I/O controllers and any communication interconnection implemented by "software".

Note 6 'APP' values must be calculated for processor combinations containing processors specially designed to enhance performance by aggregation, operating simultaneously and sharing memory;

Technical Notes

1. Aggregate all processors and accelerators operating simultaneously and located on the same die.
2. Processor combinations share memory when any processor is capable of accessing any memory location in the system through the hardware transmission of cache lines or memory words, without the involvement of any software mechanism, which may be achieved using "electronic assemblies" specified in 8A403.c.

Note 7 A 'vector processor' is defined as a processor with built-in instructions that perform multiple calculations on floating-point vectors (one-dimensional arrays of 64-bit or larger numbers) simultaneously, having at least 2 vector functional units and at least 8 vector registers of at least 64 elements each.

8A5 PART-1 TELECOMMUNICATIONS (SYSTEMS, EQUIPMENT AND COMPONENTS)

Note 1 The status of components, test and "production" equipment and "software" therefor which are specially designed for telecommunications equipment or systems is determined in (8A501, 8B501, 8C5, 8D501 and 8E501) Part 1.

N.B. For "lasers" specially designed for telecommunications equipment or systems, see 8A505.

Note 2 "Digital computers", related equipment or "software", when essential for the operation and support of telecommunications equipment described in 8A5- Part-I, are regarded as specially designed components, provided they are the standard models customarily supplied by the manufacturer. This includes operation, administration, maintenance, engineering or billing computer systems.

8A501 Telecommunications systems, equipment, components and accessories, as follows:

- a. Any type of telecommunications equipment having any of the following characteristics, functions or features:
 1. Specially designed to withstand transitory electronic effects or electromagnetic pulse effects, both arising from a nuclear explosion;
 2. Specially hardened to withstand gamma, neutron or ion radiation;
 3. Specially designed to operate below 218 K (-55° C); or
 4. Specially designed to operate above 397 K (124° C);

Note 1 8A501.a.3 and 8A501.a.4 apply only to electronic equipment.

Note 2 8A501.a.2, 8A501.a.3 and 8A501.a.4 do not apply to equipment designed or modified for use on board satellites.

8A501 b. Telecommunication systems and equipment, and specially designed components and accessories therefor, having any of the following characteristics, functions or features:

1. Being underwater untethered communications systems having any of the following:
 - a. An acoustic carrier frequency outside the range from 20 kHz to 60 kHz;
 - b. Using an electromagnetic carrier frequency below 30 kHz;
 - c. Using electronic beam steering techniques; or
 - d. Using "lasers" or light-emitting diodes (LEDs), with an output wavelength greater than 400 nm and less than 700 nm, in a "local area network";

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2. Being radio equipment operating in the 1.5 MHz to 87.5 MHz band and having all of the following:
- a. Automatically predicting and selecting frequencies and "total digital transfer rates" per channel to optimise the transmission; and
- 8A501 b. 2. b. Incorporating a linear power amplifier configuration having a capability to support multiple signals simultaneously at an output power of 1 kW or more in the frequency range of 1.5 MHz or more but less than 30 MHz, or 250 W or
- more in the frequency range of 30 MHz or more but not exceeding 87.5 MHz, over an "instantaneous bandwidth" of one octave or more and with an output harmonic and distortion content of better than -80 dB;
- 8A501 b. 3. Being radio equipment employing "spread spectrum" techniques, including "frequency hopping" techniques, not specified by 8A501.b.4 and having any of the following:
- a. User programmable spreading codes; or
 - b. A total transmitted bandwidth which is 100 or more times the bandwidth of any one information channel and in excess of 50 kHz;
- Note 8A501.b.3.b does not apply to radio equipment specially designed for use with any of the following:
- a. Civil cellular radio-communications systems; or
 - b. Fixed or mobile satellite earth stations for commercial civil telecommunications.
- Note 8A501.b.3 does not apply to equipment designed to operate at an output power of 1 W or less.
- 8A501 b. 4. Being radio equipment employing ultra-wideband modulation techniques having user programmable channelizing codes, scrambling codes or network identification codes and having any of the following:
- a. A bandwidth exceeding 500 MHz; or
 - b. A "fractional bandwidth" of 20% or more;
- 8A501 b. 5. Being digitally controlled radio receivers having all of the following:
- a. More than 1,000 channels;
 - b. A 'channel switching time' of less than 1 ms;
 - c. Automatic searching or scanning of a part of the electromagnetic spectrum; and
 - d. Identification of the received signals or the type of transmitter; or
- Note 8A501.b.5 does not apply to radio equipment specially designed for use with civil cellular radio-communications systems.
- Technical Note
'Channel switching time': the time (i.e, delay) to change from one receiving frequency to another, to arrive at or within $\pm 0.05\%$ of the final specified receiving frequency. Items having a specified frequency range of less than $\pm 0.05\%$ around their centre frequency are defined to be incapable of channel frequency switching.
- 8A501 b. 6. Employing functions of digital "signal processing" to provide 'voice coding' output at rates of less than 700 bit/s.
- Technical Notes
1. For variable rate 'voice coding', 8A501.b.6 applies to the 'voice coding' output of continuous speech.
 2. For the purpose of 8A501.b.6, 'voice coding' is defined as the technique to take samples of human voice and then convert these samples into a digital signal, taking into account specific characteristics of human speech.
- 8A501 c. Optical fibres of more than 500 m in length and specified by the manufacturer as being capable of withstanding a 'proof test' tensile stress of 2×10^9 N/m² or more;

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N.B. For underwater umbilical cables, see 8A802.a.3.

Technical Note

'Proof Test': on-line or off-line production screen testing that dynamically applies a prescribed tensile stress over a 0.5 to 3 m length of fibre at a running rate of 2 to 5 m/s while passing between capstans approximately 150 mm in diameter. The ambient temperature is a nominal 293 K (20 °C) and relative humidity 40%.

- 8A501 d. 'Electronically steerable phased array antennae' as follows:
1. Rated for operation above 31.8 GHz, but not exceeding 57 GHz, and having an Effective Radiated Power (ERP) equal to or greater than +20 dBm (22.15 dBm Effective Isotropic Radiated Power (EIRP));
 2. Rated for operation above 57 GHz, but not exceeding 66 GHz, and having an ERP equal to or greater than +24 dBm (26.15 dBm EIRP);
 3. Rated for operation above 66 GHz, but not exceeding 90 GHz, and having an ERP equal to or greater than +20 dBm (22.15 dBm EIRP);
 4. Rated for operation with a peak saturated power output greater than 0.1 nW (-70 dBm) at any frequency exceeding 31.8 GHz up to and including 37 GHz;
- Note 1: 8A501.d does not apply to 'electronically steerable phased array antennae' for landing systems with instruments meeting ICAO standards covering Microwave Landing Systems (MLS).
- Note 2 8A501.d does not apply to antennae specially designed for any of the following:
- a. Civil cellular or WLAN radio-communications systems;
 - b. IEEE 802.15 or wireless HDMI; or
 - c. Fixed or mobile satellite earth stations for commercial civil telecommunications.

Technical Note

For the purposes of 8A501.d 'electronically steerable phased array antenna' is an antenna which forms a beam by means of phase coupling, (i.e., the beam direction is controlled by the complex excitation coefficients of the radiating elements) and the direction of that beam can be varied (both in transmission and reception) in azimuth or in elevation, or both, by application of an electrical signal.

- 8A501 e. Radio direction finding equipment operating at frequencies above 30 MHz and having all of the following, and specially designed components therefor:
1. "Instantaneous bandwidth" of 10 MHz or more; and
 2. Capable of finding a Line of Bearing (LOB) to non-cooperating radio transmitters with a signal duration of less than 1 ms;
- 8A501 f. Mobile telecommunications interception or jamming equipment, and monitoring equipment therefor, as follows, and specially designed components therefor:
1. Interception equipment designed for the extraction of voice or data, transmitted over the air interface;
 2. Interception equipment not specified in 8A501.f.1, designed for the extraction of client device or subscriber identifiers (e.g. IMSI, TIMSI or IMEI), signalling, or other metadata transmitted over their interface;
- 8A501 f. 3. Jamming equipment specially designed or modified to intentionally and selectively interfere with, deny, inhibit, degrade or seduce mobile telecommunication services and performing any of the following:
- a. Simulate the functions of Radio Access Network (RAN) equipment;
 - b. Detect and exploit specific characteristics of the mobile telecommunications protocol employed (e.g. GSM); or
 - c. Exploit specific characteristics of the mobile telecommunications protocol employed (e.g. GSM);
4. RF monitoring equipment designed or modified to identify the operation of items specified in 8A501.f.1, 8A501.f.2 or 8A501.f.3;

Note 8A501.f.1 and 8A501.f.2 do not apply to any of the following:

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- a. *Equipment specially designed for the interception of analogue Private Mobile Radio (PMR), IEEE 802.11 WLAN;*
- b. *Equipment designed for mobile telecommunications network operators; or*
- c. *Equipment designed for the "development" or "production" of mobile telecommunications equipment or systems.*

N.B.1. *See also 6A010.*

N.B.2. *For radio receivers see 8A501.b.5.*

- 8A501 g. Passive Coherent Location (PCL) systems or equipment, specially designed for detecting and tracking moving objects by measuring reflections of ambient radio frequency emissions, supplied by non-radar transmitters;

Technical Note

Non-radar transmitters may include commercial radio, television or cellular telecommunications base stations.

Note 8A501.g does not apply to any of the following:

- a. *Radio-astronomical equipment; or*
- b. *Systems or equipment, that requires any radio transmission from the target.*

- 8A501 h. Counter Improvised Explosive Device (IED) equipment and related equipment, as follows:
1. Radio Frequency (RF) transmitting equipment, not specified by 8A501.f, designed or modified for prematurely activating or preventing the initiation of Improvised Explosive Devices (IEDs);
 2. Equipment using techniques designed to enable radio communications in the same frequency channels on which co-located equipment specified by 8A501.h.1 is transmitting.

N.B. *See also Category 6.*

- 8A501 i. (Reserved)

N.B. *See 8A501.f for items specified by 8A501.i.*

- 8A501 j. IP network communications surveillance systems or equipment, and specially designed components therefor, having all of the following:
1. Performing all of the following on a carrier class IP network (e.g. national grade IP backbone):
 - a. Analysis at the application layer (e.g. Layer 7 of Open Systems Interconnection (OSI) model (ISO/IEC 7498-1));
 - b. Extraction of selected metadata and application content (e.g. voice, video, messages, attachments); and
 - c. Indexing of extracted data; and
 2. Being specially designed to carry out all of the following:
 - a. Execution of searches on the basis of "hard selectors"; and
 - b. Mapping of the relational network of an individual or of a group of people.

Note 8A501.j does not apply to systems or equipment, specially designed for any of the following:

- a. *Marketing purpose;*
- b. *Network Quality of Service (QoS); or*
- c. *Quality of Experience (QoE).*

8B5 PART 1 – TELECOMMUNICATIONS (TEST, INSPECTION AND PRODUCTION EQUIPMENT)

- 8B501 Telecommunication test, inspection and production equipment, components and accessories, as follows:
- a. Equipment and specially designed components or accessories therefor, specially designed for the "development" or "production" of equipment, functions or features, specified by 8A501;

Note 8B501.a does not apply to optical fibre characterization equipment.

- 8B501 b. Equipment and specially designed components or accessories therefor, specially designed for the "development" of any of the following telecommunication transmission or switching equipment:
1. (Reserved)
 2. Equipment employing a "laser" and having any of the following:
 - a. A transmission wavelength exceeding 1,750 nm;
 - b. (Reserved)

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	c.	(Reserved);
	d.	Employing analogue techniques and having a bandwidth exceeding 2.5 GHz; <u>or</u>
	<u>Note</u>	8B501.b.2.d. does not apply to equipment specially designed for the "development" of commercial TV systems.
8B501	b.	3. (Reserved)
		4. Radio equipment employing Quadrature-Amplitude-Modulation (QAM) techniques above level 1,024.
		5. (Reserved)
8C5	PART 1 – TELECOMMUNICATIONS (MATERIALS) – (Reserved)	
8D5	PART 1 – TELECOMMUNICATIONS (SOFTWARE)	
8D501	"Software" as follows:	
	a.	"Software" specially designed or modified for the "development", "production" or "use" of equipment, functions or features, specified by 8A501;
	b.	Reserved
	c.	Specific "software" specially designed or modified to provide characteristics, functions or features of equipment, specified by 8A501 or 8B501;
8D501	d.	"Software" specially designed or modified for the "development" of any of the following telecommunication transmission or switching equipment:
	1.	Reserved
	2.	Equipment employing a "laser" and having any of the following:
	a.	A transmission wavelength exceeding 1,750 nm; <u>or</u>
	b.	Employing analogue techniques and having a bandwidth exceeding 2.5 GHz; <u>or</u>
	<u>Note</u>	8D501.d.2.b does not apply to "software" specially designed or modified for the "development" of commercial TV systems.
	3.	(Reserved)
	4.	Radio equipment employing Quadrature-Amplitude-Modulation (QAM) techniques above level 1,024.
8D501	e.	"Software", other than that specified by 8D501.a. or 8D501.c., specially designed or modified for monitoring or analysis by law enforcement, providing all of the following:
	1.	Execution of searches on the basis of "hard selectors" of either the content of communication or metadata acquired from a communications service provider using a 'handover interface'; <u>and</u>
	<u>Technical Notes</u>	
	1.	For the purposes of 8D501.e., a 'handover interface' is a physical and logical interface, designed for use by an authorised law enforcement authority, across which targeted interception measures are requested from a communications service provider and the results of interception are delivered from a communications service provider to the requesting authority. The 'handover interface' is implemented within systems or equipment (e.g., mediation devices) that receive and validate the interception request, and deliver to the requesting authority only the results of interception that fulfil the validated request.
	2.	'Handover interfaces' may be specified by international standards (including but not limited to ETSI TS 101 331, ETSI TS 101 671, 3GPP TS 33.108) or national equivalents.
8D501	e.	2. Mapping of the relational network or tracking the movement of targeted individuals based on the results of searches on content of communication or metadata or searches as described in 8D501.e.1.
	<u>Note</u>	8D501.e. does not apply to "software" specially designed or modified for any of the following:
	a.	Billing purposes;
	b.	Network Quality of Service (QoS);
	c.	Quality of Experience (QoE);
	d.	Mediation devices; or
	e.	Mobile payment or banking use.

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8E5 PART 1 – TELECOMMUNICATIONS (TECHNOLOGY)

8E501 "Technology" as follows:

- a. "Technology" according to the General Technology Note for the "development", "production" or "use" (excluding operation) of equipment, functions or features specified by 8A501 or "software" specified by 8D501.a. or 8D501.e.;
- b. Specific "technology" as follows:
 1. "Required" "technology" for the "development" or "production" of telecommunications equipment specially designed to be used on board satellites;
 2. "Technology" for the "development" or "use" of "laser" communication techniques with the capability of automatically acquiring and tracking signals and maintaining communications through exoatmosphere or sub-surface (water) media;

8E501 b. 3. "Technology" for the "development" of digital cellular radio base station receiving equipment whose reception capabilities that allow multi-band, multi-channel, multi-mode, multi-coding algorithm or multi-protocol operation can be modified by changes in "software";

4. "Technology" for the "development" of "spread spectrum" techniques, including "frequency hopping" techniques;

Note 8E501b.4 does not apply to "technology" for the "development" of any of the following:

- a. Civil cellular radio-communications systems; or
- b. Fixed or mobile satellite earth stations for commercial civil telecommunications.

8E501 c. "Technology" according to the General Technology Note for the "development" or "production" of any of the following:

1. (Reserved)
2. Equipment employing a "laser" and having any of the following:
 - a. A transmission wavelength exceeding 1,750 nm;
 - b. (Reserved)
 - c. (Reserved)
 - d. Employing wavelength division multiplexing techniques of optical carriers at less than 100 GHz spacing; or
 - e. Employing analogue techniques and having a bandwidth exceeding 2.5 GHz;

Note 8E501.c.2.e does not apply to "technology" for commercial TV systems.

N.B. For "technology" for the "development" or "production" of non-telecommunications equipment employing a "laser", see 8E6.

8E501 c. 3. Equipment employing "optical switching" and having a switching time less than 1 ms;

8E501 c. 4. Radio equipment having any of the following:

- a. Quadrature-Amplitude-Modulation (QAM) techniques above level 1,024;

8E501 c. 4. b. Operating at input or output frequencies exceeding 31.8 GHz; or

Note 8E501.c.4.b does not apply to "technology" for equipment designed or modified for operation in any frequency band which is "allocated by the ITU" for radio-communications services, but not for radio-determination.

- c. Operating in the 1.5 MHz to 87.5 MHz band and incorporating adaptive techniques providing more than 15 dB suppression of an interfering signal; or

8E501 c. 5. (Reserved)

8E501 c. 6. Mobile equipment having all of the following:

- a. Operating at an optical wavelength greater than or equal to 200 nm and less than or equal to 400 nm; and
- b. Operating as a "local area network";

8E501 d. "Technology" according to the General Technology Note for the "development" or "production" of

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"Monolithic Microwave Integrated Circuit" ("MMIC") amplifiers specially designed for telecommunications and that are any of the following:

Technical Note

For purposes of 8E501.d, the parameter peak saturated power output may also be referred to on product data sheets as output power, saturated power output, maximum power output, peak power output, or peak envelope power output.

- | | | | |
|-------|----|----|---|
| 8E501 | d. | 1. | Rated for operation at frequencies exceeding 2.7 GHz up to and including 6.8 GHz with a "fractional bandwidth" greater than 15%, and having any of the following: <ul style="list-style-type: none"> a. A peak saturated power output greater than 75 W (48.75 dBm) at any frequency exceeding 2.7 GHz up to and including 2.9 GHz; b. A peak saturated power output greater than 55 W (47.4 dBm) at any frequency exceeding 2.9 GHz up to and including 3.2 GHz; c. A peak saturated power output greater than 40 W (46 dBm) at any frequency exceeding 3.2 GHz up to and including 3.7 GHz; <u>or</u> d. A peak saturated power output greater than 20 W (43 dBm) at any frequency exceeding 3.7 GHz up to and including 6.8 GHz; |
| 8E501 | d. | 2. | Rated for operation at frequencies exceeding 6.8 GHz up to and including 16 GHz with a "fractional bandwidth" greater than 10%, and having any of the following: <ul style="list-style-type: none"> a. A peak saturated power output greater than 10 W (40 dBm) at any frequency exceeding 6.8 GHz up to and including 8.5 GHz; <u>or</u> b. A peak saturated power output greater than 5 W (37 dBm) at any frequency exceeding 8.5 GHz up to and including 16 GHz; |
| 8E501 | d. | 3. | Rated for operation with a peak saturated power output greater than 3 W (34.77 dBm) at any frequency exceeding 16 GHz up to and including 31.8 GHz, and with a "fractional bandwidth" of greater than 10%; |
| 8E501 | d. | 4. | Rated for operation with a peak saturated power output greater than 0.1n W (-70 dBm) at any frequency exceeding 31.8 GHz up to and including 37 GHz; |
| 8E501 | d. | 5. | Rated for operation with a peak saturated power output greater than 1 W (30 dBm) at any frequency exceeding 37 GHz up to and including 43.5 GHz, and with a "fractional bandwidth" of greater than 10%; |
| | | 6. | Rated for operation with a peak saturated power output greater than 31.62 mW (15 dBm) at any frequency exceeding 43.5 GHz up to and including 75 GHz, and with a "fractional bandwidth" of greater than 10%; |
| | | 7. | Rated for operation with a peak saturated power output greater than 10 mW (10 dBm) at any frequency exceeding 75 GHz up to and including 90 GHz, and with a "fractional bandwidth" of greater than 5%; <u>or</u> |
| | | 8. | Rated for operation with a peak saturated power output greater than 0.1 nW (-70 dBm) at any frequency exceeding 90 GHz; |
| 8E501 | e. | | "Technology" according to the General Technology Note for the "development" or "production" of electronic devices and circuits, specially designed for telecommunications and containing components manufactured from "superconductive" materials, specially designed for operation at temperatures below the "critical temperature" of at least one of the "superconductive" constituents and having any of the following: <ul style="list-style-type: none"> 1. Current switching for digital circuits using "superconductive" gates with a product of delay time per gate (in seconds) and power dissipation per gate (in watts) of less than 10^{-14} J; <u>or</u> 2. Frequency selection at all frequencies using resonant circuits with Q-values exceeding 10,000. |

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8A5 Part 2 - "INFORMATION SECURITY"(SYSTEMS, EQUIPMENT AND COMPONENTS)

Note 1 (Reserved)

Note 2 (8A502, 8A503, 8A504, 8B502, 8C5, 8D502 and 8E502)-Part 2 do not apply to products when accompanying their user for the user's personal use.

Note 3 Cryptography Note

8A502, 8D502.a.1, 8D502.b and 8D502.c.1, do not apply to items as follows:

- a. Items meeting all of the following:
 1. Generally available to the public by being sold, without restriction, from stock at retail selling points by means of any of the following:
 - a. Over-the-counter transactions;
 - b. Mail order transactions;
 - c. Electronic transactions; or
 - d. Telephone call transactions;
 2. The cryptographic functionality cannot easily be changed by the user;
 3. Designed for installation by the user without further substantial support by the supplier; and
 4. When necessary, details of the items are accessible and will be provided, upon request, to the appropriate authority in the exporter's country in order to ascertain compliance with conditions described in 1. to 3. above;
- b. Hardware components or 'executable software', of existing items described at a. of this Note, that have been designed for these existing items, and meeting all of the following:
 1. "Information security" is not the primary function or set of functions of the component or 'executable software';
 2. The component or 'executable software' does not change any cryptographic functionality of the existing items, or add new cryptographic functionality to the existing items;
 3. The feature set of the component or 'executable software' is fixed and is not designed or modified to customer specification; and
 4. When necessary as determined by the appropriate authority in the exporter's country, details of the component or 'executable software', and details of relevant end-items are accessible and will be provided to the authority upon request, in order to ascertain compliance with conditions described above.

Technical Note

For the purpose of the Cryptography Note, 'executable software' means "software" in executable form, from an existing hardware component excluded from 8A502 by the Cryptography Note.

Note 'Executable software' does not include complete binary images of the "software" running on an end-item.

Note to the Cryptography Note:

1. To meet paragraph a. of Note 3, all of the following must apply:
 - a. The item is of potential interest to a wide range of individuals and businesses; and
 - b. The price and information about the main functionality of the item are available before purchase without the need to consult the vendor or supplier. A simple price enquiry is not considered to be a consultation.
2. In determining eligibility of paragraph a. of Note 3, national authorities may take into account relevant factors such as quantity, price, required technical skill, existing sales channels, typical customers, typical use or any exclusionary practices of the supplier.

CRYPTOGRAPHIC "INFORMATION SECURITY"

8A502 "Information security" systems, equipment and components, as follows:

N.B. For "satellite navigation system" receiving equipment containing or employing decryption see 8A705., and for related decryption "software" and "technology" see 8D705., and 8E701.

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- a. Designed or modified to use 'cryptography for data confidentiality' having a 'described security algorithm', where that cryptographic capability is usable, has been activated, or can be activated by any means other than secure "cryptographic activation", as follows:
1. Items having "information security" as a primary function;
 2. Digital communication or networking systems, equipment or components, not specified in paragraph 8A502.a.1.;
 3. Computers, other items having information storage or processing as a primary function, and components therefor, not specified in paragraphs 8A502.a.1. or 8A502.a.2.;

N.B. For operating systems, see also 8D502.a.1. and 8D502.c.1.

4. Items, not specified in paragraphs 8A502.a.1. to a.3., where the 'cryptography for data confidentiality' having 'described security algorithm' meets all of the following:
 - a. It supports a non-primary function of the item; and
 - b. It is performed by incorporated equipment or "software" that would, as a standalone item, be specified by (8A502, 8A503, 8A504, 8B502, 8C5, 8D502 and 8E502)-Part 2.

Technical Notes

1. For the purposes of 8A502.a., 'cryptography for data confidentiality' means "cryptography" that employs digital techniques and performs any cryptographic function other than any of the following:
 - a. "Authentication";
 - b. Digital signature;
 - c. Data integrity;
 - d. Non-repudiation;
 - e. Digital rights management, including the execution of copy-protected "software";
 - f. Encryption or decryption in support of entertainment, mass commercial broadcasts or medical records management; or
 - g. Key management in support of any function described in paragraph a. to f. above.
2. For the purposes of 8A502.a., 'described security algorithm' means any of the following:
 - a. A "symmetric algorithm" employing a key length in excess of 56 bits, not including parity bits;
 - b. An "asymmetric algorithm" where the security of the algorithm is based on any of the following:
 1. Factorisation of integers in excess of 512 bits (e.g., RSA);
 2. Computation of discrete logarithms in a multiplicative group of a finite field of size greater than 512 bits (e.g., Diffie-Hellman over $\mathbb{Z}/p\mathbb{Z}$); or
 3. Discrete logarithms in a group other than mentioned in paragraph b.2. in excess of 112 bits (e.g., Diffie-Hellman over an elliptic curve); or
 - c. An "asymmetric algorithm" where the security of the algorithm is based on any of the following:
 1. Shortest vector or closest vector problems associated with lattices (e.g., NewHope, Frodo, NTRUEncrypt, Kyber, Titanium);
 2. Finding isogenies between Supersingular elliptic curves (e.g., Supersingular Isogeny Key Encapsulation); or
 3. Decoding random codes (e.g., McEliece, Niederreiter).

Technical Note

An algorithm described by Technical Note 2.c. may be referred to as being post-quantum, quantum-safe or quantum-resistant.

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Note 1 When necessary as determined by the appropriate authority in the exporter's country, details of items must be accessible and provided to the authority upon request, in order to establish any of the following:

- a. Whether the item meets the criteria of 8A502.a.1. to a.4.; or
- b. Whether the cryptographic capability for data confidentiality specified by 8A502.a. is usable without "cryptographic activation".

Note 2 8A502.a. does not apply to any of the following items, or specially designed "information security" components therefor:

- a. Smart cards and smart card 'readers/writers' as follows:
 1. A smart card or an electronically readable personal document (e.g., token coin, e-passport) that meets any of the following:
 - a. The cryptographic capability meets all of the following:
 1. It is restricted for use in any of the following:
 - a. Equipment or systems not described by 8A502.a.1. to a.4.;
 - b. Equipment or systems not using 'cryptography for data confidentiality' having 'described security algorithm'; or
 - c. Equipment or systems excluded from 8A502.a by entries b. to f. of this Note; and
 2. It cannot be reprogrammed for any other use; or
 - b. Having all of the following:
 1. It is specially designed and limited to allow protection of 'personal data' stored within;
 2. Has been, or can only be, personalized for public or commercial transactions or individual identification; and
 3. Where the cryptographic capability is not user-accessible;

Technical Note

'Personal data' includes any data specific to a particular person or entity, such as the amount of money stored and data necessary for "authentication".

2. 'Readers/writers' specially designed or modified, and limited, for items specified by paragraph a.1. of this Note;

Technical Note

'Readers/writers' include equipment that communicates with smart cards or electronically readable documents through a network.

- b. Cryptographic equipment specially designed and limited for banking use or 'money transactions';

Technical Note

'Money transactions' in 8A502 Note 2.b. includes the collection and settlement of fares or credit functions.

- c. Portable or mobile radiotelephones for civil use (e.g., for use with commercial civil cellular radio communication systems) that are not capable of transmitting encrypted data directly to another radiotelephone or equipment (other than Radio Access Network (RAN) equipment), nor of passing encrypted data through RAN equipment (e.g., Radio Network Controller (RNC) or Base Station Controller (BSC));
- d. Cordless telephone equipment not capable of end-to-end encryption where the maximum effective range of unboosted cordless operation (i.e., a single, unrelayed hop between terminal and home base station) is less than 400 metres according to

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the manufacturer's specifications;

- e. *Portable or mobile radiotelephones and similar client wireless devices for civil use, that implement only published or commercial cryptographic standards (except for anti-piracy functions, which may be non-published) and also meet the provisions of paragraphs a.2. to a.4. of the Cryptography Note (Note 3 in Category 5 – Part 2), that have been customised for a specific civil industry application with features that do not affect the cryptographic functionality of these original non-customised devices;*
- f. *Items, where the "information security" functionality is limited to wireless "personal area network" functionality implementing only published or commercial cryptographic standards;*
- g. *Mobile telecommunications Radio Access Network (RAN) equipment designed for civil use, which also meet the provisions of paragraphs a.2. to a.4. of the Cryptography Note (Note 3 in Category 5 – Part 2), having an RF output power limited to 0.1W (20 dBm) or less, and supporting 16 or fewer concurrent users;*
- h. *Routers, switches, gateways or relays, where the "information security" functionality is limited to the tasks of "Operations, Administration or Maintenance" ("OAM") implementing only published or commercial cryptographic standards;*
- i. *General purpose computing equipment or servers, where the "information security" functionality meets all of the following:*
 - 1. *Uses only published or commercial cryptographic standards; and*
 - 2. *Is any of the following:*
 - a. *Integral to a CPU that meets the provisions of Note 3 in (8A502, 8A503, 8A504, 8B502, 8C5, 8D502 and 8E502)-Part 2;*
 - b. *Integral to an operating system that is not specified by 5.D.2.; or*
 - c. *Limited to "OAM" of the equipment.*
- j. *Items specially designed for a 'connected civil industry application', meeting all of the following:*
 - 1. *Being any of the following:*
 - a. *A network-capable endpoint device meeting any of the following:*
 - 1. *The "information security" functionality is limited to securing 'non-arbitrary data' or the tasks of "Operations, Administration or Maintenance" ("OAM"); or*
 - 2. *The device is limited to a specific 'connected civil industry application'; or*
 - b. *Networking equipment meeting all of the following:*
 - 1. *Being specially designed to communicate with the devices specified by paragraph j.1.a. above; and*
 - 2. *The "information security" functionality is limited to supporting the 'connected civil industry application' of devices specified by paragraph j.1.a. above, or the tasks of "OAM" of this networking equipment or of other items specified by paragraph j. of this Note; and*
 - 2. *Where the "information security" functionality implements only published or commercial cryptographic standards, and the cryptographic functionality cannot easily be changed by the user.*

Technical Notes

- 1. *'Connected civil industry application' means a network-connected consumer or civil industry application other than "information security", digital communication, general purpose networking or computing.*
- 2. *'Non-arbitrary data' means sensor or metering data directly related to the stability, performance or physical measurement of a system (e.g., temperature, pressure, flow rate, mass, volume, voltage, physical location etc.), that cannot be changed by the user of the device..*

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- 8A502 b. Being a 'cryptographic activation token';
- Technical Note
A 'cryptographic activation token' is an item designed or modified for any of the following:
1. Converting, by means of "cryptographic activation", an item not specified by (8A502, 8A503, 8A504, 8B502, 8C5, 8D502 and 8E502)-Part 2 into an item specified by 8A502.a. or 8D502.c.1., and not released by the Cryptography Note (Note 3 in Category 5 – Part 2), or
2. Enabling, by means of "cryptographic activation", additional functionality specified by 8A502.a. of an item already specified by (8A502, 8A503, 8A504, 8B502, 8C5, 8D502 and 8E502)-Part 2;
- 8A502 c. Designed or modified to use or perform "quantum cryptography";
- Technical Note
"Quantum cryptography" is also known as Quantum Key Distribution (QKD).
- 8A502 d. Designed or modified to use cryptographic techniques to generate channelising codes, scrambling codes or network identification codes, for systems using ultra-wideband modulation techniques and having any of the following:
1. A bandwidth exceeding 500 MHz; or
2. A "fractional bandwidth" of 20% or more;
- 8A502 e. Designed or modified to use cryptographic techniques to generate the spreading code for "spread spectrum" systems, not specified by 5.A.2.d., including the hopping code for "frequency hopping" systems.

NON-CRYPTOGRAPHIC "INFORMATION SECURITY"

- 8A503 Systems, equipment and components, for non-cryptographic "information security", as follows:
- a. Communications cable systems designed or modified **to use** mechanical, electrical or electronic means to detect surreptitious intrusion;
- Note 8A503.a applies only to physical layer security. For the purpose of 8A503.a., the physical layer includes Layer 1 of the Reference Model of Open Systems Interconnection (OSI) (ISO/IEC 7498-1).
- b. Specially designed or modified to reduce the compromising emanations of information-bearing signals beyond what is necessary for health, safety or electromagnetic interference standards.

DEFEATING, WEAKENING OR BYPASSING "INFORMATION SECURITY"

- 8A504 Systems, equipment and components for defeating, weakening or bypassing "information security", as follows:
- a. Designed or modified to perform 'cryptanalytic functions'.
- Note 8A504.a includes systems or equipment, designed or modified to perform 'cryptanalytic functions' by means of reverse engineering.
- Technical Note
'Cryptanalytic functions' are functions designed to defeat cryptographic mechanisms in order to derive confidential variables or sensitive data, including clear text, passwords or cryptographic keys.
- 8A504 b. Items, not specified by 8A405 or 8A504.a., designed to perform all of the following:
1. 'Extract raw data' from a computing or communications device; and
2. Circumvent "authentication" or authorisation controls of the device, in order to perform the function described in 8A504.b.1.
- Technical Note
'Extract raw data' from a computing or communications device means to retrieve binary data from a storage medium, e.g. RAM, flash or hard disk, of the device without interpretation by the device's operating system or filesystem.

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Note 1 8A504.b. does not apply to systems or equipment specially designed for the "development" or "production" of a computing or communications device.

Note 2 8A504.b. does not include:
a. Debuggers, hypervisors;
b. Items limited to logical data extraction;
c. Data extraction items using chip-off or JTAG; or
d. Items specially designed and limited to jail-breaking or rooting.

8B5PART 2 - "INFORMATION SECURITY"(TEST, INSPECTION AND PRODUCTION EQUIPMENT)

8B502 "Information security" test, inspection and "production" equipment, as follows:

- a. Equipment specially designed for the "development" or "production" of equipment specified by 8A502, 8A503, 8A504 or 8B502.b;
- b. Measuring equipment specially designed to evaluate and validate the "information security" functions of equipment specified by 8A502, 8A503 or 8A504, or of "software" specified by 8D502.a or 8D502.c.

8C5 PART 2 - "INFORMATION SECURITY"(MATERIALS)- (Reserved)

8D5 PART 2 - "INFORMATION SECURITY"(SOFTWARE)

8D502 "Software" as follows:

- a. "Software" specially designed or modified for the "development", "production" or "use" of any of the following:
 1. Equipment specified by 8A502 or "software" specified by 8D502.c.1;
 2. Equipment specified by 8A503 or "software" specified by 8D502.c.2; or
 3. Equipment or "software", as follows:
 - a. Equipment specified by 8A504.a. or "software" specified by 8D502.c.3.a.;
 - b. Equipment specified by 8A504.b. or "software" specified by 8D502.c.3.b.;
- b. "Software" having the characteristics of a 'cryptographic activation token' specified by 8A502.b.
- c. "Software" having the characteristics of, or performing or simulating the functions of, any of the following:
 1. Equipment specified by 8A502.a., 8A502.c., 8A502.d. or 8A502.e.;

Note 8D502.c.1. does not apply to "software" limited to the tasks of "OAM" implementing only published or commercial cryptographic standards.

 2. Equipment specified by 8A503; or
 3. Equipment, as follows:
 - a. Equipment specified by 8A504.a.;
 - b. Equipment specified by 8A504.b.

Note 8D502.c.3.b. does not apply to "intrusion software".
- d. (Reserved)

8E5PART 2 - "INFORMATION SECURITY"(TECHNOLOGY)

8E502 "Technology" as follows:

- a. "Technology" according to the General Technology Note for the "development", "production" or "use" of equipment specified by 8A502, 8A503, 8A504 or 8B502, or of "software" specified by 8D502.a or 8D502.c.;
- Note 8E502.a. does not apply to "technology" for items specified by 8A504.b., 8D502.a.3.b. or 8D502.c.3.b.
- b. "Technology" having the characteristics of a 'cryptographic activation token' specified by 8A502.b.;

Note 8E502. includes "information security" technical data resulting from procedures carried out to evaluate or determine the implementation of functions, features or techniques specified in Category 8A5 Part 2.

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8A6 SENSORS AND LASERS (SYSTEMS, EQUIPMENT AND COMPONENTS)

ACOUSTICS

8A601 Acoustic systems, equipment and components, as follows:

8A601 a. Marine acoustic systems, equipment and specially designed components therefor, as follows:

1. Active (transmitting or transmitting-and-receiving) systems, equipment and specially designed components therefor, as follows:

Note 8A601.a.1 does not apply to equipment as follows:

- a. Depth sounders operating vertically below the apparatus, not including a scanning function exceeding $\pm 20^\circ$, and limited to measuring the depth of water, the distance of submerged or buried objects or fish finding;
- b. Acoustic beacons, as follows:
 1. Acoustic emergency beacons;
 2. Pingers specially designed for relocating or returning to an underwater position.

8A601 a. 1. a. Acoustic seabed survey equipment as follows:

1. Surface vessel survey equipment designed for seabed topographic mapping and having all of the following:
 - a. Designed to take measurements at an angle exceeding 20° from the vertical;
 - b. Designed to measure seabed topography at seabed depths exceeding 600 m;
 - c. 'Sounding resolution' less than 2; and
 - d. 'Enhancement' of the depth "accuracy" through compensation for all the following:
 1. Motion of the acoustic sensor;
 2. In-water propagation from sensor to the seabed and back; and
 3. Sound speed at the sensor;

Technical Notes

1. 'Sounding resolution' is the swath width (degrees) divided by the maximum number of soundings per swath.
2. 'Enhancement' includes the ability to compensate by external means.

8A601 a. 1. a. 2. Underwater survey equipment designed for seabed topographic mapping and having any of the following:

Technical Note

The acoustic sensor pressure rating determines the depth rating of the equipment specified by 8A601.a.1.a.2.

8A601 a. 1. a. 2. a. Having all of the following:

1. Designed or modified to operate at depths exceeding 300 m; and
2. 'Sounding rate' greater than 3,800 m/s; or

Technical Note

'Sounding rate' is the product of the maximum speed (m/s) at which the sensor can operate and the maximum number of soundings per swath assuming 100% coverage. For systems that produce soundings in two directions (3D sonars), the maximum of the 'sounding rate' in either direction should be used.

8A601 a. 1. a. 2. b. Survey equipment, not specified by 8A601.a.1.a.2.a, having all of the following:

1. Designed or modified to operate at depths exceeding 100 m;
2. Designed to take measurements at an angle exceeding 20° from the vertical;
3. Having any of the following:
 - a. Operating frequency below 350 kHz; or
 - b. Designed to measure seabed topography at a range exceeding 200 m from the acoustic sensor; and

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4. 'Enhancement' of the depth "accuracy" through compensation of all of the following:
 - a. Motion of the acoustic sensor;
 - b. In-water propagation from sensor to the seabed and back; and
 - c. Sound speed at the sensor.
- 8A601 a. 1. a. 3. Side Scan Sonar (SSS) or Synthetic Aperture Sonar (SAS), designed for seabed imaging and having all of the following, and specially designed transmitting and receiving acoustic arrays therefor:
- a. Designed or modified to operate at depths exceeding 500 m;
 - b. An 'area coverage rate' of greater than 570 m²/s while operating at the maximum range that it can operate with an 'along track resolution' of less than 15 cm; and
 - c. An 'across track resolution' of less than 15 cm;
- Technical Notes
1. 'Area coverage rate' (m²/s) is twice the product of the sonar range (m) and the maximum speed (m/s) at which the sensor can operate at that range.
 2. 'Along track resolution' (cm), for SSS only, is the product of azimuth (horizontal) beamwidth (degrees) and sonar range (m) and 0.873.
 3. 'Across track resolution' (cm) is 75 divided by the signal bandwidth (kHz).
- 8A601 a. 1. b. Systems or transmitting and receiving arrays, designed for object detection or location, having any of the following:
1. A transmitting frequency below 10 kHz;
 2. Sound pressure level exceeding 224 dB (reference 1 µPa at 1 m) for equipment with an operating frequency in the band from 10 kHz to 24 kHz inclusive;
 3. Sound pressure level exceeding 235 dB (reference 1 µPa at 1 m) for equipment with an operating frequency in the band between 24 kHz and 30 kHz;
 4. Forming beams of less than 1° on any axis and having an operating frequency of less than 100 kHz;
 5. Designed to operate with an unambiguous display range exceeding 5,120 m; or
 6. Designed to withstand pressure during normal operation at depths exceeding 1,000 m and having transducers with any of the following:
 - a. Dynamic compensation for pressure; or
 - b. Incorporating other than lead zirconate titanate as the transduction element;
- 8A601 a. 1. c. Acoustic projectors (including transducers), incorporating piezoelectric, magnetostrictive, electrostrictive, electrodynamic or hydraulic elements operating individually or in a designed combination, and having any of the following:
- Note 1 The status of acoustic projectors, including transducers, specially designed for other equipment not specified by 8A601 is determined by the status of the other equipment.
- Note 2 8A601.a.1.c does not apply to electronic sources which direct the sound vertically only, or mechanical (e.g. air gun or vapour-shock gun) or chemical (e.g. explosive) sources.
- Note 3 Piezoelectric elements specified in 8A601.a.1.c include those made from lead-magnesium-niobate/lead-titanate ($Pb(Mg_{1/3}Nb_{2/3})O_3$ - $PbTiO_3$, or PMN-PT) single crystals grown from solid solution or lead-indium-niobate/lead-magnesium niobate/lead-titanate ($Pb(In_{1/2}Nb_{1/2})O_3$ - $Pb(Mg_{1/3}Nb_{2/3})O_3$ - $PbTiO_3$, or PIN-PMN-PT) single crystals grown from solid solution.
- 8A601 a. 1. c. 1. Operating at frequencies below 10 kHz and having any of the following:
- a. Not designed for continuous operation at 100% duty cycle and having a radiated 'free-field Source Level (SL_{RMS})' exceeding (10log(f) + 169.77)dB (reference 1 µPa at 1 m) where f is the frequency in Hertz of maximum Transmitting Voltage Response (TVR) below 10 kHz; or
 - b. Designed for continuous operation at 100% duty cycle and having a continuously radiated 'free-field Source Level (SL_{RMS})' at 100% duty cycle exceeding (10log(f)

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+ 159.77)dB (reference 1 μ Pa at 1 m) where f is the frequency in Hertz of maximum Transmitting Voltage Response (TVR) below 10 kHz; or

Technical Note

The 'free-field Source Level (SL_{RMS})' is defined along the maximum response axis and in the far field of the acoustic projector. It can be obtained from the Transmitting Voltage Response using the following equation: $SL_{RMS} = (TVR + 20\log V_{RMS})$ dB (ref 1 μ Pa at 1 m), where SL_{RMS} is the source level, TVR is the Transmitting Voltage Response and V_{RMS} is the Driving Voltage of the Projector.

- | | | | | |
|-------|----|----|----|--|
| 8A601 | a. | 1. | c. | 2. (Reserved)
<u>N.B.</u> See 8A601.a.1.c.1 for items specified in 8A601.a.1.c.2.
3. Side-lobe suppression exceeding 22 dB; |
| 8A601 | a. | 1. | d. | Acoustic systems and equipment, designed to determine the position of surface vessels or underwater vehicles and having all of the following, and specially designed components therefor:
1. Detection range exceeding 1,000 m; and
2. Determined position error of less than 10 m rms (root mean square) when measured at a range of 1,000 m;
<u>Note</u> 8A601.a.1.d includes:
a. Equipment using coherent "signal processing" between two or more beacons and the hydrophone unit carried by the surface vessel or underwater vehicle;
b. Equipment capable of automatically correcting speed-of-sound propagation errors for calculation of a point. |
| 8A601 | a. | 1. | e. | Active individual sonars, specially designed or modified to detect, locate and automatically classify swimmers or divers, having all of the following, and specially designed transmitting and receiving acoustic arrays therefor:
1. Detection range exceeding 530 m;
2. Determined position error of less than 15 m rms (root mean square) when measured at a range of 530 m; and
3. Transmitted pulse signal bandwidth exceeding 3 kHz;
<u>N.B.</u> For diver detection systems specially designed or modified for military use, see Category 6
<u>Note</u> For 8A601.a.1.e, where multiple detection ranges are specified for various environments, the greatest detection range is used. |
| 8A601 | a. | 2. | | Passive systems, equipment and specially designed components therefor, as follows:
<u>Note</u> 8A601.a.2. also applies to receiving equipment, whether or not related in normal application to separate active equipment, and specially designed components therefor. |
| 8A601 | a. | 2. | a. | Hydrophones having any of the following:
<u>Note</u> The status of hydrophones specially designed for other equipment is determined by the status of the other equipment.
<u>Technical Notes</u>
1. Hydrophones consist of one or more sensing elements producing a single acoustic output channel. Those that contain multiple elements can be referred to as a hydrophone group.
2. For the purposes of 8A601.a.2.a., underwater acoustic transducers designed to operate as passive receivers are hydrophones. |
| 8A601 | a. | 2. | a. | 1. Incorporating continuous flexible sensing elements;
2. Incorporating flexible assemblies of discrete sensing elements with either a diameter or length less than 20 mm and with a separation between elements of less than 20 mm;
3. Having any of the following sensing elements:
a. Optical fibres; |

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- b. 'Piezoelectric polymer films' other than polyvinylidene-fluoride (PVDF) and its co-polymers {P(VDF-TrFE) and P(VDF-TFE)};
- c. 'Flexible piezoelectric composites';
- d. Lead-magnesium-niobate/lead-titanate (i.e, $\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3$ - PbTiO_3 , or PMN-PT) piezoelectric single crystals grown from solid solution; or
- e. Lead-indium-niobate/lead-magnesium niobate/lead-titanate (i.e, $\text{Pb}(\text{In}_{1/2}\text{Nb}_{1/2})\text{O}_3$ - $\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3$ - PbTiO_3 , or PIN-PMN-PT) piezoelectric single crystals grown from solid solution;
- 4. A 'hydrophone sensitivity' better than -180 dB at any depth with no acceleration compensation;
- 5. Designed to operate at depths exceeding 35 m with acceleration compensation; or
- 6. Designed for operation at depths exceeding 1,000 m and having a 'hydrophone sensitivity' better than -230 dB below 4 kHz;

Technical Notes

1. 'Piezoelectric polymer film' sensing elements consist of polarized polymer film that is stretched over and attached to a supporting frame or spool (mandrel).
2. 'Flexible piezoelectric composite' sensing elements consist of piezoelectric ceramic particles or fibres combined with an electrically insulating, acoustically transparent rubber, polymer or epoxy compound, where the compound is an integral part of the sensing elements.
3. 'Hydrophone sensitivity' is defined as twenty times the logarithm to the base 10 of the ratio of rms output voltage to a 1 V rms reference, when the hydrophone sensor, without a pre-amplifier, is placed in a plane wave acoustic field with an rms pressure of 1 μPa . For example, a hydrophone of -160 dB (reference 1 V per μPa) would yield an output voltage of 10^{-8} V in such a field, while one of -180 dB sensitivity would yield only 10^{-9} V output. Thus, -160 dB is better than -180 dB.

- 8A601 a. 2. b. Towed acoustic hydrophone arrays having any of the following:

Technical Note

Hydrophone arrays consist of a number of hydrophones providing multiple acoustic output channels.

- 8A601 a. 2. b. 1. Hydrophone group spacing of less than 12.5 m or 'able to be modified' to have hydrophone group spacing of less than 12.5 m;
3. Designed or 'able to be modified' to operate at depths exceeding 35 m;

Technical Note

'Able to be modified' in 8A601.a.2.b means having provisions to allow a change of the wiring or interconnections to alter hydrophone group spacing or operating depth limits. These provisions are: spare wiring exceeding 10% of the number of wires, hydrophone group spacing adjustment blocks or internal depth limiting devices that are adjustable or that control more than one hydrophone group.

- 8A601 a. 2. b. 3. Heading sensors specified by 8A601.a.2.d;
4. Longitudinally reinforced array hoses;
5. An assembled array of less than 40 mm in diameter;
6. (Reserved)
7. Hydrophone characteristics specified by 8A601.a.2.a; or
8. Accelerometer-based hydro-acoustic sensors specified by 8A601.a.2.g;

- 8A601 a. 2. c. Processing equipment, specially designed for towed acoustic hydrophone arrays, having "user-accessible programmability" and time or frequency domain processing and correlation, including spectral analysis, digital filtering and beamforming using Fast Fourier or other transforms or processes;

- 8A601 a. 2. d. Heading sensors having all of the following:
1. An "accuracy" of better than 0.5°; and
 2. Designed to operate at depths exceeding 35 m or having an adjustable or removable depth sensing device in order to operate at depths exceeding 35 m;

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N.B. For inertial heading systems, see 8A703.c.

- 8A601 a. 2. e. Bottom or bay-cable hydrophone arrays having any of the following:
1. Incorporating hydrophones specified by 8A601.a.2.a;
 2. Incorporating multiplexed hydrophone group signal modules having all of the following characteristics:
 - a. Designed to operate at depths exceeding 35 m or having an adjustable or removable depth sensing device in order to operate at depths exceeding 35 m; and
 - b. Capable of being operationally interchanged with towed acoustic hydrophone array modules; or
 3. Incorporating accelerometer-based hydro-acoustic sensors specified by 8A601.a.2.g;
- 8A601 a. 2. f. Processing equipment, specially designed for bottom or bay cable systems, having "user-accessible programmability" and time or frequency domain processing and correlation, including spectral analysis, digital filtering and beamforming using Fast Fourier or other transforms or processes;
- 8A601 a. 2. g. Accelerometer-based hydro-acoustic sensors having all of the following:
1. Composed of three accelerometers arranged along three distinct axes;
 2. Having an overall 'acceleration sensitivity' better than 48 dB (reference 1,000 mV rms per 1g);
 3. Designed to operate at depths greater than 35 meters; and
 4. Operating frequency below 20 kHz.

Note 8A601.a.2.g does not apply to particle velocity sensors or geophones.

Note 8A601.a.2 also applies to receiving equipment, whether or not related in normal application to separate active equipment, and specially designed components therefor.

Technical Notes

1. Accelerometer-based hydro-acoustic sensors are also known as vector sensors.
2. 'Acceleration sensitivity' is defined as twenty times the logarithm to the base 10 of the ratio of rms output voltage to a 1 V rms reference, when the hydro-acoustic sensor, without a preamplifier, is placed in a plane wave acoustic field with an rms acceleration of 1 g (i.e. 9.81 m/s²).

- 8A601 b. Correlation-velocity and Doppler-velocity sonar log equipment, designed to measure the horizontal speed of the equipment carrier relative to the sea bed, as follows:
1. Correlation-velocity sonar log equipment having any of the following characteristics:
 - a. Designed to operate at distances between the carrier and the sea bed exceeding 500 m; or
 - b. Having speed "accuracy" better than 1% of speed;
 2. Doppler-velocity sonar log equipment having speed "accuracy" better than 1% of speed.
- Note 1 8A601.b does not apply to depth sounders limited to any of the following:
- a. Measuring the depth of water;
 - b. Measuring the distance of submerged or buried objects; or
 - c. Fish finding.
- Note 2 8A601.b does not apply to equipment specially designed for installation on surface vessels.

- 8A601 c. (Reserved)

N.B. For diver deterrent acoustic systems, see 8A802.r.

OPTICAL SENSORS

- 8A602 Optical sensors or equipment and components therefor, as follows:

- a. Optical detectors as follows:

1. "Space-qualified" solid-state detectors as follows:

Note For the purpose of 8A602.a.1, solid-state detectors include "focal plane arrays".

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- 8A602 a. 1. a. "Space-qualified" solid-state detectors having all of the following:
1. A peak response in the wavelength range exceeding 10 nm but not exceeding 300 nm; and
 2. A response of less than 0.1% relative to the peak response at a wavelength exceeding 400 nm;
- b. "Space-qualified" solid-state detectors having all of the following:
1. A peak response in the wavelength range exceeding 900 nm but not exceeding 1,200 nm; and
 2. A response "time constant" of 95 ns or less;
- c. "Space-qualified" solid-state detectors having a peak response in the wavelength range exceeding 1,200 nm but not exceeding 30,000 nm;
- d. "Space-qualified" "focal plane arrays" having more than 2,048 elements per array and having a peak response in the wavelength range exceeding 300 nm but not exceeding 900 nm.
-
- 8A602 a. 2. Image intensifier tubes and specially designed components therefor, as follows:
- Note 8A602.a.2 does not apply to non-imaging photomultiplier tubes having an electron sensing device in the vacuum space limited solely to any of the following:*
- a. A single metal anode; *or*
 - b. Metal anodes with a centre to centre spacing greater than 500 µm.
- Technical Note*
'Charge multiplication' is a form of electronic image amplification and is defined as the generation of charge carriers as a result of an impact ionization gain process. 'Charge multiplication' sensors may take the form of an image intensifier tube, solid state detector or "focal plane array".
-
- 8A602 a. 2. a. Image intensifier tubes having all of the following:
1. A peak response in the wavelength range exceeding 400 nm but not exceeding 1,050 nm;
 2. Electron image amplification using any of the following:
 - a. A microchannel plate with a hole pitch (centre-to-centre spacing) of 12 µm or less; *or*
 - b. An electron sensing device with a non-binned pixel pitch of 500 µm or less, specially designed or modified to achieve 'charge multiplication' other than by a microchannel plate; *and*
 3. Any of the following photocathodes:
 - a. Multialkali photocathodes (e.g. S-20 and S-25) having a luminous sensitivity exceeding 350 µA/lm;
 - b. GaAs or GaInAs photocathodes; *or*
 - c. Other "III/V compound" semiconductor photocathodes having a maximum "radiant sensitivity" exceeding 10 mA/W;
-
- 8A602 a. 2. b. Image intensifier tubes having all of the following:
1. A peak response in the wavelength range exceeding 1,050 nm but not exceeding 1,800 nm;
 2. Electron image amplification using any of the following:
 - a. A microchannel plate with a hole pitch (centre-to-centre spacing) of 12 µm or less; *or*
 - b. An electron sensing device with a non-binned pixel pitch of 500 µm or less, specially designed or modified to achieve 'charge multiplication' other than by a microchannel plate; *and*
 3. "III/V compound" semiconductor (e.g. GaAs or GaInAs) photocathodes and transferred electron photocathodes, having a maximum "radiant sensitivity" exceeding 15 mA/W;
-
- 8A602 a. 2. c. Specially designed components as follows:
1. Microchannel plates having a hole pitch (centre-to-centre spacing) of 12 µm or less;
 2. An electron sensing device with a non-binned pixel pitch of 500 µm or less, specially designed or modified to achieve 'charge multiplication' other than by a microchannel plate;

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3. "III/V compound" semiconductor (e.g. GaAs or GaInAs) photocathodes and transferred electron photocathodes;

Note 8A602.a.2.c.3 does not apply to compound semiconductor photocathodes designed to achieve a maximum "radiant sensitivity" of any of the following:

- a. 10 mA/W or less at the peak response in the wavelength range exceeding 400 nm but not exceeding 1,050 nm; or
- b. 15 mA/W or less at the peak response in the wavelength range exceeding 1,050 nm but not exceeding 1,800 nm.

- 8A602 a. 3. Non-"space-qualified" "focal plane arrays" as follows:

N.B. 'Microbolometer' non-"space-qualified" "focal plane arrays" are only specified by 8A602.a.3.f.

Technical Note

Linear or two-dimensional multi-element detector arrays are referred to as "focal plane arrays";

Note 1 8A602.a.3 includes photoconductive arrays and photovoltaic arrays.

Note 2 8A602.a.3 does not apply to:

- a. Multi-element (not to exceed 16 elements) encapsulated photoconductive cells using either lead sulphide or lead selenide;
- b. Pyroelectric detectors using any of the following:
 1. Triglycine sulphate and variants;
 2. Lead-lanthanum-zirconium titanate and variants;
 3. Lithium tantalate;
 4. Polyvinylidene fluoride and variants; or
 5. Strontium barium niobate and variants.
- c. "Focal plane arrays" specially designed or modified to achieve 'charge multiplication' and limited by design to have a maximum "radiant sensitivity" of 10 mA/W or less for wavelengths exceeding 760 nm, having all of the following:
 1. Incorporating a response limiting mechanism designed not to be removed or modified; and
 2. Any of the following:
 - a. The response limiting mechanism is integral to or combined with the detector element; or
 - b. The "focal plane array" is only operable with the response limiting mechanism in place.

Technical Note

A response limiting mechanism integral to the detector element is designed not to be removed or modified without rendering the detector inoperable.

- d. Thermopile arrays having less than 5,130 elements;

- 8A602 a. 3. a. Non-"space-qualified" "focal plane arrays" having all of the following:

1. Individual elements with a peak response within the wavelength range exceeding 900 nm but not exceeding 1,050 nm; and
2. Any of the following:
 - a. A response "time constant" of less than 0.5 ns; or
 - b. Specially designed or modified to achieve 'charge multiplication' and having a maximum "radiant sensitivity" exceeding 10 mA/W;

- 8A602 a. 3. b. Non-"space-qualified" "focal plane arrays" having all of the following:

1. Individual elements with a peak response in the wavelength range exceeding 1,050 nm but not exceeding 1,200 nm; and
2. Any of the following:
 - a. A response "time constant" of 95 ns or less; or
 - b. Specially designed or modified to achieve 'charge multiplication' and having a maximum "radiant sensitivity" exceeding 10 mA/W;

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- 8A602 a. 3. c. Non-"space-qualified" non-linear (2-dimensional) "focal plane arrays" having individual elements with a peak response in the wavelength range exceeding 1,200 nm but not exceeding 30,000 nm;
- N.B. *Silicon and other material based 'microbolometer' non-"space-qualified" "focal plane arrays" are only specified by 8A602.a.3.f.*
- 8A602 a. 3. d. Non-"space-qualified" linear (1-dimensional) "focal plane arrays" having all of the following :
1. Individual elements with a peak response in the wavelength range exceeding 1,200 nm but not exceeding 3,000 nm; and
- 8A602 a. 3. d. 2. Any of the following:
- a. A ratio of 'scan direction' dimension of the detector element to the 'cross-scan direction' dimension of the detector element of less than 3.8; or
- b. Signal processing in the detector elements;
- Note 8A602.a.3.d does not apply to "focal plane arrays" (not to exceed 32 elements) having detector elements limited solely to germanium material.
- Technical Note
For the purposes of 8A602.a.3.d, 'cross-scan direction' is defined as the axis parallel to the linear array of detector elements and the 'scan direction' is defined as the axis perpendicular to the linear array of detector elements.
- 8A602 a. 3. e. Non-"space-qualified" linear (1-dimensional) "focal plane arrays" having individual elements with a peak response in the wavelength range exceeding 3,000 nm but not exceeding 30,000 nm;
- 8A602 a. 3. f. Non-"space-qualified" non-linear (2-dimensional) infrared "focal plane arrays" based on 'microbolometer' material having individual elements with an unfiltered response in the wavelength range equal to or exceeding 8,000 nm but not exceeding 14,000 nm;
- Technical Note
For the purposes of 8A602.a.3.f, 'microbolometer' is defined as a thermal imaging detector that, as a result of a temperature change in the detector caused by the absorption of infrared radiation, is used to generate any usable signal.
- 8A602 a. 3. g. Non-"space-qualified" "focal plane arrays" having all of the following:
1. Individual detector elements with a peak response in the wavelength range exceeding 400 nm but not exceeding 900 nm;
2. Specially designed or modified to achieve 'charge multiplication' and having a maximum "radiant sensitivity" exceeding 10 mA/W for wavelengths exceeding 760 nm; and
3. Greater than 32 elements;
- 8A602 b. "Monospectral imaging sensors" and "multispectral imaging sensors", designed for remote sensing applications and having any of the following:
1. An Instantaneous-Field-Of-View (IFOV) of less than 200 μ rad (microradians); or
2. Specified for operation in the wavelength range exceeding 400 nm but not exceeding 30,000 nm and having all the following:
- a. Providing output imaging data in digital format; and
- b. Having any of the following characteristics:
1. "Space-qualified"; or
2. Designed for airborne operation, using other than silicon detectors, and having an IFOV of less than 2.5 mrad (milliradians);
- Note 8A602.b.1 does not apply to "monospectral imaging sensors" with a peak response in the wavelength range exceeding 300 nm but not exceeding 900 nm and only incorporating any of the following non-"space-qualified" detectors or non-"space-qualified" "focal plane arrays":
- a. Charge Coupled Devices (CCD) not designed or modified to achieve 'charge multiplication'; or

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- b. Complementary Metal Oxide Semiconductor (CMOS) devices not designed or modified to achieve 'charge multiplication'.

- 8A602 c. 'Direct view' imaging equipment incorporating any of the following:
1. Image intensifier tubes having the characteristics listed in 8A602.a.2.a or 8A602.a.2.b;
 2. "Focal plane arrays" having the characteristics listed in 8A602.a.3; or
 3. Solid state detectors specified by 8A602.a.1;

Technical Note

'Direct view' refers to imaging equipment that presents a visual image to a human observer without converting the image into an electronic signal for television display, and that cannot record or store the image photographically, electronically or by any other means.

Note 8A602.c does not apply to equipment as follows, when incorporating other than GaAs or GaInAs photocathodes:

- a. Industrial or civilian intrusion alarm, traffic or industrial movement control or counting systems;
- b. Medical equipment;
- c. Industrial equipment used for inspection, sorting or analysis of the properties of materials;
- d. Flame detectors for industrial furnaces;
- e. Equipment specially designed for laboratory use.

- 8A602 d. Special support components for optical sensors, as follows:
1. "Space-qualified" cryocoolers;
 2. Non-"space-qualified" cryocoolers having a cooling source temperature below 218 K (-55°C), as follows:
 - a. Closed cycle type with a specified Mean-Time-To-Failure (MTTF) or Mean-Time-Between-Failures (MTBF), exceeding 2,500 hours;
 - b. Joule-Thomson (JT) self-regulating minicoolers having bore (outside) diameters of less than 8 mm;
 3. Optical sensing fibres specially fabricated either compositionally or structurally, or modified by coating, to be acoustically, thermally, inertially, electromagnetically or nuclear radiation sensitive.

Note 8A602.d.3 does not apply to encapsulated optical sensing fibres specially designed for bore hole sensing applications.

- 8A602 e. (Reserved)
- 8A602 f. 'Read-out integrated circuits' ('ROIC') specially designed for "focal plane arrays" specified by 8A602.a.3.

Note: 8A602.f does not apply to 'read-out integrated circuits' specially designed for civil automotive applications.

Technical Note

A 'Read-Out Integrated Circuit' ('ROIC') is an integrated circuit designed to underlie or be bonded to a "focal plane array" ("FPA") and used to read-out (i.e., extract and register) signals produced by the detector elements. At a minimum the 'ROIC' reads the charge from the detector elements by extracting the charge and applying a multiplexing function in a manner that retains the relative spatial position and orientation information of the detector elements for processing inside or outside the 'ROIC'

CAMERAS

- 8A603 Cameras, systems or equipment, and components therefor, as follows:

- a. Instrumentation cameras and specially designed components therefor, as follows:

Note Instrumentation cameras, specified by 8A603.a.3 to 8A603.a.5, with modular structures should be evaluated by their maximum capability, using plug-ins available according to the camera manufacturer's specifications.

- 8A603 a. 1. (Reserved)

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2. (Reserved);
 3. Electronic streak cameras having temporal resolution better than 50 ns;
 4. Electronic framing cameras having a speed exceeding 1,000,000 frames/s;
 5. Electronic cameras having all of the following:
 - a. An electronic shutter speed (gating capability) of less than 1 μ s per full frame; and
 - b. A read out time allowing a framing rate of more than 125 full frames per second;
 6. Plug-ins having all of the following characteristics:
 - a. Specially designed for instrumentation cameras which have modular structures and which are specified by 8A603.a; and
 - b. Enabling these cameras to meet the characteristics specified by 8A603.a.3, 8A603.a.4 or 8A603.a.5, according to the manufacturer's specifications;
- 8A603 b. Imaging cameras as follows:
- Note 8A603.b does not apply to television or video cameras, specially designed for television broadcasting.
1. Video cameras incorporating solid state sensors, having a peak response in the wavelength range exceeding 10 nm, but not exceeding 30,000 nm and having all of the following:
- 8A603 b. 1. a. Having any of the following:
1. More than 4×10^6 "active pixels" per solid state array for monochrome (black and white) cameras;
 2. More than 4×10^6 "active pixels" per solid state array for colour cameras incorporating three solid state arrays; or
 3. More than 12×10^6 "active pixels" for solid state array colour cameras incorporating one solid state array; and
- b. Having any of the following:
1. Optical mirrors specified by 8A604.a;
 2. Optical control equipment specified by 8A604.d; or
 3. The capability for annotating internally generated 'camera tracking data';
- Technical Notes
1. For the purpose of this entry, digital video cameras should be evaluated by the maximum number of "active pixels" used for capturing moving images.
 2. For the purpose of this entry, 'camera tracking data' is the information necessary to define camera line of sight orientation with respect to the earth. This includes: 1) the horizontal angle the camera line of sight makes with respect to the earth's magnetic field direction and; 2) the vertical angle between the camera line of sight and the earth's horizon.
- 8A603 b. 2. Scanning cameras and scanning camera systems, having all of the following:
- a. A peak response in the wavelength range exceeding 10 nm, but not exceeding 30,000 nm;
 - b. Linear detector arrays with more than 8,192 elements per array; and
 - c. Mechanical scanning in one direction;
- Note 8A603.b.2 does not apply to scanning cameras and scanning camera systems, specially designed for any of the following:
- a. Industrial or civilian photocopiers;
 - b. Image scanners specially designed for civil, stationary, close proximity scanning applications (e.g. reproduction of images or print contained in documents, artwork or photographs); or
 - c. Medical equipment.
- 8A603 b. 3. Imaging cameras incorporating image intensifier tubes having the characteristics listed in 8A602.a.2.a or 8A602.a.2.b;
- 8A603 b. 4. Imaging cameras incorporating "focal plane arrays" having any of the following:
- a. Incorporating "focal plane arrays" specified by 8A602.a.3.a to 8A602.a.3.e;
 - b. Incorporating "focal plane arrays" specified by 8A602.a.3.f; or
 - c. Incorporating "focal plane arrays" specified by 8A602.a.3.g;

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Note 1 Imaging cameras specified by 8A603.b.4 include "focal plane arrays" combined with sufficient "signal processing" electronics, beyond the read out integrated circuit, to enable as a minimum the output of an analogue or digital signal once power is supplied.

Note 2 8A603.b.4.a does not apply to imaging cameras incorporating linear "focal plane arrays" with 12 elements or fewer, not employing time-delay-and-integration within the element and designed for any of the following:

- a. Industrial or civilian intrusion alarm, traffic or industrial movement control or counting systems;
- b. Industrial equipment used for inspection or monitoring of heat flows in buildings, equipment or industrial processes;
- c. Industrial equipment used for inspection, sorting or analysis of the properties of materials;
- d. Equipment specially designed for laboratory use; or
- e. Medical equipment.

Note 3 8A603.b.4.b does not apply to imaging cameras having any of the following:

- a. A maximum frame rate equal to or less than 9 Hz;
- b. Having all of the following:
 1. 'Having all of the following:
Having a minimum horizontal or vertical 'Instantaneous-Field-of-View (IFOV)' of at least 2 mrad (milliradians);
 2. Incorporating a fixed focal-length lens that is not designed to be removed;
 3. Not incorporating a 'direct view' display; and

Technical Note

'Direct view' refers to an imaging camera operating in the infrared spectrum that presents a visual image to a human observer using a near-to-eye micro display incorporating any light-security mechanism.

4. Having any of the following:
 - a. No facility to obtain a viewable image of the detected field-of-view; or
 - b. The camera is designed for a single kind of application and designed not to be user modified; or

Technical Note

'Instantaneous Field of View (IFOV)' specified in Note 3.b. is the lesser figure of the 'Horizontal IFOV' or the 'Vertical IFOV'.

'Horizontal IFOV' = horizontal Field of View (FOV)/number of horizontal detector elements

'Vertical IFOV' = vertical Field of View (FOV)/number of vertical detector elements.

- c. The camera is specially designed for installation into a civilian passenger land vehicle and having all of the following:
 1. The placement and configuration of the camera within the vehicle are solely to assist the driver in the safe operation of the vehicle;
 2. Is only operable when installed in any of the following:
 - a. The civilian passenger land vehicle for which it was intended and the vehicle weighs less than 4,500 kg (gross vehicle weight); or
 - b. A specially designed, authorized maintenance test facility; and
 3. Incorporates an active mechanism that forces the camera not to function when it is removed from the vehicle for which it was intended.

Note When necessary, details of the item will be provided, upon request, to the appropriate authority in the exporter's country in order to ascertain compliance with the conditions described in Note 3.b.4 and Note 3.c. above.

Note 4 8A603.b.4.c does not apply to imaging cameras having any of the following characteristics:

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- a. Having all of the following:
 - 1. Where the camera is specially designed for installation as an integrated component into indoor and wall-plug-operated systems or equipment, limited by design for a single kind of application, as follows:
 - a. Industrial process monitoring, quality control, or analysis of the properties of materials;
 - b. Laboratory equipment specially designed for scientific research;
 - c. Medical equipment;
 - d. Financial fraud detection equipment; and
 - 2. Is only operable when installed in any of the following:
 - a. The system(s) or equipment for which it was intended; or
 - b. A specially designed, authorised maintenance facility; and
 - 3. Incorporates an active mechanism that forces the camera not to function when it is removed from the system(s) or equipment for which it was intended;
- b. Where the camera is specially designed for installation into a civilian passenger land vehicle or passenger and vehicle ferries, and having all of the following:
 - 1. The placement and configuration of the camera within the vehicle or ferry is solely to assist the driver or operator in the safe operation of the vehicle or ferry;
 - 2. Is only operable when installed in any of the following:
 - a. The civilian passenger land vehicle for which it was intended and the vehicle weighs less than 4,500 kg (gross vehicle weight);
 - b. The passenger and vehicle ferry for which it was intended and having a length overall (LOA) 65 m or greater; or
 - c. A specially designed, authorised maintenance test facility; and
 - 3. Incorporates an active mechanism that forces the camera not to function when it is removed from the vehicle for which it was intended;
- c. Limited by design to have a maximum "radiant sensitivity" of 10 mW or less for wavelengths exceeding 760 nm, having all of the following:
 - 1. Incorporating a response limiting mechanism designed not to be removed or modified;
 - 2. Incorporates an active mechanism that forces the camera not to function when the response limiting mechanism is removed; and
 - 3. Not specially designed or modified for underwater use; or
- d. Having all of the following:
 - 1. Not incorporating a 'direct view' or electronic image display;
 - 2. Has no facility to output a viewable image of the detected field of view;
 - 3. The "focal plane array" is only operable when installed in the camera for which it was intended; and
 - 4. The "focal plane array" incorporates an active mechanism that forces it to be permanently inoperable when removed from the camera for which it was intended.

Note When necessary, details of the item will be provided, upon request, to the appropriate authority in the exporter's country in order to ascertain compliance with the conditions described in Note 4 above.

8A603 b. 5. Imaging cameras incorporating solid-state detectors specified by 8A602.a.1.

OPTICS

8A604 Optical equipment and components, as follows:

- a. Optical mirrors (reflectors) as follows:

Technical Note

For the purpose of 8A604.a, Laser Induced Damage Threshold (LIDT) is measured according to ISO 21254-1:2011.

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8A604 a. 1. "Deformable mirrors" having an active optical aperture greater than 10 mm and having any of the following, and specially designed components therefor:

- a. Having all the following:
 - 1. A mechanical resonant frequency of 750 Hz or more; and
 - 2. More than 200 actuators; or
- b. A Laser Induced Damage Threshold (LIDT) being any of the following:
 - 1. Greater than 1 kW/cm² using a "CW laser"; or
 - 2. Greater than 2 J/cm² using 20 ns "laser" pulses at 20 Hz repetition rate;

Technical Note

'Deformable mirrors' are mirrors having any of the following:

- a. *A single continuous optical reflecting surface which is dynamically deformed by the application of individual torques or forces to compensate for distortions in the optical waveform incident upon the mirror; or*
- b. *Multiple optical reflecting elements that can be individually and dynamically repositioned by the application of torques or forces to compensate for distortions in the optical waveform incident upon the mirror.*

'Deformable mirrors' are also known as adaptive optic mirrors;

8A604 a. 2. Lightweight monolithic mirrors having an average "equivalent density" of less than 30 kg/m² and a total mass exceeding 10 kg;

8A604 a. 3. Lightweight "composite" or foam mirror structures having an average "equivalent density" of less than 30 kg/m² and a total mass exceeding 2 kg;

Note 8A604.a.2 and 8A604.a.3 do not apply to mirrors specially designed to direct solar radiation for terrestrial heliostat installations.

8A604 a. 4. Mirrors specially designed for beam steering mirror stages specified in 8A604.d.2.a with a flatness of $\lambda/10$ or better (λ is equal to 633 nm) and having any of the following:

- a. Diameter or major axis length greater than or equal to 100 mm; or
- b. Having all of the following:
 - 1. Diameter or major axis length greater than 50 mm but less than 100 mm; and
 - 2. A Laser Induced Damage Threshold (LIDT) being any of the following:
 - a. Greater than 10 kW/cm² using a "CW laser"; or
 - b. Greater than 20 J/cm² using 20 ns "laser" pulses at 20 Hz repetition rate;

N.B. For optical mirrors specially designed for lithography equipment, see 8B301.

8A604 b. Optical components made from zinc selenide (ZnSe) or zinc sulphide (ZnS) with transmission in the wavelength range exceeding 3,000 nm but not exceeding 25,000 nm and having any of the following:

- 1. Exceeding 100 cm³ in volume; or
- 2. Exceeding 80 mm in diameter or length of major axis and 20 mm in thickness (depth);

8A604 c. "Space-qualified" components for optical systems, as follows:

- 1. Components lightweighted to less than 20% "equivalent density" compared with a solid blank of the same aperture and thickness;
- 2. Raw substrates, processed substrates having surface coatings (single-layer or multi-layer, metallic or dielectric, conducting, semiconducting or insulating) or having protective films;
- 3. Segments or assemblies of mirrors designed to be assembled in space into an optical system with a collecting aperture equivalent to or larger than a single optic 1 m in diameter;
- 4. Components manufactured from "composite" materials having a coefficient of linear thermal expansion, in any coordinate direction, equal to or less than $5 \times 10^{-6}/K$;

8A604 d. Optical control equipment as follows:

- 1. Equipment specially designed to maintain the surface figure or orientation of the "space-qualified" components specified by 8A604.c.1 or 8A604.c.3;
- 2. Steering, tracking, stabilisation and resonator alignment equipment as follows:

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- a. Beam steering mirror stages designed to carry mirrors having diameter or major axis length greater than 50 mm and having all of the following, and specially designed electronic control equipment therefor:
 - 1. A maximum angular travel of ± 26 mrad or more;
 - 2. A mechanical resonant frequency of 500 Hz or more; and
 - 3. An angular "accuracy" of 10 μ rad (microradians) or less (better);
 - b. Resonator alignment equipment having bandwidths equal to or more than 100 Hz and an "accuracy" of 10 μ rad or less (better);
- 8A604 d. 3. Gimbals having all of the following:
- a. A maximum slew exceeding 5° ;
 - b. A bandwidth of 100 Hz or more;
 - c. Angular pointing errors of 200 μ rad (microradians) or less; and
 - d. Having any of the following:
 - 1. Exceeding 0.15 m but not exceeding 1 m in diameter or major axis length and capable of angular accelerations exceeding 2 rad (radians)/s²; or
 - 2. Exceeding 1 m in diameter or major axis length and capable of angular accelerations exceeding 0.5 rad (radians)/s²;
- 8A604 d. 4. (Reserved)
- 8A604 e. 'Aspheric optical elements' having all of the following:
- 1. Largest dimension of the optical-aperture greater than 400 mm;
 - 2. Surface roughness less than 1 nm (rms) for sampling lengths equal to or greater than 1 mm; and
 - 3. Coefficient of linear thermal expansion's absolute magnitude less than $3 \times 10^{-6}/K$ at 25°C.
- Technical Notes
- 1. An 'aspheric optical element' is any element used in an optical system whose imaging surface or surfaces are designed to depart from the shape of an ideal sphere.
 - 2. Manufacturers are not required to measure the surface roughness listed in 8A604.e.2 unless the optical element was designed or manufactured with the intent to meet, or exceed, the specified parameter.
- Note 8A604.e does not apply to 'aspheric optical elements' having any of the following:
- a. Largest optical-aperture dimension less than 1 m and focal length to aperture ratio equal to or greater than 4.5:1;
 - b. Largest optical-aperture dimension equal to or greater than 1 m and focal length to aperture ratio equal to or greater than 7:1;
 - c. Designed as Fresnel, flyeye, stripe, prism or diffractive optical elements;
 - d. Fabricated from borosilicate glass having a coefficient of linear thermal expansion greater than $2.5 \times 10^{-6}/K$ at 25 °C; or
 - e. An x-ray optical element having inner mirror capabilities (e.g, tube-type mirrors).
- N.B. For 'aspheric optical elements' specially designed for lithography equipment, see 8B301.
- 8A604 f. Dynamic wavefront measuring equipment having all of the following:
- 1. 'Frame rates' equal to or more than 1 kHz; and
 - 2. A wavefront accuracy equal to or less (better) than $\lambda/20$ at the designed wavelength.

Technical Note

For the purposes of 8A604.f, 'frame rate' is a frequency at which all "active pixels" in the "focal plane array" are integrated for recording images projected by the wavefront sensor optics'.

LASERS

- 8A605 "Lasers", components and optical equipment, as follows:

Note 1 Pulsed "lasers" include those that run in a continuous wave (CW) mode with pulses superimposed.

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Note 2 Excimer, semiconductor, chemical, CO, CO₂, and 'non-repetitive pulsed' Nd:glass "lasers" are only specified by 8A605.d.

Technical Note

'Non-repetitive pulsed' refers to "lasers" that produce either a single output pulse or that have a time interval between pulses exceeding one minute.

Note 3 8A605 includes fibre "lasers".

Note 4 The status of "lasers" incorporating frequency conversion (i.e, wavelength change) by means other than one "laser" pumping another "laser" is determined by applying the specified parameters for both the output of the source "laser" and the frequency-converted optical output.

Note 5 8A605 does not apply to "lasers" as follows:

- a. Ruby with output energy below 20 J;
- b. Nitrogen;
- c. Krypton.

Note 6 'For the purposes of 8A605.a. and 8A605. b., 'single transverse mode' refers to "lasers" with a beam profile having an M2-factor of less than 1.3, while 'multiple transverse mode' refers to "lasers" with a beam profile having an M2-factor of 1.3 or higher

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| 8A605 | <ol style="list-style-type: none"> a. Non-"tunable" continuous wave "(CW) lasers" having any of the following: <ol style="list-style-type: none"> 1. Output wavelength less than 150 nm and output power exceeding 1 W; 2. Output wavelength of 150 nm or more but not exceeding 510 nm and output power exceeding 30 W; <p><u>Note</u> 8A605.a.2. does not apply to Argon "lasers" having an output power equal to or less than 50W.</p> 3. Output wavelength exceeding 510 nm but not exceeding 540 nm and any of the following: <ol style="list-style-type: none"> a. 'Single transverse mode' output and output power exceeding 50 W; or b. 'Multiple transverse mode' output and output power exceeding 150 W; 4. Output wavelength exceeding 540 nm but not exceeding 800 nm and output power exceeding 30W; |
| 8A605 | <ol style="list-style-type: none"> a. 5. Output wavelength exceeding 800 nm but not exceeding 975 nm and any of the following: <ol style="list-style-type: none"> a. 'Single transverse mode' output and output power exceeding 50 W; <u>or</u> b. 'Multiple transverse mode' output and output power exceeding 80 W; |
| 8A605 | <ol style="list-style-type: none"> a. 6. Output wavelength exceeding 975 nm but not exceeding 1,150 nm and any of the following: <ol style="list-style-type: none"> a. 'Single transverse mode' output and any of the following: <ol style="list-style-type: none"> 1. Output power exceeding 1,000 W; or 2. Having all of the following: <ol style="list-style-type: none"> a. Output power exceeding 500 W; and b. Spectral bandwidth less than 40 GHz; <u>or</u> b. 'Multiple transverse mode' output and any of the following: <ol style="list-style-type: none"> 1. 'Wall-plug efficiency' exceeding 18% and output power exceeding 1000 W; <u>or</u> 2. Output power exceeding 2 kW; |

Note 1 8A605.a.6.b. does not apply to 'multiple transverse mode', industrial "lasers" with output power exceeding 2 kW and not exceeding 6 kW with a total mass greater than 1,200 kg. For the purpose of this note, total mass includes all components required to operate the "laser", e.g., "laser", power supply, heat exchanger, but excludes external optics for beam conditioning or delivery.

Note 2 8A605.a.6.b. does not apply to 'multiple transverse mode', industrial "lasers" having any of the following:

- a. (Reserved)
- b. Output power exceeding 1 kW but not exceeding 1.6 kW and having a BPP exceeding 1.25 mm•mrad;

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- c. Output power exceeding 1.6 kW but not exceeding 2.5 kW and having a BPP exceeding 1.7 mm•mrad;
- d. Output power exceeding 2.5 kW but not exceeding 3.3 kW and having a BPP exceeding 2.5 mm•mrad;
- e. Output power exceeding 3.3 kW but not exceeding 6 kW and having a BPP exceeding 3.5 mm•mrad;
- f. (Reserved)
- g. (Reserved)
- h. Output power exceeding 6 kW but not exceeding 8 kW and having a BPP exceeding 12 mm•mrad; or
- i. Output power exceeding 8 kW but not exceeding 10 kW and having a BPP exceeding 24 mm•mrad;

Technical Note

'Wall-plug efficiency' is defined as the ratio of "laser" output power (or "average output power") to total electrical input power required to operate the "laser", including the power supply/conditioning and thermal conditioning/heat exchanger.

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| 8A605 | a. 7. Output wavelength exceeding 1,150 nm but not exceeding 1,555 nm and any of the following:
a. 'Single transverse mode' and output power exceeding 50 W; <u>or</u>
b. 'Multiple transverse mode' and output power exceeding 80 W; <u>or</u> |
| 8A605 | a. 8. Output wavelength exceeding 1,555 nm but not exceeding 1,850 nm, and output power exceeding 1 W; |
| 8A605 | a. 9. Output wavelength exceeding 1,850 nm but not exceeding 2,100 nm, and any of the following:
a. 'Single transverse mode' and output power exceeding 1 W; <u>or</u>
b. 'Multiple transverse mode' output and output power exceeding 120 W; <u>or</u> |
| 8A605 | a.10. Output wavelength exceeding 2,100 nm and output power exceeding 1 W; |
| 8A605 | b. Non-"tunable" "pulsed lasers" having any of the following:

1. Output wavelength less than 150 nm and any of the following:
a. Output energy exceeding 50 mJ per pulse and "peak power" exceeding 1 W; <u>or</u>
b. "Average output power" exceeding 1 W;

2. Output wavelength of 150 nm or more but not exceeding 510 nm and any of the following:
a. Output energy exceeding 1.5 J per pulse and "peak power" exceeding 30W; <u>or</u>
b. "Average output power" exceeding 30 W; |

Note 8A605.b.2.b does not apply to Argon "lasers" having an "average output power" equal to or less than 50 W.

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| 8A605 | b. 3. Output wavelength exceeding 510 nm but not exceeding 540 nm and any of the following:
a. 'Single transverse mode' output and any of the following:
1. Output energy exceeding 1.5 J per pulse and "peak power" exceeding 50 W; <u>or</u>
2. "Average output power" exceeding 80 W; <u>or</u>
b. 'Multiple transverse mode' output and any of the following:
1. Output energy exceeding 1.5 J per pulse and "peak power" exceeding 150 W; <u>or</u>
2. "Average output power" exceeding 150 W; |
| 8A605 | b. 4. Output wavelength exceeding 540 nm but not exceeding 800 nm and any of the following:
a. "Pulse duration" less than 1 ps and any of the following: |

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- 1. Output energy exceeding 0.005 J per pulse and "peak power" exceeding 5 GW; or
 - 2. "Average output power" exceeding 20 W; or
 - b. "Pulse duration" equal to or exceeding 1 ps and any of the following:
 - 1. Output energy exceeding 1.5 J per pulse and "peak power" exceeding 30 W; or
 - 2. "Average output power" exceeding 30 W;
- 8A605
- b. 5. Output wavelength exceeding 800 nm but not exceeding 975 nm and any of the following:
 - a. "Pulse duration" less than 1 ps and any of the following:
 - 1. Output energy exceeding 0.005 J per pulse and "peak power" exceeding 5 GW; or
 - 2. 'Single transverse mode' output and "average output power" exceeding 20 W;
 - b. "Pulse duration" equal to or exceeding 1 ps and not exceeding 1 μ s and any of the following:
 - 1. Output energy exceeding 0.5 J per pulse and "peak power" exceeding 50 W;
 - 2. 'Single transverse mode' output and "average output power" exceeding 20 W; or
 - 3. Multiple transverse mode' output and "average output power" exceeding 50 W; or
 - c. "Pulse duration" exceeding 1 μ s and any of the following:
 - 1. Output energy exceeding 2 J per pulse and "peak power" exceeding 50 W;
 - 2. Single transverse mode' output and "average output power" exceeding 50 W; or
 - 3. 'Multiple transverse mode' output and "average output power" exceeding 80 W;
- 8A605
- b. 6. Output wavelength exceeding 975 nm but not exceeding 1,150 nm and any of the following:
 - a. "Pulse duration" of less than 1 ps, and any of the following:
 - 1. Output "peak power" exceeding 2 GW per pulse;
 - 2. "Average output power" exceeding 30 W; or
 - 3. Output energy exceeding 0.002 J per pulse;
 - b. "Pulse duration" equal to or exceeding 1 ps and less than 1 ns, and any of the following:
 - 1. Output "peak power" exceeding 5 GW per pulse;
 - 2. "Average output power" exceeding 50 W; or
 - 3. Output energy exceeding 0.1 J per pulse;
 - c. "Pulse duration" equal to or exceeding 1 ns but not exceeding 1 ms and any of the following:
 - 1. 'Single transverse mode' output and any of the following:
 - a. "Peak power" exceeding 100 MW;
 - b. "Average output power" exceeding 20 W limited by design to a maximum pulse repetition frequency less than or equal to 1 kHz;
 - c. 'Wall-plug efficiency' exceeding 12%, "average output power" exceeding 100 W and capable of operating at a pulse repetition frequency greater than 1 kHz;
 - d. "Average output power" exceeding 150 W and capable of operating at a pulse repetition frequency greater than 1 kHz; or
 - e. Output energy exceeding 2 J per pulse; or
- 8A605
- b. 6. c. 2. 'Multiple transverse mode' output and any of the following:
 - a. "Peak power" exceeding 400 MW;
 - b. 'Wall-plug efficiency' exceeding 18% and "average output power" exceeding 500 W;
 - c. "Average output power" exceeding 2 kW; or
 - d. Output energy exceeding 4 J per pulse; or
- 8A605
- b. 6. d. "Pulse duration" exceeding 1 μ s and any of the following:

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1. 'Single transverse mode' output and any of the following:
 - a. "Peak power" exceeding 500 kW;
 - b. 'Wall-plug efficiency' exceeding 12% and "average output power" exceeding 100 W; or
 - c. "Average output power" exceeding 150 W; or
 2. 'Multiple transverse mode' output and any of the following:
 - a. "Peak power" exceeding 1 MW;
 - b. 'Wall-plug efficiency' exceeding 18% and "average output power" exceeding 500 W; or
 - c. "Average output power" exceeding 2 kW;
- 8A605 b. 7. Output wavelength exceeding 1,150 nm but not exceeding 1,555 nm, and any of the following:
- a. "Pulse duration" not exceeding 1 μ s and any of the following:
 1. Output energy exceeding 0.5 J per pulse and "peak power" exceeding 50 W;
 2. 'Single transverse mode' output and "average output power" exceeding 20 W; or
 3. 'Multiple transverse mode' output and "average output power" exceeding 50 W; or
 - b. "Pulse duration" exceeding 1 μ s and any of the following:
 1. Output energy exceeding 2 J per pulse and "peak power" exceeding 50 W;
 2. 'Single transverse mode' output and "average output power" exceeding 50 W; or
 3. 'Multiple transverse mode' output and "average output power" exceeding 80 W; or
- 8A605 b. 8. Output wavelength exceeding 1,555 nm but not exceeding 1,850 nm, and any of the following:
- a. Output energy exceeding 100 mJ per pulse and "peak power" exceeding 1 W; or
 - b. "Average output power" exceeding 1 W;
- 8A605 b. 9. Output wavelength exceeding 1,850 nm but not exceeding 2,100 nm, and any of the following:
- a. 'Single transverse mode' and any of the following:
 1. Output energy exceeding 100 mJ per pulse and "peak power" exceeding 1 W; or
 2. "Average output power" exceeding 1 W; or
 - b. 'Multiple transverse mode' and any of the following:
 1. Output energy exceeding 100 mJ per pulse and "peak power" exceeding 10 kW; or
 2. "Average output power" exceeding 120 W; or
- 8A605 b. 10. Output wavelength exceeding 2,100 nm and any of the following:
- a. Output energy exceeding 100 mJ per pulse and "peak power" exceeding 1 W; or
 - b. "Average output power" exceeding 1 W;
- 8A605 c. "Tunable" "lasers" having any of the following:
1. Output wavelength less than 600 nm and any of the following:
 - a. Output energy exceeding 50 mJ per pulse and "peak power" exceeding 1 W; or
 - b. Average or CW output power exceeding 1 W;
- Note 8A605.c.1 does not apply to dye "lasers" or other liquid "lasers", having a multimode output and a wavelength of 150 nm or more but not exceeding 600 nm and all of the following:*
1. Output energy less than 1.5 J per pulse or a "peak power" less than 20 W; and
 2. Average or CW output power less than 20 W.

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- | | |
|-------|--|
| 8A605 | <ul style="list-style-type: none"> c. 2. Output wavelength of 600 nm or more but not exceeding 1,400 nm, and any of the following: <ul style="list-style-type: none"> a. Output energy exceeding 1 J per pulse and "peak power" exceeding 20 W; <u>or</u> b. Average or CW output power exceeding 20 W; <u>or</u> 3. Output wavelength exceeding 1,400 nm and any of the following: <ul style="list-style-type: none"> a. Output energy exceeding 50 mJ per pulse and "peak power" exceeding 1 W; <u>or</u> b. Average or CW output power exceeding 1 W; |
| 8A605 | <ul style="list-style-type: none"> d. Other "lasers", not specified by 8A605.a, 8A605.b or 8A605.c as follows: <ul style="list-style-type: none"> 1. Semiconductor "lasers" as follows: <p style="margin-left: 40px;"><u>Note 1</u> 8A605.d.1 includes semiconductor "lasers" having optical output connectors (e.g. fibre optic pigtails).</p> <p style="margin-left: 40px;"><u>Note 2</u> The status of semiconductor "lasers" specially designed for other equipment is determined by the status of the other equipment.</p> |
| 8A605 | <ul style="list-style-type: none"> d. 1. a. Individual single-transverse mode semiconductor "lasers" having any of the following: <ul style="list-style-type: none"> 1. Wavelength equal to or less than 1,570 nm and average or CW output power, exceeding 2 W; <u>or</u> 2. Wavelength greater than 1,570 nm and average or CW output power, exceeding 500 mW; |
| 8A605 | <ul style="list-style-type: none"> d. 1. b. Individual, multiple-transverse mode semiconductor "lasers" having any of the following: <ul style="list-style-type: none"> 1. Wavelength of less than 1,400 nm and average or CW output power, exceeding 25 W; 2. Wavelength equal to or greater than 1,400 nm and less than 1,900 nm and average or CW output power, exceeding 2.5 W; <u>or</u> 3. Wavelength equal to or greater than 1,900 nm and average or CW output power, exceeding 1 W; |
| 8A605 | <ul style="list-style-type: none"> d. 1. c. Individual semiconductor "laser" 'bars' having any of the following: <ul style="list-style-type: none"> 1. Wavelength of less than 1,400 nm and average or CW output power, exceeding 100 W; 2. Wavelength equal to or greater than 1,400 nm and less than 1,900 nm and average or CW output power, exceeding 25 W; <u>or</u> 3. Wavelength equal to or greater than 1,900 nm and average or CW output power, exceeding 10 W; |
| 8A605 | <ul style="list-style-type: none"> d. 1. d. Semiconductor "laser" 'stacked arrays' (two-dimensional arrays) having any of the following: <ul style="list-style-type: none"> 1. Wavelength less than 1,400 nm and having any of the following: <ul style="list-style-type: none"> a. Average or CW total output power less than 3 kW and having average or CW output 'power density' greater than 500 W/cm²; b. Average or CW total output power equal to or exceeding 3 kW but less than or equal to 5 kW, and having average or CW output 'power density' greater than 350W/cm²; c. Average or CW total output power exceeding 5 kW; d. Peak pulsed 'power density' exceeding 2,500 W/cm²; <u>or</u> <p style="margin-left: 40px;"><u>Note</u> 8A605.d.1.d.1.d does not apply to epitaxially-fabricated monolithic devices.</p> e. Spatially coherent average or CW total output power, greater than 150 W; |
| 8A605 | <ul style="list-style-type: none"> d. 1. d. 2. Wavelength greater than or equal to 1,400 nm but less than 1,900 nm, and having any of the following: <ul style="list-style-type: none"> a. Average or CW total output power less than 250 W and average or CW output 'power density' greater than 150 W/cm²; b. Average or CW total output power equal to or exceeding 250 W but less than or equal to 500 W, and having average or CW output 'power density' greater than 50W/cm²; c. Average or CW total output power exceeding 500 W; d. Peak pulsed 'power density' exceeding 500 W/cm²; <u>or</u> |

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Note 8A605.d.1.d.2.d does not apply to epitaxially-fabricated monolithic devices.

- e. Spatially coherent average or CW total output power, exceeding 15 W;

- 8A605 d. 1. d. 3. Wavelength greater than or equal to 1,900 nm and having any of the following:
a. Average or CW output 'power density' greater than 50 W/cm²;
b. Average or CW output power greater than 10 W; or
c. Spatially coherent average or CW total output power, exceeding 1.5 W; or

- 8A605 d. 1. d. 4. At least one "laser" 'bar' specified by 8A605.d.1.c;

Technical Note

For the purposes of 8A605.d.1.d, 'power density' means the total "laser" output power divided by the emitter surface area of the 'stacked array'.

- 8A605 d. 1. e. Semiconductor "laser" 'stacked arrays', other than those specified by 8A605.d.1.d, having all of the following:

1. Specially designed or modified to be combined with other 'stacked arrays' to form a larger 'stacked array'; and
2. Integrated connections, common for both electronics and cooling;

Note 1 'Stacked arrays', formed by combining semiconductor "laser" 'stacked arrays' specified by 8A605.d.1.e, that are not designed to be further combined or modified are specified by 8A605.d.1.d.

Note 2 'Stacked arrays', formed by combining semiconductor "laser" 'stacked arrays' specified by 8A605.d.1.e, that are designed to be further combined or modified are specified by 8A605.d.1.e.

Note 3 8A605.d.1.e does not apply to modular assemblies of single 'bars' designed to be fabricated into end-to-end stacked linear arrays.

Technical Notes

1. Semiconductor "lasers" are commonly called "laser" diodes.
2. A 'bar' (also called a semiconductor "laser" 'bar', a "laser" diode 'bar' or diode 'bar') consists of multiple semiconductor "lasers" in a one-dimensional array.
3. A 'stacked array' consists of multiple 'bars' forming a two-dimensional array of semiconductor "lasers".

- 8A605 d. 2. Carbon monoxide (CO) "lasers" having any of the following:
a. Output energy exceeding 2 J per pulse and "peak power" exceeding 5 kW; or
b. Average or CW output power exceeding 5 kW;

- 8A605 d. 3. Carbon dioxide (CO₂) "lasers" having any of the following:
a. CW output power exceeding 15 kW;
b. Pulsed output with a "pulse duration" exceeding 10 µs and any of the following:
 1. "Average output power" exceeding 10 kW; or
 2. "Peak power" exceeding 100 kW; or
c. Pulsed output with a "pulse duration" equal to or less than 10 µs and any of the following:
 1. Pulse energy exceeding 5 J per pulse; or
 2. "Average output power" exceeding 2.5 kW;

- 8A605 d. 4. Excimer "lasers" having any of the following:
a. Output wavelength not exceeding 150 nm and any of the following:
 1. Output energy exceeding 50 mJ per pulse; or
 2. "Average output power" exceeding 1 W;
b. Output wavelength exceeding 150 nm but not exceeding 190 nm and any of the following:
 1. Output energy exceeding 1.5 J per pulse; or
 2. "Average output power" exceeding 120 W;

- 8A605 d. 4. c. Output wavelength exceeding 190 nm but not exceeding 360 nm and any of the following:
 1. Output energy exceeding 10 J per pulse; or

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2. "Average output power" exceeding 500 W; or
- 8A605 d. 4. d. Output wavelength exceeding 360 nm and any of the following:
1. Output energy exceeding 1.5 J per pulse; or
2. "Average output power" exceeding 30 W;
- N.B. For excimer "lasers" specially designed for lithography equipment, see 8B301.
- 8A605 d. 5. "Chemical lasers" as follows:
- a. Hydrogen Fluoride (HF) "lasers";
- b. Deuterium Fluoride (DF) "lasers";
- c. "Transfer lasers" as follows:
1. Oxygen Iodine (O₂-I) "lasers";
2. Deuterium Fluoride-Carbon dioxide (DF-CO₂) "lasers";
- Technical Note
- 'Transfer lasers' are "lasers" in which the lasing species are excited through the transfer of energy by collision of a non-lasing atom or molecule with a lasing atom or molecule species.
- 8A605 d. 6. 'Non-repetitive pulsed' Nd: glass "lasers" having any of the following:
- a. "Pulse duration" not exceeding 1 µs and output energy exceeding 50 J per pulse; or
- b. "Pulse duration" exceeding 1 µs and output energy exceeding 100 J per pulse;
- 8A605 e. Components as follows:
1. Mirrors cooled either by 'active cooling' or by heat pipe cooling;
- Technical Notes
- 'Active cooling' is a cooling technique for optical components using flowing fluids within the subsurface (nominally less than 1 mm below the optical surface) of the optical component to remove heat from the optic.
2. Optical mirrors or transmissive or partially transmissive optical or electro-optical components, other than fused tapered fibre combiners and Multi-Layer Dielectric gratings (MLDs), specially designed for use with specified "lasers";
- Note Fibre combiners and MLDs are specified by 8A605.e.3.
- 8A605 e. 3. Fibre "laser" components as follows:
- a. Multimode to multimode fused tapered fibre combiners having all of the following:
1. An insertion loss better (less) than or equal to 0.3 dB maintained at a rated total average or CW output power (excluding output power transmitted through the single mode core if present) exceeding 1,000 W; and
2. Number of input fibres equal to or greater than 3;
- b. Single mode to multimode fused tapered fibre combiners having all of the following:
1. An insertion loss better (less) than 0.5 dB maintained at a rated total average or CW output power exceeding 4,600 W;
2. Number of input fibres equal to or greater than 3; and
- 8A605 e. 3. b. 3. Having any of the following:
- a. A Beam Parameter Product (BPP) measured at the output not exceeding 1.5 mm mrad for a number of input fibres less than or equal to 5; or
- b. A BPP measured at the output not exceeding 2.5 mm mrad for a number of input fibres greater than 5;
- 8A605 e. 3. c. MLDs having all of the following:
1. Designed for spectral or coherent beam combination of 5 or more fibre "lasers"; and
2. CW "Laser" Induced Damage Threshold (LIDT) greater than or equal to 10 kW/cm².
- 8A605 f. Optical equipment as follows:
- N.B. For shared aperture optical elements, capable of operating in "Super-High Power Laser" ("SHPL") applications, see 6A019. Note 2.d.
1. (Reserved)

N.B. For items previously specified by 8A605.f.1, see 8A604.f.

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2. "Laser" diagnostic equipment specially designed for dynamic measurement of "SHPL" system angular beam steering errors and having an angular "accuracy" of 10 μ rad (microradians) or less (better);
 3. Optical equipment and components, specially designed for coherent beam combination in a phased-array "SHPL" system and having any of the following:
 - a. An "accuracy" of 0.1 μ m or less, for wavelengths greater than 1 μ m; or
 - b. An "accuracy" of $\lambda/10$ or less (better) at the designed wavelength, for wavelengths equal to or less than 1 μ m.
 4. Projection telescopes specially designed for use with "SHPL" systems;
- 8A605 g. 'Laser acoustic detection equipment' having all of the following:
1. CW "laser" output power equal to or exceeding 20 mW;
 2. "Laser" frequency stability equal to or better (less) than 10 MHz;
 3. "Laser" wavelengths equal to or exceeding 1,000 nm but not exceeding 2,000 nm;
 4. Optical system resolution better (less) than 1 nm; and
 5. Optical Signal to Noise ratio equal to or exceeding 10^3 .

Technical Note

'Laser acoustic detection equipment' is sometimes referred to as a "Laser" Microphone or Particle Flow Detection Microphone.

MAGNETIC AND ELECTRIC FIELD SENSORS

- 8A606 "Magnetometers", "magnetic gradiometers", "intrinsic magnetic gradiometers", underwater electric field sensors, "compensation systems", and specially designed components therefor, as follows:
- Note 8A606 does not apply to instruments specially designed for fishery applications or biomagnetic measurements for medical diagnostics.
- 8A606 a. "Magnetometers" and subsystems, as follows:
1. "Magnetometers" using "superconductive" (SQUID) "technology" and having any of the following:
 - a. SQUID systems designed for stationary operation, without specially designed subsystems designed to reduce in-motion noise, and having a 'sensitivity' equal to or lower (better) than 50 fT (rms) per square root Hz at a frequency of 1 Hz; or
 - b. SQUID systems having an in-motion-magnetometer 'sensitivity' lower (better) than 20 pT (rms) per square root Hz at a frequency of 1 Hz and specially designed to reduce in-motion noise;
 2. "Magnetometers" using optically pumped or nuclear precession (proton/Overhauser) "technology" having a 'sensitivity' lower (better) than 20 pT (rms) per square root Hz at a frequency of 1 Hz;
 3. "Magnetometers" using fluxgate "technology" having a 'sensitivity' equal to or lower (better) than 10 pT (rms) per square root Hz at a frequency of 1 Hz;
- 8A606 a. 4. Induction coil "magnetometers" having a 'sensitivity' lower (better) than any of the following:
- a. 0.05 nT (rms)/square root Hz at frequencies of less than 1 Hz;
 - b. 1×10^{-3} nT (rms)/square root Hz at frequencies of 1 Hz or more but not exceeding 10 Hz; or
 - c. 1×10^{-4} nT (rms)/square root Hz at frequencies exceeding 10 Hz;
5. Fibre optic "magnetometers" having a 'sensitivity' lower (better) than 1 nT (rms) per square root Hz;
- 8A606 b. Underwater Electric Field Sensors having a 'sensitivity' lower (better) than 8 nanovolt per meter per square root Hz when measured at 1 Hz;
- 8A606 c. "Magnetic gradiometers" as follows:
1. "Magnetic gradiometers" using multiple "magnetometers" specified by 8A606.a;

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- 2. Fibre optic "intrinsic magnetic gradiometers" having a magnetic gradient field 'sensitivity' lower (better) than 0.3 nT/m (rms) per square root Hz;
- 3. "Intrinsic magnetic gradiometers", using "technology" other than fibre-optic "technology", having a magnetic gradient field 'sensitivity' lower (better) than 0.015 nT/m (rms) per square root Hz;
- 8A606 d. "Compensation systems" for magnetic or underwater electric field sensors resulting in a performance equal to or better than the specified parameters of 8A606.a, 8A606.b, or 8A606.c;
- 8A606 e. Underwater electromagnetic receivers incorporating magnetic field sensors specified by 8A606.a or underwater electric field sensors specified by 8A606.b.

Technical Note

For the purposes of 8A606, 'sensitivity' (noise level) is the root mean square of the device limited noise floor which is the lowest signal that can be measured.

GRAVIMETERS

- 8A607 Gravity meters (gravimeters) and gravity gradiometers, as follows:
 - a. Gravity meters designed or modified for ground use and having a static "accuracy" of less (better) than 10 μ Gal;

Note 8A607.a does not apply to ground gravity meters of the quartz element (Worden) type.
 - b. Gravity meters designed for mobile platforms and having all of the following:
 - 1. A static "accuracy" of less (better) than 0.7 mGal; and
 - 2. An in-service (operational) "accuracy" of less (better) than 0.7 mGal having a 'time-to-steady-state registration' of less than 2 minutes under any combination of attendant corrective compensations and motional influences;

Technical Note
For the purposes of 8A607.b, 'time-to-steady-state registration' (also referred to as the gravimeter's response time) is the time over which the disturbing effects of platform induced accelerations (high frequency noise) are reduced.
 - c. Gravity gradiometers.

RADAR

- 8A608 Radar systems, equipment and assemblies, having any of the following, and specially designed components therefor:

Note 8A608 does not apply to:

 - Secondary Surveillance Radar (SSR);
 - Civil Automotive Radar;
 - Displays or monitors used for Air Traffic Control (ATC);
 - Meteorological (weather) Radar;
 - Precision Approach Radar (PAR) equipment conforming to ICAO standards and employing electronically steerable linear (1-dimensional) arrays or mechanically positioned passive antennae.

Technical Notes

 - 1. For the purposes of 8A608., 'marine radar' is a radar that is designed to navigate safely at sea, inland waterways or near-shore environments.
 - 2. For the purposes of 8A608., 'vessel traffic service' is a vessel traffic monitoring and control service similar to air traffic control for "aircraft".
- 8A608 a. Operating at frequencies from 40 GHz to 230 GHz and having any of the following:
 - 1. An average output power exceeding 100 mW; or
 - 2. Locating "accuracy" of 1 m or less (better) in range and 0.2 degree or less (better) in azimuth;
- b. A tunable bandwidth exceeding $\pm 6.25\%$ of the 'centre operating frequency';

Technical Note
The 'centre operating frequency' equals one half of the sum of the highest plus the lowest specified operating frequencies.
- c. Capable of operating simultaneously on more than two carrier frequencies;

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- d. Capable of operating in synthetic aperture (SAR), inverse synthetic aperture (ISAR) radar mode, or sidelooking airborne (SLAR) radar mode;
 - e. Incorporating electronically scanned array antennae;
Technical Note
Electronically scanned array antennae are also known as electronically steerable array antennae.
 - f. Capable of heightfinding non-cooperative targets;
 - g. Specially designed for airborne (balloon or airframe mounted) operation and having Doppler "signal processing" for the detection of moving targets;
- 8A608 h. Employing processing of radar signals and using any of the following:
- 1. "Radar spread spectrum" techniques; or
 - 2. "Radar frequency agility" techniques;
- 8A608 i. Providing ground-based operation with a maximum 'instrumented range' exceeding 185 km;
- Note 8A608.i does not apply to:
- a. Fishing ground surveillance radar;
 - b. Ground radar equipment specially designed for enroute air traffic control and having all of the following:
 - 1. A maximum 'instrumented range' of 500 km or less;
 - 2. Configured so that radar target data can be transmitted only one way from the radar site to one or more civil ATC centres;
 - 3. Contains no provisions for remote control of the radar scan rate from the enroute ATC centre; and
 - 4. Permanently installed.
 - c. Weather balloon tracking radars.
- Technical Note
 For the purposes of 8A608.i., 'instrumented range' is the specified unambiguous display range of a radar.
- 8A608 j. Being "laser" radar or Light Detection and Ranging (LIDAR) equipment and having any of the following:
- 1. "Space-qualified";
 - 2. Employing coherent heterodyne or homodyne detection techniques and having an angular resolution of less (better) than 20 µrad (microradians); or
 - 3. Designed for carrying out airborne bathymetric littoral surveys to International Hydrographic Organization (IHO) Order 1a Standard (5th Edition February 2008) for Hydrographic Surveys or better, and using one or more "lasers" with a wavelength exceeding 400 nm but not exceeding 600 nm;
- Note 1 LIDAR equipment specially designed for surveying is only specified by 8A608.j.3.
- Note 2 8A608.j does not apply to LIDAR equipment specially designed for meteorological observation.
- Note 3 Parameters in the IHO Order 1a Standard 5th Edition February 2008 are summarized as follows:
- Horizontal Accuracy (95% Confidence Level) = 5 m + 5% of depth.
- Depth Accuracy for Reduced Depths (95% confidence level)
 = $\pm\sqrt{a^2 + (b*d)^2}$ where:
- a = 0.5 m = constant depth error, i.e. the sum of all constant depth errors
 b = 0.013 = factor of depth dependent error
 b*d = depth dependent error, i.e. the sum of all depth dependent errors
 d = depth
- Feature Detection
 = Cubic features > 2 m in depths up to 40 m;
 10% of depth beyond 40 m.
- 8A608 k. Having "signal processing" sub-systems using "pulse compression" and having any of the following:
- 1. A "pulse compression" ratio exceeding 150; or

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2. A compressed pulse width of less than 200 ns; or
- Note 8A608.k.2 does not apply to two dimensional 'marine radar' or 'vessel traffic service' radar, having all of the following:
- "Pulse compression" ratio not exceeding 150;
 - Compressed pulse width of greater than 30 ns;
 - Single and rotating mechanically scanned antenna;
 - Peak output power not exceeding 250 W; and
 - Not capable of "frequency hopping".

- 8A608 1. Having data processing sub-systems and having any of the following:
- 'Automatic target tracking' providing, at any antenna rotation, the predicted target position beyond the time of the next antenna beam passage; or

Note: 8A608.l.1 does not apply to conflict alert capability in ATC systems, or 'marine radar'.

Technical Note

'Automatic target tracking' is a processing technique that automatically determines and provides as output an extrapolated value of the most probable position of the target in real time.

- (Reserved) ;
- (Reserved) ;
- Configured to provide superposition and correlation, or fusion, of target data within six seconds from two or more 'geographically dispersed' radar sensors to improve the aggregate performance beyond that of any single sensor specified by 8A608.f or 8A608.i.

Technical Note

Sensors are considered 'geographically dispersed' when each location is distant from any other more than 1,500 m in any direction. Mobile sensors are always considered 'geographically dispersed'.

N.B. See also 6A005.b.

Note: 8A608.l does not apply to systems, equipment and assemblies **designed** for 'vessel traffic services'

8B6 SENSORS AND LASERS (TEST, INSPECTION AND PRODUCTION EQUIPMENT)

8B601 ACOUSTICS – (Reserved)

. OPTICAL SENSORS –

8B602 Masks and reticles, specially designed for optical sensors specified by 8A602.a.1.b or 8A602.a.1.d

8B603. CAMERAS – (Reserved)

OPTICS

8B604 Optical equipment as follows:

- Equipment for measuring absolute reflectance to an "accuracy" of equal to or better than 0.1% of the reflectance value;
- Equipment other than optical surface scattering measurement equipment, having an unobscured aperture of more than 10 cm, specially designed for the non-contact optical measurement of a non-planar optical surface figure (profile) to an "accuracy" of 2 nm or less (better) against the required profile.

Note 8B604 does not apply to microscopes.

8B605 LASERS – (Reserved)

8B606 MAGNETIC AND ELECTRIC FIELD SENSORS – (Reserved)

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GRAVIMETERS

8B607 Equipment to produce, align and calibrate land-based gravity meters with a static "accuracy" of **less (better) than** 0.1 mGal.

RADAR

8B608 Pulse radar cross-section measurement systems having transmit pulse widths of 100 ns or less, and specially designed components therefor.

8C6 SENSORS AND LASERS (MATERIALS)

8C601 ACOUSTICS – (Reserved)

OPTICAL SENSORS

8C602 Optical sensor materials as follows:

- a. Elemental tellurium (Te) of purity levels of 99.9995% or more;
- b. Single crystals (including epitaxial wafers) of any of the following:
 1. Cadmium zinc telluride (CdZnTe) with zinc content of less than 6% by 'mole fraction';
 2. Cadmium telluride (CdTe) of any purity level; or
 3. Mercury cadmium telluride (HgCdTe) of any purity level.

Technical Note

'Mole fraction' is defined as the ratio of moles of ZnTe to the sum of the moles of CdTe and ZnTe present in the crystal.

8C603 CAMERAS – (Reserved)

OPTICS

8C604 Optical materials as follows:

- a. Zinc selenide (ZnSe) and zinc sulphide (ZnS) "substrate blanks", produced by the chemical vapour deposition process and having any of the following:
 1. A volume greater than 100 cm³; or
 2. A diameter greater than 80 mm and a thickness of 20 mm or more;
- b. Electro-optic materials and non-linear optical materials, as follows:
 1. Potassium titanyl arsenate (KTA) (CAS 59400-80-5);
 2. Silver gallium selenide (AgGaSe₂, also known as AGSE) (CAS 12002-67-4);
 3. Thallium arsenic selenide (Tl₃AsSe₃, also known as TAS) (CAS 16142-89-5);
 4. Zinc germanium phosphide (ZnGeP₂, also known as ZGP, zinc germanium biphosphide or zinc germanium diphosphide); or
 5. Gallium selenide (GaSe) (CAS 12024-11-2);

- 8C604
- c. Non-linear optical materials, other than those specified by 8C604.b, having any of the following:
 1. Having all of the following:
 - a. Dynamic (also known as non-stationary) third order non-linear susceptibility ($\chi^{(3)}$, chi 3) of 10⁻⁶m²/V² or more; and
 - b. Response time of less than 1 ms; or
 2. Second order non-linear susceptibility ($\chi^{(2)}$, chi 2) of 3.3×10⁻¹¹ m/V or more;

8C604 d. "Substrate blanks" of silicon carbide or beryllium beryllium (Be/Be) deposited materials, exceeding 300 mm in diameter or major axis length;

8C604 e. Glass, including fused silica, phosphate glass, fluorophosphate glass, zirconium fluoride (ZrF₄) (CAS

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7783-64-4) and hafnium fluoride (HfF₄) (CAS 13709-52-9) and having all of the following:

1. A hydroxyl ion (OH⁻) concentration of less than 5 ppm;
2. Integrated metallic purity levels of less than 1 ppm; and
3. High homogeneity (index of refraction variance) less than 5×10^{-6} ;

- 8C604 f. Synthetically produced diamond material with an absorption of less than 10^{-5} cm^{-1} for wavelengths exceeding 200 nm but not exceeding 14,000 nm.

LASERS

8C605 "Laser" materials as follows:

- a. Synthetic crystalline "laser" host material in unfinished form as follows:
 1. Titanium doped sapphire.
 2. (Reserved)
- b. Rare-earth-metal doped double-clad fibres having any of the following:
 1. Nominal "laser" wavelength of 975 nm to 1,150 nm and having all of the following:
 - a. Average core diameter equal to or greater than 25 μm ; and
 - b. Core 'Numerical Aperture' ('NA') less than 0.065; or

Note 8C605.b.1 does not apply to double-clad fibres having an inner glass cladding diameter exceeding 150 μm and not exceeding 300 μm .
 2. Nominal "laser" wavelength exceeding 1,530 nm and having all of the following:
 - a. Average core diameter equal to or greater than 20 μm ; and
 - b. Core 'NA' less than 0.1.

Technical Notes

1. For the purposes of 8C605, the core 'Numerical Aperture' ('NA') is measured at the emission wavelengths of the fibre.
2. 8C605.b. includes fibres assembled with end caps.

8C606 MAGNETIC AND ELECTRIC FIELD SENSORS – (Reserved)

8C607 GRAVIMETERS – (Reserved)

8C608 RADAR – (Reserved)

8D6 SENSORS AND LASERS (SOFTWARE)

8D601. "Software" specially designed for the "development" or "production" of equipment specified by 8A604, 8A605, 8A608 or 8B608.

8D602. "Software" specially designed for the "use" of equipment specified by 8A602.b, 8A608 or 8B608.

8D603 Other "software" as follows:

ACOUSTICS

- 8D603 a. "Software" as follows:
1. "Software" specially designed for acoustic beam forming for the "real-time processing" of acoustic data for passive reception using towed hydrophone arrays;
 2. "Source code" for the "real-time processing" of acoustic data for passive reception using towed hydrophone arrays;
 3. "Software" specially designed for acoustic beam forming for the "real-time processing" of acoustic data for passive reception using bottom or bay cable systems;
 4. "Source code" for the "real-time processing" of acoustic data for passive reception using bottom or bay cable systems;
 5. "Software" or "source code", specially designed for all of the following:
 - a. "Real-time processing" of acoustic data from sonar systems specified by 8A601.a.1.e; and
 - b. Automatically detecting, classifying and determining the location of divers or

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swimmers;

N.B. For diver detection "software" or "source code", specially designed or modified for military use, see Category 6.

8D603 b. OPTICAL SENSORS – (Reserved)

CAMERAS

8D603 c. "Software" designed or modified for cameras incorporating "focal plane arrays" specified by 8A602.a.3.f and designed or modified to remove a frame rate restriction and allow the camera to exceed the frame rate specified in 8A603.b.4 Note 3.a.

OPTICS

8D603 d. "Software" specially designed to maintain the alignment and phasing of segmented mirror systems consisting of mirror segments having a diameter or major axis length equal to or larger than 1 m;

8D603 e. LASERS – (Reserved)

MAGNETIC AND ELECTRIC FIELD SENSORS

8D603 f. "Software" as follows:

1. "Software" specially designed for magnetic and electric field "compensation systems" for magnetic sensors designed to operate on mobile platforms;
2. "Software" specially designed for magnetic and electric field anomaly detection on mobile platforms;
3. "Software" specially designed for "real-time processing" of electromagnetic data using underwater electromagnetic receivers specified by 8A606.e;
4. "Source code" for "real-time processing" of electromagnetic data using underwater electromagnetic receivers specified by 8A606.e;

GRAVIMETERS

8D603 g. "Software" specially designed to correct motional influences of gravity meters or gravity gradiometers;

RADAR

8D603 h. 1. **Air Traffic Control (ATC) "software" designed to be hosted on general purpose computers located at Air Traffic Control centres and capable of accepting radar target data from more than four primary radars.**

2. "Software" for the design or "production" of radomes having all of the following:

- a. Specially designed to protect the electronically scanned array antennae specified by 8A608.e; and
- b. Resulting in an antenna pattern having an 'average side lobe level' more than 40 dB below the peak of the main beam level.

Technical Note

Average side lobe level' in 8D603.h.2.b is measured over the entire array excluding the angular extent of the main beam and the first two side lobes on either side of the main beam.

8E6 **SENSORS AND LASERS (TECHNOLOGY)**

8E601 "Technology" according to the General Technology Note for the "development" of equipment, materials or "software" specified by 8A6, 8B6, 8C6 or 8D6.

8E602 "Technology" according to the General Technology Note for the "production" of equipment or materials specified by 8A6, 8B6 or 8C6.

8E603 Other "technology" as follows:

a. ACOUSTICS – (Reserved)

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b. OPTICAL SENSORS – (Reserved)

c. CAMERAS – (Reserved)

OPTICS

8E603 d. "Technology" as follows:

1. "Technology" "required" for the coating and treatment of optical surfaces to achieve an 'optical thickness' uniformity of 99.5% or better for optical coatings 500 mm or more in diameter or major axis length and with a total loss (absorption and scatter) of less than 5×10^{-3} ;

N.B. See also 8E203.f.

Technical Note

'Optical thickness' is the mathematical product of the index of refraction and the physical thickness of the coating.

2. "Technology" for the fabrication of optics using single point diamond turning techniques to produce surface finish "accuracies" of better than 10 nm rms on non-planar surfaces exceeding 0.5 m²;

LASERS

8E603 e. "Technology" "required" for the "development", "production" or "use" of specially designed diagnostic instruments or targets in test facilities for "SHPL" testing or testing or evaluation of materials irradiated by "SHPL" beams;

8E603 f. MAGNETIC AND ELECTRIC FIELD SENSORS - (Reserved)

8E603 g. GRAVIMETERS – (Reserved)

8E603 h. RADAR – (Reserved).

8A7 Navigation and Avionics (SYSTEMS, EQUIPMENT AND COMPONENTS)

N.B. For automatic pilots for underwater vehicles, see 8A8,8B8,8C8,8D8,8E.
For radar, see 8A6,8B,8C6,8D6,8E6.

8A701 Accelerometers as follows and specially designed components therefor:

N.B. For angular or rotational accelerometers, see 8A701.b.

a. Linear accelerometers having any of the following:

1. Specified to function at linear acceleration levels less than or equal to 15 g and having any of the following:
 - a. A "bias" "stability" of less (better) than 130 micro g with respect to a fixed calibration value over a period of one year; or
 - b. A "scale factor" "stability" of less (better) than 130 ppm with respect to a fixed calibration value over a period of one year;
2. Specified to function at linear acceleration levels exceeding 15 g but less than or equal to 100 g and having all of the following:
 - a. A "bias" "repeatability" of less (better) than 1,250 micro g over a period of one year; and
 - b. A "scale factor" "repeatability" of less (better) than 1,250 ppm over a period of one year; or
3. Designed for use in inertial navigation or guidance systems and specified to function at linear acceleration levels exceeding 100 g;

Note 8A701.a.1 and 8A701.a.2 do not apply to accelerometers limited to measurement of only vibration or shock.

8A701 b. Angular or rotational accelerometers, specified to function at linear acceleration levels exceeding 100 g.

8A702 Gyros or angular rate sensors, having any of the following and specially designed components therefor:

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N.B.

For angular or rotational accelerometers, see 8A701.b.

- a. Specified to function at linear acceleration levels less than or equal to 100 g and having any of the following:
1. An angular rate range of less than 500 degrees per second and having any of the following:
 - a. A "bias" "stability" of less (better) than 0.5 degree per hour, when measured in a 1 g environment over a period of one month, and with respect to a fixed calibration value; or
 - b. An "angle random walk" of less (better) than or equal to 0.0035 degree per square root hour; or

Note 8A702.a.1.b. does not apply to "spinning mass gyros".
- 8A702. a. 2. An angular rate range greater than or equal to 500 degrees per second and having any of the following:
- a. A "bias" "stability" of less (better) than 4 degrees per hour, when measured in a 1 g environment over a period of three minutes, and with respect to a fixed calibration value; or
- 8A702. a. 2. b. An "angle random walk" of less (better) than or equal to 0.1 degree per square root hour; or
- Note 8A702.a.2.b. does not apply to "spinning mass gyros".
- 8A702. b. Specified to function at linear acceleration levels exceeding 100 g.
- 8A703 'Inertial measurement equipment or systems', having any of the following:
- Note 8A703 does not apply to 'inertial measurement equipment or systems' which are certified for use on "civil aircraft" by civil aviation authorities of one or more Wassenaar Arrangement Participating States.
- Technical Notes
1. For the purposes of 8A703., 'inertial measurement equipment or systems' incorporate accelerometers or gyroscopes to measure changes in velocity and orientation in order to determine or maintain heading or position without requiring an external reference once aligned. 'Inertial measurement equipment or systems' include:
 - Attitude and Heading Reference Systems (AHRSS);
 - Gyrocompasses;
 - Inertial Measurement Units (IMUs);
 - Inertial Navigation Systems (INSs);
 - Inertial Reference Systems (IRSs);
 - Inertial Reference Units (IRUs).
 2. For the purposes of 8A703, 'positional aiding references' independently provide position, and include:
 - a. "Satellite navigation system";
 - b. "Data-Based Referenced Navigation" ("DBRN").
- 8A703 a. Designed for "aircraft", land vehicles or vessels, providing position without the use of 'positional aiding references', and having any of the following "accuracies" subsequent to normal alignment:
1. 0.8 nautical miles per hour (nm/hr) "Circular Error Probable" ("CEP") rate or less (better);
 2. 0.5% distanced travelled "CEP" or less (better); or
 3. Total drift of 1 nautical mile "CEP" or less (better) in a 24 hr period;
- Technical Note
- The performance parameters in 8A703.a.1, 8A703.a.2 and 8A703.a.3 typically apply to 'inertial measurement equipment or systems' designed for "aircraft", vehicles and vessels, respectively. These parameters result from the utilisation of specialised non-'positional aiding references' (e.g. altimeter, odometer, velocity log). As a consequence, the specified performance values cannot be readily converted between these parameters. Equipment designed for multiple platforms are evaluated against each applicable entry namely 8A703.a.1, 8A703.a.2, or 8A703.a.3.
- 8A703 b. Designed for "aircraft", land vehicles or vessels, with an embedded 'positional aiding reference' and providing position after loss of all 'positional aiding references' for a period of up to 4 minutes, having an "accuracy" of less (better) than 10 meters "CEP";

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Technical Note

8A703.b refers to systems in which 'inertial measurement equipment or systems' and other independent 'positional aiding references' are built into a single unit (i.e., embedded) in order to achieve improved performance.

- 8A703 c. Designed for "aircraft", land vehicles or vessels, providing heading or True North determination and having any of the following:
1. A maximum operating angular rate less (lower) than 500 deg/s and a heading "accuracy" without the use of 'positional aiding references' equal to or less (better) than 0.07 deg sec(Lat) (equivalent to 6 arc minutes rms at 45 degrees latitude); or
 2. A maximum operating angular rate equal to or greater (higher) than 500 deg/s and a heading "accuracy" without the use of 'positional aiding references' equal to or less (better) than 0.2 deg sec(Lat) (equivalent to 17 arc minutes rms at 45 degrees latitude);

- 8A703 d. Providing acceleration measurements or angular rate measurements, in more than one dimension, and having any of the following:
1. Performance specified by 8A701 or 8A702 along any axis, without the use of any aiding references; or
 2. Being "space-qualified" and providing angular rate measurements having an "angle random walk" along any axis of less (better) than or equal to 0.1 degree per square root hour.

Note 8A703.d.2 does not apply to 'inertial measurement equipment or systems' that contain "spinning mass gyros" as the only type of gyro.

- 8A704 'Star trackers' and components therefor, as follows:
- a. 'Star trackers' with a specified azimuth "accuracy" of equal to or less (better) than 20 seconds of arc throughout the specified lifetime of the equipment;
 - b. Components specially designed for equipment specified in 8A704.a as follows:
 1. Optical heads or baffles;
 2. Data processing units.

Technical Note

'Star trackers' are also referred to as stellar attitude sensors or gyro-astro compasses.

- 8A705 Global Navigation Satellite Systems (GNSS) receiving equipment having any of the following and specially designed components therefor:

N.B. For equipment specially designed for military use, see 6A011.

- a. Employing a decryption algorithm specially designed or modified for government use to access the ranging code for position and time; or
- b. Employing 'adaptive antenna systems'.

Note

8A705.b does not apply to GNSS receiving equipment that only uses components designed to filter, switch, or combine signals from multiple omni-directional antennae that do not implement adaptive antenna techniques.

Technical Note

For the purposes of 8A705.b 'adaptive antenna systems' dynamically generate one or more spatial nulls in an antenna array pattern by signal processing in the time domain or frequency domain.

- 8A706 Airborne altimeters operating at frequencies other than 4.2 to 4.4 GHz inclusive and having any of the following:
- a. 'Power management'; or
 - b. Using phase shift key modulation.

Technical Note

'Power management' is changing the transmitted power of the altimeter signal so that received power at the "aircraft" altitude is always at the minimum necessary to determine the altitude.

- 8A707 (Reserved)

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- 8A708 Underwater sonar navigation systems using doppler velocity or correlation velocity logs integrated with a heading source and having a positioning "accuracy" of equal to or less (better) than 3% of distance travelled "Circular Error Probable" ("CEP") and specially designed components therefor.
- Note 8A708 does not apply to systems specially designed for installation on surface vessels or systems requiring acoustic beacons or buoys to provide positioning data.
- N.B. See 8A601.a for acoustic systems, and 8A601.b for correlation-velocity and Doppler-velocity sonar log equipment.
- See 8A802 for other marine systems.
- 8B7 NAVIGATION AND AVIONICS (TEST, INSPECTION AND PRODUCTION EQUIPMENT)
- 8B701 Test, calibration or alignment equipment, specially designed for equipment specified by 8A7.
- Note 8B701 does not apply to test, calibration or alignment equipment for 'Maintenance Level I' or 'Maintenance Level II'.
- Technical Notes
1. 'Maintenance Level I'
The failure of an inertial navigation unit is detected on the "aircraft" by indications from the Control and Display Unit (CDU) or by the status message from the corresponding sub-system. By following the manufacturer's manual, the cause of the failure may be localised at the level of the malfunctioning Line Replaceable Unit (LRU). The operator then removes the LRU and replaces it with a spare.
 2. 'Maintenance Level II'
The defective LRU is sent to the maintenance workshop (the manufacturer's or that of the operator responsible for level II maintenance). At the maintenance workshop, the malfunctioning LRU is tested by various appropriate means to verify and localise the defective Shop Replaceable Assembly (SRA) module responsible for the failure. This SRA is removed and replaced by an operative spare. The defective SRA (or possibly the complete LRU) is then shipped to the manufacturer. 'Maintenance Level II' does not include the disassembly or repair of specified accelerometers or gyro sensors.
- 8B702 Equipment specially designed to characterize mirrors for ring "laser" gyros, as follows:
- a. Scatterometers having a measurement "accuracy" of 10 ppm or less (better);
 - b. Profilometers having a measurement "accuracy" of 0.5 nm (5 angstrom) or less (better).
- 8B703 Equipment specially designed for the "production" of equipment specified by 8A7.
- Note 8B703 includes:
- Gyro tuning test stations;
 - Gyro dynamic balance stations;
 - Gyro run-in/motor test stations;
 - Gyro evacuation and fill stations;
 - Centrifuge fixtures for gyro bearings;
 - Accelerometer axis align stations;
 - Fibre optic gyro coil winding machines.
- 8C7 NAVIGATION AND AVIONICS (MATERIALS) – (Reserved)
- 8D7 NAVIGATION AND AVIONICS (SOFTWARE)
- 8D701 "Software" specially designed or modified for the "development" or "production" of equipment specified by 8A7 or 8B7.
- 8D702 "Source code" for the operation or maintenance of any inertial navigation equipment, including inertial equipment not specified by 8A703 or 8A704, or Attitude and Heading Reference Systems ('AHRS').
- Note 8D702 does not apply to "source code" for the operation or maintenance of gimballed 'AHRS'.
- Technical Note
'AHRS' generally differ from Inertial Navigation Systems (INS) in that an 'AHRS' provides attitude and heading information and normally does not provide the acceleration, velocity and position information associated with an INS.

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- 8D703 Other "software" as follows:
- a. "Software" specially designed or modified to improve the operational performance or reduce the navigational error of systems to the levels specified by 8A703, 8A704 or 8A708;
 - b. "Source code" for hybrid integrated systems which improves the operational performance or reduces the navigational error of systems to the level specified by 8A703 or 8A708 by continuously combining heading data with any of the following:
 1. Doppler radar or sonar velocity data;
 2. Global Navigation Satellite Systems (GNSS) reference data; or
 3. Data from "Data-Based Referenced Navigation" ("DBRN") systems;
 - c. (Reserved)
 - d. (Reserved)
- N.B. For flight control "source code", see 8D704.
- 8D703 e. Computer-Aided-Design (CAD) "software" specially designed for the "development" of "active flight control systems", helicopter multi-axis fly-by-wire or fly-by-light controllers or helicopter "circulation-controlled anti-torque or circulation-controlled direction control systems", whose "technology" is specified by 8E704.b.1, 8E704.b.3 to 8E704.b.5, 8E704.b.7, 8E704.b.8, 8E704.c.1 or 8E704.c.2.
- 8D704 "Source code" incorporating "development" "technology" specified by 8E704.a.2, 8E704.a.3, 8E704.a.5, 8E704.a.6 or 8E704.b, for any of the following:
- a. Digital flight management systems for "total control of flight";
 - b. Integrated propulsion and flight control systems;
 - c. "Fly-by-wire systems" or "fly-by-light systems";
 - d. Fault-tolerant or self-reconfiguring "active flight control systems";
 - e. (Reserved)
 - f. Air data systems based on surface static data; or
 - g. Three dimensional displays.
- Note 8D704 does not apply to "source code" associated with common computer elements and utilities (e.g, input signal acquisition, output signal transmission, computer "program" and data loading, built-in test, task scheduling mechanisms) not providing a specific flight control system function.
- 8D705 "Software" specially designed to decrypt "satellite navigation system" ranging code designed for government use.'
- 8E7NAVIGATION AND AVIONICS (TECHNOLOGY)
- 8E701 "Technology" according to the General Technology Note for the "development" of equipment or "software", specified by 8A7, 8B7, 8D701, 8D702, 8D703 or 8D705.
- Note 8E701 includes key management "technology" exclusively for equipment specified in 8A705.a.
- 8E702 "Technology" according to the General Technology Note for the "production" of equipment specified by 8A7 or 8B7.
- 8E703 "Technology" according to the General Technology Note for the repair, refurbishing or overhaul of equipment specified by 8A701 to 8A704.
- Note 8E703 . does not apply to "technology" for maintenance, directly associated with calibration, removal or replacement of damaged or unserviceable LRUs and SRAs of a "civil aircraft" as described in 'Maintenance Level I' or 'Maintenance Level II'.
- N.B. See Technical Notes to 8B701.
- 8E704 Other "technology" as follows:
- a. "Technology" for the "development" or "production" of any of the following:
 1. (Reserved) ;

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2. Air data systems based on surface static data only, i.e, which dispense with conventional air data probes;
 3. Three dimensional displays for "aircraft";
 4. (Reserved)
- 8E704 a. 5. Electric actuators (i.e., electromechanical, electrohydrostatic and integrated actuator package) specially designed for 'primary flight control';

Technical Note

Primary flight control' is "aircraft" stability or manoeuvring control using force/moment generators, i.e. aerodynamic control surfaces or propulsive thrust vectoring;

6. "Flight control optical sensor array" specially designed for implementing "active flight control systems"; or

Technical Note

A 'flight control optical sensor array' is a network of distributed optical sensors, using "laser" beams, to provide real-time flight control data for on-board processing.

7. "DBRN" systems designed to navigate underwater, using sonar or gravity databases, that provide a positioning "accuracy" equal to or less (better) than 0.4 nautical miles;
- 8E704 b. "Development" "technology", as follows, for "active flight control systems" (including "fly-by-wire systems" or "fly-by-light systems"):
1. Photonic-based "technology" for sensing "aircraft" or flight control component state, transferring flight control data, or commanding actuator movement, "required" for "fly-by-light systems" "active flight control systems";
 2. (Reserved) ;
 3. Real-time algorithms to analyze component sensor information to predict and preemptively mitigate impending degradation and failures of components within an "active flight control system";
- Note 8E704.b.3 does not include algorithms for the purpose of off-line maintenance.
4. Real-time algorithms to identify component failures and reconfigure force and moment controls to mitigate "active flight control system" degradations and failures;
- Note 8E704.b.4 does not include algorithms for the elimination of fault effects through comparison of redundant data sources, or off-line pre-planned responses to anticipated failures.
5. Integration of digital flight control, navigation and propulsion control data, into a digital flight management system for "total control of flight";

Note 8E704.b.5 does not apply to:

1. "Technology" for integration of digital flight control, navigation and propulsion control data, into a digital flight management system for 'flight path optimisation';
2. "Technology" for "aircraft" flight instrument systems integrated solely for VOR, DME, ILS or MLS navigation or approaches.

Technical Note

'Flight path optimisation' is a procedure that minimises deviations from a four-dimensional (space and time) desired trajectory based on maximising performance or effectiveness for mission tasks.

- 8E704 b. 6. (Reserved)

- 8E704 b. 7. "Technology" "required" for deriving the functional requirements for "fly-by-wire systems" having all of the following:

- a. 'Inner-loop' airframe stability controls requiring loop closure rates of 40 Hz or greater; and

Technical Note

'Inner-loop' refers to functions of "active flight control systems" that automate airframe stability controls.

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- 8E704 b. 7. b. Having any of the following:
1. Corrects an aerodynamically unstable airframe, measured at any point in the design flight envelope, that would lose recoverable control if not corrected within 0.5 seconds;
 2. Couples controls in two or more axes while compensating for 'abnormal changes in aircraft state';
Technical Note
'Abnormal changes in aircraft state' include in-flight structural damage, loss of engine thrust, disabled control surface, or destabilizing shifts in cargo load.
 3. Performs the functions specified in 8E704.b.5; or
Note 8E704 .b.7.b.3 does not apply to autopilots.
 4. Enables "aircraft" to have stable controlled flight, other than during take-off or landing, at greater than 18 degrees angle of attack, 15 degrees side slip, 15 degrees/second pitch or yaw rate, or 90 degrees/second roll rate;
- 8E704 b. 8. "Technology" "required" for deriving the functional requirements for "fly-by-wire systems" to achieve all of the following:
- a. No loss of control of the "aircraft" in the event of a consecutive sequence of any two individual faults within the "fly-by-wire system"; and
 - b. Probability of loss of control of the "aircraft" being less (better) than 1×10^{-9} failures per flight hour;
- Note* 8E704.b does not apply to "technology" associated with common computer elements and utilities (e.g, input signal acquisition, output signal transmission, computer "program" and data loading, built-in test, task scheduling mechanisms) not providing a specific flight control system function.
- 8E704 c. "Technology" for the "development" of helicopter systems, as follows:
1. Multi-axis fly-by-wire or fly-by-light controllers, which combine the functions of at least two of the following into one controlling element:
 - a. Collective controls;
 - b. Cyclic controls;
 - c. Yaw controls;
 2. "Circulation-controlled anti-torque or circulation-controlled direction control systems";
 3. Rotor blades incorporating 'variable geometry **aerofoils**', for use in systems using individualblade control.

Technical Note

Variable geometry airfoils' use trailing edge flaps or tabs, or leading edge slats or pivoted nose droop, the position of which can be controlled in flight.

8A8 MARINE (SYSTEMS, EQUIPMENT AND COMPONENTS)

8A801 Submersible vehicles and surface vessels, as follows:

N.B. For the status of equipment for submersible vehicles, see:

- 8A601 to 8E603 for sensors;
- 8A701 to 8E704 and for navigation equipment;
- 8A801, 8E802 for underwater equipment.

8A801 a. Manned, tethered submersible vehicles designed to operate at depths exceeding 1,000 m;

8A801 b. Manned, untethered submersible vehicles having any of the following:

1. Designed to 'operate autonomously' and having a lifting capacity of all the following:
 - a. 10% or more of their weight in air; and
 - b. 15 kN or more;

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2. Designed to operate at depths exceeding 1,000 m; or
3. Having all of the following:
 - a. Designed to continuously 'operate autonomously' for 10 hours or more; and
 - b. 'Range' of 25 nautical miles or more;

Technical Notes

1. For the purposes of 8A801.b, 'operate autonomously' means fully submerged, without snorkel, all systems working and cruising at minimum speed at which the submersible can safely control its depth dynamically by using its depth planes only, with no need for a support vessel or support base on the surface, sea-bed or shore, and containing a propulsion system for submerged or surface use.
2. For the purposes of 8A801.b, 'range' means half the maximum distance a submersible vehicle can 'operate autonomously'.

- 8A801 c. Unmanned submersible vehicles, as follows:
1. Unmanned submersible vehicles having any of the following:
 - a. Designed for deciding a course relative to any geographical reference without real time human assistance;
 - b. Acoustic data or command link; or
 - c. **Wireless** data or command link exceeding 1,000 m;
 2. Unmanned submersible vehicles, not specified in 8A801c.1., having all of the following:
 - a. Designed to operate with a tether;
 - b. Designed to operate at depths exceeding 1,000 m; and
 - c. Having any of the following:
 1. Designed for self-propelled manoeuvre using propulsion motors or thrusters specified by 8A802.a.2.; or
 2. Fibre optic data link;
- 8A801 d. (Reserved)
- 8A801 e. Ocean salvage systems with a lifting capacity exceeding 5 MN for salvaging objects from depths exceeding 250 m and having any of the following:
1. Dynamic positioning systems capable of position keeping within 20 m of a given point provided by the navigation system; or
 2. Seafloor navigation and navigation integration systems, for depths exceeding 1,000 m and with positioning "accuracies" to within 10 m of a predetermined point.
- 8A802 Marine systems, equipment and components, as follows:
- N.B. For underwater communications systems, see 8A5 part 1 Telecommunications.
- a. Systems, equipment and components, specially designed or modified for submersible vehicles and designed to operate at depths exceeding 1,000 m, as follows:
 1. Pressure housings or pressure hulls with a maximum inside chamber diameter exceeding 1.5 m;
 2. Direct current propulsion motors or thrusters;
 3. Umbilical cables, and connectors therefor, using optical fibre and having synthetic strength members;
 4. Components manufactured from material specified by 8C801;

Technical Note
The objective of 8A802.a.4 should not be defeated by the export of 'syntactic foam' specified by 8C801 when an intermediate stage of manufacture has been performed and it is not yet in its final component form.
 - b. Systems specially designed or modified for the automated control of the motion of submersible vehicles specified by 8A801, using navigation data, having closed loop servo-controls and having any of the following:
 1. Enabling a vehicle to move within 10 m of a predetermined point in the water column;
 2. Maintaining the position of the vehicle within 10 m of a predetermined point in the water column; or
 3. Maintaining the position of the vehicle within 10 m while following a cable on or under the seabed;
 - c. Fibre optic pressure hull penetrators;
- 8A802

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- 8A802 d. Underwater vision systems having all of the following:
1. Specially designed or modified for remote operation with an underwater vehicle; and
 2. Employing any of the following techniques to minimise the effects of back scatter:
 - a. Range-gated illuminators; or
 - b. Range-gated laser systems;
- 8A802 e.. (Reserved)
- 8A802 f (Reserved)
- N.B. For electronic imaging systems specially designed or modified for underwater use incorporating "focal plane arrays" specified by 8A602.a.3.g, see 8A603.b.4.c.
- 8A802 g. Light systems specially designed or modified for underwater use, as follows:
1. Stroboscopic light systems capable of a light output energy of more than 300 J per flash and a flash rate of more than 5 flashes per second;
 2. Argon arc light systems specially designed for use below 1,000 m;
- 8A802 h. "Robots" specially designed for underwater use, controlled by using a dedicated computer and having any of the following:
1. Systems that control the "robot" using information from sensors which measure force or torque applied to an external object, distance to an external object, or tactile sense between the "robot" and an external object; or
 2. The ability to exert a force of 250 N or more or a torque of 250 Nm or more and using titanium based alloys or "composite" "fibrous or filamentary materials" in their structural members;
- 8A802 i. Remotely controlled articulated manipulators specially designed or modified for use with submersible vehicles and having any of the following:
1. Systems which control the manipulator using information from sensors which measure any of the following:
 - a. Torque or force applied to an external object; or
 - b. Tactile sense between the manipulator and an external object; or
 2. Controlled by proportional master-slave techniques and having 5 degrees of 'freedom of movement' or more;
- Technical Note
Only functions having proportionally related motion control using positional feedback are counted when determining the number of degrees of 'freedom of movement'.
- 8A802 j. Air independent power systems specially designed for underwater use, as follows:
1. Brayton or Rankine cycle engine air independent power systems having any of the following:
 - a. Chemical scrubber or absorber systems, specially designed to remove carbon dioxide, carbon monoxide and particulates from recirculated engine exhaust;
 - b. Systems specially designed to use a monoatomic gas;
 - c. Devices or enclosures, specially designed for underwater noise reduction in frequencies below 10 kHz, or special mounting devices for shock mitigation; or
 - d. Systems having all of the following:
 1. Specially designed to pressurise the products of reaction or for fuel reformation;
 2. Specially designed to store the products of the reaction; and
 3. Specially designed to discharge the products of the reaction against a pressure of 100 kPa or more;
- 8A802 j. 2. Diesel cycle engine air independent systems having all of the following:
- a. Chemical scrubber or absorber systems, specially designed to remove carbon dioxide, carbon monoxide and particulates from recirculated engine exhaust;
 - b. Systems specially designed to use a monoatomic gas;
 - c. Devices or enclosures, specially designed for underwater noise reduction in frequencies below 10 kHz, or special mounting devices for shock mitigation; and
 - d. Specially designed exhaust systems that do not exhaust continuously the products of combustion;
- 8A802 j. 3. "Fuel cell" air independent power systems with an output exceeding 2 kW and having any of the following:

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- | | | |
|-------|----|---|
| | a. | Devices or enclosures, specially designed for underwater noise reduction in frequencies below 10 kHz, or special mounting devices for shock mitigation; <u>or</u> |
| | b. | Systems having all of the following: <ol style="list-style-type: none"> 1. Specially designed to pressurise the products of reaction or for fuel reformation; 2. Specially designed to store the products of the reaction; and 3. Specially designed to discharge the products of the reaction against a pressure of 100 kPa or more; |
| 8A802 | j. | 4. Stirling cycle engine air independent power systems having all of the following: <ol style="list-style-type: none"> a. Devices or enclosures, specially designed for underwater noise reduction in frequencies below 10 kHz, or special mounting devices for shock mitigation; <u>and</u> b. Specially designed exhaust systems which discharge the products of combustion against a pressure of 100 kPa or more; |
| 8A802 | k. | (Reserved) |
| 8A802 | l. | (Reserved) |
| 8A802 | m. | (Reserved) |
| 8A802 | n. | (Reserved) |
| 8A802 | o. | Propellers, power transmission systems, power generation systems and noise reduction systems, as follows: <ol style="list-style-type: none"> 1. (Reserved); 2. Water-screw propeller, power generation systems or transmission systems, designed for use on vessels, as follows: <ol style="list-style-type: none"> a. Controllable-pitch propellers and hub assemblies, rated at more than 30 MW; b. Internally liquid-cooled electric propulsion motors with a power output exceeding 2.5 MW; c. "Superconductive" propulsion motors, with a power output exceeding 0.1 MW; d. Power transmission shaft systems incorporating "composite" material components and capable of transmitting more than 2 MW; e. Ventilated or base-ventilated propeller systems, rated at more than 2.5 MW; |
| 8A802 | 3. | Noise reduction systems designed for use on vessels of 1,000 tonnes displacement or more, as follows: <ol style="list-style-type: none"> a. Systems that attenuate underwater noise at frequencies below 500 Hz and consist of compound acoustic mounts for the acoustic isolation of diesel engines, diesel generator sets, gas turbines, gas turbine generator sets, propulsion motors or propulsion reduction gears, specially designed for sound or vibration isolation and having an intermediate mass exceeding 30% of the equipment to be mounted; b. 'Active noise reduction or cancellation systems' or magnetic bearings, specially designed for power transmission systems; <p><u>Technical Note</u>
 <i>'Active noise reduction or cancellation systems' incorporate electronic control systems capable of actively reducing equipment vibration by the generation of anti-noise or anti- vibration signals directly to the source.</i></p> <p>4. Permanent magnet electric propulsion motors specially designed for submersible vehicles, having a power output exceeding 0.1 MW.
 <i>Note 8A802.o.4. includes rim-driven propulsion systems.</i></p> |
| 8A802 | p. | Pumpjet propulsion systems having all of the following: <ol style="list-style-type: none"> 1. Power output exceeding 2.5 MW; <u>and</u> 2. Using divergent nozzle and flow conditioning vane techniques to improve propulsive efficiency or reduce propulsion-generated underwater-radiated noise; |

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- 8A802 q. Underwater swimming and diving equipment as follows:
1. Closed circuit rebreathers;
 2. Semi-closed circuit rebreathers;
- Note 8A802.q does not apply to individual rebreathers for personal use when accompanying their users.
- N.B. For equipment and devices specially designed for military use, see 6A017.a.
- 8A802 r. Diver deterrent acoustic systems specially designed or modified to disrupt divers and having a sound pressure level equal to or exceeding 190 dB (reference 1 μ Pa at 1 m) at frequencies of 200 Hz and below.
- Note 1 8A802.r does not apply to diver deterrent systems based on underwater explosive devices, air guns or combustible sources.
- Note 2 8A802.r includes diver deterrent acoustic systems that use spark gap sources, also known as plasma sound sources.

8B8MARINE (TEST, INSPECTION AND PRODUCTION EQUIPMENT)

- 8B801 Water tunnels designed to have a background noise of less than 100 dB (reference 1 μ Pa, 1 Hz) within the frequency range exceeding 0 Hz but not exceeding 500 Hz and designed for measuring acoustic fields generated by a hydro-flow around propulsion system models.

8C8MARINE (MATERIALS)

- 8C801 'Syntactic foam' designed for underwater use and having all of the following:
- a. Designed for marine depths exceeding 1,000 m; and
 - b. A density less than 561 kg/m³.
- Technical Note
'Syntactic foam' consists of hollow spheres of plastic or glass embedded in a resin "matrix".
- N.B. See also 8A802.a.4.

8D8 MARINE (SOFTWARE)

- 8D801 "Software" specially designed or modified for the "development", "production" or "use" of equipment or materials, specified by 8A8, 8B8 or 8C8.
- 8D802 Specific "software" specially designed or modified for the "development", "production", repair, overhaul or refurbishing (re-machining) of propellers specially designed for underwater noise reduction.

8E8MARINE (TECHNOLOGY)

- 8E801 "Technology" according to the General Technology Note for the "development" or "production" of equipment or materials, specified by 8A8, 8B8 or 8C8.
- 8E802 Other "technology" as follows:
- a. "Technology" for the "development", "production", repair, overhaul or refurbishing (re-machining) of propellers specially designed for underwater noise reduction;
 - b. "Technology" for the overhaul or refurbishing of equipment specified by 8A801, 8A802.b, 8A802.j, 8A802.o or 8A802.p.
- 8E802 c. "Technology" according to the General Technology Note for the "development" or "production" of any of the following:
1. Surface-effect vehicles (fully skirted variety) having all of the following:
 - a. Maximum design speed, fully loaded, exceeding 30 knots in a significant wave height of 1.25 m or more;
 - b. Cushion pressure exceeding 3,830 Pa; and

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- c. Light-ship-to-full-load displacement ratio of less than 0.70;
2. Surface-effect vehicles (rigid sidewalls) with a maximum design speed, fully loaded, exceeding 40 knots in a significant wave height of 3.25 m or more;
3. Hydrofoil vessels with active systems for automatically controlling foil systems, with a maximum design speed, fully loaded, of 40 knots or more in a significant wave height of 3.25 m or more; or
4. 'Small waterplane area vessels' having any of the following:
 - a. Full load displacement exceeding 500 tonnes with a maximum design speed, fully loaded, exceeding 35 knots in a significant wave height of 3.25 m or more; or
 - b. Full load displacement exceeding 1,500 tonnes with a maximum design speed, fully loaded, exceeding 25 knots in a significant wave height of 4 m or more.

Technical Note

A 'small waterplane area vessel' is defined by the following formula: waterplane area at an operational design draft less than $2x$ (displaced volume at the operational design draft)^{2/3}.

8A9 AEROSPACE AND PROPULSION (SYSTEMS, EQUIPMENT AND COMPONENTS)

N.B. For propulsion systems designed or rated against neutron or transient ionizing radiation, see Category 6.

8A901 Aero gas turbine engines having any of the following:

- a. Incorporating any of the "technologies" specified by 8E903.a, 8E903.h or 8E903.i; or

Note 1 8 A901.1.a does not apply to aero gas turbine engines which meet all of the following:

- a. Certified by civil aviation authorities of India; and
- b. Intended to power non-military manned "aircraft" for which any of the following has been issued by civil aviation authorities of India for the "aircraft" with this specific engine type:
 1. A civil type certificate; or
 2. An equivalent document recognised by the International Civil Aviation Organisation (ICAO).

Note 2 8 A901.1.a does not apply to aero gas turbine engines designed for Auxiliary Power Units (APUs) approved by the civil aviation authority of India.

- 8A901 b. (Reserved)
- 8A902 'Marine gas turbine engines' designed to use liquid fuel and having all of the following, and specially designed assemblies and components therefor:

- a. Maximum continuous power when operating in "steady state mode" at standard reference conditions specified by ISO 3977-2:1997 (or national equivalent) of 24,245 kW or more; and
- b. 'Corrected specific fuel consumption' not exceeding 0.219 kg/kWh at 35% of the maximum continuous power when using liquid fuel.

Note: The term 'marine gas turbine engines' includes those industrial, or aero-derivative, gas turbine engines adapted for a ship's electric power generation or propulsion.

Technical Note

For the purposes of 8A902, 'corrected specific fuel consumption' is the specific fuel consumption of the engine corrected to a marine distillate liquid fuel having a net specific energy (i.e., net heating value) of 42MJ/kg (ISO 3977-2:1997).

8A903 Specially designed assemblies or components, incorporating any of the "technologies" specified by 8E903.a, 8E903.h. or 8E903.i, **8E903.k** for any of the following aero gas turbine engines:

- a. Specified by 8A901.1 or;
- b. Whose design or production origins are either not from India or unknown to the manufacturer.

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- 8A904 Space launch vehicles, "spacecraft", "spacecraft buses", "spacecraft payloads", "spacecraft" on-board systems or equipment, terrestrial equipment, and air-launch platforms, **as under:-**
- a. Space launch vehicles;
 - b. "Spacecraft";
 - c. "Spacecraft buses";
 - d. "Spacecraft payloads" incorporating items specified by 8A301.b.1.a.4., 8A302.g., 8A501.a.1., 8A501.b.3., 8A502.c., 8A502.e., 8A602.a.1., 8A602.a.2., 8A602.2.b., 8A602.2.d., 8A603.b., 8A604.4.c., 8A604.e., 8A608.d., 8A608.e., 8A608.k., 8A608.l. or 8A910.c.;
 - e. On-board systems or equipment, specially designed for "spacecraft" and having any of the following functions:
 1. 'Command and telemetry data handling';
Note For the purpose of 8A904.e.1., 'command and telemetry data handling' includes bus data management, storage, and processing.
 2. 'Payload data handling'; or
Note For the purpose of 8A904.e.2., 'payload data handling' includes payload data management, storage, and processing.
 3. 'Attitude and orbit control';
Note For the purpose of 8A904.e.3., 'attitude and orbit control' includes sensing and actuation to determine and control the position and orientation of a "spacecraft".

N.B. For equipment specially designed for military use, see 6A011.c.
- 8A904 e. On-board systems or equipment, specially designed for "spacecraft" and having any of the following functions:
1. 'Command and telemetry data handling';
Note For the purpose of 8A904.e.1., 'command and telemetry data handling' includes bus data management, storage, and processing. 8A904
 2. 'Payload data handling'; or
Note For the purpose of 8A904.e.2., 'payload data handling' includes payload data management, storage, and processing.
 3. 'Attitude and orbit control';
Note For the purpose of 8A904.e.3., 'attitude and orbit control' includes sensing and actuation to determine and control the position and orientation of a "spacecraft".
- N.B. For equipment specially designed for military use, see 6A011.c.
- 8A904 f. Terrestrial equipment specially designed for "spacecraft", as follows:
1. Telemetry and telecommand equipment specially designed for any of the following data processing functions:
 - a. Telemetry data processing of frame synchronisation and error corrections, for monitoring of operational status (also known as health and safe status) of the "spacecraft bus"; or
 - b. Command data processing for formatting command data being sent to the "spacecraft" to control the "spacecraft bus";
 2. Simulators specially designed for 'verification of operational procedures' of "spacecraft".
Technical Note
 For the purposes of 8A904.f.2., 'verification of operational procedures' is any of the following:
 1. Command sequence confirmation;
 2. Operational training;
 3. Operational rehearsals; or
 4. Operational analysis
- 8A904 g. "Aircraft" specially designed or modified to be air-launch platforms for space launch vehicles or "sub-orbital craft";
- 8A904 h. "Sub-orbital craft".
- 8A905 Liquid rocket propulsion systems containing any of the systems or components, specified by 8A906.

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- 8A906 Systems and components, specially designed for liquid rocket propulsion systems, as follows:
- Cryogenic refrigerators, flightweight dewars, cryogenic heat pipes or cryogenic systems, specially designed for use in space vehicles and capable of restricting cryogenic fluid losses to less than 30% per year;
 - Cryogenic containers or closed-cycle refrigeration systems, capable of providing temperatures of 100 K (-173°C) or less for "aircraft" capable of sustained flight at speeds exceeding Mach 3, launch vehicles or "spacecraft";
 - Slush hydrogen storage or transfer systems;
 - High pressure (exceeding 17.5 MPa) turbo pumps, pump components or their associated gas generator or expander cycle turbine drive systems;
 - High-pressure (exceeding 10.6 MPa) thrust chambers and nozzles therefor;
 - Propellant storage systems using the principle of capillary containment or positive expulsion (i.e. with flexible bladders);
 - Liquid propellant injectors with individual orifices of 0.381 mm or smaller in diameter (an area of $1.14 \times 10^{-3} \text{ cm}^2$ or smaller for non-circular orifices) and specially designed for liquid rocket engines;
 - One-piece carbon-carbon thrust chambers or one-piece carbon-carbon exit cones, with densities exceeding 1.4 g/cm^3 and tensile strengths exceeding 48 MPa.
- 8A907 Solid rocket propulsion systems having any of the following:
- Total impulse capacity exceeding 1.1 MNs;
 - Specific impulse of 2.4 kNs/kg or more, when the nozzle flow is expanded to ambient sea level conditions for an adjusted chamber pressure of 7 MPa;
 - Stage mass fractions exceeding 88% and propellant solid loadings exceeding 86%;
 - Components specified by 8A908; or
 - Insulation and propellant bonding systems, using direct-bonded motor designs to provide a 'strong mechanical bond' or a barrier to chemical migration between the solid propellant and case insulation material.
- Technical Note
A 'strong mechanical bond' means bond strength equal to or more than propellant strength.
- 8A908 Components specially designed for solid rocket propulsion systems, as follows:
- Insulation and propellant bonding systems, using liners to provide a 'strong mechanical bond' or a barrier to chemical migration between the solid propellant and case insulation material;
 - Filament-wound "composite" motor cases exceeding 0.61 m in diameter or having 'structural efficiency ratios (PV/W)' exceeding 25 km;
- Technical Note
'Structural efficiency ratio (PV/W)' is the burst pressure (P) multiplied by the vessel volume (V) divided by the total pressure vessel weight (W).
- Nozzles with thrust levels exceeding 45 kN or nozzle throat erosion rates of less than 0.075 mm/s;
 - Movable nozzle or secondary fluid injection thrust vector control systems, capable of any of the following:
 - Omni-axial movement exceeding $\pm 5^\circ$;
 - Angular vector rotations of $20^\circ/\text{s}$ or more; or
 - Angular vector accelerations of $40^\circ/\text{s}^2$ or more.
- 8A909 Hybrid rocket propulsion systems having any of the following:
- Total impulse capacity exceeding 1.1 MNs; or
 - Thrust levels exceeding 220 kN in vacuum exit conditions.
- 8A910 Specially designed components, systems and structures, for launch vehicles, launch vehicle propulsion systems or "spacecraft", as follows:
- Components and structures, each exceeding 10 kg and specially designed for launch vehicles manufactured using any of the following:
 - "Composite" materials consisting of "fibrous or filamentary materials" specified by 8C110.e

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- and resins specified by 8C008 or 8C008b;
2. Metal "matrix" "composites" reinforced by any of the following:
 - a. Materials specified by 8C107;
 - b. "Fibrous or filamentary materials" specified by 8C110; or
 - c. Aluminides specified by 8C102.a; or
 3. Ceramic "matrix" "composite" materials specified by 8C107;
- Note *The weight cut-off is not relevant for nose cones.*
- 8A910 b. Components and structures, specially designed for launch vehicle propulsion systems specified by 8A5 to 8A9 manufactured using any of the following:
1. "Fibrous or filamentary materials" specified by 8C110.e and resins specified by 8C108 or 8C108b;
 2. Metal "matrix" "composite" materials reinforced by any of the following:
 - a. Materials specified by 8C107;
 - b. "Fibrous or filamentary materials" specified by 8C110; or
 - c. Aluminides specified by 8C102.a; or
 3. Ceramic "matrix" "composite" materials specified by 8C107;
- c. Structural components and isolation systems, specially designed to control actively the dynamic response or distortion of "spacecraft" structures;
- d. Pulsed liquid rocket engines with thrust-to-weight ratios equal to or more than 1 kN/kg and a 'responsetime' of less than 30 ms.
- Technical Note
For the purposes of 8A910.d., 'response time' is the time required to achieve 90% of total rated thrust from start-up'
- 8A911 Ramjet, scramjet or 'combined cycle engines', and specially designed components therefor.
- Technical Note
For the purposes of 8A911., 'combined cycle engines' combine two or more of the following types of engines:
- Gas turbine engine (turbojet, turboprop and turbofan);
 - Ramjet or scramjet;
 - Rocket motor or engine (liquid/gel/solid-propellant and hybrid).
- 8A912 "Unmanned Aerial Vehicles" ("UAVs"), unmanned **"airships"**, related equipment and components, as follows:
- N.B. *For "UAVs" that are "sub-orbital craft", see 8A904.h.*
- a. "UAVs" or unmanned "airships", designed to have controlled flight out of the direct 'natural vision' of the 'operator' and having any of the following:
 1. Having all of the following:
 - a. A maximum 'endurance' greater than or equal to 30 minutes but less than 1 hour; and
 - b. Designed to take-off and have stable controlled flight in wind gusts equal to or exceeding 46.3 km/h (25 knots); or
 2. A maximum 'endurance' of 1 hour or greater;
- Technical Notes
1. *For the purposes of 8A912.a, 'operator' is a person who initiates or commands the "UAV" or unmanned "airship" flight.*
 2. *For the purposes of 8A912.a, 'endurance' is to be calculated for ISA conditions (ISO 2533:1975) at sea level in zero wind.*
 3. *For the purposes of 8A912.a, 'natural vision' means unaided human sight, with or without corrective lenses.*
- 8A912 b. Related equipment and components, as follows:
1. (Reserved) ;
 2. (Reserved) ;
 3. Equipment or components, specially designed to convert a manned "aircraft" or a manned "airship" to a "UAV" or unmanned "airship", specified by 8A912.a;
 4. Air breathing reciprocating or rotary internal combustion type engines, specially designed or modified to propel "UAVs" or unmanned "airships", at altitudes above 15,240 meters (50,000 feet).

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8B9 AEROSPACE AND PROPULSION (TEST, INSPECTION AND PRODUCTION EQUIPMENT)

- 8B901 Manufacturing equipment, tooling or fixtures, as follows:
- a. Directional solidification or single crystal casting equipment designed for "superalloys";
 - b. Casting tooling, specially designed for manufacturing gas turbine engine blades, vanes or "tip shrouds", manufactured from refractory metals or ceramics, as follows:
 1. Cores;
 2. Shells (moulds);
 3. Combined core and shell (mould) units;
 - c. Directional-solidification or single-crystal additive-manufacturing equipment designed for "superalloys".
- 8B902 On-line (real time) control systems, instrumentation (including sensors) or automated data acquisition and processing equipment, having all of the following:
- a. Specially designed for the "development" of gas turbine engines, assemblies or components; and
 - b. Incorporating any of the "technologies" specified by 8E903.h or 8E903.i.
- 8B903 Equipment specially designed for the "production" or test of gas turbine brush seals designed to operate at tip speeds exceeding 335 m/s and temperatures in excess of 773 K (500°C), and specially designed components or accessories therefor.
- 8B904 Tools, dies or fixtures, for the solid state joining of "superalloy", titanium or intermetallic **aerofoil**-to-disk combinations described in 8E903.a.3 or 8E903.a.6 for gas turbines.
- 8B905 On-line (real time) control systems, instrumentation (including sensors) or automated data acquisition and processing equipment, specially designed for use with any of the following:
- a. Wind tunnels designed for speeds of Mach 1.2 or more;
- Note 8B905.a does not apply to wind tunnels specially designed for educational purposes and having a 'test section size' (measured laterally) of less than 250 mm.
- Technical Note
'Test section size' means the diameter of the circle, or the side of the square, or the longest side of the rectangle, at the largest test section location.
- 8B905
- b. Devices for simulating flow-environments at speeds exceeding Mach 5, including hot-shot tunnels, plasma arc tunnels, shock tubes, shock tunnels, gas tunnels and light gas guns; or
 - c. Wind tunnels or devices, other than two-dimensional sections, capable of simulating Reynolds number flows exceeding 25×10^6 .
- 8B906 Acoustic vibration test equipment capable of producing sound pressure levels of 160 dB or more (referenced to 20 Pa) with a rated output of 4 kW or more at a test cell temperature exceeding 1,273 K (1,000°C), and specially designed quartz heaters therefor.
- 8B907 Equipment specially designed for inspecting the integrity of rocket motors and using Non-Destructive Test (NDT) techniques other than planar x-ray or basic physical or chemical analysis.
- 8B908 Direct measurement wall skin friction transducers specially designed to operate at a test flow total (stagnation) temperature exceeding 833 K (560°C).
- 8B909 Tooling specially designed for producing gas turbine engine powder metallurgy rotor components having all of the following:
- a. Designed to operate at stress levels of 60% of Ultimate Tensile Strength (UTS) or more measured at a temperature of 873 K (600°C); and
 - b. Designed to operate at 873 K (600°C) or more.
- Note 8B909 does not specify tooling for the production of powder.
- 8B910 Equipment specially designed for the production of items specified by 8A912.
- 8C9 AEROSPACE AND PROPULSION (MATERIALS) – (Reserved)
- 8D9 AEROSPACE AND PROPULSION (SOFTWARE)

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- 8D901 "Software", not specified in 8D903 or 8D904, specially designed or modified for the "development" of equipment or "technology", specified by 8A9, 8B9 or 8E903.
- 8D902 "Software", not specified in 8D903 or 8D904, specially designed or modified for the "production" of equipment specified by 8A9 or 8B9.
- 8D903 "Software" incorporating "technology" specified by 8E903.h and used in "FADEC Systems" for systems specified by 8A901 to 8A912 or equipment specified by 8B901 to 8B910.
- 8D904 Other "software" as follows:
- a. 2D or 3D viscous "software", validated with wind tunnel or flight test data required for detailed engine flow modelling;
 - b. "Software" for testing aero gas turbine engines, assemblies or components, having all of the following:
 1. Specially designed for testing any of the following:
 - a. Aero gas turbine engines, assemblies or components, incorporating "technology" specified by 8E903.a, 8E903.h or 8E903.i; or
 - b. Multi-stage compressors providing either bypass or core flow, specially designed for aero gas turbine engines incorporating "technology" specified by 8E903.a or 8E903.h; and
 2. Specially designed for all of the following:
 - a. Acquisition and processing of data, in real time; and
 - b. Feedback control of the test article or test conditions (e.g., temperature, pressure, flow rate) while the test is in progress;
- Note: 8D904.b does not specify software for operation of the test facility or operator safety (e.g., overspeed shutdown, fire detection and suppression), or production, repair or maintenance acceptance-testing limited to determining if the item has been properly assembled or repaired.*
- c. "Software" specially designed to control directional-solidification or single-crystal material growth in equipment specified by 8B901.a or 8B901.c;
- 8D904 d. (Reserved) ;
- e. "Software" specially designed or modified for the operation of items specified by 8A912;
- f. "Software" specially designed to design the internal cooling passages of aero gas turbine engine blades, vanes and "tip shrouds";
- g. "Software" having all of the following:
 1. Specially designed to predict aero thermal, aeromechanical and combustion conditions in aero gas turbine engines; and
 2. Theoretical modelling predictions of the aero thermal, aeromechanical and combustion conditions, which have been validated with actual aero gas turbine engine (experimental or production) performance data.
- 8D905 "Software" specially designed or modified for the operation of items specified by 8A904.e or 8A904.f.
N.B. For "software" for items listed in 8A904.d. that are incorporated into "spacecraft payloads", see the appropriate Categories.
- 8E9 AEROSPACE AND PROPULSION (TECHNOLOGY)
- Note "Development" or "production" "technology" specified by 8E9 for gas turbine engines remains specified by 8E9 when used for repair or overhaul. Excluded from 8E9 are: technical data, drawings or documentation for maintenance activities directly associated with calibration, removal or replacement of damaged or unserviceable line replaceable units, including replacement of whole engines or engine modules.*
- 8E901 "Technology" according to the General Technology Note for the "development" of equipment or "software", specified by 8A904 to 8A912, 8B901 to 8B910 or 8D901 to 8D905.

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8E902 "Technology" according to the General Technology Note for the "production" of equipment specified by 8A904 to 8A911 or 8B901 to 8B910.

N.B. For "technology" for the repair of specified structures, laminates or materials, see 8E902.f.

8E903 Other "technology" as follows:

a. "Technology" "required" for the "development" or "production" of any of the following gas turbine engine components or systems:

1. Gas turbine blades, vanes or "tip shrouds", made from directionally solidified (DS) or single crystal (SC) alloys and having (in the 001 Miller Index Direction) a stress-rupture life exceeding 400 hours at 1,273 K (1,000°C) at a stress of 200 MPa, based on the average property values;

Technical Note

For the purposes of 8E903.a.1., stress-rupture life testing is typically conducted on a test specimen.

2. Combustors having any of the following:
 - a. 'Thermally decoupled liners' designed to operate at 'combustor exit temperature' exceeding 1,883K (1,610°C);
 - b. Non-metallic liners;
 - c. Non-metallic shells; or
 - d. Liners designed to operate at 'combustor exit temperature' exceeding 1,883K (1,610°C) and having holes that meet the parameters specified by 8E903.c;

Note The "required" "technology" for holes in 8B903.a.2 is limited to the derivation of the geometry and location of the holes.

- e. Utilising 'pressure gain combustion';

Technical Note

1. 'Thermally decoupled liners' are liners that feature at least a support structure designed to carry mechanical loads and a combustion facing structure designed to protect the support structure from the heat of combustion. The combustion facing structure and support structure have independent thermal displacement (mechanical displacement due to thermal load) with respect to one another, i.e. they are thermally decoupled.
2. 'Combustor exit temperature' is the bulk average gas path total (stagnation) temperature between the combustor exit plane and the leading edge of the turbine inlet guide vane (i.e., measured at engine station T40 as defined in SAE ARP 755A) when the engine is running in a "steady state mode" of operation at the certificated maximum continuous operating temperature.
3. In 'pressure gain combustion' the bulk average stagnation pressure at the combustor outlet is greater than the bulk average stagnation pressure at the combustor inlet due primarily to the combustion process, when the engine is running in a "steady state mode" of operation.

N.B. See 8B903.c for "technology" "required" for manufacturing cooling holes.

- 8E903
- a. 3. Components that are any of the following:
 - a. Manufactured from organic "composite" materials designed to operate above 588 K (315°C);
 - b. Manufactured from any of the following:
 1. Metal "matrix" "composites" reinforced by any of the following:
 - a. Materials specified by 8C007;
 - b. "Fibrous or filamentary materials" specified by 8C110; or
 - c. Aluminides specified by 8C102.a; or
 2. Ceramic "matrix" "composites" specified by 8C107; or

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- c. Stators, vanes, blades, tip seals (shrouds), rotating blings, rotating blisks, or 'splitter ducts', that are all of the following:
1. Not specified in 8E303.a.3.a;
 2. Designed for compressors or fans; and
 3. Manufactured from material specified by 8C110.e with resins specified by 8C108;
- Technical Note
A 'splitter duct' performs the initial separation of the air-mass flow between the bypass and core sections of the engine.
- 8E903 a. 4. Uncooled turbine blades, vanes or "tip shrouds", designed to operate at a 'gas path temperature' of 1,373 K (1,100°C) or more;
- 8E903 a. 5. Cooled turbine blades, vanes, "tip shrouds" other than those described in 8E903.a.1, designed to operate at a 'gas path temperature' of 1,693 K (1,420°C) or more;
- Technical Note
1. 'Gas path temperature' is the bulk average gas path total (stagnation) temperature at the leading edge plane of the turbine component when the engine is running in a 'steady state mode' of operation at the certificated or specified maximum continuous operating temperature.
- 8E903 a. 6. **Aerofoil**-to-disk blade combinations using solid state joining;
7. (Reserved)
8. 'Damage tolerant' gas turbine engine rotor components using powder metallurgy materials specified by 8C102.b; or
- Technical Note
'Damage tolerant' components are designed using methodology and substantiation to predict and limit crack growth.
9. (Reserved)
- N.B. For "FADEC systems", see 8E903.h.
10. (Reserved)
- N.B. For adjustable flow path geometry, see 8E903.i.
11. 'Fan blades' having all of the following:
- a. 20% or more of the total volume being one or more closed cavities containing vacuum or gas only; and
 - b. One or more closed cavities having a volume of 5 cm³ or larger;
- Technical Note
For the purposes of 8E903.a.11., a 'fan blade' is the aerofoil portion of the rotating stage or stages, which provide both compressor and bypass flow in a gas turbine engine.
- 8E903 b. "Technology" "required" for the "development" or "production" of any of the following:
1. Wind tunnel aero-models equipped with non-intrusive sensors capable of transmitting data from the sensors to the data acquisition system; or
 2. "Composite" propeller blades or prop fans, capable of absorbing more than 2,000 kW at flight speeds exceeding Mach 0.55;
- 8E903 c. "Technology" "required" for manufacturing cooling holes in gas turbine engine components incorporating any of the "technologies" specified by 8E903.a.1, 8E903.a.2 or 8E903.a.5, and having any of the following:
1. Having all of the following:
 - a. Minimum 'cross-sectional area' less than 0.45 mm²;
 - b. 'Hole shape ratio' greater than 4.52; and
 - c. 'Incidence angle' equal to or less than 25°; or
 2. Having all of the following:
 - a. Minimum 'cross-sectional area' less than 0.12 mm²;
 - b. 'Hole shape ratio' greater than 5.65; and
 - c. 'Incidence angle' more than 25°;
- Note 8E903.c does not apply to "technology" for manufacturing constant radius cylindrical holes that are straight through and enter and exit on the external surfaces of the component.

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Technical Notes

1. For the purposes of 8E903.c, the 'cross-sectional area' is the area of the hole in the plane perpendicular to the hole axis.
2. For the purposes of 8E903.c, 'hole shape ratio' is the nominal length of the axis of the hole divided by the square root of its minimum 'cross-sectional area'.
3. For the purposes of 8E903.c, 'incidence angle' is the acute angle measured between the plane tangential to the aerofoil surface and the hole axis at the point where the hole axis enters the aerofoil surface.
4. Methods for manufacturing holes in 8E903.c include "laser" beam machining, water jet machining, Electro-Chemical Machining (ECM) or Electrical Discharge Machining (EDM).

- 8E903 d. "Technology" "required" for the "development" or "production" of helicopter power transfer systems or tilt rotor or tilt wing "aircraft" power transfer systems;
- 8E903 e. "Technology" for the "development" or "production" of reciprocating diesel engine ground vehicle propulsion systems having all of the following:
1. 'Box volume' of 1.2 m³ or less;
 2. An overall power output of more than 750 kW based on 80/1269/EEC, ISO 2534; and
 3. Power density of more than 700 kW/m³ of 'box volume';

Technical Note

'Box volume' is the product of three perpendicular dimensions measured in the following way:

Length: The length of the crankshaft from front flange to flywheel face;

Width: The widest of any of the following:

- a. The outside dimension from valve cover to valve cover;
- b. The dimensions of the outside edges of the cylinder heads; or
- c. The diameter of the flywheel housing;

Height: The largest of any of the following:

- a. The dimension of the crankshaft centre-line to the top plane of the valve cover (or cylinder head) plus twice the stroke; or
- b. The diameter of the flywheel housing.

- 8E903 f. "Technology" "required" for the "production" of specially designed components for high output diesel engines, as follows:
1. "Technology" "required" for the "production" of engine systems having all of the following components employing ceramics materials specified by 8C107:
 - a. Cylinder liners;
 - b. Pistons;
 - c. Cylinder heads; and
 - d. One or more other components (including exhaust ports, turbochargers, valve guides, valve assemblies or insulated fuel injectors);
- 8E903 f. 2. "Technology" "required" for the "production" of turbocharger systems with single-stage compressors and having all of the following:
- a. Operating at pressure ratios of 4:1 or higher;
 - b. Mass flow in the range from 30 to 130 kg per minute; and
 - c. Variable flow area capability within the compressor or turbine sections;
- 8E903 f. 3. "Technology" "required" for the "production" of fuel injection systems with a specially designed multifuel (e.g. diesel or jet fuel) capability covering a viscosity range from diesel fuel (2.5 cSt at 310.8 K (37.8°C)) down to gasoline fuel (0.5 cSt at 310.8 K (37.8°C)) and having all of the following:
- a. Injection amount in excess of 230 mm³ per injection per cylinder; and
 - b. Electronic control features specially designed for switching governor characteristics automatically depending on fuel property to provide the same torque characteristics by using the appropriate sensors;
- 8E903 g. "Technology" "required" for the "development" or "production" of 'high output diesel engines' for solid, gas phase or liquid film (or combinations thereof) cylinder wall lubrication and permitting operation to temperatures exceeding 723 K (450°C), measured on the cylinder wall at the top limit of travel of the top ring of the piston;

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Technical Note

'High output diesel engines' are diesel engines with a specified brake mean effective pressure of 1.8 MPa or more at a speed of 2,300 r.p.m, provided the rated speed is 2,300 r.p.m. or more.

- 8E903 h. "Technology" for gas turbine engine "FADEC systems" as follows:
1. "Development" "technology" for deriving the functional requirements for the components necessary for the "FADEC system" to regulate engine thrust or shaft power (e.g, feedback sensor time constants and accuracies, fuel valve slew rate);
 2. "Development" or "production" "technology" for control and diagnostic components unique to the "FADEC system" and used to regulate engine thrust or shaft power;
 3. "Development" "technology" for the control law algorithms, including "source code", unique to the "FADEC system" and used to regulate engine thrust or shaft power;

Note 8E903.h does not apply to technical data related to engine-"aircraft" integration required by civil aviation authorities of India to be published for general airline use (e.g, installation manuals, operating instructions, instructions for continued airworthiness) or interface functions (e.g, input/output processing, airframe thrust or shaft power demand).

- 8E903 i. "Technology" for adjustable flow path systems designed to maintain engine stability for gas generator turbines, fan or power turbines, or propelling nozzles, as follows:
1. "Development" "technology" for deriving the functional requirements for the components that maintain engine stability;
 2. "Development" or "production" "technology" for components unique to the adjustable flow path system and that maintain engine stability;
 3. "Development" "technology" for the control law algorithms, including "source code", unique to the adjustable flow path system and that maintain engine stability.

Note 8E903.i. does not apply to "technology" for any of the following:

- a. Inlet guide vanes;
- b. Variable pitch fans or prop-fans;
- c. Variable compressor vanes;
- d. Compressor bleed valves; or
- e. Adjustable flow path geometry for reverse thrust.

- 8E903 j. "Technology" "required" for the "development" of wing-folding systems designed for fixed-wing "aircraft" powered by gas turbine engines.

N.B. For "technology" "required" for the "development" of wing-folding systems designed for fixed-wing "aircraft" specified in 6A010, see 6A022.

- 8E903 k. "Technology", not specified in 8E903.a., 8E903.h., or 8E903.i., "required" for the "development" of any of the following components or systems, specially designed for aero gas turbine engines to enable "aircraft" to cruise at Mach 1 or greater for more than 30 minutes:

1. Propulsion inlet systems;
2. Propulsion exhaust systems;
3. 'Reheat systems';
4. 'Active thermal management systems' to condition fluids used to lubricate or cool 'engine rotor supports';
5. Oil-free 'engine rotor supports'; or
6. Systems to remove heat from 'compression system' core gas path flow.

Technical Notes

For the purposes of 8E903.k.:

1. Propulsion inlet systems include core flow pre-coolers.
2. 'Reheat systems' provide additional thrust by combusting fuel in exhaust and/or bypass flow downstream of the last turbomachinery stage. 'Reheat systems' are also referred to as afterburners.
3. 'Active thermal management systems' employ methods other than passive oil-to-air cooling or oil-to-fuel cooling, such as vapour cycle systems.
4. 'Compression system' is any stage or combination of stages between the engine inlet face and the combustor that increases gas path pressure through mechanical work.
5. An 'engine rotor support' is the bearing supporting the main engine shaft that drives the compression system or turbine rotors.

N.B. 1 See 8E903.h. for engine control technology.

N.B. 2 See 8E903.i. for adjustable flow path systems technology.

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ANNEXURE – 1

PROFORMA FOR INFORMATION TO BE SUBMITTED BY SUPPLIER IN DOMESTIC TARIFF AREA (DTA) TO DEVELOPMENT COMMISSIONER, SEZ FOR SUPPLY OF SCOMET ITEMS TO SEZ UNITS
(Report to be submitted within 1(one) week of the supplies getting effected)

1. Name and address of the Development Commissioner, SEZ:
2. Name and address of the Supplier:
3. Importer – Exporter Code (IEC) of the Supplier:
4. Details of SCOMET items supplied:

Sl. No.	Description of items	SCOMET Category	ITC (HS) Code, if available	Quantity	SEZ Unit to which supplied	Date of Supply	FOB Value	
							Rs.	US\$

5. I hereby declare that I am authorized to verify and sign this declaration.

Date:

Signature:

Place:

Name:

Designation:

Telephone:

Fax:

Email address:

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ANNEXURE – 2

PROFORMA FOR INFORMATION TO BE SUBMITTED ANNUALLY (*) BY DEVELOPMENT COMMISSIONER, SEZ TO SCOMET CELL, DGFT (HQRS), DEPARTMENT OF COMMERCE, VANIYA BHAWAN, AKBAR ROAD, NEW DELHI – 110011

1. Name of the SEZ:
2. Details of SCOMET items supplied from DTA:

Sl. No.	Description of items	SCOMET Category	ITC (HS) Code, if available	Quantity	Name of supplier	Importer-Exporter Code (IEC)	SEZ Unit to which supplied	Date of Supply	FOB Value	
									Rs.	US\$

Official Seal/Stamp:

Date:

Signature of Development Commissioner /
authorized officer of the SEZ

Name:

Designation:

Telephone:

Fax:

Email address:

*Notes Report to be submitted by 15th May of every financial year for the supplies effected during the preceding financial year. Soft copy of the report should also be sent to scomet-dgft@nic.in